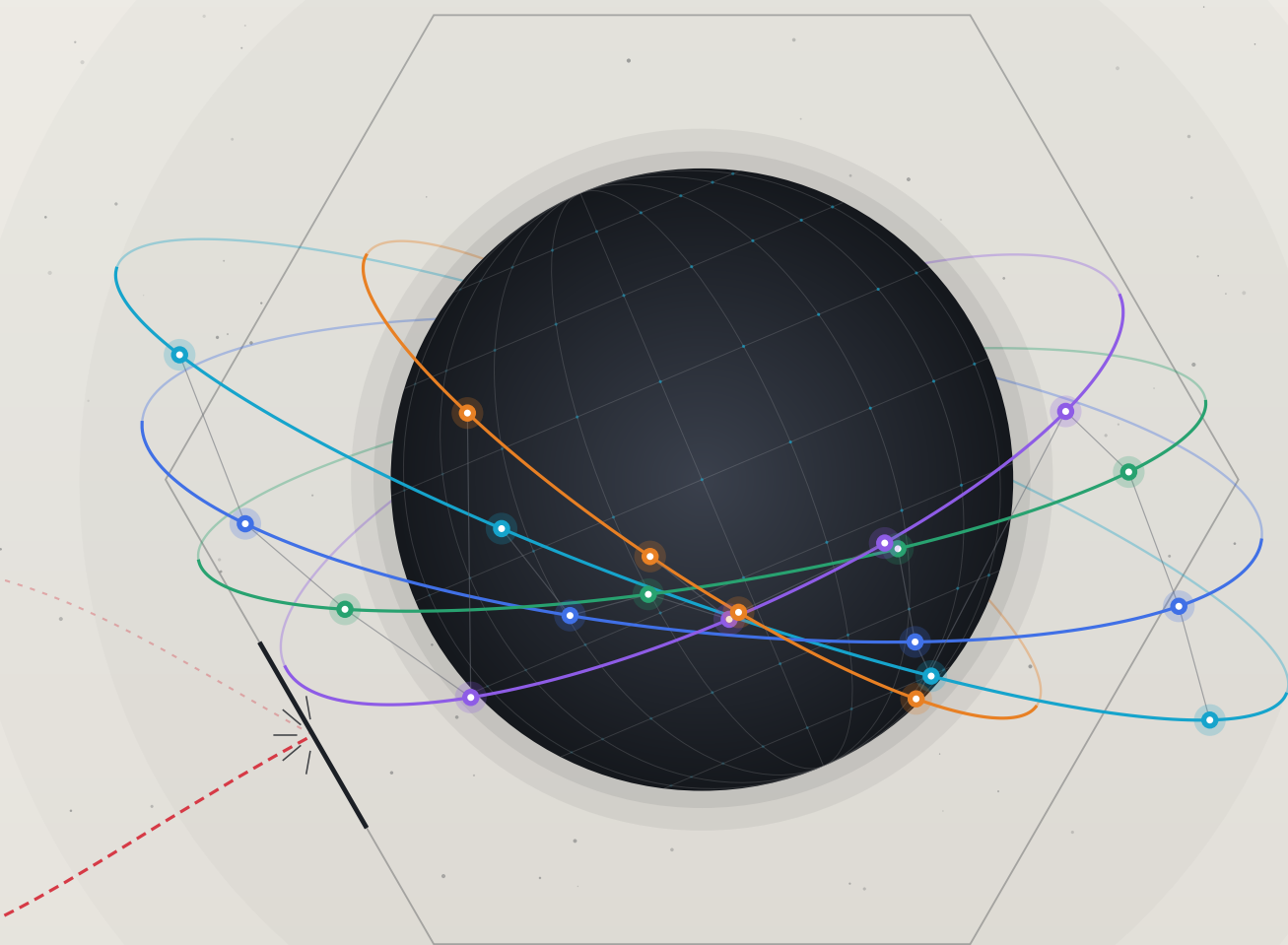


Introduction to

Space Security and Resilience



Frameworks, Models, and Quantitative Methods



Quanyan Zhu

New York University

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Sample for publisher review

This excerpt contains the front matter, the complete table of contents of the full manuscript, and Part I (Chapters 1–4). Page numbers and cross-references are those of the full manuscript (1080 pages), and the bibliography lists the works cited in this excerpt.

Contents

Preface	xiii
Notation	xvii
I Foundations	1
1 Why Space Security and Resilience?	3
1.1 Space as Critical Infrastructure	4
1.2 From Sanctuary to Contested Domain	7
1.3 What Makes Space Different	10
1.4 Security Is Multi-Layered: The Space Security Stack	12
1.5 Security and Resilience	17
1.6 The Security-Resilience Cube	19
1.7 Adversaries in Time: The Space Kill Chain	20
1.8 The HELIOS Running Example	22
1.9 A Systems-Science and Quantitative Perspective	26
1.10 How to Use This Book	27
1.11 Summary	29
Exercises	30
Notes and Further Reading	32
2 Space Systems Primer for Security Analysis	35
2.1 Orbits and Orbital Regimes	38
2.2 Visibility and Access Geometry	44
2.3 Constellation Design	47
2.4 Segment Architecture	49
2.5 RF Communications Fundamentals	52
2.6 The Space Communication and Network Stack	58
2.7 Ground and Onboard Computing	61
2.8 The Onboard Control Loop	65
2.9 The Space Mission Lifecycle	67
2.10 Summary	69
Exercises	69
Notes and Further Reading	71
3 Security and Resilience: Concepts and Metrics	73
3.1 Security, Safety, Sustainability, and Mission Assurance	75
3.2 Reliability, Robustness, Survivability, Fault Tolerance, Resilience, and Reconstitution	77

3.3	Security Properties by Layer	82
3.4	Space Systems as Cyber-Physical Systems	84
3.5	Resilience Goals, Phases, and Attributes	86
3.6	Defense in Depth Across Layers and Zero Trust	88
3.7	The Resilience Curve and Performance Functions	90
3.8	Stochastic Resilience Models	94
3.9	Network Resilience Metrics	98
3.10	Risk-Averse and Worst-Case Resilience	101
3.11	Summary	104
	Exercises	105
	Notes and Further Reading	108
4	Threats, Risk, and the Space Kill Chain	111
4.1	A Layered Threat Taxonomy	113
4.2	Actors and Capabilities	118
4.3	The Architecture Threat Map	120
4.4	The Space Kill Chain	122
4.5	The Kill Chain as a Stochastic Process	127
4.6	Adversary Models	134
4.7	Strategic Adversaries and Game Models: A First Look	137
4.8	Attack Trees and Attack Graphs	140
4.9	Probabilistic Risk Quantification	143
4.10	STPA-Sec: A Control-Theoretic View of Security	147
4.11	Risk Frameworks	149
4.12	Two Case Studies	150
4.13	Summary	152
	Exercises	153
	Notes and Further Reading	155
II	Communication and Network Security	157
5	Physical-Layer (Signal) Security	159
	Notation and Link-Budget Background	161
5.1	Channel and Interference Models	162
5.2	Jamming	164
5.3	Anti-Jam Techniques	169
5.4	Jamming Games	174
5.5	Eavesdropping and Physical-Layer Secrecy	179
5.6	Covert (LPI/LPD) Communication	182
5.7	Spoofing Detection and RF Fingerprinting	183
5.8	Interference Geolocation	187
5.9	Resilient Communications under Interference	189
5.10	Summary	193
	Exercises	194
	Notes and Further Reading	196

6	Link Security and Space Cryptography	199
	Notation and Cryptographic Background	201
6.1	Link Security Requirements	202
6.2	CCSDS SDLS and Link-Layer Protocols	205
6.3	Key Management for Space Links	212
6.4	Cryptography under SWaP and Radiation Constraints	217
6.5	Crypto Agility and Post-Quantum Migration	221
6.6	Satellite Quantum Key Distribution	226
6.7	Formal Verification of Security Protocols	230
6.8	Summary	233
	Exercises	234
	Notes and Further Reading	236
7	Space Network Security	239
7.1	Constellations, ISLs, and Onboard Routing	242
7.2	Space-Terrestrial Networks and NTN	246
7.3	Network Threats	248
7.4	Time-Varying Graph Models and Percolation	251
7.5	Secure Routing and Moving-Target Defense	253
7.6	Distributed Trust and Byzantine Consensus	257
7.7	Robust Traffic Engineering and Interdiction	260
7.8	Network Resilience: Graceful Degradation and Self-Healing	263
7.9	Summary	267
	Exercises	268
	Notes and Further Reading	270
III	Physical Security	273
8	Physical Threats and Protection of Space Assets	275
8.1	A Taxonomy of Physical Threats	278
8.2	Kinetic Threats: Reach and Closing Speed	280
8.3	Co-orbital Approach and Debris Consequences	283
8.4	Co-orbital Approach as Pursuit and Evasion	288
8.5	Directed-Energy Threats	295
8.6	Nuclear Effects in Space	298
8.7	Physical Threats to Ground and Launch Infrastructure	300
8.8	Vulnerability and Damage Modeling	302
8.9	Physical Protection and Hardening	305
8.10	Resilience to Physical Attack: Degradation, Spares, and Reconstitution	309
8.11	Summary	312
	Exercises	313
	Notes and Further Reading	315
9	Space Environment, Debris, and Sustainability	319
9.1	Space Weather Effects	323
9.2	Conjunction Assessment and Collision Probability	329

9.3	Debris Population Dynamics	335
9.4	Space Traffic Management	340
9.5	Orbits as a Commons	342
9.6	Coupling Security and Sustainability	345
9.7	Summary	347
	Exercises	348
	Notes and Further Reading	350
10	Space Domain Awareness, Proximity Operations, and Orbital Games	353
10.1	From Space Situational Awareness to Space Domain Awareness	355
10.2	Tracking and Catalog Maintenance	359
10.3	Maneuver Detection and Intent Inference	363
10.4	Relative Motion and Proximity Operations	367
10.5	Orbital Pursuit-Evasion Games	371
10.6	Beyond Two-Player Pursuit-Evasion	375
10.7	Sensor Tasking, Concealment, and Deception	385
10.8	Attribution under Uncertainty	388
10.9	Case Studies: Publicly Reported Proximity Operations	391
10.10	Summary	393
	Exercises	394
	Notes and Further Reading	396
IV	Cyber and Cyber-Physical Security	399
11	Cyber Security of the Ground and User Segments	401
11.1	Ground Architectures and IT/OT Convergence	404
11.2	User Terminal Security	408
11.3	Defensive Architectures: Zero Trust and Segmentation	412
11.4	Detection Analytics and the Space-Ops SOC	415
11.5	Optimal Hardening and Monitor Placement	419
11.6	Intrusion Response as a Stochastic Game	423
11.7	Cyber Resilience and Recovery of Ground Operations	427
11.8	Case Studies	430
11.9	Summary	432
	Exercises	433
	Notes and Further Reading	435
12	Onboard Computing and Software Security	437
12.1	Onboard Computing and Flight Software	439
12.2	Roots of Trust, Secure Boot, and Secure Updates	444
12.3	Formal Verification of Flight Software	448
12.4	Onboard Intrusion Detection	452
12.5	FDIR Meets IDS: Faults versus Attacks	455
12.6	Security Mode Management and Fallback	460
12.7	Reliability of Onboard Security Functions	465
12.8	Onboard Cyber Resilience	470

12.9 Summary	473
Exercises	475
Notes and Further Reading	477
13 Cross-Layer Cyber-Physical Security: Models and Frameworks	479
13.1 Seams Between Layers: Cross-Layer Attack Pathways	481
13.2 Modeling Cyber-Physical Space Systems	485
13.3 Multi-Layer and Interdependent Networks	489
13.4 Cross-Layer Attack Graphs and Kill Chains	493
13.5 Games-in-Games: Multi-Layer Strategic Models	497
13.6 Cross-Layer Detection and Fusion	502
13.7 Cross-Layer Co-Design: Security versus Performance	505
13.8 Cross-Layer Resilience: Metrics, Containment, and Coordinated Recovery	510
13.9 Summary	514
Exercises	516
Notes and Further Reading	518
14 Resilient Estimation, Navigation, and Timing	521
14.1 Estimation for Orbit Determination and PNT	523
14.2 Measurement Attack Models	527
14.3 Residual-Based Detection and Stealthy Attacks	531
14.4 Secure State Estimation	536
14.5 Robust and Resilient Filtering	539
14.6 Resilient PNT and Timing Security	543
14.7 Summary	547
Exercises	548
Notes and Further Reading	550
15 Resilient Guidance, Control, and Autonomy	553
15.1 Spacecraft Dynamics and Control	555
15.2 Attacks on Control Loops	559
15.3 DoS-Resilient and Event-Triggered Control	563
15.4 Robust and Resilient Control	566
15.5 Reachability and Safety Guarantees	570
15.6 Evasive Maneuvering as a Safety Filter	573
15.7 Resilient Multi-Agent Formations	582
15.8 Autonomy for Resilience	592
15.9 Summary	597
Exercises	598
Notes and Further Reading	600
V Architecture, Lifecycle, and Strategy	603
16 Supply Chain, Insider, and Lifecycle Security	605
16.1 Security by Design Across the Lifecycle	607

16.2	Hardware, Firmware, and Software Supply-Chain Threats	611
16.3	Insider Threat Models	616
16.4	Correlated and Common-Mode Failures	620
16.5	Assurance, Verification, and Certification	626
16.6	Incentives and Contracts	630
16.7	Case Studies	635
16.8	Summary	637
	Exercises	638
	Notes and Further Reading	640
17	Resilient Architecture Design and Reconstitution	643
17.1	Architectural Levers Across Layers	645
17.2	Tradespace Exploration	650
17.3	Designing Against Adversaries: Bilevel Models	656
17.4	Reconstitution and Responsive Space	661
17.5	Economics of Resilience	668
17.6	Summary	672
	Exercises	674
	Notes and Further Reading	676
18	Deterrence, Escalation, Law, and Governance	679
18.1	International Space Law	681
18.2	Norms and Responsible Behaviors	687
18.3	National and Alliance Strategies	690
18.4	Thresholds and Attribution Across Layers	693
18.5	Deterrence by Denial and Punishment; Resilience as Deterrence	698
18.6	Formal Models of Escalation and Signaling	701
18.7	Summary	710
	Exercises	712
	Notes and Further Reading	713
VI	Integration and Frontiers	715
19	Integrated Case Studies, Wargames, and Digital Twins	717
19.1	A Method for Case Analysis	719
19.2	Comparative Case Studies	722
19.3	Wargaming and Tabletop Exercises	729
19.4	Digital Twins, Cyber Ranges, and Testbeds	734
19.5	Capstone: A Multi-Layer HELIOS Campaign	741
19.6	Summary	748
	Exercises	750
	Notes and Further Reading	752
20	Frontiers and Open Problems	755
20.1	Artificial Intelligence and Machine Learning	757
20.2	Quantum Technologies	763

20.3	Cislunar Security	767
20.4	On-Orbit Servicing and Dual Use	772
20.5	Commercial Integration and Sovereignty	775
20.6	A Research Agenda	777
20.7	Summary	782
	Exercises	783
	Notes and Further Reading	785
A	Probability and Stochastic Processes	789
A.1	Probability Review	790
A.2	Tail Risk and Dependence	794
A.3	Discrete-Time Markov Chains	798
A.4	Continuous-Time Markov Chains	802
A.5	Poisson, Renewal, and Semi-Markov Processes	805
A.6	Hypothesis Testing and Sequential Detection	807
A.7	Further Reading	810
B	Optimization and Game Theory	811
B.1	Convex Optimization and Duality	812
B.2	Linear, Integer, and Combinatorial Optimization	816
B.3	Bilevel Optimization	822
B.4	Zero-Sum Games and the Minimax Theorem	824
B.5	Stackelberg and Security Games	828
B.6	Bayesian and Signaling Games	830
B.7	Repeated Games	832
B.8	Stochastic, Dynamic, and Potential Games	835
B.9	Computing Equilibria	840
B.10	Further Reading	841
C	Estimation and Control	843
C.1	Linear State-Space Systems	843
C.2	State Estimation	847
C.3	Stability	854
C.4	Optimal and Predictive Control	861
C.5	Safety and Reachability	866
	Further Reading	871
D	Orbital Mechanics and RF Communications	873
D.1	The Two-Body Problem	873
D.2	Perturbations, Frames, and Time	877
D.3	Relative Motion and Maneuvers	880
D.4	Access Geometry and Link Budgets	882
D.5	Modulation, Coding, and Capacity	885
D.6	Detection and Information Theory	887
	Further Reading	891

E	Standards and Frameworks	893
E.1	How the Pieces Fit	893
E.2	Cyber-Risk and Cyber-Resilience Frameworks	894
E.3	Space-Specific Cybersecurity Guidance and Knowledge Bases	898
E.4	CCSDS Security Standards and Cryptography	901
E.5	Software Assurance, Supply Chain, and Space Sustainability	906
E.6	Chapter Crosswalk and Status Caveats	909
F	Tools and Data	913
F.1	Orbital and Space-Environment Data	913
F.2	Security Knowledge Bases	915
F.3	Simulation and Analysis Tools	918
F.4	Companion Code	921
F.5	Reproducibility Practices	923
G	Glossary	925

Preface

Space has become one of the most consequential domains of modern life. Positioning, navigation, and timing signals synchronize financial networks and power grids; satellite communications connect ships, aircraft, remote communities, and military forces; Earth observation informs agriculture, disaster response, and climate science. Over the past decade the pace of change has accelerated dramatically. Reusable launch vehicles have lowered the cost of access to orbit, and proliferated low-Earth-orbit constellations such as SpaceX's Starlink now place thousands of satellites in orbit and deliver broadband service directly to consumer terminals. Commercial ground-station services, software-defined payloads, optical inter-satellite links, and cloud-hosted mission operations have followed. These innovations have opened space to new operators and new applications, and they have also opened it to new adversaries.

Securing space systems is a distinctive problem. A space system is a cyber-physical system in the fullest sense: its software commands physical assets that obey orbital mechanics, its links are radio or optical signals that a third party within view can observe or interfere with, and its failures propagate across a ground segment, a link segment, a space segment, and a user segment that are often owned by different organizations. Satellites are expensive, long-lived, and largely unreachable for repair, so design decisions made years before launch constrain the defenses available for a decade or more. Propellant, power, and bandwidth are scarce, which limits how much security can be added after the fact. Attribution is difficult, intent is ambiguous, and many capabilities are inherently dual-use. The same maneuver can inspect, service, or threaten a satellite. The orbital environment is a shared commons in which an attack on one system can create debris that endangers all. Security in this setting therefore cannot be treated as a single discipline. It spans the signal, the network, the physical and orbital, the cyber, and the mission and strategic layers, and the most consequential attacks are those that cross from one layer to another. Because disruptions cannot be prevented in every case, resilience (the ability to withstand, recover, and adapt) is as important as protection.

My own journey into this subject was shaped by two excellent doctoral students at New York University, Yuhan Zhao and Linan Huang. Yuhan's doctoral dissertation addressed the planning and coordination of multi-agent systems through game theory, optimization, and machine learning, and some of his earliest work in the program was motivated by the resilience of satellite constellations. He approached constellation coverage under space threats from a systems and control perspective, treating satellites as agents that plan and reconfigure autonomously when parts of the constellation are compromised. That viewpoint was rare at the time, and it reappears in several chapters of this book. Linan's dissertation

developed AI-powered, system-scientific defenses for high-confidence cyber-physical systems, combining game theory, learning, and human factors. After completing his doctorate he turned to the security and resilience of satellite networks at Tsinghua University. Working with both of them convinced me that a unified treatment of this topic was needed, one that could educate a much wider audience than the handful of specialists who work at the intersection of space and security.

Over the past several years I have also attended many workshops and sessions devoted to space security, at venues such as DEF CON, the American Control Conference, the IEEE Conference on Decision and Control, the ACM Conference on Computer and Communications Security, and the USENIX Security Symposium, as well as meetings organized by funding agencies. A recurring impression from these gatherings is how hard the field is to grasp as a whole. The literature is vast and scattered across communications, aerospace engineering, control engineering, automation, robotics, cybersecurity, networking, economics, and policy. Each community uses its own vocabulary, models, and assumptions, and few sources connect them. For students and early-career researchers this fragmentation makes it difficult both to learn the fundamentals and to identify research problems worth pursuing. This book is an attempt to address that gap, and I believe the timing is right: the systems are being built now, the threats are no longer hypothetical, and the next generation of engineers and researchers will need a common foundation.

The book is organized around the layers of security rather than around a single discipline. Part I lays the foundations: why space security and resilience matter, a primer on space systems, precise definitions and metrics for security and resilience, and models of threats, risk, and adversaries. Part II treats communication and network security, from the physical layer through links and constellation networks. Part III turns to physical security, including kinetic and non-kinetic threats, the space environment, and space domain awareness with orbital games. Part IV addresses cyber and cyber-physical security, covering ground and onboard systems, the cross-layer framework that ties the layers together, secure estimation, and resilient control. Part V considers architecture, lifecycle, supply chain, governance, and strategy, and Part VI integrates the material through case studies and a discussion of research frontiers. Six appendices collect the background in probability, games, control, orbital mechanics, standards, and software tools on which the chapters draw.

Throughout, the emphasis is on frameworks and quantitative methods. Concepts are made precise through definitions, results are derived rather than asserted, and each chapter works through numerical examples on HELIOS, a notional proliferated low-Earth-orbit constellation used as a running example. Every chapter opens with learning objectives and a layer map that locates its material in the book's framework, and closes with a summary, exercises at conceptual, analytical, and project levels, and notes on further reading. The chapters are written to be as self-contained as possible, so that instructors can assemble courses from different subsets. [Chapter 1](#) suggests several reading paths, for example a course emphasizing communications and networks, one emphasizing control and cyber-physical systems, and one emphasizing policy and strategy. The discussion

throughout remains at the level of the open literature and is oriented toward defense.

The intended readers include graduate and advanced undergraduate students in electrical, aerospace, and computer engineering and in computer science, researchers entering the field from each of its contributing disciplines, and practitioners in industry and government who want a structured, quantitative view of the problem. I hope the book will serve both as a classroom text and as a reference for researchers, and that it will help build the community that the security and resilience of space systems increasingly require.

Quanyan Zhu
New York University
September 22, 2026

Notation

This list collects the symbols that are used in more than one chapter with a single meaning. Each chapter lists only the additional symbols that are specific to it, together with each local meaning that a symbol below takes there. A symbol is also defined where it first appears in the text, so a chapter can be read on its own.

Typefaces. Scalars are italic (x , P_t). Vectors are bold lower case ($\mathbf{x}$, $\boldsymbol{\omega}$), matrices bold upper case ($\mathbf{A}$, $\boldsymbol{\Sigma}$), and sets calligraphic ($\mathcal{A}$, $\mathcal{G}$). Number sets are blackboard bold ($\mathbb{R}$). Subscripts that are labels are upright (L_{fs} , P_{FA}); subscripts that are indices are italic (x_k , p_{ij}). Cryptographic algorithms are set in sans serif (Enc, MAC).

Operators and decorations. $\mathbf{M}^\top$ is a transpose and $\|\cdot\|$ the Euclidean norm unless a subscript says otherwise. A hat marks an estimate ($\hat{x}_{k|k-1}$), a bar a bound or mean ($\bar{w}$), a tilde an attacked or corrupted signal ($\tilde{y}_k$), and a star an optimal, equilibrium, or critical value (x^* , δ^*). Distributions over the states of a Markov chain are row vectors.

Time and indices. Continuous time is t and the discrete time index is k , with sampling period Δt ; T is the evaluation horizon of a resilience metric, except in orbital mechanics, where it is the orbital period. The indices i and j label components, satellites, players, or states, and ℓ labels a layer of the Space Security Stack.

Units and decibels. Units are SI, and angles are in degrees unless stated otherwise. For a power ratio x , $[x]_{\text{dB}} = 10 \log_{10} x$. Absolute levels are written dBW (power), dBi (antenna gain), dB/K (G/T), and dB-Hz (C/N_0). In link-budget equations written in decibels, Boltzmann's constant enters as $k_B = -228.6 \text{ dBW}/(\text{K Hz})$.

Context tags. A few symbols carry a second meaning by long convention in a particular field. Those entries name the context in italics, and the local meaning does not appear in the same section as the primary one.

Mathematics and probability

Symbol	Meaning	Unit
$\mathbb{P}(\cdot), \mathbb{E}[\cdot]$	probability; expectation	
Var, Cov	variance; covariance	
$\mathbb{1}\{E\}$	indicator of the event E	
$\mathbf{1}, \mathbf{I}, \mathbf{0}$	all-ones vector; identity matrix; zero vector or matrix	
$\mathbb{R}, \mathbb{R}^n$	real numbers; real n -vectors (n a dimension)	
$(z)^+$	positive part $\max\{z, 0\}$	
$\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$	
μ, σ	mean; standard deviation (<i>statistics</i>)	as variable
$\boldsymbol{\Sigma}$	covariance matrix	as variable
$\Phi(\cdot), Q(x)$	standard normal distribution function; Gaussian tail probability $1 - \Phi(x)$	
$\chi_m^2, \chi_m^2(\lambda_{\text{nc}})$	central and noncentral chi-square laws with m degrees of freedom	
$\rho(\mathbf{M})$	spectral radius of $\mathbf{M}$	
$\lambda_i(\mathbf{M}), \lambda_{\min}, \lambda_{\max}$	eigenvalues of $\mathbf{M}$; smallest and largest eigenvalue	
$\Delta t, \Delta v, \dots$	increment of the quantity that follows	as variable
λ	rate of a Poisson or failure process; arrival rate	h^{-1}
μ	service or repair rate (<i>queueing and Markov models</i>)	h^{-1}
c, ρ, W_q	number of servers; utilization $\lambda/(c\mu)$; mean waiting time in queue (<i>queueing</i>)	none, none, h
$\mathbf{P} = [p_{ij}]$	transition matrix of a discrete-time Markov chain; written $\boldsymbol{\Pi} = [\pi_{ij}]$ where $\mathbf{P}$ denotes a covariance	
$\mathbf{p}_k, \mathbf{p}(t)$	state distribution of a Markov chain (row vector)	
$\boldsymbol{\pi}^*$	stationary distribution	
$\mathbf{G}$	generator matrix of a continuous-time Markov chain	h^{-1}
$\mathbf{Q}, \mathbf{R}$	transient and absorbing blocks of an absorbing chain (<i>absorbing chains</i>)	
$\mathbf{N}, \mathbf{B}$	fundamental matrix $(\mathbf{I} - \mathbf{Q})^{-1}$; absorption probabilities $\mathbf{NR}$	
$R(t), \text{MTTF}$	reliability function (written with its argument throughout); mean time to failure	none, h

Resilience and risk

Symbol	Meaning	Unit
$\mathcal{A}$	architecture: the design decisions of a space system	
$\mathcal{W}, w \in \mathcal{W}$	admissible set of attacks and disturbances; one attack	
π	defender policy (estimation, control, and response)	
$Q(t), Q_0$	mission performance at time t ; its nominal value	mission units
T	evaluation horizon	h
$R(\mathcal{A}, \pi; \mathcal{W})$	resilience functional: worst-case time-averaged normalized performance $\inf_{w \in \mathcal{W}} \frac{1}{T} \int_0^T Q/Q_0 \, dt$	
R_T, R	resilience of a single scenario over $[0, T]$, $R_T = 1 - L_T$	
L_T	normalized performance loss over $[0, T]$	
A_L	loss area $\int_0^T (1 - Q/Q_0)^+ \, dt$	h
Δ	depth of the performance drop	
TTR	time to recover	h
ℓ	layer of the Space Security Stack	
$\sigma_\ell(t), \sigma(t)$	health index of layer ℓ ; vector of layer health indices	
ϕ	composition function from layer health indices to mission performance	
L	loss, as a random variable	\$, satellites
α	confidence level of a risk measure	
$\text{VaR}_\alpha, \text{CVaR}_\alpha$	value-at-risk; conditional value-at-risk at level α	as L
R_α^{CVaR}	CVaR resilience $1 - \text{CVaR}_\alpha(L_T)$	
β	common-cause (beta) factor (<i>redundancy and site diversity</i>)	
CER	cost-exchange ratio	

Orbital mechanics and geometry

Symbol	Meaning	Unit
μ	Earth gravitational parameter	km^3/s^2
$R_{\oplus}$	Earth equatorial radius	km
$\omega_{\oplus}$	Earth rotation rate	rad s^{-1}
J_2	second zonal harmonic of the geopotential	
$\mathbf{r}, \mathbf{v}$	inertial position and velocity	km, km s^{-1}
r, h	orbital radius; altitude $r - R_{\oplus}$	km
a, e, i	semi-major axis; eccentricity; inclination	km, none, $^{\circ}$
Ω, ω, ν	right ascension of the ascending node (RAAN); argument of perigee; true anomaly	$^{\circ}$
u	argument of latitude $\omega + \nu$	$^{\circ}$
$\dot{\Omega}$	nodal drift rate	$^{\circ}/\text{day}$
n	mean motion $\sqrt{\mu/a^3}$ (<i>orbital motion</i>)	rad s^{-1}
T	orbital period (<i>orbital motion</i>)	min
$v, \Delta v$	orbital speed; velocity change	$\text{km s}^{-1}, \text{m s}^{-1}$
a_t	thrust acceleration	m s^{-2}
B_c, C_D	ballistic coefficient $m/(C_D A)$; drag coefficient	kg m^{-2} , none
$\varepsilon, \varepsilon_{\min}$	elevation angle; minimum elevation	$^{\circ}$
η	nadir angle at the satellite	$^{\circ}$
$\lambda, \lambda_{\max}$	Earth central angle; footprint half-angle (<i>access geometry</i>)	$^{\circ}$
d	slant range	km
x, y, z	radial, in-track, and cross-track coordinates in the Hill frame	m, km
v_{rel}	relative velocity at closest approach	km s^{-1}
r_{KOS}	keep-out sphere radius	km
$i:t/p/f$	Walker-delta pattern: inclination, total satellites, planes, phasing	

Radio-frequency links and communications

Symbol	Meaning	Unit
f, λ_w	carrier frequency; wavelength c_0/f	GHz, m
c_0	speed of light in vacuum	m s^{-1}
P_t	transmit power	W, dBW
G, G_t, G_r	antenna gain; transmit and receive antenna gains	dBi
D, η_a	aperture diameter; aperture efficiency	m, none
$\theta_{3\text{dB}}$	half-power beamwidth	$^\circ$
EIRP	effective isotropic radiated power $P_t G_t$	dBW
L_{fs}, L_a	free-space path loss $(4\pi d/\lambda_w)^2$; additional atmospheric and pointing losses	dB
k_B	Boltzmann constant, $1.380\,649 \times 10^{-23} \text{ J K}^{-1}$	dBW/(K Hz)
T_{sys}	system noise temperature	K
G/T	receiver figure of merit	dB/K
C, N, N_0	received carrier power; noise power; noise power spectral density $k_B T_{\text{sys}}$	W, W, W Hz^{-1}
$C/N_0, C/N$	carrier-to-noise density ratio; carrier-to-noise ratio in the bandwidth B	dB-Hz, dB
E_b, E_s	energy per bit; energy per symbol	J
$E_b/N_0, E_s/N_0$	energy per bit and per symbol to noise density	dB
B	occupied (receiver) bandwidth	MHz
R_b, R_s	bit rate; symbol rate	bit/s, sym/s
P_b	bit error probability	
$J, J_0, J/S$	jammer power; jamming power spectral density; jamming-to-signal ratio	W, W Hz^{-1} , dB
SNR	signal-to-noise ratio	none, dB
C_{Sh}, C_s	Shannon capacity; secrecy capacity	bit/use

Detection and information theory

Symbol	Meaning	Unit
H_0, H_1	null (nominal) and alternative (attack or fault) hypotheses	
P_D, P_{FA}	probability of detection; probability of false alarm	
Λ, τ	likelihood ratio; detection threshold	
S_n, h	cumulative-sum (CUSUM) statistic; CUSUM threshold	
ARL_0	average run length to false alarm	samples
D	Kullback-Leibler divergence per sample	nat
g_k	normalized innovation squared (chi-square test statistic)	
λ_{nc}	noncentrality parameter of a chi-square law	
$H(\cdot), I(\cdot; \cdot)$	entropy; mutual information	bit

Estimation and control

Symbol	Meaning	Unit
x_k, u_k, y_k	state; control input; measurement or output at step k	
w_k, v_k	process noise or disturbance; measurement noise	
A, B, C	system (state-transition), input, and output matrices	
W, V	covariance matrices of w_k and v_k	
$h(\cdot)$	nonlinear measurement function	
$\hat{x}_{k j}$	estimate of x_k given the measurements $y_{1:j}$	
P	estimation-error covariance; solution of a Riccati or Lyapunov equation	
K	Kalman gain or state-feedback gain	
r_k, S_k	innovation (residual); innovation covariance	
a_k	attack signal injected into a model	
a_k^u, a_k^y	actuator and sensor attack signals	
γ_k^u, γ_k^y	actuator and sensor packet-delivery indicators (0 during denial of service)	
q	maximum number of attacked sensors	
Q, R	state and input weights of a quadratic cost (<i>optimal control</i>)	
γ, γ^*	H^∞ disturbance-attenuation level; its smallest achievable value	
$V(x)$	Lyapunov function (<i>stability</i>)	
$h(x)$	control barrier function, with safe set $\{x : h(x) \geq 0\}$ (<i>safety</i>)	
$\mathcal{R}_t$	forward reachable set at time t	
$\theta_k, \Pi = [\pi_{ij}]$	mode of a Markov jump system or hidden Markov model; mode transition matrix	
ω_c, PM	crossover frequency; phase margin	rad s ⁻¹ , °
Δt	sampling period	s
O	observability matrix	

Optimization and games

Symbol	Meaning	Unit
A	defender's loss matrix of a zero-sum matrix game (<i>matrix games</i>)	
x, y	mixed strategies of the defender (row player) and the attacker (column player)	
Δ_m	probability simplex in $\mathbb{R}^m$	
$\text{val}(A)$	value of the matrix game A	
$s, \mathcal{S}$	state; state set of a stochastic game	
$\ell(s, a, b)$	stage cost paid by the defender for actions a (defender) and b (attacker)	
$P(s' \mid s, a, b)$	state-transition probabilities of a stochastic game	
$V(s)$	value of state s	
δ	discount factor of a stochastic game, repeated game, or Markov reward model	
U_d, U_a	defender and attacker utilities	
B	budget of the defender or the attacker	\$, elements

Networks and graphs

Symbol	Meaning	Unit
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	graph with node set $\mathcal{V}$ and edge set $\mathcal{E}$; also an attack graph	
$L, \lambda_2(L)$	graph Laplacian; algebraic connectivity	
$S(f)$	fraction of nodes in the giant component after a fraction f of nodes is removed	

Cyber operations and cryptography

Symbol	Meaning	Unit
$N, I(t), I_0$	size of a terminal fleet; compromised terminals at time t ; initial compromise	terminals
β	spread rate of a logistic compromise (<i>compromise spread</i>)	h^{-1}
f_{OTA}	fraction of compromised terminals recoverable over the air	
x, y, z	Mosca's confidentiality lifetime, migration time, and time until a cryptographically relevant quantum computer (<i>post-quantum migration</i>)	yr
Enc, Dec	authenticated encryption and decryption	
MAC, Sign, Vrfy	message authentication code; signature generation and verification	

Part I

Foundations

Why Space Security and Resilience?

Learning Objectives

- ▶ Explain why space-based positioning, navigation, and timing (PNT), communications, and Earth observation function as critical infrastructure, and describe how their disruption cascades into other sectors.
- ▶ Trace the historical arc by which space went from a perceived sanctuary to a congested, contested, and competitive domain, and identify the features that make security in space different from security on the ground.
- ▶ Recognize that space security spans several distinct layers (signal, link and network, physical and orbital, cyber, and mission) and that many consequential attacks cross between them.
- ▶ Distinguish security from resilience, read a resilience curve, and state the resilience functional R used throughout the book.
- ▶ Use the book's organizing frameworks (the Space Security Stack, the Security-Resilience Cube, the space kill chain, and the HELIOS running example) and choose a reading path through the book.

This chapter argues that space security must be studied as a layered, quantitative, and resilience-oriented discipline, and it introduces the frameworks with which the rest of the book organizes that discipline. We begin with the dependence of modern economies on space services and with the history that turned orbit into a contested domain. We then identify what makes space different from terrestrial security, introduce the Space Security Stack and cross-layer attacks, distinguish security from resilience, and locate problems in a three-axis cube and along an eight-stage kill chain. The chapter closes with the HELIOS running example, the systems-science perspective that unifies the book, and guidance on how to read it.

The layer map in [Fig. 1.1](#) is the first instance of a device used at the start of every chapter. Its rows are the five layers of the Space Security Stack, defined in [Section 1.4](#), and its columns are the four segments of a space system: the ground segment, the links that connect it to spacecraft and users, the space segment, and the user segment.

	Ground	Link	Space	User
Mission/strategic				
Cyber				
Physical/orbital				
Link & network				
Signal/RF				

Figure 1.1. Layer map for [Chapter 1](#). The chapter’s emphasis is the mission/strategic layer across all four segments (dark cells): why disruptions matter to the people and economies that depend on space services. Every other layer and segment is touched at the level of motivation and vocabulary (light cells).

HELIOS Running Example

HELIOS is a notional constellation created for this book and is not modeled on a real operator; [Section 1.8](#) introduces it.

At 04:10 UTC the HELIOS network operations center receives trouble reports from aviation customers flying near a conflict zone. Business jets using the HELIOS L-band timing augmentation see position jumps of several kilometers, because their receivers blend it with GNSS signals that are being jammed and apparently spoofed.

At 05:25 UTC the security operations center flags a login to the portal of the ground-station-as-a-service (GSaaS) provider that leases HELIOS six of its eight telemetry, tracking, and command (TT&C) sites. The credentials are genuine, the source network is unfamiliar, and the session requests twelve hours of contact schedules. No command has been sent.

At 06:40 UTC flight dynamics receives a conjunction warning: a Shell A satellite at 540 km has a collision probability above the maneuver threshold for an approach 30 hours ahead. The next two suitable passes run through the same provider’s antennas.

Each event is manageable alone. Together they pose a cross-layer decision: upload the maneuver through a possibly compromised ground path, wait for an owned site and accept more collision risk, or relay the command over inter-satellite links.

1.1 Space as Critical Infrastructure

Few people have seen a satellite, yet most people use one many times a day. A card payment, a mobile phone call, a weather forecast, a container ship’s arrival time, a farmer’s fertilizer application map, and a power grid’s synchronized measurements all draw, directly or indirectly, on services delivered from orbit.

The dependence is deep precisely because it is invisible: space services have become utilities, and utilities are noticed mainly when they fail.

1.1.1 Three families of space services

The civil and commercial services delivered from space fall naturally into three families. The first is *positioning, navigation, and timing (PNT)*. Global navigation satellite systems (GNSS), namely GPS, Galileo, GLONASS, and BeiDou, broadcast signals from which a receiver computes its position, velocity, and time. The timing function is often more important economically than the positioning function. A GNSS receiver with a clear view of the sky recovers Coordinated Universal Time (UTC) to within tens of nanoseconds, cheaply and over nearly the whole globe, and this has made GNSS the default source of synchronization for cellular networks, data centers, financial trading systems, and electric-grid phasor measurement units.

The second family is *satellite communications (SATCOM)*. Geostationary (GEO) satellites have long provided broadcast television, maritime and aeronautical connectivity, and backhaul for remote regions. Proliferated low-Earth-orbit (LEO) broadband constellations now add low-latency internet access to ships, aircraft, rural households, emergency responders, and military users. SATCOM is frequently the communication layer of last resort when terrestrial networks are damaged, which is exactly when it is most needed.

The third family is *Earth observation and environmental monitoring*. Weather satellites feed numerical weather prediction, and optical and radar imaging satellites support agriculture, insurance, disaster response, climate science, and intelligence. Space-based sensing also underpins space domain awareness itself.

1.1.2 Sectoral dependencies

The dependence of terrestrial sectors on these services is well documented. Among the most careful quantitative studies is a 2019 report prepared by RTI International and sponsored by the U.S. National Institute of Standards and Technology (NIST) [1]. Working with nearly 200 domain experts, the study estimated that GPS had generated roughly \$1.4 trillion (2017 dollars) in economic benefits to the U.S. private sector since civilian use began in the 1980s, with a stated range of about \$0.9 trillion to \$1.8 trillion and a recommendation to treat the figure as an order of magnitude. The largest contributions came from telecommunications (where precise timing enables efficient use of spectrum in wireless networks), telematics and fleet logistics, and location-based services; the electricity, agriculture, mining, oil and gas, and surveying sectors also showed substantial benefits. About 90% of the estimated benefits accrued after 2010, a reminder that dependence has grown rapidly and continues to grow.

The same study asked a harder question: what would a 30-day loss of GPS cost? Its answer, under the assumption that other GNSS constellations were also unavailable, was on the order of \$1 billion per day, with impacts potentially 50% higher if the outage struck during the spring planting season, when precision agriculture is most intensively used [1]. A comparable study for the United

Kingdom estimated that a five-day GNSS disruption would cost the UK economy about £5.2 billion [2].

Example 1.1 (Back-of-envelope outage cost). A planner wants a rough sense of the economic exposure of the United States to a sustained loss of GNSS. The given value is the RTI estimate of roughly \$1 billion per day [1], with a 50% uplift during the planting season. A 30-day outage outside the planting season implies losses on the order of

$$30 \text{ days} \times \$1 \text{ billion/day} \approx \$30 \text{ billion},$$

and an outage in April and May gives roughly $30 \times 1.5 = \$45$ billion. Two cautions apply. First, these figures are order-of-magnitude estimates built on expert elicitation and counterfactual modeling, and they should not be quoted to more than one significant figure. Second, the per-day cost is not constant: many timing users *hold over* on local oscillators for hours to days before drifting out of tolerance, so losses typically grow with outage duration. The shape of that growth, and not only its average, is what resilience engineering tries to change, which is the subject of [Section 1.5](#).

Behind these aggregate figures lie concrete technical dependencies in each sector. In telecommunications, cellular base stations use GNSS-derived time and frequency for synchronization, so a loss of timing forces holdover on local clocks and, eventually, degraded handover and throughput. Trading venues and banks time-stamp transactions for ordering, auditing, and regulatory compliance, frequently with GNSS-disciplined clocks. In electric power, phasor measurement units and protection systems compare voltage and current phase angles across wide areas and therefore need precise common time. Aviation uses GNSS for navigation and surveillance, maritime shipping for navigation and port operations, and road and rail logistics for fleet management. Precision agriculture relies on high-accuracy positioning for guidance and variable-rate application, while weather satellites inform planting and harvest decisions. Forecasters depend on satellite observations, and emergency responders use SATCOM when terrestrial networks fail and GNSS for dispatch. Military forces, finally, depend on all three service families for navigation, precision guidance, command and control, missile warning, and intelligence.

1.1.3 Cascading dependency

These dependencies are not independent. Space services feed *intermediate* infrastructures, chiefly timing distribution and terrestrial telecommunications, that in turn feed many end sectors. A disruption therefore propagates along paths in a dependency graph, as sketched in [Fig. 1.2](#). Two structural properties of such graphs matter for security. First, *common-cause* nodes such as GNSS timing sit upstream of many sectors, so a single disruption can produce correlated losses in finance, telecommunications, and energy simultaneously. Second, some dependencies are *hidden*: a grid operator may know that its phasor measurement units use GNSS time, but may not know that its telecommunications carrier, on which its control center depends, does as well.

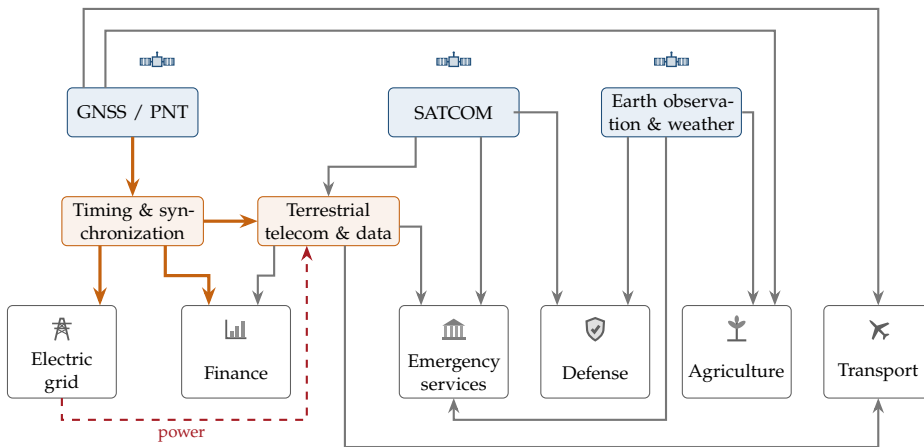


Figure 1.2. A simplified dependency graph from space services (blue) through intermediate infrastructures (orange) to terrestrial sectors. Thick orange arrows mark a common-cause path: GNSS timing feeds telecommunications, finance, and the electric grid simultaneously. The dashed feedback edge from the grid to telecommunications, which needs power, closes a cycle, so disruptions can reinforce each other.

A small, real incident illustrates how a space-segment error becomes a terrestrial one. On 25 and 26 January 2016, a software fault triggered while a GPS satellite was being removed from service caused 15 GPS satellites to broadcast incorrect UTC offset parameters. Many GPS-controlled clocks were in error by roughly 13 microseconds for periods of several hours [3]. Thirteen microseconds is invisible to a driver, but it exceeds the tolerance of some telecommunications and broadcast systems, and operators reported service anomalies. No system was attacked. The lesson is that tiny errors in a common-cause service propagate widely, and that the dependency graph, not a single node in isolation, determines the size of the consequence. [Chapter 13](#) formalizes graphs of this kind as interdependent networks.

1.2 From Sanctuary to Contested Domain

For much of the space age, national security planners in the major spacefaring states thought of orbit as a *sanctuary*: a place where strategic reconnaissance and early-warning satellites operated without serious threat, partly because of technical difficulty and partly because the superpowers had a mutual interest in keeping each other’s “national technical means” of verification intact. That picture was only partly accurate, and it has now been abandoned. [Figure 1.3](#) collects some of the milestones along the way, marked by type of event.

Several threads run through the timeline. The first is that *counterspace capabilities are old*. Both superpowers explored nuclear and conventional anti-satellite (ASAT) concepts in the 1960s, and the Starfish Prime test showed as early as 1962 that a single high-altitude nuclear detonation could harm satellites that were not its target. The second thread is *debris as an externality*. The Kessler-Cour-Palais analysis of 1978 framed collisional cascading as a long-term risk, and the FY-1C

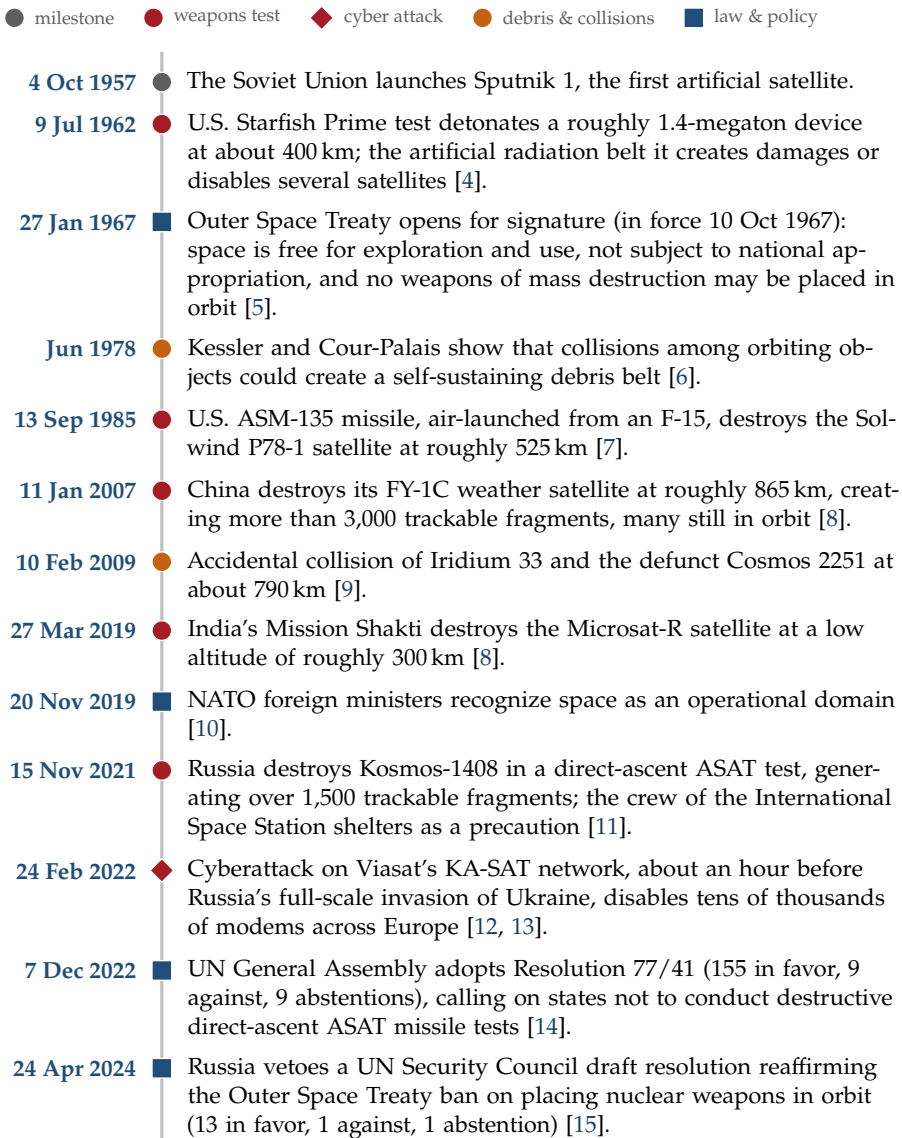


Figure 1.3. Selected milestones in the evolution of space security, with dates as reported by the cited sources. Destructive weapons tests (red circles) recur across six decades, debris and law develop in response, and the most recent attack in the sequence (red diamond) did not touch a spacecraft at all.

test gave it a vivid demonstration: a single event at a long-lived altitude produced a debris population that still threatens other operators. Mission Shakti's low target altitude and Kosmos-1408's debris cloud crossing the orbit of a crewed station sharpened the debate over whether destructive testing is ever responsible, culminating in the 2022 General Assembly resolution. The third thread is the *broadening of the threat beyond the kinetic*. The KA-SAT attack used a ground-network intrusion to disable user terminals, with effects spilling far beyond the intended military targets; [Section 1.4](#) examines it in detail.

1.2.1 Congested, contested, competitive

In 2011 the U.S. National Security Space Strategy summarized the strategic environment in three words: space was becoming increasingly *congested*, *contested*, and *competitive* [16]. The phrase has since become common shorthand well beyond the United States.

Congested refers to the physical and spectral crowding of orbit. ESA's 2025 Space Environment Report, using data through the end of 2024, counted about 40,000 objects tracked by surveillance networks, of which about 11,000 were active payloads, and estimated more than 1.2 million debris objects larger than 1 cm [17]. Independent tallies of active satellites have continued to rise quickly since then, driven mainly by LEO broadband constellations, and by mid-2026 exceeded 15,000 [18]. ESA also reported that, around 550 km, the density of debris large enough to be dangerous is now of the same order as the density of active satellites [17]. Congestion is not only physical. Spectrum is a finite, shared resource coordinated through the International Telecommunication Union, and interference, whether accidental or deliberate, is a daily operational reality. The environment adds its own stresses: in February 2022 a moderate geomagnetic storm increased thermospheric drag enough that 38 of 49 newly launched Starlink satellites re-entered before they could raise their orbits [19].

Contested refers to the development and use of counterspace capabilities. Open-source assessments by the Secure World Foundation and the Center for Strategic and International Studies document a broad spectrum of capabilities across many countries, including direct-ascent and co-orbital systems, electronic warfare, directed energy, and cyber operations [8, 20]. These assessments emphasize that the capabilities actually being *used* in current conflicts are overwhelmingly non-destructive and often reversible: GNSS jamming and spoofing, SATCOM jamming, and cyber operations. European aviation regulators, for example, have repeatedly warned operators about GNSS jamming and spoofing affecting aircraft around conflict zones and in regions including the eastern Mediterranean, the Black Sea, the Middle East, the Baltic Sea, and the Arctic [21].

Competitive refers to the economics and geopolitics of access. Launch costs have fallen, commercial operators field constellations that rival or exceed government systems in size, and many more states operate satellites than in earlier decades. Governments increasingly buy commercial services rather than build dedicated systems. This diversifies supply, but it also places commercial operators, their ground networks, and their supply chains inside the security perimeter of national missions.

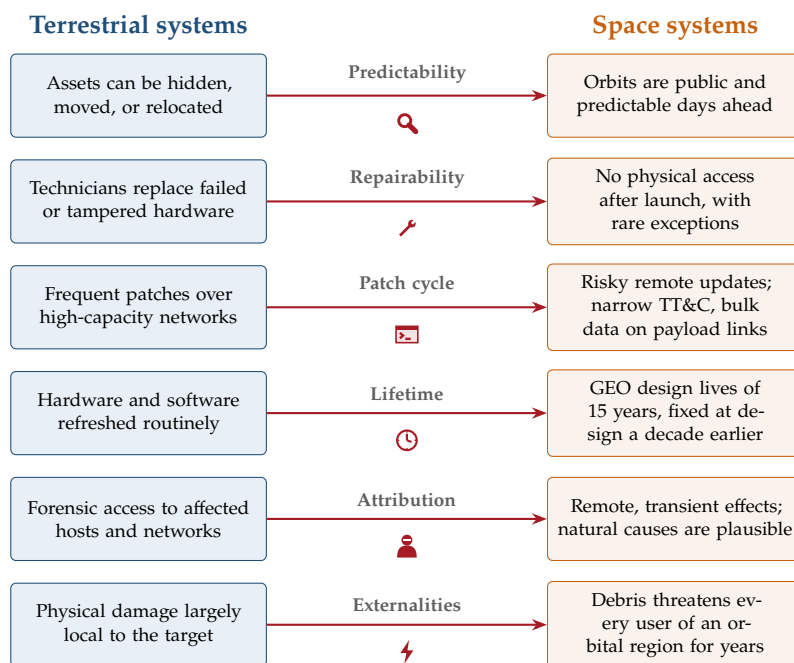


Figure 1.4. What makes space different. For six security-relevant dimensions, the left column describes the typical terrestrial situation and the right column its space counterpart. In every row the shift makes defense harder: the adversary can plan against a predictable, unrepairable, remotely patched, long-lived target whose attackers are hard to name and whose destruction harms third parties.

1.3 What Makes Space Different

Many principles of security engineering transfer directly to space: least privilege, defense in depth, authentication, monitoring, and incident response apply to a mission operations center just as they apply to a bank. But space systems have properties that change the balance between threats and defenses, and that make some terrestrial practices impractical. [Figure 1.4](#) contrasts six of them with their terrestrial counterparts; this section discusses ten, grouped into four families, and each recurs throughout the book.

The first family concerns the physics of operating in orbit. A satellite follows dynamics that are well understood and publicly modeled, so once an adversary has an orbit estimate it can predict when the satellite will be overhead, when it will be in view of a particular ground station, and where it will be in a week. Concealment in space is hard, and maneuvers cost propellant. With rare exceptions, a spacecraft also cannot be touched after launch: a hardware fault, a radiation-damaged component, or a hardware trojan cannot be replaced, only worked around. On-orbit servicing may change this for some missions, but servicing is itself a dual-use capability. Finally, every kilogram, watt, and processor cycle on board is expensive. These size, weight, and power (SWaP) limits mean that cryptography, intrusion detection, and redundancy compete

with the payload for resources, and radiation-tolerant processors lag commercial ones by generations.

The second family concerns time. GEO satellites commonly have design lives of 15 years, and a system may be designed a decade before launch, so algorithms and threat assumptions chosen at design time must remain adequate for 20 years or more. This is a major concern for cryptographic agility and for the migration to post-quantum cryptography. Flight software can be updated over the air, and on software-defined satellites in-orbit reconfiguration can be frequent. The TT&C link is too narrow for large files, so bulk reconfiguration data typically travels over high-rate payload or feeder links, while TT&C carries the commands that authorize and monitor the change and fall back if it fails [22]. Updates remain risky: a failed update to the platform software can lose the spacecraft, and every path that can deliver code to the spacecraft is a path an adversary may try to use. Ground and user-segment software patches face the operational constraints of services that run around the clock and of terminal fleets numbering in the tens or hundreds of thousands.

The third family concerns strategic ambiguity. Most space technologies, including rendezvous and proximity operations, high-power lasers, powerful transmitters, on-orbit servicing, and even debris removal, have both benign and hostile applications, so capability alone rarely reveals intent. Attribution is correspondingly difficult. Jamming can be reversible and transient, cyber operations can be routed through third parties, and a satellite failure can plausibly be blamed on space weather. The KA-SAT attack of 2022 was publicly attributed by the European Union, the United States, and the United Kingdom about eleven weeks after it occurred [13]. Delayed or uncertain attribution weakens deterrence by punishment, because a response that arrives late or rests on contestable evidence is a weaker threat.

The fourth family concerns the shared environment. Destroying a satellite kinetically can impose costs on every operator using the same orbital region for years or decades, and the attacker shares the commons it pollutes. Frequencies and orbital slots are coordinated internationally, and physical interference, radio interference, and debris do not respect borders, so security choices by one operator affect the environment for all. Government and military missions, meanwhile, increasingly rely on commercial satellites, commercial ground-station networks, cloud-hosted operations, and multinational supply chains. Many actors who are not parties to a conflict are therefore inside its reach.

These ten properties interact. Predictable motion combined with long patch cycles means that an adversary can study a target at leisure and act at a moment of its choosing, while the defender must fight with the configuration it launched. SWaP limits combined with long lifecycles mean that security mechanisms must be chosen early and cheaply, yet last a long time. Dual use combined with attribution difficulty means that many hostile acts sit below clear thresholds of armed attack. Much of the rest of the book can be read as a set of responses to these properties.

1.4 Security Is Multi-Layered: The Space Security Stack

“Space security” is used to mean very different things. To a policy analyst it may mean arms control and deterrence; to an RF engineer, anti-jam waveforms; to a flight dynamicist, collision avoidance; to a cyber analyst, hardening a mission operations network. Each of these communities is right, and each sees a different part of the problem. This book organizes the field into *layers*, each with its own assets, threats, physics, and mathematics. We call this organization the *Space Security Stack*.

Definition 1.1 (Space Security Stack). The *Space Security Stack* partitions the security problems of a space system into five layers: (i) signal/RF, (ii) link and network, (iii) physical/orbital, (iv) cyber, and (v) mission/strategic. A threat is characterized by the layer at which it *enters* the system and the layer or layers at which it produces *effects*; a threat whose entry and effect layers differ is called *cross-layer*.

Figure 1.5 draws the stack. For each layer it names what is attacked or protected, representative threats, and the main mathematical tools, together with the Part of the book that treats the layer. The signal/RF layer concerns waveforms, spectrum, and PNT signals, which are threatened by jamming, spoofing, eavesdropping, and meaconing (the rebroadcast of captured signals) and analyzed with information theory, detection and estimation theory, stochastic geometry, and jamming games. The link and network layer concerns protocols, cryptographic keys, inter-satellite link (ISL) routing, and integrated space-ground-terrestrial networks. Its threats include replay, key compromise, routing manipulation, link flooding, topology inference, and Byzantine (arbitrarily misbehaving) nodes, and its tools are protocol verification, graph theory, network interdiction, robust flow, and consensus. The physical/orbital layer treats spacecraft, orbits, ground facilities, and hardware as physical objects exposed to direct-ascent ASAT (DA-ASAT) weapons, co-orbital systems performing rendezvous and proximity operations (RPO), directed energy, nuclear effects, debris, space weather, and tampering; its mathematics is orbital mechanics, engagement kinematics, damage and stochastic hazard models, and differential games. The cyber layer covers ground information and operational technology (IT/OT), mission operations, flight software, and user terminals, threatened by intrusion, malware, malicious commands and updates, and insiders, and studied with attack graphs, Markov games, anomaly and sequential detection, and formal verification. The mission/strategic layer, finally, concerns services, deterrence stability, and governance; its threats are denial of service to users, coercion, and escalation, and its tools are resilience metrics, decision and game theory, and bargaining models.

1.4.1 Cross-layer attacks

Many of the most consequential attacks exploit the seams between layers. A defense designed for one layer, such as a robust anti-jam waveform, a well-segmented network, or a hardened spacecraft bus, can be bypassed by an attack that enters elsewhere and propagates. Figure 1.6 shows seven representative pathways, each drawn from its entry layer (red dot) to its effect layer (arrowhead).

		Attacked or protected	Representative threats	Main mathematical tools
	Mission / strategic Part I, V	Services, deterrence stability, governance	Denial of service to users, coercion, escalation	Resilience metrics, decision and game theory, bargaining
	Cyber Part IV	Ground IT/OT, mission operations, flight software, terminals	Intrusion, malware, malicious commands and updates, insiders	Attack graphs, Markov games, anomaly detection, formal verification
	Physical / orbital Part III	Spacecraft, orbits, ground facilities, hardware	DA-ASAT, co-orbital RPO, directed energy, nuclear effects, debris	Orbital mechanics, damage and hazard models, differential games
	Link & network Part II	Protocols, keys, ISL routing, integrated networks	Replay, key compromise, routing attacks, flooding, Byzantine nodes	Protocol verification, graphs, interdiction, consensus
	Signal / RF Part II	Waveforms, spectrum, PNT signals	Jamming, spoofing, eavesdropping, meaconing	Information and detection theory, stochastic geometry, jamming games
				

Figure 1.5. The Space Security Stack. Each layer has its own assets, threats (red), and mathematics (blue), and the Parts of the book follow the layers from the signal upward. Effects at each layer ultimately matter through their consequences at the mission layer, and supply-chain and insider threats cut across all five.

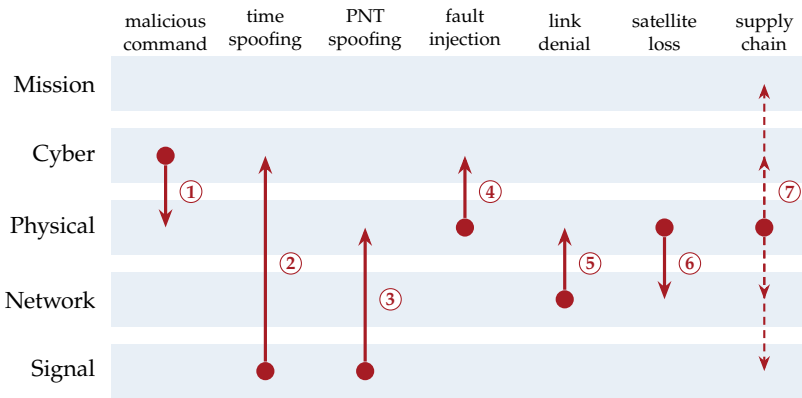


Figure 1.6. Representative cross-layer pathways. Each column is one pathway, drawn from the layer where it enters (red dot) to the layer where it takes effect (arrowhead): (1) cyber to physical, (2) signal to cyber, (3) signal to physical, (4) physical to cyber, (5) network to physical, (6) physical to network, and (7) a supply-chain compromise that can surface at every layer. A defense confined to a single lane cannot see an attack whose arrow crosses into it.

In the first, a malicious TT&C command fires thrusters, tumbling the spacecraft or placing it on a collision course that creates debris, so a cyber intrusion ends as a physical event. In the second, a spoofed GNSS time signal causes authentication, certificate-validity, and synchronization failures in ground and user networks. In the third, spoofed PNT or dazzled sensors corrupt state estimates, and the guidance, navigation, and control system steers on bad data. The fourth runs from the physical layer to the cyber layer: laser or radiation fault injection into cryptographic hardware leaks keys, and single-event upsets (bit flips caused by energetic particles) corrupt security functions. In the fifth, denial of service on ISL or TT&C links prevents a collision-avoidance maneuver from being uploaded in time. In the sixth, the loss of satellites to an ASAT weapon or to debris partitions the constellation graph and forces traffic to reroute. The seventh is the least visible: a hardware trojan or compromised firmware inserted through the supply chain creates a latent common-mode failure that can surface at one or more layers.

The opening vignette contains a network-to-physical pathway in embryo: a possible compromise of the ground path to the satellite (cyber and network layers) threatens the timely upload of a collision-avoidance maneuver (physical layer), with consequences for service (mission layer). No single-layer analysis captures the decision the HELIOS operators face. [Chapter 13](#) builds the models needed to analyze such pathways quantitatively.

Space Security in Practice: The KA-SAT / AcidRain Attack (2022)

On 24 February 2022, beginning at about 03:02 UTC and roughly an hour before Russia's full-scale invasion of Ukraine, a cyberattack disrupted service on Viasat's KA-SAT network, a GEO Ka-band broadband system serving

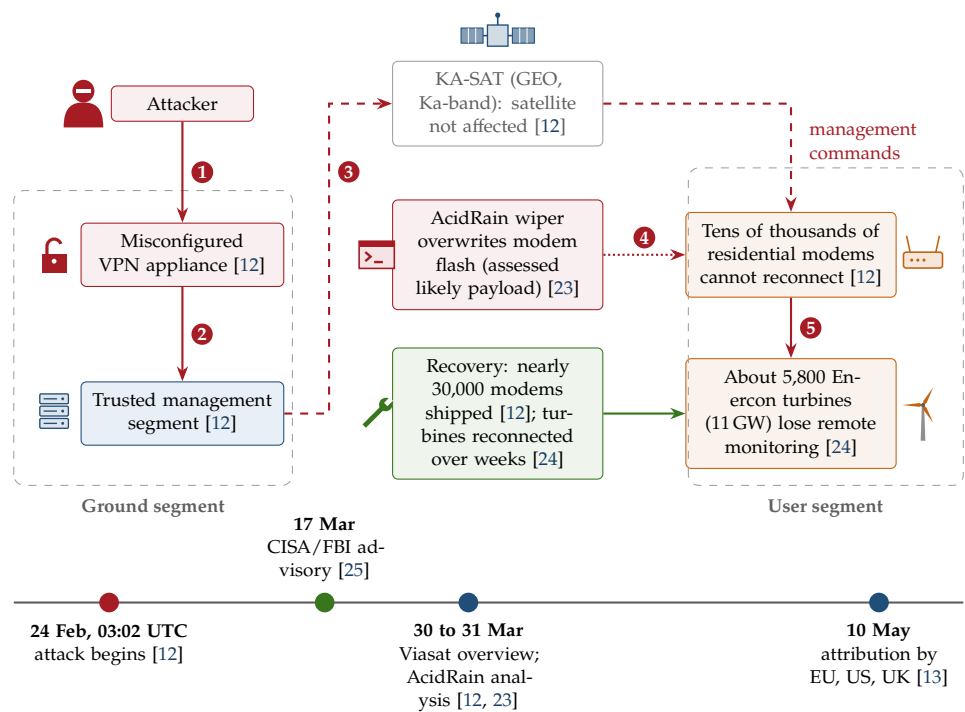


Figure 1.7. Annotated reconstruction of the KA-SAT/AcidRain attack of 24 February 2022, based on the cited public sources. Numbered steps trace the attack from initial access (steps 1 and 2) through a trusted management path relayed via the satellite (3) to the destructive payload (4) and third-party effects (5); the strip below marks the public response on a time axis. The satellite itself was not touched, and recovery required physical replacement of terminals.

Europe and Ukraine (Fig. 1.7). According to Viasat's public overview, the attacker exploited a misconfigured VPN appliance to gain remote access to the trusted management segment of the network, then issued legitimate, targeted management commands to a large number of residential modems simultaneously. Destructive commands overwrote key data in the modems' flash memory, rendering tens of thousands of them unable to reconnect. The satellite itself was not affected. Viasat and its distributors shipped nearly 30,000 replacement modems to restore service [12].

On 31 March 2022, researchers at SentinelLabs published an analysis of a wiper they named *AcidRain*, a MIPS ELF binary designed to erase modems and routers by overwriting storage devices, which they assessed as the likely destructive payload. They also reported, with medium confidence, developmental similarities to a destructive plugin of the earlier VPNFilter malware [23]. On 10 May 2022, the European Union, the United States, and the United Kingdom publicly attributed the attack to Russia [13]. CISA and the FBI had already issued a joint advisory urging SATCOM providers and customers to harden their networks [25].

The effects spread far beyond military users in Ukraine. The most widely reported knock-on effect was in Germany: about 5,800 Enercon wind turbines, with a combined capacity of roughly 11 GW, lost remote monitoring and control because their SCADA links used KA-SAT terminals. The turbines kept generating in automatic mode, but faults could no longer be handled remotely, and restoring connectivity to most of them took weeks [24, 23].

Read through the Space Security Stack, the attack *entered* at the cyber layer in the ground segment and produced *effects* in the user segment, and its mission-layer consequences reached third parties in countries not party to the conflict. The initial access exploited a trusted management path rather than a cryptographic weakness in the RF link, which is why link encryption did not stop it. Recovery depended on physically replacing or reflashing terminals at scale, which made it a reconstitution problem as much as a cyber one. The incident recurs throughout the book as a reference case for the kill chain, for ground and user-segment security, and for integrated case analysis.

1.4.2 Why the book covers the layers in this order

The layers could be presented in several orders. We cover them as *communication and network* (Part II), then *physical* (Part III), then *cyber and cyber-physical* (Part IV). Communication links connect all four segments of a space system, and most day-to-day interference, including jamming, spoofing, and unauthorized access over the air, happens there, so the signal and network layers are the natural starting point once the foundations are in place. Physical security follows. It treats the spacecraft and ground facilities as physical objects in a hazardous environment, and it introduces the orbital dynamics and damage models on

which later control-theoretic material relies. Cybersecurity comes last, together with cross-layer cyber-physical security, because by then the reader has seen both the communication paths along which cyber effects travel and the physical dynamics in which they can end. The cross-layer framework at the start of the second half of Part IV is the hinge of the book: it composes models from all earlier layers into a single framework.

1.5 Security and Resilience

1.5.1 An intuitive distinction

Security is concerned with preventing adversaries from achieving their objectives: keeping unauthorized parties from reading, altering, or denying use of a system and its data, and keeping hostile actions from succeeding. Security asks how an attack can be stopped. *Resilience* accepts that some attacks, failures, and environmental events will succeed at least partially, and asks how much of the mission can be preserved and how quickly the rest can be restored. NIST's systems security engineering guidance defines cyber resiliency in terms of four abilities: to *anticipate*, *withstand*, *recover from*, and *adapt to* adverse conditions, stresses, attacks, or compromises [26]. We adopt these four phases throughout the book.

The two notions are complementary, not competing. A system with strong security but no resilience is brittle: when its defenses are eventually breached, as the history of cybersecurity suggests they will be, it fails badly. A system with resilience but weak security invites repeated attacks and spends its life recovering. Space adds a further consideration. Because spacecraft cannot be repaired and replacement takes months or years, resilience in space often has to be designed into the *architecture*, through proliferation, diversity, disaggregation, and spare capacity, rather than achieved through rapid repair. [Chapter 3](#) makes these distinctions precise and compares resilience with related notions such as reliability, robustness, survivability, and reconstitution. The distinction between withstanding uncertainty anticipated at design time and recovering from disruptions that exceed design assumptions is examined, with game-theoretic formulations of both, by Zhu and Başar [27].

1.5.2 The resilience curve

A common way to visualize resilience, borrowed from infrastructure engineering [28, 29], is to plot a measure of mission performance $Q(t)$ over time, as in [Fig. 1.8](#). Performance might be the fraction of users served, network throughput, positioning accuracy, or revisit rate, whichever best captures what the mission is for. Before a disruptive event, performance sits at its nominal level Q_0 . At the event it drops, and the depth of the drop reflects how well the system *withstands* the event. Performance then recovers over some time, and the system may *adapt* so that the next event of the same kind does less damage. Preparation before the event, such as monitoring, threat intelligence, spares, and rehearsed playbooks, is *anticipation*.

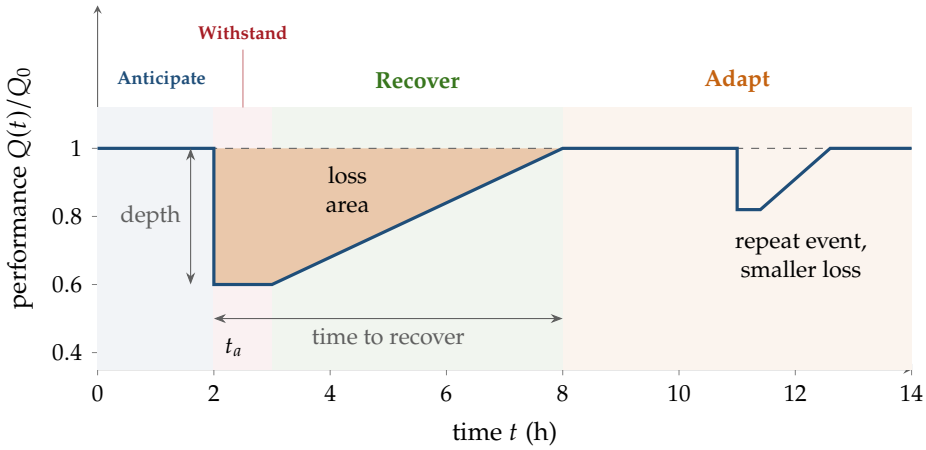


Figure 1.8. A stylized resilience curve with the four resilience phases shaded. A disruption at $t_a = 2$ h cuts performance to 60% of nominal. The system withstands the event (performance holds at 60% while operators diagnose it), recovers linearly by $t = 8$ h, and adapts, so a similar event at $t = 11$ h causes a shallower and shorter loss. The orange region is the performance lost to the first event.

1.5.3 The resilience functional

The curve suggests natural scalar metrics: the depth of the drop, the time to recover, and the area between the nominal and actual curves. The book’s organizing metric combines three further ideas. First, performance depends on the *architecture* $\mathcal{A}$ (how many satellites, which orbits, which ground stations, which links and protocols) and on the defender’s *policy* π (how it detects, responds, reroutes, and recovers). Second, the event is an attack w drawn from an *attack set* $\mathcal{W}$ that decomposes by layer,

$$\mathcal{W} = \mathcal{W}_{\text{sig}} \times \mathcal{W}_{\text{net}} \times \mathcal{W}_{\text{phys}} \times \mathcal{W}_{\text{cyb}},$$

so that a multi-layer campaign is a tuple of attacks at the signal, network, physical, and cyber layers. Third, a defender who does not know which attack will occur should evaluate its design against the worst case in the set.

Definition 1.2 (Resilience functional, preview). Let $Q(t; \mathcal{A}, \pi, w)$ denote mission performance at time t under architecture $\mathcal{A}$, defender policy π , and attack $w \in \mathcal{W}$, and let $Q_0 > 0$ be nominal performance. Over a horizon $[0, T]$, the *worst-case normalized resilience* is

$$R(\mathcal{A}, \pi; \mathcal{W}) = \inf_{w \in \mathcal{W}} \frac{1}{T} \int_0^T \frac{Q(t; \mathcal{A}, \pi, w)}{Q_0} dt. \quad (1.1)$$

If performance stays at nominal throughout, $R = 1$; if the mission is lost for the whole horizon under some attack, $R = 0$. The functional rewards both a shallow drop and a fast recovery, because both shrink the loss area. It is deliberately simple. A full treatment must address when the infimum is attained, how to replace the worst case with a risk measure such as conditional value-at-risk when

attacks are better described probabilistically, how to model $Q(t)$ with Markov degradation-recovery processes, and how to handle performance that can exceed nominal after adaptation. Later chapters compute $Q(t)$ for specific layers and then compose the layers.

Example 1.2 (Resilience of the vignette’s morning). Consider a single attack w on HELIOS’s aviation PNT service, with performance measured as the fraction of aviation users receiving usable positioning. The given values are $T = 24$ h and $Q_0 = 1$. Performance drops to 0.6 at $t_a = 2$ h, holds at 0.6 for one hour while operators diagnose the problem, then recovers linearly to 1 by $t = 8$ h, as in the first event of Fig. 1.8. The loss area is a rectangle plus a triangle:

$$\underbrace{0.4 \times 1}_{\text{withstand}} + \underbrace{\frac{1}{2} \times 0.4 \times 5}_{\text{recover}} = 1.4 \text{ h.}$$

Hence the normalized performance integral for this attack is $1 - 1.4/24 \approx 0.942$, and if $\mathcal{W}$ contained only this attack, $R \approx 0.942$. A second attack that caused a smaller drop (to 0.8) but took 16 hours to recover would have loss area $\frac{1}{2} \times 0.2 \times 16 = 1.6$ h and normalized integral ≈ 0.933 . The infimum in (1.1) would then select this slower attack, even though its instantaneous impact was milder. Time to recover therefore matters as much as depth.

1.6 The Security-Resilience Cube

The Space Security Stack identifies *which kind* of problem a defender faces. Two further questions locate the problem precisely: *where* in the system it arises, and *when* in the life of an incident the defender acts. Combining the three gives the *Security-Resilience Cube* (Fig. 1.9).

Definition 1.3 (Security-Resilience Cube). The *Security-Resilience Cube* is the product of three axes. The *layer* axis takes the values signal/RF, link & network, physical/orbital, cyber, and mission/strategic. The *segment* axis takes the values ground, link (TT&C, feeder, user, ISL, and PNT links), space, and user. The *resilience phase* axis takes the values anticipate, withstand, recover, and adapt [26]. A problem instance is placed in the cell, or set of cells, given by the layer of the threat or defense, the segment in which it acts, and the phase of the defender’s response. Cross-layer problems occupy more than one cell.

Four problems from this chapter show how placement in the cube works. The recovery of user terminals after the KA-SAT intrusion, marked (a) in Fig. 1.9, sits at the cyber layer and spans two segments: the ground segment where the attack entered and the user segment where it took effect. Its phase is recover, with lessons for anticipation in the form of hardened management paths. GNSS spoofing that affects aviation, marked (b), is a signal/RF problem whose effects can become physical through navigation errors. It occupies the link segment (the PNT signal) and the user segment (the receiver), and its phase is withstand, since the task is to detect the spoofing and fall back on other navigation sources.

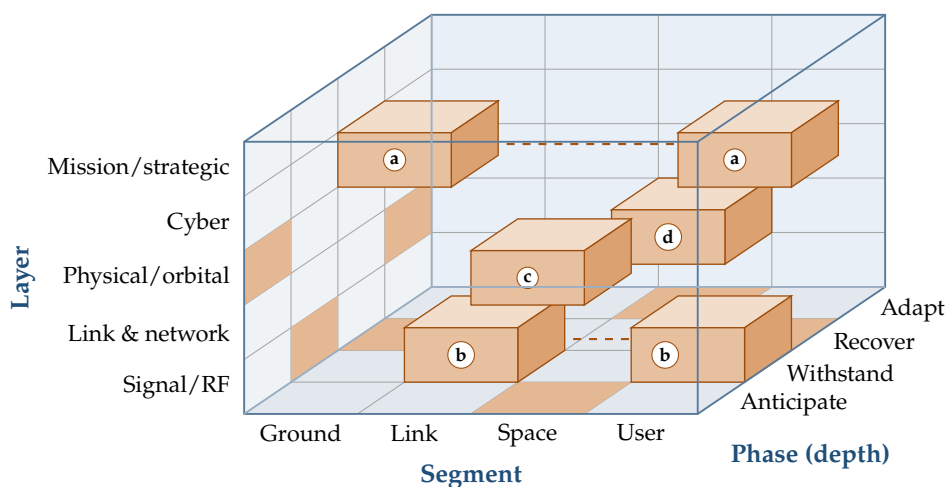


Figure 1.9. The Security-Resilience Cube. The front face shows Layer $\times$ Segment, and depth runs through the four resilience phases, from anticipate at the front to adapt at the back. Each example problem discussed in the text appears as an orange block in the cell it occupies: (a) fleet-wide recovery of user terminals after a ground-network intrusion; (b) aviation receivers withstanding GNSS spoofing; (c) anticipating a debris conjunction; (d) adapting ISL routing after satellite loss. Dashed lines join blocks belonging to the same cross-segment problem, and the light shadows on the floor and left wall help read off each block's segment, layer, and phase.

The HELIOS conjunction warning of the opening vignette, marked (c), is a physical/orbital problem in the space segment, and its phase is anticipate: the operators must assess collision probability and plan the maneuver before the event. The rerouting of ISL traffic after a satellite is lost, marked (d), is a link and network problem triggered by a physical event in the space segment. Its phase is adapt, because the constellation must reconfigure routing tables in the short term and replenishment plans over time.

The cube is not a taxonomy for its own sake. It serves first as a completeness check: a security program that fills only the “cyber $\times$ ground $\times$ withstand” cells is visibly incomplete, however well it fills them. It also serves as a map of the book, since every chapter opens with a layer map that shows the front face of the cube and marks the cells the chapter covers.

1.7 Adversaries in Time: The Space Kill Chain

The cube is static. Real adversaries act over time, moving through stages that the defender can observe and interrupt. Borrowing the “kill chain” idea from intrusion analysis [30] and from space-specific threat frameworks such as SPARTA [31], the book models a campaign against a space system as an eight-stage *space kill chain*, drawn in Fig. 1.10.

In the first stage, *reconnaissance*, the adversary builds a target picture from orbital elements, RF emissions, ground-site locations, and information about personnel and suppliers. In *resource development* it acquires or builds the tools,

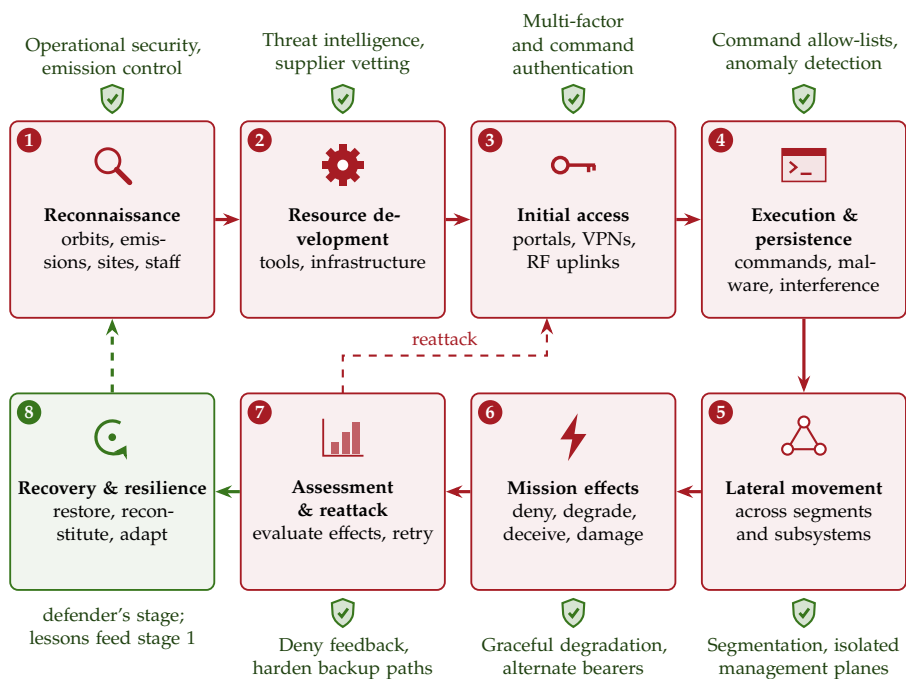


Figure 1.10. The eight-stage space kill chain. Red boxes are adversary stages, read left to right on the top row and right to left on the bottom row; green shields mark representative defender opportunities to interrupt each stage. The dashed red arrow shows reattack after assessment, and the final stage belongs to the defender, whose lessons feed back into anticipation of the next campaign.

infrastructure, and capabilities the campaign needs. *Initial access* gains a foothold in a trusted path, such as a ground-station portal, a remote-access appliance, or an RF uplink. *Execution and persistence* turns that foothold into action, whether malicious commands, malware, or interference, and arranges to keep it. *Lateral movement* expands control across segments and subsystems, for example from enterprise networks into mission operations or from the ground into a spacecraft bus or payload. *Mission effects* are the purpose of the campaign: to deny, degrade, deceive, or damage a space-enabled service. In *assessment and reattack* the adversary evaluates what it achieved and returns to an earlier stage if it must. The final stage, *recovery and resilience*, belongs to the defender, who restores service, reconstitutes lost capacity, and adapts.

Two features distinguish the book's treatment from the intrusion-analysis original. First, stages may *switch layers*: reconnaissance of a satellite's RF emissions can enable a cyber intrusion into its ground segment, which in turn produces a physical effect. The chain is therefore drawn over the whole Space Security Stack, not only over computer networks. Second, the chain is modeled as a *stochastic process that the defender can interrupt*, so that the value of detection at each stage, and the time the adversary dwells undetected, can be quantified.

A simple calculation shows why interruption at early stages is so valuable. Suppose, purely for illustration, that the defender detects and stops the campaign at each of the five stages before mission effects independently with probability p . The campaign then reaches stage 6 with probability $(1-p)^5$, which is $0.7^5 \approx 0.17$ for $p = 0.3$ and $0.5^5 \approx 0.03$ for $p = 0.5$. Modest detection probabilities compound into strong protection when they are layered along the chain, which is the quantitative content of defense in depth. The independence assumption is optimistic, because a capable adversary adapts to the defenses it observes, and relaxing it is one of the reasons game-theoretic adversary models are needed.

The GSaaS portal login in the opening vignette is a candidate reconnaissance or initial-access event, and whether the defender treats it as such is a detection problem under uncertainty. An aggressive response to a false alarm delays a necessary maneuver, whereas a relaxed response to a real intrusion may hand the adversary the command path. [Chapter 4](#) develops the kill chain as an absorbing Markov chain, maps it to established frameworks, and introduces the risk-assessment and adversary models used throughout the book.

1.8 The HELIOS Running Example

1.8.1 Why a running example?

A book on a field as broad as space security faces a fragmentation risk: a jamming model in one chapter, a debris model in another, and an intrusion model in a third, with no way to see how they fit together. We counter this with a single notional system, *HELIOS*, used at every layer. A shared example has three benefits. It keeps numbers consistent, so that the link budget computed for *HELIOS* is the one later jammed and the orbit later attacked. It lets readers accumulate intuition about one system rather than re-learning a new one in each chapter. And it makes cross-layer composition concrete, because models built for different layers can be

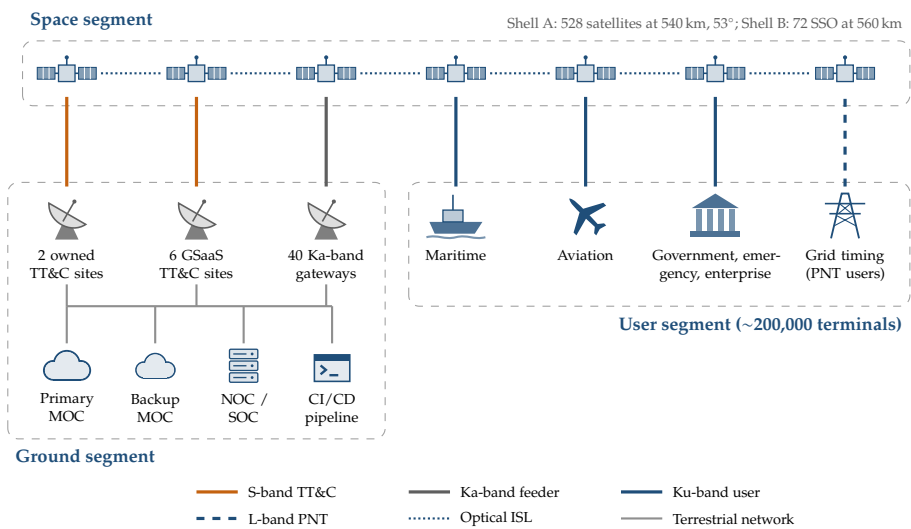


Figure 1.11. HELIOS architecture (notional). Seven representative satellites stand for the two shells, linked by optical inter-satellite links. S-band TT&C links (orange) run from two owned and six leased ground-station-as-a-service (GSaaS) sites, so the command path that is the subject of the opening vignette relies mostly on a third-party provider. Ka-band feeder links serve 40 gateways, Ku-band links serve maritime, aviation, government, emergency, and enterprise users, and an L-band signal provides PNT augmentation for timing users; the ground network connects the TT&C sites and gateways to the primary and backup MOCs, the core network, and the signed-update pipeline.

assembled into a single multi-layer HELIOS model.

HELIOS is notional. It is a composite created for teaching; it is not modeled on a real operator, and its parameters were chosen to be plausible and internally consistent, not to describe an existing system.

1.8.2 HELIOS at a glance

HELIOS is a proliferated LEO constellation providing broadband connectivity and a PNT augmentation service. It has optical inter-satellite links, a TT&C network that mixes owned sites with leased ground-station-as-a-service antennas, a cloud-hosted mission operations center (MOC), and a diverse user base. Table 1.1 lists its canonical parameters, and Fig. 1.11 sketches its four segments.

The table uses a few standard abbreviations that recur throughout the book. EIRP (effective isotropic radiated power) is the transmitter power multiplied by antenna gain, expressed in decibels relative to one watt (dBW). DVB-S2X is the broadcast-satellite waveform standard used on the user downlink. CCSDS is the Consultative Committee for Space Data Systems, whose Space Data Link Security (SDLS) protocol provides authenticated encryption of commands, and cFS is NASA’s open-source core Flight System framework for flight software.

The architecture was chosen to expose problems at every layer. Its ISLs create a dynamic space network whose routing can be attacked and defended. Its reliance on leased GSaaS antennas and a cloud-hosted MOC puts third parties inside

Table 1.1. Canonical HELIOS parameters (notional).

Parameter	Value
<i>Space segment</i>	
Shell A (primary)	528 satellites; circular, 540 km, inclination 53°; Walker-delta 528/24/1 (24 planes × 22 satellites); period 95.44 min
Shell B (polar)	72 satellites; 560 km, sun-synchronous ($i \approx 97.6^\circ$); 6 planes × 12 satellites; period 95.86 min
Satellite bus	300 kg wet mass; 3 kW end-of-life power; Hall-effect electric propulsion; 5-year design life
Onboard computing	Radiation-tolerant ARM-based onboard computer (OBC) with cold spare; NASA cFS-based flight software; separate payload processor
Inter-satellite links	4 optical terminals per satellite (2 intra-plane, 2 cross-plane), 20 Gbps each
Replenishment	About 120 satellites per year, launched in batches of about 30
<i>Links</i>	
User links	Ku-band: downlink 10.7 GHz to 12.7 GHz (250 MHz channels, 34 dBW EIRP per beam, DVB-S2X); uplink 14.0 GHz to 14.5 GHz
Feeder links	Ka-band: downlink 17.8 GHz to 19.3 GHz; uplink 27.5 GHz to 30.0 GHz
TT&C	S-band: uplink 2025 MHz to 2110 MHz, downlink 2200 MHz to 2290 MHz; CCSDS protocols with SDLS authenticated encryption (AES-GCM) on commands; 64 kbps telemetry
PNT augmentation	L-band timing and ranging broadcast with authenticated navigation messages
<i>Ground segment</i>	
Operations	Primary MOC in a commercial cloud region; warm-standby backup MOC at a separate site
TT&C sites	2 owned high-latitude sites plus 6 sites leased from a GSaaS provider
Gateways and core	40 Ka-band gateways connected to a terrestrial core network hosting the network and security operations centers (NOC/SOC)
Software pipeline	Continuous integration and delivery (CI/CD) pipeline builds signed updates: platform flight software uploaded over TT&C, payload-processor images over the feeder link under TT&C authorization, and terminal firmware over the user link
<i>User segment</i>	
Terminals	About 200,000: maritime, aviation, government and defense, emergency services, enterprise backhaul; firmware updated over the air
PNT users	A subset (aviation, grid timing) relies on the HELIOS timing augmentation

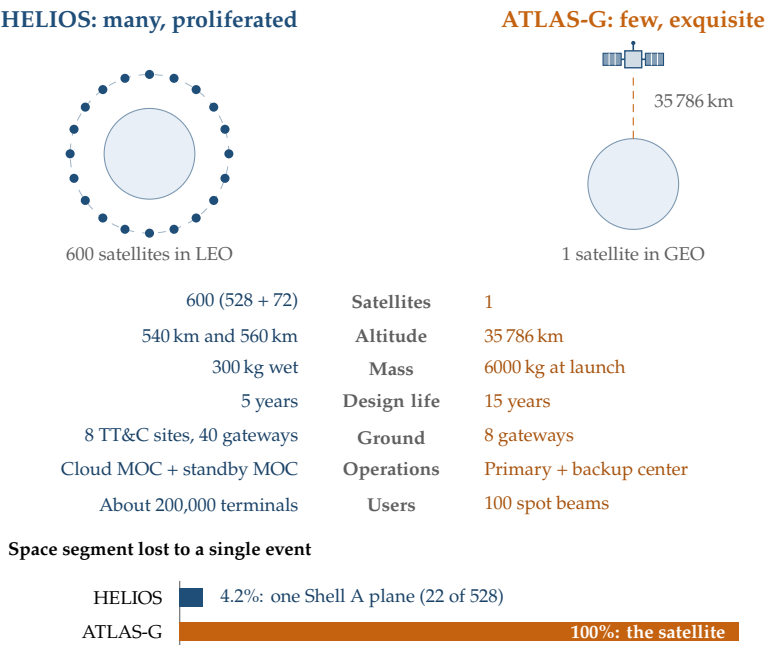


Figure 1.12. HELIOS and ATLAS-G side by side (both notional). The proliferated constellation loses about 4% of its primary shell when a whole orbital plane is lost, whereas the single GEO satellite takes the whole space segment with it. The attribute rows show the other side of the trade: HELIOS has far more satellites, sites, and terminals to secure.

the command path. Its signed over-the-air updates for both flight software and terminal firmware make the software supply chain a target. Its PNT augmentation connects the signal layer to aviation safety and grid timing. And its size makes resilience through proliferation a live design question, which the comparison with a very different architecture brings into focus.

1.8.3 ATLAS-G: the exquisite contrast

To make architectural trade-offs visible, the book pairs HELIOS with *ATLAS-G*, a notional high-throughput GEO SATCOM system, likewise not modeled on a real operator. ATLAS-G is a single 6000 kg Ka-band satellite with 100 spot beams and a 15-year design life, in one orbital slot at 35 786 km altitude, supported by 8 gateways and a primary and a backup operations center. ATLAS-G represents the “few, exquisite assets” architecture, and HELIOS represents the “many, proliferated assets” architecture. [Figure 1.12](#) sets the two side by side.

The comparison is instructive even at the level of arithmetic. Losing one full orbital plane of HELIOS Shell A removes 22 satellites, or $22/528 \approx 4.2\%$ of that shell, fewer than a single replenishment launch of about 30. Losing the ATLAS-G spacecraft removes the whole space segment for the years it takes to build and launch a replacement. The picture is not one-sided, however. HELIOS has 600 spacecraft, 8 TT&C sites, 40 gateways, and 200,000 terminals to secure, and

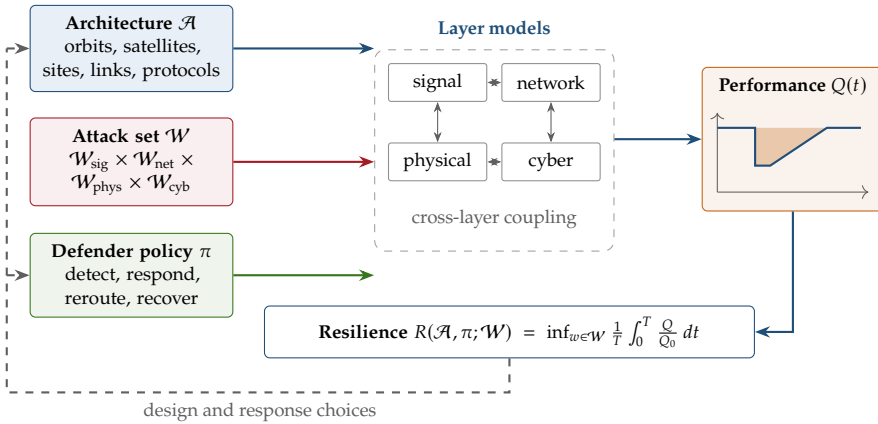


Figure 1.13. The systems-science scheme of the book as a block diagram. Architecture, attack set, and defender policy drive coupled layer models whose output is mission performance over time; the resilience functional condenses that curve into a single figure of merit, which in turn guides the choice of architecture and policy (dashed feedback).

its short satellite lives demand a continuous, high-tempo launch and software-update pipeline, each element of which is an attack surface. Coverage geometry, capacity, and cost all complicate the comparison, and [Chapter 17](#) develops it as a resilience-cost tradespace.

1.9 A Systems-Science and Quantitative Perspective

The literature on space security has tended to divide into two groups: descriptive policy and threat surveys, which cover the whole field but rarely quantify, and rigorous technical papers on narrow problems, which quantify carefully but rarely connect to mission outcomes. This book aims to connect them. Its unifying scheme is simple to state:

$$\underbrace{\mathcal{A}}_{\text{architecture}} + \underbrace{\mathcal{W}}_{\text{attack set}} + \underbrace{\pi}_{\text{defender policy}} \longrightarrow \underbrace{Q(t)}_{\text{mission performance over time}}. \quad (1.2)$$

Each chapter builds a piece of the map from $(\mathcal{A}, \mathcal{W}, \pi)$ to $Q(t)$ for one layer or one interaction between layers, and the resilience functional (1.1) turns that map into a figure of merit for design and comparison. [Figure 1.13](#) draws the scheme as a block diagram.

This perspective is *systems-scientific* in three senses. It treats a space system as a system-of-systems whose behavior emerges from the interaction of physical dynamics, communication networks, software, and human decisions. It treats the adversary as a strategic, adaptive agent rather than as a fixed failure rate. And it insists on metrics tied to mission outcomes, so that a defense at one layer can be traded against a defense at another. The feedback loop in [Fig. 1.13](#) captures the last point: the defender chooses $\mathcal{A}$ and π to make R large, while the adversary's choice within $\mathcal{W}$ tries to make it small.

Filling in the blocks requires several families of mathematics. Probability and stochastic processes answer how often failures and attacks occur and how long recovery and kill-chain stages take. Detection and information theory answer whether spoofing can be detected, how much an eavesdropper can learn, and what secrecy capacity a link offers. Game theory answers how a defender should allocate anti-jam or monitoring resources against an adaptive adversary. Estimation and control answer whether orbit determination or attitude control can remain safe when measurements or commands are manipulated. Network science and graph theory answer how satellite losses or routing attacks fragment a constellation network and how failures propagate between interdependent networks. Optimization, finally, answers which architecture, sourcing strategy, or defense allocation maximizes resilience per unit cost.

A method is taught the first time the book needs it, and later material refers back to it instead of re-deriving it. Readers who want background in advance can consult the appendices on probability and stochastic processes, optimization and game theory, estimation and control, and orbital mechanics and RF communications.

1.10 How to Use This Book

1.10.1 Structure

The book has twenty chapters in six Parts, followed by six appendices. [Figure 1.14](#) shows every chapter, colored by Part, together with the dependencies among them.

Part I, *Foundations* (Chapters 1 to 4), sets up the common language. After this motivating chapter, Chapter 2 is a primer on orbits, segment architecture, RF link budgets, and the communication and computing stacks of space systems, each mapped onto the security layers. Chapter 3 defines security, resilience, and related notions precisely and develops the quantitative resilience metrics previewed here. Chapter 4 presents the layered threat taxonomy, the space kill chain, adversary models (including a first look at game-theoretic ones), and risk-assessment methods.

Part II, *Communication and Network Security* (Chapters 5 to 7), covers physical-layer signal security (jamming, spoofing, eavesdropping, and covert communication), then link security and space cryptography, and then the security of space networks and constellation routing. Part III, *Physical Security* (Chapters 8 to 10), covers kinetic, directed-energy, and nuclear threats and protection; the space environment, debris, and sustainability; and space domain awareness (SDA), proximity operations, and orbital games.

Part IV, *Cyber and Cyber-Physical Security* (Chapters 11 to 15), covers cybersecurity of the ground and user segments and of onboard computing, then the cross-layer cyber-physical framework that integrates all earlier layers, followed by resilient estimation and timing and resilient guidance, control, and autonomy. Part V, *Architecture, Lifecycle, and Strategy* (Chapters 16 to 18), covers supply-chain, insider, and lifecycle security; resilient architecture design and reconstitution; and deterrence, escalation, law, and governance. Part VI, *Integration and Frontiers* (Chapters 19 and 20), applies the frameworks to integrated case studies, wargames,

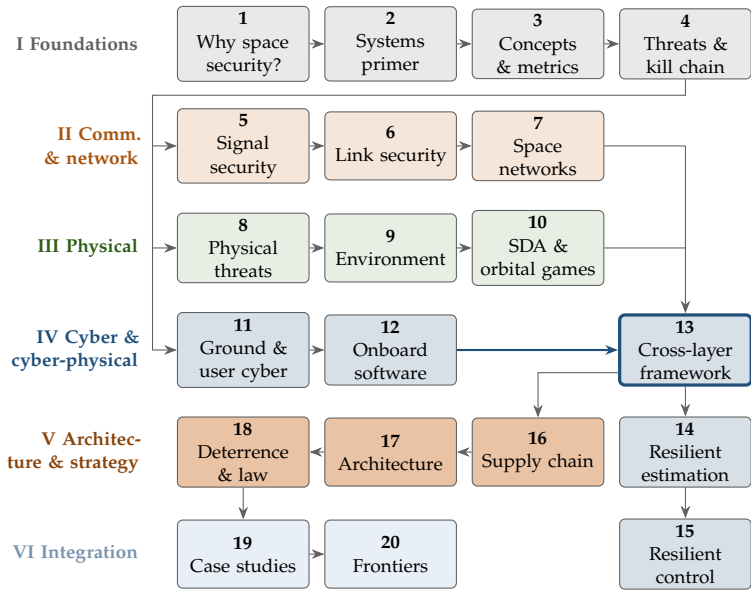


Figure 1.14. The Parts and chapters of the book and their dependencies. Part I is prerequisite to all later Parts. The three layer-specific tracks (Part II, Part III, and the first two chapters of Part IV) are independent of one another and converge on the cross-layer framework of Chapter 13 (bold outline), which leads to resilient estimation and control and to Parts V and VI.

and digital twins, and closes with open problems. The appendices cover probability and stochastic processes (A), optimization and game theory (B), estimation and control (C), orbital mechanics and RF communications (D), standards and frameworks (E), and tools, data, notation, and a glossary (F).

1.10.2 Reading paths

The book supports several courses. Table 1.2 lists suggested paths; instructors will adapt them to their students’ backgrounds. Readers from security studies and policy can read Chapters 1, 3, 4, 8 to 10, 18, and 19 with only light mathematics, and the quantitative sections can be skimmed on a first reading without losing the argument.

1.10.3 Prerequisites and pedagogical features

We assume undergraduate probability, linear algebra, and calculus. No prior knowledge of orbital mechanics, communications theory, cybersecurity, or game theory is assumed; the primer in Chapter 2 and the appendices supply what is needed. Readers with a strong background in one area will find some material familiar, and the layered organization is designed to let them move quickly through their own specialty and slow down elsewhere.

Every chapter follows a common template, of which this chapter is the first instance. It opens with learning objectives and a layer map showing which cells of the Security-Resilience Cube the chapter covers, followed by a HELIOS vignette.

Table 1.2. Suggested course paths through the book.

Course	Chapters
Graduate: space security engineering (overview)	1 to 5, 7, 8, 11, 13, 17
Graduate: satellite communication and network security	1 to 7, 11, 13
Graduate: space cybersecurity	1 to 4, 6, 11, 12, 13, 16
Graduate: cyber-physical and control security	1 to 4, 10, 12, 13 to 15, App. B and C
Security studies / policy	1, 3, 4, 8 (light), 9, 10 (light), 18, 19
Professional short course	1, 3, 4, 5, 8, 11, 13, 19
Research seminar	3, 7, 10, 13 to 15, 17, 20

HELIOS recurs in worked examples, consistent with the canonical parameters of Table 1.1, and worked examples use real numbers so that every step can be checked. “Space Security in Practice” boxes summarize real, well-documented incidents or programs and read them through the book’s frameworks. Each chapter closes with a prose summary, exercises, and notes on further reading. Exercises are tagged [C] for conceptual, [A] for analytical (derivations and calculations by hand), and [P] for computational (short Python tasks). One [P] exercise per chapter is designated the chapter lab; the labs build toward a multi-layer HELIOS model and a capstone case study, and companion code is kept outside the text.

1.10.4 A note on scope and stance

The book takes the defensive and analytical stance that is standard in the field. Threat physics and attack models are presented at the level of the open, peer-reviewed, and textbook literature, with the aim of understanding exposure and designing defenses; the book does not provide operational attack procedures. When discussing the behavior of states, we describe what authoritative public sources report and flag contested claims as such.

1.11 Summary

Space services, namely PNT, communications, and Earth observation, have become critical infrastructure. Their value is large, on the order of \$1 trillion in cumulative U.S. benefits from GPS alone, and their disruption cascades through intermediate infrastructures such as timing and telecommunications into finance, energy, transport, and defense. Over six decades space has moved from a perceived sanctuary to a congested, contested, and competitive domain. Destructive ASAT tests have left long-lived debris, while the capabilities most used in current conflicts are non-destructive: jamming, spoofing, and cyber operations. Space also differs from terrestrial domains in ways that matter for security, including predictable orbits, the absence of physical repair, SWaP limits, long lifecycles and patch cycles, dual use, difficult attribution, debris externalities, commercial dependence, and shared commons.

Space security is multi-layered. The Space Security Stack separates signal/RF, link and network, physical/orbital, cyber, and mission/strategic problems, each

with its own threats and mathematics, and many consequential attacks cross between layers. The KA-SAT attack of 2022 is the reference example: it entered through a ground network, disabled user terminals, and reached third parties far from the conflict, while the satellite itself was untouched.

Security aims to prevent attacks from succeeding, and resilience aims to preserve and restore mission performance when they do. The resilience curve and the functional $R(\mathcal{A}, \pi; \mathcal{W})$ make the latter measurable, rewarding both shallow drops and fast recovery and evaluating a design against the worst attack in a layered attack set. The Security-Resilience Cube (Layer $\times$ Segment $\times$ Phase) locates every problem in the book, and the eight-stage space kill chain adds the time dimension of an adversary campaign together with the defender's opportunities to interrupt it.

Finally, the notional HELIOS constellation, contrasted with the notional GEO system ATLAS-G, provides consistent numbers and a single system in which models from all layers can be composed. The systems-science scheme that maps architecture, attack set, and defender policy to mission performance over time is the thread that ties the rest of the book together.

Exercises

Exercise 1.1 [C] Choose one sector not treated in detail in [Section 1.1.2](#) (for example, broadcasting, ports, water utilities, or hospitals). Build a space-dependency map for it in the style of [Fig. 1.2](#): identify the space services it uses directly, the intermediate infrastructures through which it uses them indirectly, and at least one hidden dependency. Mark the node whose loss would affect the most downstream functions.

Exercise 1.2 [C] Place each of the following in the Security-Resilience Cube (layer, segment, phase), noting where a problem occupies more than one cell: (a) the FY-1C test, from the perspective of an operator whose satellite shares its orbital region; (b) the January 2016 GPS UTC offset anomaly, from the perspective of a telecommunications operator; (c) the February 2022 Starlink losses to a geomagnetic storm; (d) the Kosmos-1408 test, from the perspective of the International Space Station; (e) the release of a signed firmware update to all HELIOS user terminals.

Exercise 1.3 [C] Of the ten properties discussed in [Section 1.3](#), pick the two that you believe most constrain defenders of a proliferated LEO constellation, and the two that most constrain defenders of a single GEO satellite. Justify your choices, and explain why the lists differ.

Exercise 1.4 [C] Using the KA-SAT box, identify which of the eight kill-chain stages of [Section 1.7](#) are supported by public information and which are not. For two of the stages, propose a defensive measure that could have interrupted the campaign at that stage, and state which layer and segment the measure acts on.

Exercise 1.5 [C] Give an example of a design choice for HELIOS that improves security but reduces resilience, and one that improves resilience but may reduce

security. (Hint: consider authentication requirements during safe mode, or adding more GSaaS providers.)

Exercise 1.6 [C] Return to the opening vignette. List the options available to the HELIOS operators for uploading the collision-avoidance maneuver, and for each option identify (a) the layers at which it carries risk, and (b) what additional information would most change your choice. Do not attempt to compute probabilities; the goal is to structure the decision.

Exercise 1.7 [C] Explain, with reference to at least two properties from [Section 1.3](#), why reversible counterspace actions such as jamming may be more destabilizing for deterrence than their reversibility alone suggests.

Exercise 1.8 [A] A GNSS-disciplined clock is in error by $13\ \mu\text{s}$, as in the 2016 UTC offset anomaly. (a) What range error does this correspond to if the clock were used for ranging at the speed of light? (b) A cellular network requires phase alignment between neighboring base stations to within $\pm 1.5\ \mu\text{s}$. If one base station's clock drifts during holdover at a constant rate of $0.5\ \mu\text{s}$ per hour after losing GNSS, how long does it remain within tolerance? (c) Discuss why the answer to (b) makes holdover duration a natural resilience metric for timing users.

Exercise 1.9 [A] For the resilience functional (1.1) with $T = 24\ \text{h}$ and $Q_0 = 1$, consider a family of attacks parameterized by drop depth $d \in (0, 1]$ and linear recovery time $\tau > 0$, with no withstand plateau, so that the loss area is $d\tau/2$ (assume $\tau \leq 24\ \text{h}$). (a) Show that the normalized integral is $1 - d\tau/48$. (b) If the adversary can choose (d, τ) freely subject to $d\tau \leq c$, what is R ? (c) Suppose the defender can halve recovery times at a cost, or halve drop depths at the same cost. Under this model, which is better, and why is the answer different if the attack set is not of the form $d\tau \leq c$?

Exercise 1.10 [A] HELIOS Shell A has 528 satellites and is replenished at about 120 satellites per year across both shells. (a) If an event destroys k Shell A satellites and replacements are taken from the normal production flow, estimate how many days of production are needed to replace $k = 22$ and $k = 66$ satellites. (b) Treating Shell A capacity as proportional to the number of working satellites (a strong simplification), compute the loss area in satellite-days of capacity for each case if satellites are replaced at a uniform rate over the replacement period. (c) Compare with a notional ATLAS-G loss with a 3-year replacement time. State at least two reasons why the proportional-capacity assumption overstates or understates HELIOS's resilience.

Exercise 1.11 [P] **Lab (resilience curves).** Write a short Python program that takes a piecewise-linear performance profile $Q(t)$ (a list of breakpoints) and returns the depth, time to recover, loss area, and normalized integral over a horizon T . Use it to reproduce the numbers in [Example 1.2](#). Then generate 1,000 random attacks with drop depth uniform on $[0.1, 0.7]$, plateau uniform on $[0, 3]\ \text{h}$, and recovery time uniform on $[1, 12]\ \text{h}$, and report both the worst case (the empirical infimum) and the mean normalized integral. Comment on how much the two differ, and what this suggests about worst-case versus risk-averse resilience measures such as conditional value-at-risk.

Exercise 1.12 [P] Encode the dependency graph of Fig. 1.2 as a directed graph in Python. (a) For each space-service node, compute the set of sectors reachable from it. (b) Assume each edge independently transmits a disruption with probability p ; estimate by Monte Carlo simulation the expected number of disrupted sectors when GNSS/PNT fails, for $p \in \{0.2, 0.5, 0.8\}$. (c) Remove the timing node's edges to finance and the grid (representing independent holdover clocks) and repeat. Interpret the difference.

Notes and Further Reading

Space as critical infrastructure. The RTI/NIST study of the economic benefits of GPS [1] is the most detailed sector-by-sector analysis of U.S. dependence on satellite navigation and timing; its chapters on telecommunications, electricity, and agriculture are especially relevant to cascading-dependency analysis. The London Economics study [2] provides a comparable analysis for the United Kingdom. The NIST analysis of the 2016 UTC offset anomaly [3] is a compact, well-documented account of how a small space-segment software error reached terrestrial timing users.

History, threats, and the strategic environment. The Secure World Foundation's annual *Global Counterspace Capabilities* report [8] and the CSIS *Space Threat Assessment* [20] are the standard open-source references on counterspace capabilities by country; both are updated regularly and should be consulted in their latest editions. The 2011 U.S. National Security Space Strategy [16] introduced the "congested, contested, competitive" framing. The Outer Space Treaty [5] and UNGA Resolution 77/41 [14] are the primary texts for the legal thread, which Chapter 18 develops in depth. On debris, the original Kessler-Cour-Palais paper [6] remains readable and instructive; ESA's annual Space Environment Report [17] is the authoritative source on the current population. Kelso's analysis of the Iridium-Cosmos collision [9] and Berger et al. on the 2022 Starlink losses [19] show how congestion and the environment create security-relevant risk in the absence of an adversary. On nuclear effects on satellites, see the DTRA assessment of Conrad et al. [4].

Space cybersecurity. Falco [32] made an early, influential case that commercial space cybersecurity was badly neglected. Pavur and Martinovic [33] survey satellite cybersecurity research and organize threat models across disciplines; their survey is an excellent entry point for readers from computer security. Government guidance includes Space Policy Directive-5 [34], NIST IR 8270 on commercial satellite operations [35], and NIST IR 8401 on applying the Cybersecurity Framework to the satellite ground segment [36]. The Aerospace Corporation's SPARTA matrix [31] catalogs space-cyber tactics, techniques, and countermeasures. For KA-SAT, the primary sources are Viasat's overview [12], the SentinelLabs analysis [23], the EU attribution statement [13], and the CISA/FBI advisory [25].

Resilience. The four-phase view of resilience used in this book comes from NIST SP 800-160 Vol. 2 [26], which also catalogs cyber-resiliency techniques applicable to space systems. The resilience-curve (“resilience triangle”) tradition originates in infrastructure engineering with Bruneau et al. [28]; Linkov and Kott [29] give an accessible overview of how these ideas carry over to cyber systems. Zhu [37] gives a systems-theoretic account of cyber resilience that combines game, control, and learning theories. The kill-chain concept is due to Hutchins, Cloppert, and Amin [30]. [Chapters 3](#) and [4](#) provide the full treatment and further references.

Space systems background. Readers new to space systems will find Wertz, Everett, and Puschell [38] a comprehensive reference on mission design, including orbits, link budgets, and ground systems; [Chapter 2](#) gives the security-oriented subset needed for this book.

Space Systems Primer for Security Analysis

Learning Objectives

- ▶ Describe the main orbital regimes and compute periods, sun-synchronous inclinations, ground tracks, and access windows from first principles.
- ▶ Compute slant range, footprint, and pass duration for a satellite-ground-station pair, and explain what these mean for command access and adversary access.
- ▶ Name the components of the ground, link, space, and user segments of a space system and the trust boundaries between them.
- ▶ Compute a satellite link budget (EIRP, free-space path loss, G/T , C/N_0 , E_b/N_0) and interpret link margin as a security quantity.
- ▶ Map the space communication stack (CCSDS, SDLS, DVB-S2X, DTN, 3GPP NTN) onto the layers of the Space Security Stack.
- ▶ Describe ground and onboard computing, the onboard control loop, and the mission lifecycle, and locate the attack surfaces in each.

This chapter supplies the engineering baseline on which security analysis of a space system rests. It proceeds from the physics of orbits, visibility, and constellations (Sections 2.1 to 2.3) to the architecture of a space system and the radio links that join its parts (Sections 2.4 and 2.5), and then to protocols, computing, attitude control, and the mission lifecycle (Sections 2.6 to 2.9). Each section closes by stating what its engineering content implies for security.

Throughout the book, security questions are organized by the *Space Security Stack*, which crosses five layers (mission/strategic, cyber, physical/orbital, link and network, and signal/RF) with four segments (ground, link, space, and user). A primer must touch every cell of that grid, but it develops some cells quantitatively and others only descriptively. Figure 2.1 shows the distinction: the orbital and radio-frequency rows are treated with equations and worked examples, while the cyber and mission rows are introduced at the level of architecture.

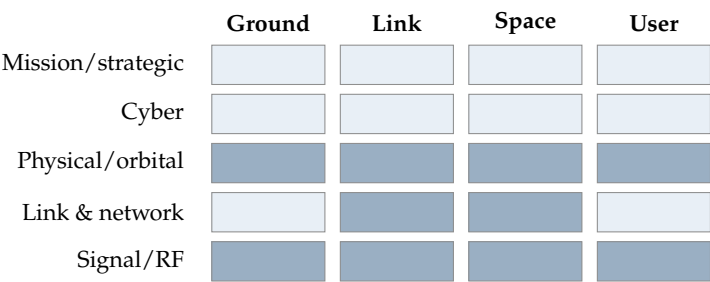


Figure 2.1. Layer map for this chapter. Dark cells (the physical/orbital and signal/RF layers, and the link-and-network layer on the link and in space) are developed quantitatively. Light cells are introduced descriptively, so that every cell of the Space Security Stack has at least a primer-level treatment.

HELIOS Running Example

HELIOS is a *notional* proliferated low-Earth-orbit (LEO) broadband and PNT-augmentation constellation, created for teaching and not modeled on a real operator. It has 600 satellites in two shells, optical inter-satellite links, a cloud-hosted mission operations center, a partly leased TT&C network, 40 Ka-band gateways, and about 200,000 user terminals.

On a routine morning the duty analyst at the HELIOS security operations center receives three reports. A maritime customer in the North Atlantic sees throughput on one vessel fall from about 1 Gbps to roughly 700 Mbps several times an hour before recovering. A leased TT&C antenna in the southern hemisphere has missed a scheduled 7 min pass, which delays a software patch for satellite A-07-13 to the next contact. A polar-shell satellite reports an unexpected reaction-wheel speed excursion over the South Atlantic.

Each report could indicate an attack, and each could be ordinary behavior: a satellite sinking toward the coverage edge, a scheduling error, a radiation upset. Separating these cases requires a quantitative model of normal operation. [Example 2.5](#), for instance, shows that the clear-sky capacity of a HELIOS user beam falls by a third between nadir and the 25° coverage edge with no adversary present.

The vignette illustrates a methodological principle that runs through the book: *security analysis of a space system starts from a quantitative model of its nominal behavior*. Detection needs a baseline, and attribution must first rule out physics. Resilience analysis, which tracks a mission performance function $Q(t)$ as it degrades under attack and recovers, needs the nominal value Q_0 from which degradation is measured. This chapter computes those baselines for HELIOS. The notation follows the list at the front of the book; [Table 2.1](#) restates the HELIOS parameters on which the worked examples draw.

Table 2.1. HELIOS and ATLAS-G parameters used in this chapter. Both systems are notional; values follow the book’s reference specification.

Element	Value
Shell A	528 satellites, circular, 540 km, $i = 53^\circ$, Walker-delta 53°: 528/24/1
Shell B	72 satellites, 560 km, sun-synchronous ($i \approx 97.6^\circ$), 6 planes of 12 (Walker-star)
Satellite	300 kg, 3 kW end of life, Hall-effect propulsion, 5-year life; ARM-based OBC with cold spare, cFS flight software, separate payload processor
Inter-satellite links	4 optical terminals (2 intra-plane, 2 cross-plane), 20 Gbps each
User downlink	Ku band, 10.7 GHz to 12.7 GHz (11.7 GHz used), 250 MHz channels, EIRP 34 dBW per beam, DVB-S2X
User terminal	$G/T = 9$ dB/K (0.5 m-class phased array); Ku losses $L_a = 1.5$ dB; $\varepsilon_{\min} = 25^\circ$
Feeder links	Ka band, 17.8 GHz to 19.3 GHz down, 27.5 GHz to 30.0 GHz up; 40 gateways
TT&C	S band, 2025 MHz to 2110 MHz up, 2200 MHz to 2290 MHz down; CCSDS with SDLS (AES-GCM) on commands; satellite EIRP -3 dBW; ground $G/T = 15$ dB/K; 64 kbps telemetry; $\varepsilon_{\min} = 10^\circ$
Ground segment	Cloud-hosted MOC with warm-standby backup; 2 owned and 6 leased (GSaaS) TT&C sites; NOC/SOC in the core network; CI/CD pipeline for signed updates
User segment	About 200,000 terminals (maritime, aviation, government, enterprise); over-the-air firmware; L-band PNT augmentation
Replenishment	About 120 satellites per year in batches of about 30
ATLAS-G (contrast)	One GEO Ka-band high-throughput satellite, 100 spot beams, 15-year life; 8 gateways

2.1 Orbits and Orbital Regimes

Orbits are the most distinctive feature of space systems. A satellite cannot stop, hide behind a hill, or change its path quickly. It follows a trajectory set mostly by gravity, which an observer with a telescope or radar and a textbook can predict days ahead. This section starts from the two-body problem, adds the dominant perturbation (J_2), describes ground tracks, and surveys the orbital regimes. The textbooks of Vallado [39] and Curtis [40] give full derivations.

2.1.1 The two-body problem

Treat the Earth as a point mass (or a spherically symmetric body) of gravitational parameter $\mu = GM_\oplus \approx 398\,600.4418 \text{ km}^3/\text{s}^2$ and the satellite as a test mass. Newton's law gives

$$\ddot{\mathbf{r}} = -\frac{\mu}{\|\mathbf{r}\|^3} \mathbf{r}, \quad (2.1)$$

where $\mathbf{r}$ is the satellite's position vector from the Earth's center. The solutions are conic sections. Bound orbits are ellipses with the Earth's center at one focus. Two conserved quantities, specific energy and specific angular momentum, give the two relations we use most.

Proposition 2.1 (Vis-viva and Kepler's third law). *For an elliptical orbit with semi-major axis a , the speed v at radius r and the orbital period T satisfy*

$$v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right), \quad T = 2\pi \sqrt{\frac{a^3}{\mu}}. \quad (2.2)$$

For a circular orbit ($r = a$) the speed is $v_c = \sqrt{\mu/a}$.

Example 2.1 (HELIOS periods and speeds). Take the equatorial radius $R_\oplus = 6378.137 \text{ km}$. Shell A orbits at $h = 540 \text{ km}$, so $a = 6918.137 \text{ km}$ and

$$T_A = 2\pi \sqrt{\frac{(6918.137)^3}{398600.4418}} \text{ s} = 5726.6 \text{ s} = 95.44 \text{ min},$$

$$v_A = \sqrt{\frac{398600.4418}{6918.137}} \text{ km s}^{-1} = 7.59 \text{ km s}^{-1}.$$

Shell B at 560 km has $T_B = 95.86 \text{ min}$ and $v_B = 7.58 \text{ km s}^{-1}$. These values match the HELIOS specification (Table 2.1). Each satellite completes about 15.1 orbits per day. A geostationary satellite needs T equal to one sidereal day ($86\,164.1 \text{ s}$). Inverting Kepler's law gives $a = 42\,164 \text{ km}$, or an altitude of $35\,786 \text{ km}$, where $v = 3.07 \text{ km s}^{-1}$.

2.1.2 Orbital elements and element sets

Six numbers fix a Keplerian orbit and the satellite's place on it (Fig. 2.2). The *semi-major axis* a and *eccentricity* e give size and shape. The *inclination* i is the tilt of the orbit plane relative to the equator. The *right ascension of the ascending node* (RAAN) Ω is the angle, measured in the equatorial plane from the vernal

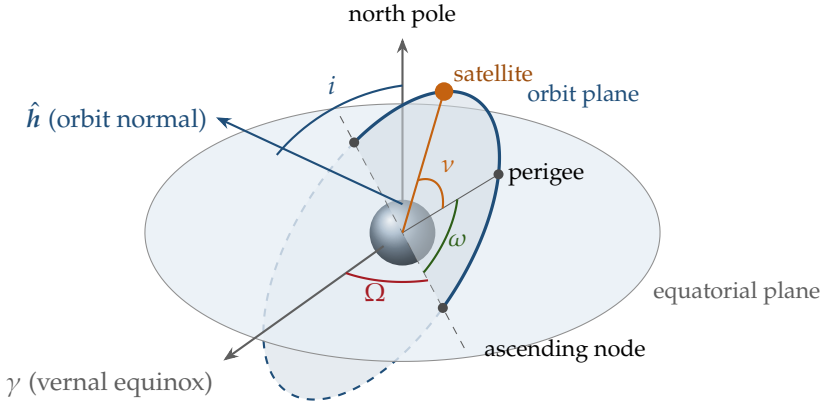


Figure 2.2. Classical orbital elements. The RAAN Ω locates the ascending node from the vernal equinox direction γ , the inclination i tilts the orbit plane about the line of nodes, and the argument of perigee ω and true anomaly ν locate the perigee and the satellite within the plane. The orbit plane is shaded, dashed where it lies below the equator, and its normal $\hat{h}$ makes the angle i with the north axis. The eccentricity is exaggerated; for the circular HELIOS orbits only $u = \omega + \nu$ is meaningful.

equinox direction, at which the satellite crosses the equator northbound. The *argument of perigee* ω orients the ellipse within its plane, and the *true anomaly* ν , or equivalently the mean anomaly M at an epoch, locates the satellite along the orbit. For the circular orbits of HELIOS, $e \approx 0$ and ω is undefined, so it is conventional to use the argument of latitude $u = \omega + \nu$ instead. In a constellation, satellites in the same plane share (a, e, i, Ω) and differ only in u , while different planes differ in Ω .

Most unclassified orbit information is distributed as *two-line element sets* (TLEs). These are fixed-format, 69-character lines that the U.S. Space Force publishes through Space-Track and that are redistributed by services such as CelesTrak. A TLE holds *mean* elements, so it must be propagated with the model used to fit it, the Simplified General Perturbations model SGP4 (with its deep-space extension SDP4). SGP4 was documented in Spacetrack Report No. 3 [41] and later revised and released as open reference code [42]. It includes secular and periodic effects of J_2 through J_4 , a simple atmospheric drag term (the B^* coefficient), and, for deep-space orbits, lunar-solar terms. TLE accuracy depends on the object, regime, and age of the element set. Errors are typically on the order of a kilometer near epoch and grow over days, especially for low, drag-dominated orbits [42]. Operators that need better accuracy, for conjunction screening for example, use their own GNSS-derived ephemerides or special-perturbations catalogs [43].

2.1.3 The J_2 perturbation and sun-synchronous orbits

The Earth is oblate: its equatorial radius exceeds its polar radius by about 21 km. The dominant term in the resulting gravity field is the second zonal harmonic, $J_2 \approx 1.08263 \times 10^{-3}$. Averaged over an orbit, J_2 causes no secular change in a , e , or i , but it makes the orbit plane precess about the Earth's polar axis and the line

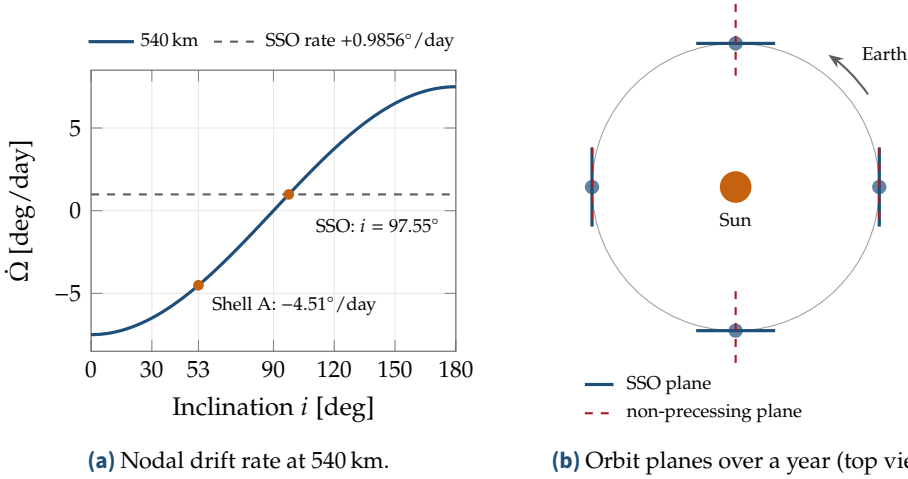


Figure 2.3. The J_2 perturbation. (a) Prograde orbits regress and retrograde orbits advance; the SSO inclination is where the curve meets the solar rate. (b) An SSO plane (solid, shown edge-on) keeps a fixed angle to the Sun as the Earth orbits, whereas a plane without precession (dashed) stays fixed in inertial space and sweeps through all Sun angles in a year.

of apsides rotate within the plane [39].

Proposition 2.2 (Secular nodal precession). *Under J_2 , the RAAN drifts at the mean rate*

$$\dot{\Omega} = -\frac{3}{2} J_2 \left(\frac{R_{\oplus}}{p} \right)^2 n \cos i, \quad n = \sqrt{\mu/a^3}, \quad p = a(1 - e^2). \quad (2.3)$$

Prograde orbits ($i < 90^\circ$) regress westward. Retrograde orbits ($i > 90^\circ$) advance eastward.

For HELIOS Shell A ($h = 540$ km, $i = 53^\circ$), (2.3) gives $\dot{\Omega} \approx -4.51^\circ$ per day. All 24 planes precess at the same rate, so the constellation pattern of Section 2.3 is preserved. The planes move relative to the Sun, however, completing a full cycle relative to the Sun in about $360^\circ / (4.51^\circ + 0.99^\circ) \approx 65$ days. A *sun-synchronous orbit* (SSO) chooses i so that $\dot{\Omega}$ equals the Earth's mean motion around the Sun, $360^\circ / 365.2422$ days = 0.9856° per day = 1.991×10^{-7} rad s $^{-1}$. The orbit plane then keeps a fixed orientation relative to the Sun, and the satellite crosses each latitude at the same local solar time every day. Figure 2.3 shows both the dependence of $\dot{\Omega}$ on inclination and the resulting geometry over a year.

Example 2.2 (Sun-synchronous inclination for HELIOS Shell B). Shell B is circular at 560 km, so $a = p = 6938.137$ km and

$$n = \sqrt{\frac{398600.4418}{6938.137^3}} = 1.09246 \times 10^{-3} \text{ rad s}^{-1},$$

$$\frac{3}{2} J_2 \left(\frac{R_{\oplus}}{a} \right)^2 n = 1.49926 \times 10^{-6} \text{ rad s}^{-1}.$$

Setting $\dot{\Omega} = +1.99106 \times 10^{-7} \text{ rad s}^{-1}$ in (2.3) gives

$$\cos i = -\frac{1.99106 \times 10^{-7}}{1.49926 \times 10^{-6}} = -0.13280 \implies i = 97.63^\circ.$$

This confirms the specification value $i \approx 97.6^\circ$. At 540 km the same calculation gives 97.55° , the point marked in Fig. 2.3. The SSO inclination rises slowly with altitude, reaching about 98.6° at 800 km.

2.1.4 Ground tracks

The *ground track* is the path of the sub-satellite point on the rotating Earth. Its maximum latitude equals i for prograde orbits and $180^\circ - i$ for retrograde orbits. HELIOS Shell A therefore does not pass directly over latitudes above 53° , although, as Section 2.2 shows, it can still be seen from somewhat higher latitudes. While the satellite completes one orbit, the Earth turns eastward by $\omega_\oplus T$, where $\omega_\oplus = 7.2921 \times 10^{-5} \text{ rad s}^{-1}$. Each successive equator crossing is therefore displaced westward by

$$\Delta\lambda \approx \omega_\oplus T + |\dot{\Omega}|T, \quad (2.4)$$

where the plus sign applies to prograde (westward-regressing) orbits. For Shell A this is $23.93^\circ + 0.30^\circ \approx 24.2^\circ$ per revolution. Figure 2.4 plots three consecutive revolutions computed from two-body motion with J_2 nodal regression, together with one revolution of a Shell B satellite. If the ratio of the Earth's rotation period relative to the node to T is rational, the track eventually repeats. *Repeat-ground-track* orbits are common for Earth observation, because they revisit the same sites under the same geometry.

2.1.5 Orbital regimes

Table 2.2 summarizes the regimes used throughout the book. The boundaries are conventions and differ slightly between sources [38]. Figure 2.5 plots period and propagation delay as functions of altitude for circular orbits, and it makes the central trade visible: moving from LEO to GEO multiplies the period by about fifteen and the propagation delay by about seventy.

Space Security in Practice: Launch Into a Geomagnetic Storm (Starlink, February 2022)

On 3 February 2022, SpaceX launched 49 Starlink satellites into a deliberately low insertion orbit, with a perigee reported at roughly 210 km. Operators use this practice so that a satellite that fails checkout reenters quickly. A geomagnetic storm was under way at the time. It heated and expanded the thermosphere, and SpaceX reported that atmospheric drag at the deployment altitude rose by up to about 50 % over previous launches [44]. The operator commanded the satellites into a safe mode that flew them edge-on to reduce drag, but most could not raise their orbits. SpaceX initially said up to 40 would reenter, and a subsequent space-weather analysis puts the loss at 38 of the 49 satellites [45].

Table 2.2. Orbital regimes and their security-relevant properties. Periods are for circular orbits at the stated altitudes. “Latency” is the one-way user-satellite propagation delay at nadir.

Regime	Altitude	Period; latency	Typical missions	Security-relevant properties
LEO	160 to 2000 km	88 to 127 min; 1 to 7 ms	Broadband (HELIOS), Earth observation, crewed stations	Reachable by direct-ascent interceptors and ground-based lasers; short passes; drag limits lifetime at low altitude; proliferation feasible; each satellite sees a small footprint
MEO	2000 to 35 786 km (GNSS near 20 200 km)	2 to 24 h; GNSS ≈ 67 ms	GNSS, O3b-class broadband	Radiation belts stress electronics; few, high-value GNSS satellites; weak received PNT signals invite jamming and spoofing
GEO	35 786 km, $i \approx 0^\circ$	23 h 56 min; 119 ms	SATCOM (ATLAS-G), weather, missile warning	Fixed in the sky: continuous access for users <i>and</i> adversaries; valuable slots; co-orbital inspectors can loiter; uplink jamming from across the visible hemisphere
HEO	Perigee 500 to 1000 km, apogee $\approx 40\,000$ km	12 to 24 h	Molniya/Tundra high-latitude comms, early warning	Long dwell over high latitudes near apogee; passes repeatedly through radiation belts
Cislunar	Beyond GEO to lunar distance ($\approx 384\,400$ km)	Days to weeks; ≈ 1.3 s	Lunar gateways, science, relay	Sparse sensor coverage and weak space domain awareness; three-body dynamics; long delays favor autonomy and delay-tolerant networking

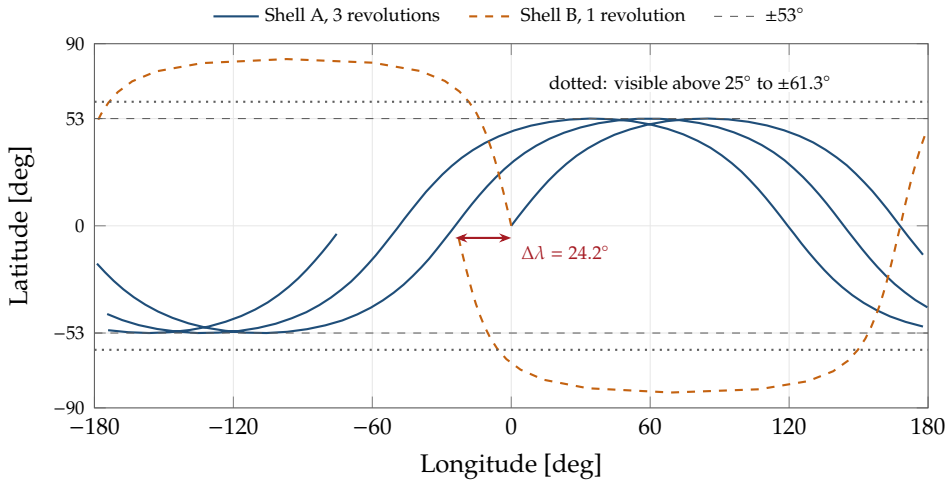


Figure 2.4. Ground tracks of a HELIOS Shell A satellite (three revolutions from an ascending node at 0° longitude) and a Shell B satellite (one revolution). Successive Shell A equator crossings shift westward by 24.2° . Shell A does not overfly latitudes above 53° , but its 25° footprint reaches 61.3° ; the retrograde SSO track reaches 82.4° and covers the polar regions.

The episode carries three lessons for security analysis. First, the orbit itself was a vulnerability during the most fragile phase of the lifecycle (launch and early orbit, [Section 2.9](#)), when satellites are low, not yet commissioned, and dependent on ground commands. Second, it shows how hard attribution is: a sudden loss of many satellites had a natural explanation, which an analyst must rule out before suspecting an adversary. Third, *drag-dominated orbits are predictable only as far as the environment is predictable*. That uncertainty complicates the defender's conjunction screening, and it equally complicates an adversary's tracking.

The security significance of this section can be stated compactly. Orbital motion is *public, predictable, and slow to change*. An observer with TLEs and SGP4 can compute when a satellite will pass over a given point, when it will be sunlit, and where it will be tomorrow, and this predictability fixes when an adversary can observe, jam, dazzle, or engage a satellite as well as when the defender can command it. Changing an orbit costs propellant (Δv), a finite resource, so maneuver is an expensive defensive option rather than a routine one. Regime choice sets exposure: LEO satellites are within reach of more counterspace systems but can be proliferated, while GEO satellites are few, valuable, and continuously in view of a hemisphere. The J_2 and drag effects above also govern how debris spreads and how long it persists. [Chapter 10](#) builds tracking, characterization, and orbital-game models on this predictability.

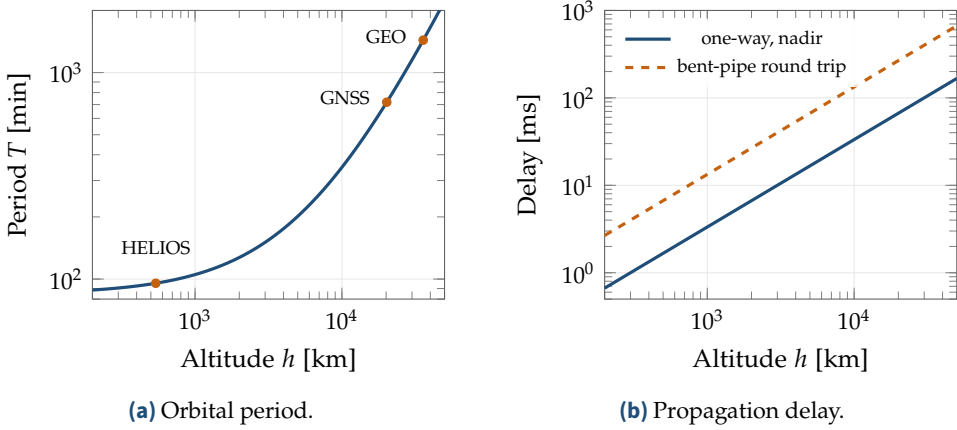


Figure 2.5. Period and propagation delay of circular orbits as functions of altitude. HELIOS at 540 km has a one-way nadir delay of 1.8 ms and a bent-pipe round trip (user-satellite-gateway and back) of about 7 ms. The same round trip via GEO exceeds 475 ms.

2.2 Visibility and Access Geometry

A ground station can exchange signals with a satellite only when the satellite is above the station's local horizon, or more precisely above a *minimum elevation angle* $\varepsilon_{\min}$. The minimum elevation is set by terrain masking, atmospheric loss, multipath, and regulatory limits. Access geometry turns the orbit into contact opportunities, and it is the same for the operator's TT&C antenna as for every other antenna nearby.

2.2.1 Elevation, nadir angle, central angle, and slant range

Consider a spherical Earth of radius $R_{\oplus}$, a satellite at altitude h , and a ground station that sees the satellite at elevation angle ε (Fig. 2.6). Three other angles describe the triangle formed by the Earth's center, the station, and the satellite. The *nadir angle* η is the off-nadir angle at the satellite. The *Earth central angle* λ is the angle at the Earth's center between station and sub-satellite point. The interior angle at the station is $90^\circ + \varepsilon$. They satisfy

$$\sin \eta = \frac{R_{\oplus} \cos \varepsilon}{R_{\oplus} + h}, \quad (2.5)$$

$$\lambda = 90^\circ - \varepsilon - \eta, \quad (2.6)$$

$$d = \sqrt{(R_{\oplus} + h)^2 - R_{\oplus}^2 \cos^2 \varepsilon} - R_{\oplus} \sin \varepsilon = \frac{R_{\oplus} \sin \lambda}{\cos \varepsilon}, \quad (2.7)$$

where d is the slant range. The first follows from the law of sines, the second from the angle sum of the triangle, and the third from the law of cosines. The set of ground points that see the satellite above $\varepsilon_{\min}$ is a spherical cap, the *footprint* or coverage circle, with angular radius $\lambda_{\max}$, surface radius $R_{\oplus} \lambda_{\max}$ (with λ in radians), and area

$$A_{\text{fp}} = 2\pi R_{\oplus}^2 (1 - \cos \lambda_{\max}). \quad (2.8)$$

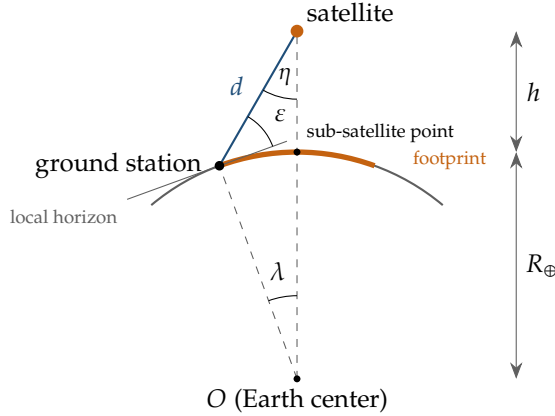


Figure 2.6. Access geometry for a satellite at altitude h seen at elevation ε from a ground station. The nadir angle η , Earth central angle λ , and elevation satisfy $\eta + \lambda + \varepsilon = 90^\circ$. The heavy arc is a cross-section of the footprint for $\varepsilon = \varepsilon_{\min}$. Not to scale.

2.2.2 Pass duration

Suppose a satellite in a circular orbit passes directly over a station, so its ground track goes through the station. It stays above $\varepsilon_{\min}$ while its central angle from the station is less than $\lambda_{\max}$, that is, for an arc of $2\lambda_{\max}$ along its orbit. Ignoring Earth rotation, the maximum pass duration is

$$t_{\text{pass,max}} \approx \frac{2\lambda_{\max}}{360^\circ} T. \quad (2.9)$$

For a pass whose ground track misses the station by a central angle λ_{off} , spherical trigonometry gives the shorter duration [38]

$$t_{\text{pass}} \approx \frac{T}{\pi} \arccos\left(\frac{\cos \lambda_{\max}}{\cos \lambda_{\text{off}}}\right). \quad (2.10)$$

Earth rotation lengthens or shortens passes by a few percent depending on direction, and a numerical propagation is needed for precise values; the chapter lab (Exercise 2.14) carries it out. Figure 2.7 evaluates (2.7) and (2.9) for the two HELIOS shells. It shows the basic tension of LEO access: lowering $\varepsilon_{\min}$ lengthens every pass, but it also doubles or triples the slant range that the link must close.

Example 2.3 (HELIOS footprint and TT&C pass). Shell A at $h = 540$ km with the user minimum elevation $\varepsilon_{\min} = 25^\circ$ gives

$$\begin{aligned} \sin \eta &= \frac{6378.137 \cos 25^\circ}{6918.137} = 0.83557 \Rightarrow \eta = 56.67^\circ, \\ \lambda_{\max} &= 90^\circ - 25^\circ - 56.67^\circ = 8.33^\circ, \\ d_{\max} &= \sqrt{6918.137^2 - (6378.137 \cos 25^\circ)^2} - 6378.137 \sin 25^\circ = 1105.2 \text{ km}. \end{aligned}$$

The footprint radius is $R_\oplus \lambda_{\max} = 6378.137 \times 0.14530 = 927$ km. By (2.8), the footprint covers $2.69 \times 10^6 \text{ km}^2$, or 0.53 % of the Earth's surface. The specification

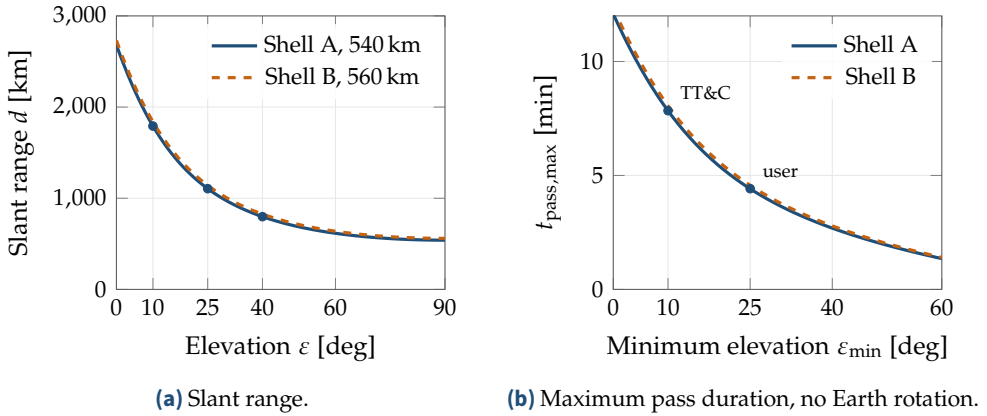


Figure 2.7. Slant range and maximum pass duration for the HELIOS shells. Marks show the TT&C ($\varepsilon_{\text{min}} = 10^\circ$) and user (25°) operating points used in Example 2.3. A 10° TT&C mask gives passes almost twice as long as the 25° user mask, at the price of a 62 % longer path.

rounds these to $\lambda \approx 8.3^\circ$ and radius ≈ 930 km. The same equations give slant ranges of 798.0 km at 40° elevation and, for a TT&C antenna with $\varepsilon_{\text{min}} = 10^\circ$, 1791.9 km with $\lambda_{\text{max}} = 14.78^\circ$.

The longest possible pass over a TT&C site with $\varepsilon_{\text{min}} = 10^\circ$ lasts about $(2 \times 14.78/360) \times 95.44$ min = 7.8 min by (2.9). A numerical propagation that includes Earth rotation gives about 8.2 min for a station directly under an ascending pass. The same calculation for a user terminal at 25° gives about 4.4 min (about 4.6 min with rotation), so a user of the 53° shell is handed over from satellite to satellite every few minutes. On average, a point on the Earth has $528 \times 0.00527 \approx 2.8$ Shell A satellites above 25° . Because the inclined shell concentrates satellites near $\pm 53^\circ$ latitude, the actual count is lower at the equator and higher at mid-latitudes.

A satellite passes over a given TT&C site only a few times a day, and most passes are much shorter than the maximum. HELIOS therefore uses eight TT&C sites (two owned, six leased) to obtain enough contact time with 600 satellites. Each pass, which lasts at most about 8 minutes, has to carry telemetry downlink, ranging, command uploads, and, occasionally, software patches, which is why the missed pass in the vignette pushed a patch to the next contact.

Access windows are *shared*, and that is their central security property. The geometry that gives HELIOS an 8-minute TT&C pass also tells a party near the station, or one with an antenna inside the satellite's footprint, when uplink interference or unauthorized commanding could be attempted, and it bounds the time available for eavesdropping on unencrypted telemetry or for observing the satellite optically or with radar. For the defender, short and scheduled passes mean that a command-path denial lasting a few minutes can cost a whole contact opportunity. If that contact was meant to carry a collision-avoidance maneuver, a network-level disruption acquires a physical consequence. Slant range at the edge of the footprint also drives the link margins of Section 2.5, and the pass

predictions computed here are the raw material for TT&C scheduling and for the contact plans of delay-tolerant networks, both of which an adversary may try to manipulate.

2.3 Constellation Design

A single LEO satellite covers each point on the Earth for a few minutes a day. Continuous service needs a *constellation* arranged so that at least one satellite (or k satellites, for PNT or diversity) is above $\varepsilon_{\min}$ at every point of the service area at all times. This section describes the symmetric patterns used for that purpose and the inter-satellite link topologies that they support.

2.3.1 Walker-delta constellations

The most widely used symmetric pattern is the Walker-delta (or Walker “rosette”) constellation [46]. All satellites share altitude and inclination.

Definition 2.1 (Walker-delta notation). A Walker-delta constellation $i:t/p/f$ has t satellites in circular orbits of inclination i , spread evenly over p planes, with $s = t/p$ satellites per plane. The planes’ ascending nodes are spaced by $\Delta\Omega = 360^\circ/p$. Within each plane the satellites are spaced by $\Delta u = 360^\circ/s$. The phasing parameter $f \in \{0, \dots, p-1\}$ sets the along-track offset between satellites in adjacent planes: a satellite in plane $j+1$ leads the corresponding satellite in plane j by $\Delta\phi = f \cdot 360^\circ/t$.

Walker-*star* (polar) constellations instead spread p near-polar planes over about 180° of RAAN. Adjacent planes then counter-rotate across a “seam” where the ascending halves of the planes meet the descending halves.

Example 2.4 (HELIOS Shell A geometry). Shell A is the Walker-delta constellation $53^\circ:528/24/1$. It therefore has $s = 528/24 = 22$ satellites per plane, node spacing $\Delta\Omega = 360^\circ/24 = 15^\circ$, in-plane spacing $\Delta u = 360^\circ/22 = 16.36^\circ$, and inter-plane phasing $\Delta\phi = 1 \times 360^\circ/528 = 0.68^\circ$. After 24 planes the accumulated phasing is $24 \times 0.68^\circ = 16.36^\circ$, exactly one in-plane slot, so the pattern closes on itself with a shift of one satellite.

The straight-line (chord) distance between neighboring satellites in the same plane is $2a \sin(\Delta u/2) = 2 \times 6918.137 \times \sin 8.18^\circ = 1969$ km. This is the range over which an intra-plane optical ISL must close, and it is constant because in-plane neighbors keep their relative positions. The distance between adjacent planes’ nodes at the equator is $2a \sin 7.5^\circ = 1806$ km, but cross-plane distances shrink toward the maximum-latitude turning points, where the planes cross, and cross-plane lines of sight sweep in angle. Shell B (72/6) is a polar SSO shell with 12 satellites per plane (30° in-plane spacing). For this book we assume the six planes are spread over 180° of RAAN in a Walker-*star* arrangement, so their nodes are 30° apart.

The *coverage* of a constellation at a point is the fraction of time that at least k satellites are visible, and *revisit time* is the gap between successive accesses. For a user terminal the *handover rate* also matters, because in LEO the terminal must switch beams and satellites every few minutes. A rough check of whether a

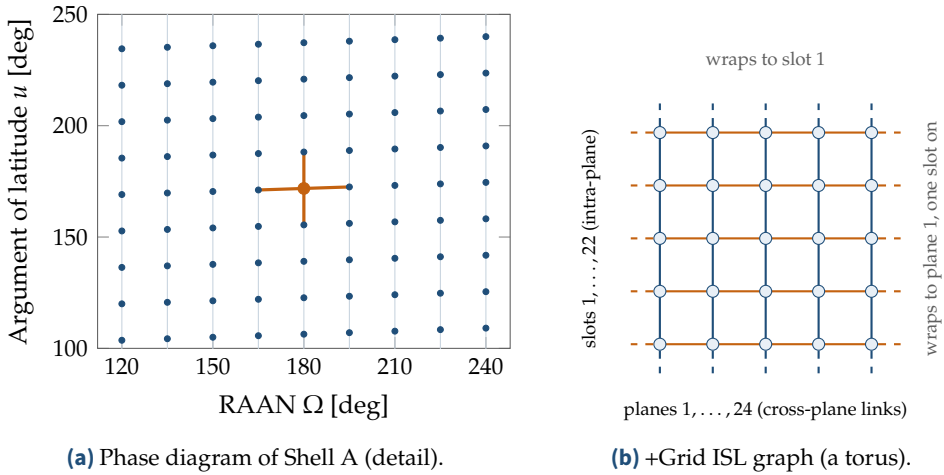


Figure 2.8. Walker-delta geometry and +Grid inter-satellite links for HELIOS Shell A. (a) A detail of the 360° -by- 360° diagram: each column is a plane, the slight upward step between columns is the 0.68° phasing, and the highlighted satellite links to its two in-plane and two cross-plane neighbors. (b) The link graph is a torus whose structure is fixed and publicly derivable from the orbital elements.

single-shell design is dense enough compares the constellation's total footprint area, $t \cdot A_{fp}$, with the Earth's area; for Shell A that ratio is 2.8 (Example 2.3). Because the footprints overlap heavily near the inclination latitude and leave gaps near the equator and poles, detailed design uses simulation or analytical streets-of-coverage methods [38]. Shell B covers the polar regions that Shell A cannot reach.

2.3.2 Inter-satellite link topologies

Inter-satellite links (ISLs) let traffic stay in space between the uplink and the downlink. They reduce dependence on gateways near the user, can lower latency over long distances [47], and let an operator reach a satellite through its neighbors when the satellite is not over a ground station. The standard topology for a Walker constellation is the +Grid, in which each satellite keeps four links: fore and aft in its plane, and left and right to the nearest satellites in the adjacent planes. HELIOS's four optical terminals (two intra-plane, two cross-plane, 20 Gbps each) implement exactly this pattern.

Figure 2.8 shows the pattern in two ways. In the phase diagram of panel (a), every satellite of Shell A is a point at its RAAN and argument of latitude; because all planes precess together and all satellites move at the same rate, the whole diagram simply translates in time, and the +Grid neighbors of each satellite stay the same. Panel (b) abstracts the same structure as a graph. The result is a 24×22 torus in which the seam between plane 24 and plane 1 is closed with a shift of one slot, the signature of $f = 1$.

Intra-plane links are stable. Cross-plane links change length and pointing angle over each orbit, and in polar shells they must be broken near the poles

and across the seam, where relative velocities are too high for the terminals to track [48]. Routing therefore operates over a graph that varies in time but does so *predictably*. Huang et al. [49] describe how ISLs and onboard packet switching let operators run operations and management (O&M), including telecommanding, for satellites outside ground-station coverage, and how this widens the attack surface.

Constellation geometry is what gives a proliferated system *graceful degradation*. In a 528-satellite Walker shell, losing one satellite opens a short coverage gap that moves along its plane, whereas losing a GEO satellite loses service to a hemisphere; proliferation also raises the number of targets an adversary must engage to achieve a given effect. The same regularity has a cost. A torus has a known structure, so its minimum cuts, its most central nodes, and the effect of removing a given satellite are all computable, by the defender and by the attacker alike. If ISLs carry telecommands, a compromised satellite also becomes a possible relay into its neighbors, turning a single intrusion into a path across the network. [Chapter 7](#) develops routing and interdiction models on this graph.

2.4 Segment Architecture

A space system is more than a satellite. Following standard practice [38] and the convention used throughout this book, we divide it into four segments (ground, link, space, and user) and examine the interfaces at which they meet. [Figure 2.9](#) shows the arrangement for HELIOS.

2.4.1 The four segments

The *ground segment* operates the fleet and connects it to terrestrial networks. Its center is the *mission operations center* (MOC), which holds the command and control (C2) software, the telemetry archive, the flight dynamics system, and the planning and scheduling system described in [Section 2.7](#). *TT&C stations* are antennas and baseband equipment that send telecommands, receive telemetry, and measure range and range rate for orbit determination. *Gateways* (Earth stations) carry user traffic between the satellites and the operator's terrestrial core network over feeder links. The *network operations center* (NOC) manages service and capacity, and the *security operations center* (SOC) monitors for attacks. The ground segment is increasingly delivered as a service: *ground-station-as-a-service* (GSaaS) providers rent antenna time through web APIs, and MOC software is often hosted in commercial clouds. HELIOS does both. Its primary MOC runs in a cloud region, with a warm-standby backup at a second site, and six of its eight TT&C sites are leased.

The *link segment* consists of the radio and optical paths themselves. We treat it as a segment in its own right because it crosses space that nobody physically controls. HELIOS has five link types: a low-rate, high-margin, safety-critical S-band TT&C link for telecommand and telemetry; high-capacity Ka-band feeder links between gateways and satellites; Ku-band user links to terminals; optical ISLs between satellites ([Section 2.3](#)); and a one-way L-band broadcast that carries the PNT timing and ranging augmentation.

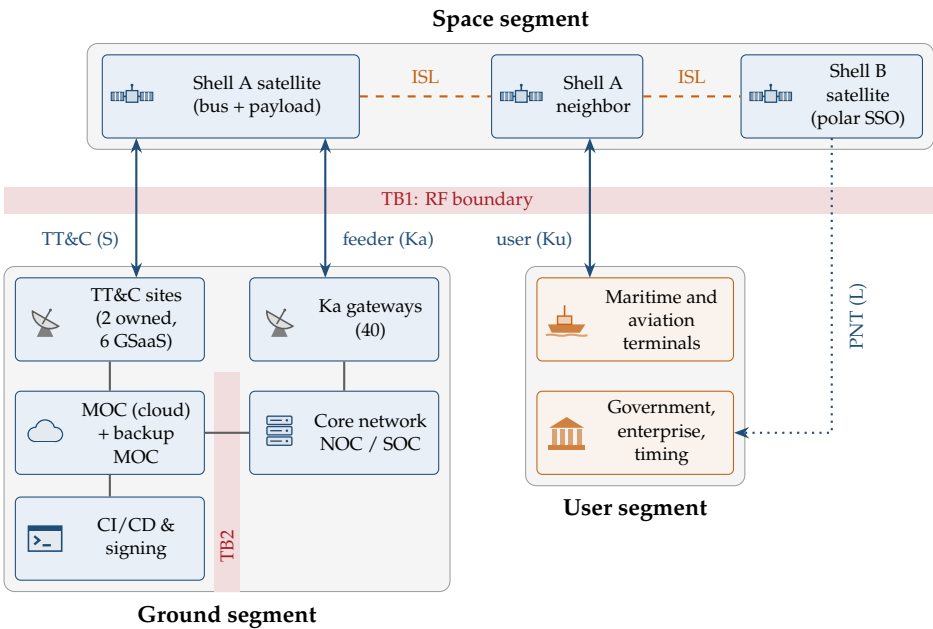


Figure 2.9. The four-segment HELIOS architecture. Solid blue arrows are RF links, dashed orange lines are optical ISLs, and the dotted arrow is the one-way PNT broadcast. Red shaded bands mark two of the trust boundaries of Table 2.3: the RF boundary (TB1), across which a party in view can transmit or listen, and the boundary between mission operations and the commercial service network (TB2).

The *space segment* is the satellites. Each has a *bus* and a *payload*. The bus provides the common services: electrical power (solar arrays, batteries, power distribution), thermal control, structure, propulsion (Hall-effect thrusters on HELIOS), attitude determination and control (ADCS), the TT&C transponder, and the onboard computer (OBC) that runs the flight software. The payload performs the mission. For HELIOS it comprises the Ku-band phased-array user beams, the Ka-band feeder antennas, the optical ISL terminals, a regenerative processor that demodulates and switches packets, and the L-band PNT transmitter. HELIOS runs the payload processor separately from the bus OBC, a separation that is both a reliability decision and a security boundary.

The *user segment* is the population of terminals. In HELIOS’s case it consists of about 200,000 flat-panel phased arrays on ships, aircraft, and fixed sites, each containing an RF front end, a modem, and a network router, with firmware updated over the air. Some users also depend on the PNT augmentation for navigation, or for timing in power-grid and telecommunications equipment.

2.4.2 Interfaces and trust boundaries

Segments meet at interfaces, and a security analysis is concerned less with the segments themselves than with what can cross between them. We call an interface a *trust boundary* when the two sides are administered by different parties or when

Table 2.3. HELIOS interfaces and trust boundaries. “Owners” names who administers each side. A boundary is *crossed* whenever the two sides have different owners or the medium is open.

Interface	Medium / protocol	Owners	Trust-boundary issue
MOC ↔ TT&C site	Internet VPN; CCSDS SLE or vendor API	HELIOS ↔ HELIOS or GSaaS	Leased sites and cloud hosting place third-party administrators in the command path
TT&C site ↔ satellite	S-band RF; CCSDS TC/TM + SDLS	HELIOS ↔ open medium	A party in view can listen or transmit; authentication and anti-replay must be cryptographic
Gateway ↔ satellite	Ka-band RF; DVB-S2X	HELIOS ↔ open medium	High-capacity trunk; interference affects many users at once
Satellite ↔ terminal	Ku-band RF; DVB-S2X / RCS2	HELIOS ↔ customer	Terminals are physically in untrusted hands and are updated over the air
Satellite ↔ satellite	Optical ISL; onboard switching	HELIOS ↔ HELIOS	Narrow beams are hard to intercept, but a compromised node can relay
Bus OBC ↔ payload processor	Onboard bus (SpaceWire, CAN)	HELIOS vendors	Different suppliers’ software; payload faults must not reach bus control
NOC/SOC ↔ MOC	Enterprise network	HELIOS IT ↔ HELIOS OT	IT/OT convergence: enterprise compromise must not reach command authority
CI/CD ↔ satellite	Signed platform update via TT&C; payload image via feeder link	Developers ↔ flight	Signing keys and build pipeline are high-value targets
L-band PNT → users	Broadcast	HELIOS → public	One-way; receivers must authenticate messages without a return channel

the medium between them is open to third parties. Table 2.3 lists the principal HELIOS interfaces. Two of them are drawn in Fig. 2.9: the RF boundary TB1, which every radio link crosses and at which a party within view can transmit or listen, and TB2, which separates mission operations from the enterprise and service network. The remaining rows show how many boundaries arise from ordinary commercial choices, such as leasing antenna time, hosting software in a cloud, or accepting firmware from several suppliers.

A system’s security properties are only as strong as the weakest crossing in Table 2.3, and experience suggests where the weak crossings tend to lie. Survey evidence indicates that most historical attacks on satellite systems have targeted the ground and user segments and the links rather than the spacecraft [50]. The attack on Viasat’s KA-SAT network on 24 February 2022 is the best-documented example: according to the operator’s account, an intruder entered the ground management network through a misconfigured VPN appliance and used legitimate management functions to push destructive commands, disabling tens of thousands of user modems in Europe without touching the satellite [12]. In the

language of this section, the attack crossed TB2-type boundaries on the ground and then exploited the trust that the user segment places in its operator. The interfaces tabulated here therefore become the edges along which later analysis traces multi-step attacks. [Chapter 11](#) develops ground- and user-segment cybersecurity in depth.

2.5 RF Communications Fundamentals

Every service a satellite provides, whether broadband, TT&C, or PNT, depends on a radio or optical link that closes with some margin. The *link budget* is the accounting that shows whether it does. It is also the first quantitative tool of the signal/RF security layer, because a jammer, a spoofer, and an eavesdropper each have a link budget of their own. We follow the notation of Maral, Bousquet, and Sun [51] and Pratt and Allnutt [52]. All quantities in decibels are $10 \log_{10}$ of power ratios.

2.5.1 Antennas, EIRP, and path loss

A parabolic or equivalent aperture of diameter D and aperture efficiency η_a at wavelength $\lambda_w = c/f$ has on-axis gain and approximate half-power beamwidth

$$G = \eta_a \left(\frac{\pi D}{\lambda_w} \right)^2, \quad \theta_{3\text{dB}} \approx 70^\circ \frac{\lambda_w}{D}. \quad (2.11)$$

For example, a 0.5 m aperture at 11.7 GHz ($\lambda_w = 2.56$ cm) with $\eta_a = 0.6$ has $G \approx 33.5$ dBi and $\theta_{3\text{dB}} \approx 3.6^\circ$. Eliminating D from (2.11) gives $G \theta_{3\text{dB}}^2 \approx \eta_a \pi^2 70^2 \approx 29,000$ square degrees for $\eta_a = 0.6$, so gain and beamwidth trade directly against each other: halving the beamwidth quadruples the gain. [Figure 2.10](#) shows both the pattern of a single aperture and this trade. A high-gain antenna collects more signal and rejects more off-axis interference, but it must be pointed or electronically steered, and its sidelobes still admit energy from directions far off the main beam.

The *effective isotropic radiated power* is $\text{EIRP} = P_t G_t$, the power an isotropic radiator would need to produce the same flux in the direction of interest. Over a slant range d , isotropic spreading and the effective aperture of an isotropic receiving antenna combine into the *free-space path loss*

$$L_{\text{fs}} = \left(\frac{4\pi d f}{c} \right)^2, \quad L_{\text{fs}} [\text{dB}] = 92.45 + 20 \log_{10} d_{[\text{km}]} + 20 \log_{10} f_{[\text{GHz}]}. \quad (2.12)$$

Proposition 2.3 (Friis transmission equation). *The carrier power received by an antenna of gain G_r is*

$$C = \frac{\text{EIRP} G_r}{L_{\text{fs}} L_a}, \quad (2.13)$$

where $L_a \geq 1$ collects additional losses: atmospheric absorption, rain, pointing error, and polarization mismatch.

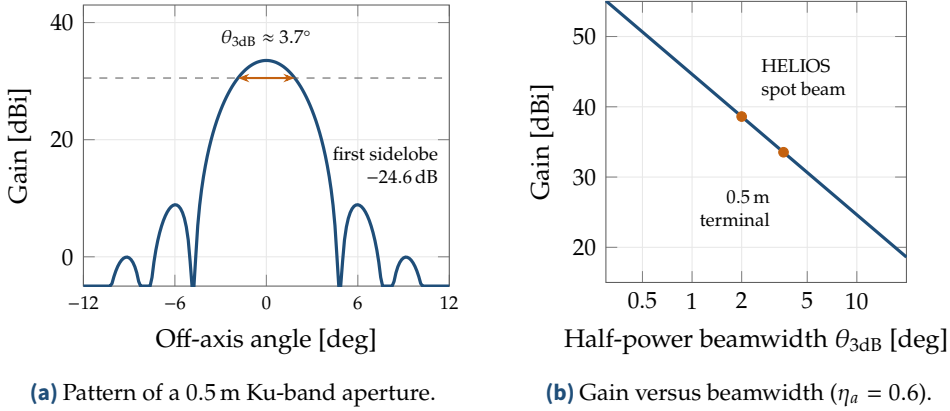


Figure 2.10. Antenna gain and beamwidth. (a) Pattern of a 0.5 m aperture at 11.7 GHz with a tapered illumination and 33.5 dBi peak gain; the sidelobes admit signals from far outside the main beam at reduced gain. (b) By (2.11), gain rises by 6 dB for every halving of beamwidth; the 2° HELIOS user spot beam corresponds to about 38.6 dBi.

2.5.2 Noise, G/T , and carrier-to-noise density

Thermal noise in a receiver has a flat one-sided power spectral density $N_0 = k_B T_{\text{sys}}$, where $k_B = 1.380649 \times 10^{-23} \text{ J K}^{-1}$ ($-228.6 \text{ dBW K}^{-1} \text{ Hz}^{-1}$) and T_{sys} is the *system noise temperature*. The system noise temperature adds the antenna temperature (sky, ground spillover, and rain) to the receiver's own noise referred to the antenna terminals. For a low-noise amplifier with noise figure F preceded by a feed loss L_f at physical temperature T_0 , a standard approximation is

$$T_{\text{sys}} \approx \frac{T_{\text{ant}}}{L_f} + T_0 \left(1 - \frac{1}{L_f} \right) + T_0(F - 1). \quad (2.14)$$

Because received carrier power scales with G_r and noise with T_{sys} , the receiving station's quality is summarized by a single figure of merit, G/T (in dB/K). Combining (2.13) with N_0 gives the central link-budget equation, with C/N_0 in dB-Hz, EIRP in dBW, losses in dB, G/T in dB/K, and k_B in dBW K⁻¹ Hz⁻¹:

$$\frac{C}{N_0} = \text{EIRP} - L_{\text{fs}} - L_a + \frac{G}{T} - k_B. \quad (2.15)$$

Two normalizations follow. In an occupied bandwidth B , $C/N = C/N_0 - 10 \log_{10} B$. At bit rate R_b , the energy per bit to noise density is

$$\frac{E_b}{N_0} = \frac{C}{N_0} - 10 \log_{10} R_b \quad [\text{dB}]. \quad (2.16)$$

The *link margin* is $M = (E_b/N_0)_{\text{achieved}} - (E_b/N_0)_{\text{required}}$, or the equivalent difference in C/N .

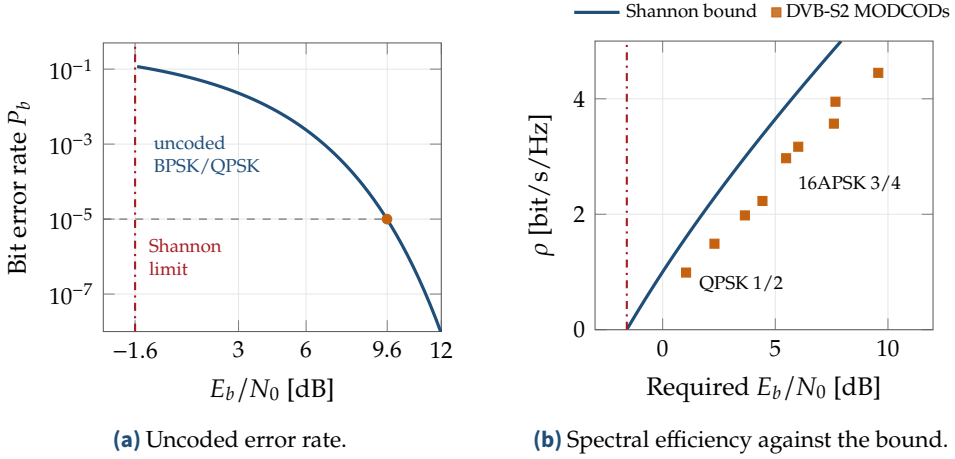


Figure 2.11. Error rate and the Shannon limit. (a) Uncoded BPSK/QPSK needs 9.6 dB for $P_b = 10^{-5}$, far from the -1.59 dB limit. (b) DVB-S2 MODCODs (ideal quasi-error-free thresholds of Table 2.4, converted to E_b/N_0) lie 1.1 dB to 2.9 dB to the right of the Shannon bound; no scheme can operate to its left.

2.5.3 Modulation, error rates, and the Shannon limit

For coherent BPSK, and for QPSK per bit with Gray coding, over an additive white Gaussian noise (AWGN) channel the bit error rate is [53]

$$P_b = Q\left(\sqrt{2E_b/N_0}\right), \quad Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-u^2/2} du. \quad (2.17)$$

Uncoded BPSK/QPSK needs $E_b/N_0 \approx 9.6$ dB for $P_b = 10^{-5}$ and about 10.5 dB for 10^{-6} . Forward error correction lowers the requirement substantially. Shannon's capacity theorem [54] bounds what a code can achieve:

$$C_{\text{Sh}} = B \log_2\left(1 + \frac{C}{N}\right), \quad \frac{E_b}{N_0} \geq \frac{2^\rho - 1}{\rho} \xrightarrow{\rho \rightarrow 0} \ln 2 = -1.59 \text{ dB}, \quad (2.18)$$

where $\rho = R_b/B$ is the spectral efficiency in bit/s/Hz. Figure 2.11 places these results side by side. The uncoded curve in panel (a) lies about 11 dB to the right of the Shannon limit at $P_b = 10^{-5}$, and panel (b) shows that practical coded modulations recover most of that gap: the DVB-S2 modulation and coding combinations introduced below operate between about 1 dB and 3 dB from the bound. Modern LDPC and turbo codes generally operate within about 1 dB of the bound at practical block lengths.

Broadband satellite downlinks use the DVB-S2 standard [55] and its extension DVB-S2X [56]. They combine LDPC/BCH coding with QPSK, 8PSK, 16APSK, and 32APSK (and higher orders in S2X). Each MODCOD, a pair of modulation and code rate, has a threshold E_s/N_0 for quasi-error-free operation and a spectral efficiency. With *adaptive coding and modulation* (ACM), the terminal reports its measured signal quality and the gateway picks the most efficient MODCOD that

Table 2.4. Selected DVB-S2 MODCODs: spectral efficiency and ideal AWGN threshold E_s/N_0 for quasi-error-free operation (normal 64 800-bit frames, no pilots), after ETSI EN 302 307-1 [55] as tabulated in CCSDS 130.12-G-2 [57]. DVB-S2X adds finer steps. Real systems add an implementation margin.

MODCOD	bit/symbol	E_s/N_0 [dB]	MODCOD	bit/symbol	E_s/N_0 [dB]
QPSK 1/2	0.99	1.00	16APSK 3/4	2.97	10.21
QPSK 3/4	1.49	4.03	16APSK 4/5	3.17	11.03
8PSK 2/3	1.98	6.62	16APSK 9/10	3.57	13.13
8PSK 3/4	2.23	7.91	32APSK 4/5	3.95	13.64
			32APSK 9/10	4.45	16.05

still closes, frame by frame. Table 2.4 lists a subset of DVB-S2 MODCODs with their ideal AWGN thresholds.

Above about 10 GHz, rain dominates the variable part of L_a . Raindrops absorb and scatter the signal, and wet atmosphere also raises T_{ant} , so rain lowers G/T at the same time as it lowers C . Rain attenuation increases with frequency, rain rate, and path length through rain, and so with lower elevation. It is predicted statistically, as the attenuation exceeded for a given percentage of an average year, using the ITU-R models: Recommendation P.618 for Earth-space propagation [58], P.838 for specific rain attenuation [59], and P.676 for gaseous absorption [60]. Ka-band links typically need several dB more rain margin than Ku-band links for the same availability. ACM, uplink power control, and gateway site diversity are the standard countermeasures. Ionospheric scintillation affects lower frequencies (L-band PNT and, at times, S-band) at low latitudes and at high latitudes during storms.

2.5.4 Worked link budgets

Example 2.5 (HELIOS Ku-band user downlink). Use the specification values of Table 2.1: EIRP = 34 dBW per beam, $f = 11.7$ GHz, terminal $G/T = 9$ dB/K, $L_a = 1.5$ dB, and $B = 250$ MHz. The slant ranges come from Example 2.3. At nadir, (2.12) gives

$$L_{\text{fs}} = 92.45 + 20 \log_{10} 540 + 20 \log_{10} 11.7 = 92.45 + 54.65 + 21.36 = 168.46 \text{ dB},$$

and (2.15) gives

$$C/N_0 = 34 - 168.46 - 1.5 + 9 + 228.6 = 101.64 \text{ dB-Hz},$$

$$C/N = 101.64 - 83.98 = 17.66 \text{ dB}.$$

Repeating the calculation at 40° and 25° gives the table below, which reproduces the HELIOS reference values ($C/N_0 \approx 101.6, 98.3$, and 95.4 dB-Hz).

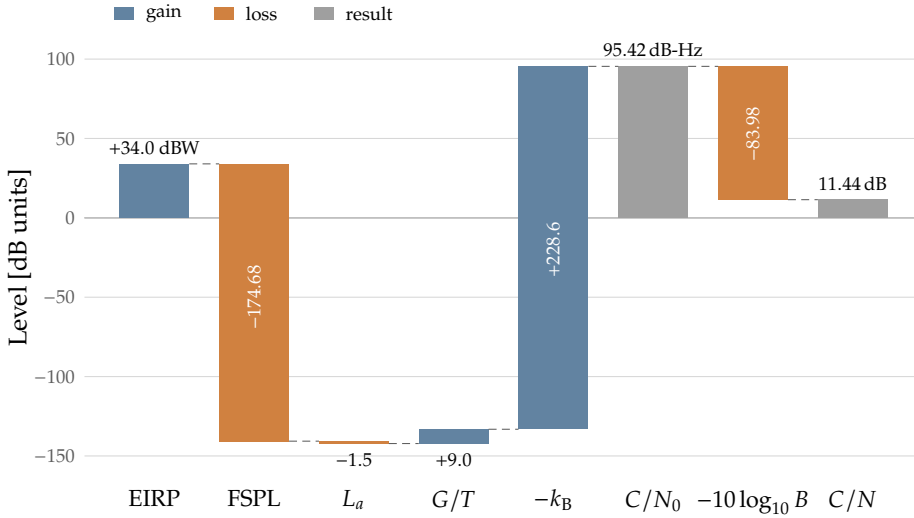


Figure 2.12. Link-budget waterfall for the HELIOS Ku-band user downlink at the 25° coverage edge. Each bar adds one term of (2.15): path loss dominates, and the 228.6 dB contribution of $-k_B$ converts the carrier-to-temperature ratio into C/N_0 . The same chain at nadir ends at 17.66 dB; the 6.2 dB difference lies wholly in the FSPL bar.

	Nadir	40°	25°
Slant range [km]	540.0	798.0	1105.2
FSPL at 11.7 GHz [dB]	168.46	171.85	174.68
C/N_0 [dB-Hz]	101.64	98.25	95.42
C/N in 250 MHz [dB]	17.66	14.27	11.44
Shannon capacity [Mbps]	1473	1198	975
MODCOD (1 dB margin)	32APSK 9/10	16APSK 9/10	16APSK 3/4
DVB-S2 throughput [Mbps]	1060	849	706

To select the MODCOD we assume a root-raised-cosine roll-off of 0.05, so the symbol rate is $R_s = 250/1.05 = 238.1$ Msym/s and $E_s/N_0 = C/N + 10 \log_{10} 1.05 = C/N + 0.21$ dB. We then require E_s/N_0 to exceed the threshold in Table 2.4 by 1 dB. Throughput is R_s times the spectral efficiency. At the 25° edge, clear-sky throughput is 33 % below nadir. This purely geometric effect is the throughput drop reported in the vignette, and Figs. 2.12 and 2.13 show where it comes from. DVB-S2 operates about 2 dB to 3 dB from the Shannon bound in this regime.

Example 2.6 (HELIOS S-band TT&C downlink at 10° elevation). The satellite TT&C EIRP is -3 dBW, the ground station G/T is 15 dB/K (typical of a dish of a few meters at S-band), $f = 2.25$ GHz, the telemetry rate is 64 kbps, and we allow

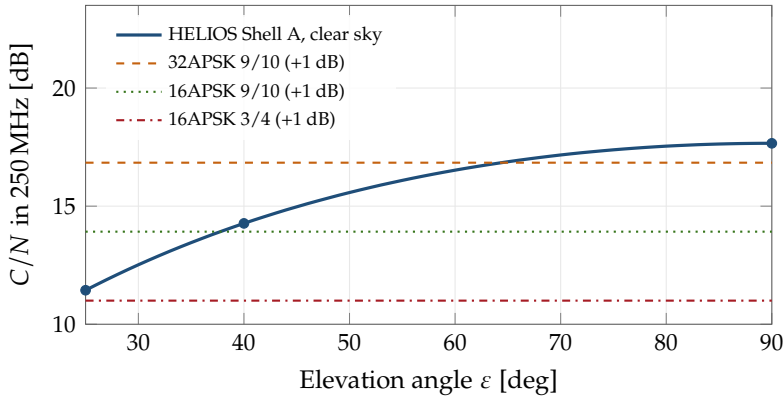


Figure 2.13. Clear-sky C/N of the HELIOS Ku-band user downlink as a function of elevation angle (EIRP 34 dBW, $G/T = 9$ dB/K, $L_a = 1.5$ dB). Horizontal lines are the C/N needed for three DVB-S2 MODCODs with 1 dB margin, and marks show the reference values of Example 2.5. The 6.2 dB variation between nadir and 25° comes from slant range alone.

$L_a = 1$ dB. At $\varepsilon = 10^\circ$, $d = 1791.9$ km (Example 2.3), so

$$\begin{aligned} L_{fs} &= 92.45 + 65.07 + 7.04 = 164.56 \text{ dB}, \\ C/N_0 &= -3 - 164.56 - 1 + 15 + 228.6 = 75.04 \text{ dB-Hz}, \\ E_b/N_0 &= 75.04 - 10 \log_{10}(64\,000) = 75.04 - 48.06 = 26.98 \text{ dB}. \end{aligned}$$

Against the 9.6 dB that uncoded BPSK needs for $P_b = 10^{-5}$, the margin is about 17.4 dB before coding gain, a factor of about 55 in power. This matches the specification ($C/N_0 \approx 75.0$ dB-Hz, $E_b/N_0 \approx 27.0$ dB). For Shell B at 560 km, the 10° slant range grows to 1839 km and the margin shrinks by only 0.2 dB.

The TT&C margin looks generous, and it is meant to be. Low-rate, high-margin TT&C is designed to work when the satellite is tumbling, when it is on an omnidirectional antenna, or when a ground antenna is at the horizon. The next example shows why geostationary systems cannot afford such margins on their broadband links.

Example 2.7 (ATLAS-G GEO comparison). ATLAS-G users see the satellite at 30° elevation. By (2.7) with $h = 35\,786$ km, $d = 38\,612$ km, and at 19.7 GHz

$$L_{fs} = 92.45 + 20 \log_{10} 38\,612 + 20 \log_{10} 19.7 = 92.45 + 91.73 + 25.89 = 210.07 \text{ dB},$$

matching the specification's 210.1 dB. At the *same* frequency, the path-loss penalty relative to HELIOS depends only on the distance ratio:

$$\begin{aligned} 20 \log_{10}(35\,786/540) &= 36.4 \text{ dB} && (\text{nadir to nadir}), \\ 20 \log_{10}(38\,612/540) &= 37.1 \text{ dB} && (30^\circ \text{ GEO user vs. LEO nadir}), \\ 20 \log_{10}(38\,612/1105.2) &= 30.9 \text{ dB} && (\text{vs. LEO } 25^\circ \text{ edge}). \end{aligned}$$

To recover roughly 36 dB, a factor of about 4000, a GEO system combines a much higher-gain spacecraft antenna (narrow spot beams), higher transmit power, and larger user terminals.

For illustration, assume an ATLAS-G spot-beam EIRP of 60 dBW, a user terminal G/T of 17 dB/K (a 0.75 m Ka-band dish, $G \approx 41.9$ dBi with $T_{\text{sys}} \approx 310$ K), and $L_a = 2$ dB in clear sky. Then $C/N_0 = 60 - 210.07 - 2 + 17 + 228.6 = 93.5$ dB-Hz and $C/N = 9.5$ dB in 250 MHz. This is lower than the HELIOS edge-of-coverage value, even with 26 dB more EIRP and a larger terminal. The geometry has other consequences too. One-way user-satellite delay is 129 ms, a user-satellite-gateway hop is 258 ms, and a round trip through the bent pipe is about 515 ms, compared with 7 to 15 ms for HELIOS. In exchange, one GEO satellite at 25° minimum elevation covers a cap of central angle 57.1° , compared with 8.3° for HELIOS.

Link margin is, in the end, the defender's budget against interference. If a jammer adds noise-like power spectral density J_0 at the receiver, the effective noise density becomes $N_0 + J_0$, and a link with margin M (in dB) fails once

$$\frac{J_0}{N_0} \geq 10^{M/10} - 1. \quad (2.19)$$

The HELIOS TT&C link of [Example 2.6](#) tolerates interference about 17 dB above its thermal noise floor. The Ku user link at the 25° edge is in a very different position: its E_s/N_0 of 11.65 dB exceeds the 16APSK 3/4 threshold by only 1.44 dB, so by (2.19) an interference density about 4 dB *below* thermal noise is enough to force a lower MODCOD. ACM therefore turns interference into graceful throughput loss, which serves resilience but can also disguise a low-level attack as bad weather or the edge of coverage, exactly the ambiguity in the vignette. An eavesdropper's link budget follows the same equations, so path loss and antenna gain decide who can listen from where. [Chapter 5](#) builds jamming, anti-jam, and spoofing-detection models on (2.15) and (2.19).

2.6 The Space Communication and Network Stack

A closed RF link carries bits. Protocols turn those bits into frames, packets, commands, files, and user traffic. Space systems use a mix of space-specific standards from the Consultative Committee for Space Data Systems (CCSDS) and broadcast and terrestrial standards (DVB, IETF, 3GPP). The CCSDS overview Green Book [61] and the survey by Kodheli et al. [62] are good entry points. Knowing *where* in the stack a protection sits is essential, because a protection at one layer offers no assurance against attacks at another.

2.6.1 CCSDS space data link protocols

CCSDS organizes the space link like the lower layers of the OSI model ([Fig. 2.14](#)). The *physical layer* is covered by the RF and modulation recommendations (CCSDS 401.0-B). The *synchronization and channel coding* sublayer adds frame synchronization markers, randomization, and error-correcting codes (CCSDS 131.0-B for telemetry, 231.0-B for telecommand [63]). Above them, the *data link protocol* sublayer defines four families of transfer frames:

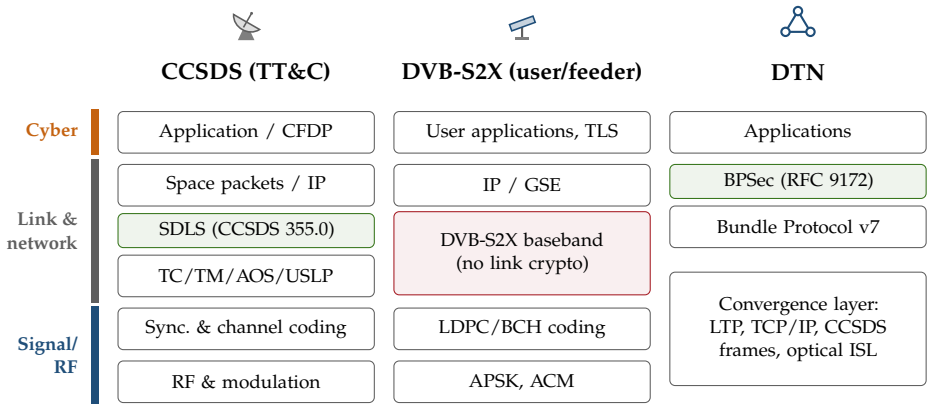


Figure 2.14. Simplified protocol stacks for the TT&C (CCSDS), broadband (DVB-S2X), and disruption-tolerant (DTN) links, aligned with the security layers on the left. Green boxes are standardized security sublayers; the red box marks a layer with no standardized protection. Protection at one layer does not cover the layers above or below it.

- ▶ the *TC Space Data Link Protocol* [64], with variable-length telecommand frames on the uplink and an optional retransmission service (the Communications Operation Procedure, COP-1) that guarantees in-order, complete delivery;
- ▶ the *TM Space Data Link Protocol* [65], with fixed-length telemetry frames on the downlink, multiplexed into virtual channels;
- ▶ the *AOS Space Data Link Protocol* [66], with fixed-length frames for high-rate bidirectional links;
- ▶ the *Unified Space Data Link Protocol* (USLP) [67], a single frame format that covers the TC, TM, and AOS use cases, including large frames for high-rate and optical links.

At the network layer, the *Space Packet Protocol* [68] carries application data in space packets identified by an 11-bit Application Process Identifier (APID), which routes a packet to an onboard application or ground processing function. IP datagrams can also be carried over CCSDS links by encapsulation. File transfer uses the CCSDS File Delivery Protocol (CFDP, CCSDS 727.0-B). Between ground facilities, CCSDS Space Link Extension (SLE) services carry frames from the TT&C antenna to the MOC over terrestrial networks. Finally, *Proximity-1* [69] is a short-range, bidirectional link protocol for spacecraft that communicate with each other, for example Mars orbiters relaying data from landers and rovers; the same functions matter for rendezvous and servicing in Earth orbit.

2.6.2 Space Data Link Security

The original CCSDS link protocols had no security. The *Space Data Link Security* (SDLS) Protocol (CCSDS 355.0-B) [70] adds an optional security sublayer inside the TC, TM, AOS, and USLP data link protocols. Figure 2.15 shows how it changes a telecommand frame. Between the frame header and the data field it inserts a security header, which carries a Security Parameter Index (SPI) selecting a *security*

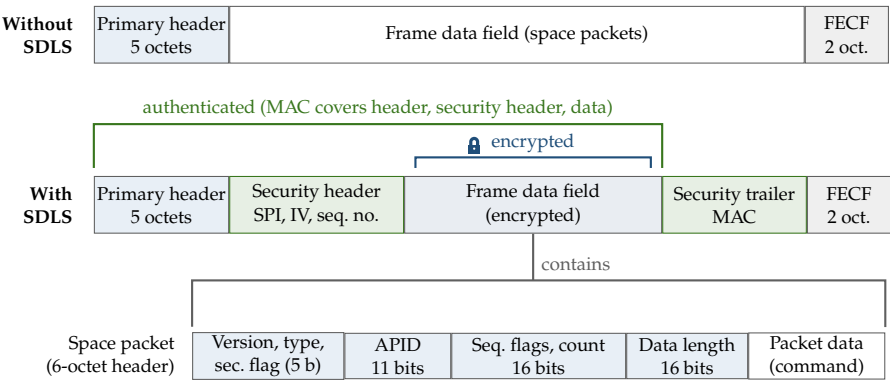


Figure 2.15. A CCSDS telecommand transfer frame without and with SDLS authenticated encryption. SDLS inserts a security header and trailer around the data field; the space packet inside, whose APID selects the onboard application, is encrypted and authenticated but not validated for meaning. Lengths of the IV, sequence number, and MAC are set by the security association; optional fields are omitted.

association (SA) and, depending on the SA, an initialization vector and an anti-replay sequence number. After the data field it adds a security trailer carrying a message authentication code (MAC). Each SA specifies authentication, encryption, or authenticated encryption, with algorithms such as AES-GCM taken from the CCSDS cryptographic algorithms recommendation (CCSDS 352.0-B). HELIOS uses SDLS authenticated encryption on commands. The *SDLS Extended Procedures* (CCSDS 355.1-B) [71] define in-band key management (including over-the-air rekeying), SA management, and security monitoring and control.

The limits of this protection follow directly from the figure. SDLS protects *frames on the space link*. It does not protect a command before it reaches the TT&C station, it does not validate what an authenticated command does once the packet is delivered, and its security depends wholly on how keys are generated, stored, and loaded. The CCSDS security Green Book [72] discusses these limits.

2.6.3 Broadband, delay-tolerant, and terrestrial-derived protocols

On broadband forward links, DVB-S2X [56] provides the physical layer and baseband framing described in Section 2.5. IP traffic is carried by Generic Stream Encapsulation (GSE), and return links from terminals often use DVB-RCS2. The DVB standards themselves specify no link encryption for user traffic, so confidentiality depends on operator-added or end-to-end encryption. Researchers have shown that unencrypted satellite broadband traffic can be received with inexpensive equipment [73].

When no single instant offers a complete end-to-end path, as with ISLs that come and go, sparse ground contacts, or cislunar and deep-space links, the assumptions of IP fail. Delay/Disruption-Tolerant Networking (DTN) [74] instead stores, carries, and forwards *bundles*. The Bundle Protocol version 7 is specified in RFC 9171 [75], and Bundle Protocol Security (BPsec) in RFC 9172 [76]. BPsec adds block-level integrity and confidentiality that can be applied by intermediate

nodes, not only by endpoints. DTN routing uses *contact plans*, schedules of future link availability computed from orbits as in [Section 2.2](#).

Starting with Release 17, 3GPP specifications support *non-terrestrial networks* (NTN): 5G New Radio and NB-IoT/LTE-M operating through satellites, with adjustments for long delays and large Doppler shifts. Release 17 builds on the Release 16 study of NR over satellites [77]. It targets transparent (bent-pipe) payloads, and later releases extend the work [78]. NTN brings the 5G security architecture into space, along with its signaling attack surface and the prospect of connecting ordinary handsets directly to satellites.

Regenerative payloads like HELIOS's demodulate traffic, route packets onboard, and forward them over ISLs or downlinks. Routing tables and beam and channel assignments are increasingly managed by software-defined networking (SDN) controllers on the ground, which push flow rules to the satellites. That centralizes network policy, and it turns the SDN control channel into another command path. [Table 2.5](#) maps all of these protocols onto the security layers.

[Table 2.5](#) is the bridge from engineering to security analysis, and three recurring facts stand out in it. First, most standardized security in space protocols is *optional*, and legacy missions often fly without it [50]. Second, each protection covers only its own layer: SDLS stops a forged frame but not a legitimately signed malicious command, and no cryptography stops a jammer. Third, the protocols that route and schedule traffic, such as contact plans, SDN rules, and ISL routing, are themselves control channels and deserve the protection given to telecommands. [Chapter 6](#) develops SDLS, key management, and the migration to post-quantum cryptography.

2.7 Ground and Onboard Computing

The links carry commands, but computers decide what the commands are and what they do. A space system has two computing domains with very different properties. The ground is large, networked, and regularly patched. The spacecraft is small, radiation-constrained, and hard to change once launched. [Figure 2.16](#) shows the two domains for HELIOS and the path that joins them.

2.7.1 Ground computing

A modern MOC contains several interacting systems [38]. *Command and control* (C2) software builds, validates, and sends telecommands, and it decommutates, displays, archives, and checks limits on telemetry; critical commands are commonly screened against a database and may require two-person approval. The *flight dynamics system* (FDS) performs orbit determination from ranging, GNSS telemetry, and external catalogs, predicts passes ([Section 2.2](#)), plans maneuvers, and screens for conjunctions. The *planning and scheduling system* allocates antenna time, payload resources, and maneuvers across the fleet; for 600 satellites and eight TT&C sites this is a large combinatorial optimization that is rerun many times a day. *Ground-station-as-a-service* integrations book antenna time and move telemetry and command data through the provider's APIs and networks, with commercial offerings that range from cloud-provider services such as AWS Ground Station to

Table 2.5. Mapping of space communication protocols to security layers, typical threats, and built-in protections. “Signal” denotes the signal/RF layer and “Link/Net” the link and network layer.

Protocol / function	Security layer	Typical threats	Built-in protection
RF & modulation (CCSDS 401.0; DVB-S2X PHY; L-band PNT)	Signal	Jamming, spoofing, meaconing, eavesdropping, signal fingerprinting	Spread spectrum (some), margin, ACM; none cryptographic
Sync. & channel coding (131.0, 231.0; LDPC)	Signal / Link/Net	Protocol-aware jamming of sync markers and headers	Robust coding only
TC / TM / AOS / USLP	Link/Net	Frame injection, replay, spoofed telemetry, virtual-channel abuse	None without SDLS; COP-1 handles errors, not adversaries
SDLS (355.0), SDLS-EP (355.1)	Link/Net	Key compromise, SA misconfiguration, downgrade to clear mode, abuse of rekeying	Authenticated encryption, anti-replay counters, key management
Space Packet Protocol, CFDP	Link/Net / Cyber	APID spoofing, malformed packets reaching flight software, file tampering	None intrinsic; relies on SDLS and onboard validation
Proximity-1	Link/Net	Relay hijacking, unauthorized session establishment near a target	Optional SDLS
DVB-S2X / GSE / RCS2	Signal / Link/Net	Traffic interception, injection, terminal impersonation	None standard; operator or end-to-end cryptography
BPv7 / BPSec	Link/Net	Bundle forgery, contact-plan manipulation, storage exhaustion	BPSec integrity and confidentiality blocks
3GPP NTN	Link/Net / Cyber	Rogue access nodes, signaling attacks, core-network intrusion	5G security architecture
SDN / onboard switching	Link/Net / Cyber	Controller compromise, rule injection, routing manipulation	Depends on control-channel security
SLE, GSaaS APIs, MOC networks	Cyber	Credential theft, man-in-the-middle between station and MOC	TLS, VPN, access control

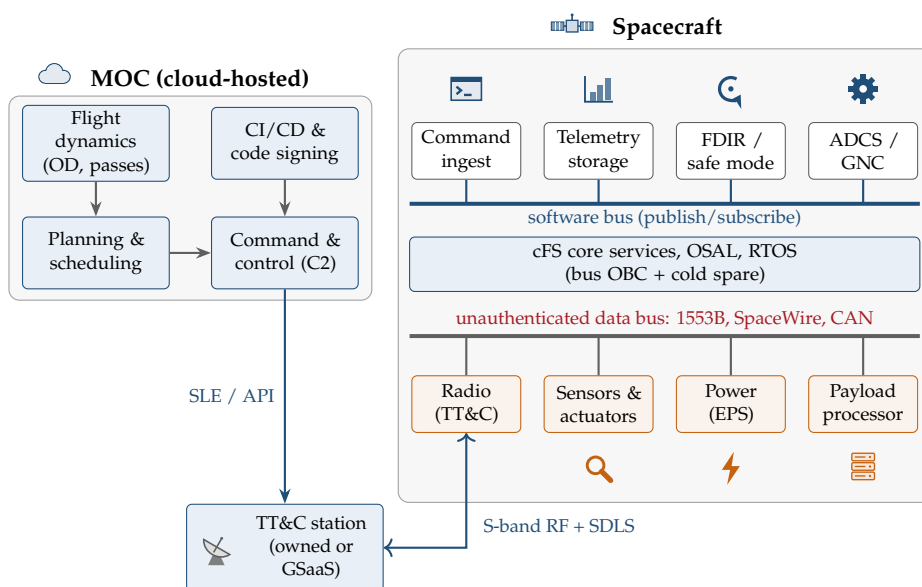


Figure 2.16. Ground and onboard computing for HELIOS. On the ground, planning, flight dynamics, and the CI/CD pipeline all feed the C2 system, which reaches the spacecraft through a TT&C station that may belong to a third party. On board, cFS applications exchange messages on a software bus, while subsystems share a hardware data bus that trusts every connected device.

dedicated antenna-network operators. Finally, *cloud hosting* and CI/CD pipelines build flight-software and terminal-firmware updates, sign them, and stage them for upload.

Automation is essential at constellation scale. Much of HELIOS's routine commanding is generated by software rather than typed by operators, which makes the integrity of the planning chain in Fig. 2.16 as important as the integrity of the command link. NIST's guidance on applying the Cybersecurity Framework to satellite command and control [79] and to commercial satellite operations [80], and Space Policy Directive-5 [34], set out baseline expectations for these systems.

Space Security in Practice: The JPL Intrusion and the Unauthorized Raspberry Pi (NASA OIG, 2019)

In June 2019 NASA's Office of Inspector General (OIG) published an audit of cybersecurity at the Jet Propulsion Laboratory [81]. It described a breach discovered in April 2018. Attackers had entered through the compromised account of an external user and had moved through the network undetected for about ten months. Along the way they targeted a Raspberry Pi computer that had been attached to the JPL network without authorization. They reached two of the laboratory's three primary networks through a gateway that was not adequately segmented, and they took about 500 MB of data in

23 files. Two of those files contained export-controlled (ITAR) information related to the Mars Science Laboratory mission.

Because the attackers had been inside the network, NASA questioned whether Deep Space Network data could be trusted. The Johnson Space Center temporarily cut its gateway connection in May 2018 and did not reconnect until November. The OIG found that JPL's asset inventory was incomplete, so the device was unknown to security staff, and that security tickets had gone unresolved for long periods.

The lesson for security analysts is *interconnection*. An unmanaged device and an unsegmented shared gateway on an enterprise network put systems that support flight operations within reach. The weaknesses were ordinary IT failures (inventory, segmentation, patch and ticket management), not exotic space-specific ones.

2.7.2 Onboard computing

Spacecraft computers must survive total ionizing dose, single-event upsets (bit flips), and latch-up. Traditional missions used radiation-hardened processors such as the PowerPC-based RAD750 or SPARC-based LEON devices, which are reliable but orders of magnitude slower than terrestrial chips. Proliferated LEO constellations like HELIOS instead use radiation-tolerant commercial or ARM-based processors with error-correcting memory, watchdogs, and redundancy (HELIOS has a cold-spare OBC). Flight software runs on a real-time operating system (RTOS) such as VxWorks, RTEMS, or FreeRTOS, or on embedded Linux for payload processors.

Reusable frameworks have replaced much mission-specific code. NASA's *core Flight System* (cFS), developed at Goddard Space Flight Center and released as open source, provides a publish/subscribe software bus, an operating-system abstraction layer, and a library of applications for commanding, telemetry, scheduling, data storage, and health and safety [82]. JPL's *F'* (F Prime) is an open-source component-based framework designed for small spacecraft and instruments, and it flew on the Ingenuity Mars Helicopter [83]. HELIOS's bus software is cFS-based. Reuse brings well-tested code and a shared understanding, but it also means that a vulnerability found in a public framework can affect many missions. Willbold et al. found exploitable weaknesses in the firmware of real satellites, including missing protection of telecommand interfaces [84].

Subsystems communicate over data buses designed for determinism and fault tolerance, not security. MIL-STD-1553B is a dual-redundant, 1 Mbit/s command/response bus with a single bus controller [85]. SpaceWire (ECSS-E-ST-50-12C) provides point-to-point, packet-switched links with routers at rates from a few to hundreds of Mbit/s [86]. Controller Area Network (CAN), common on small satellites, is a broadcast bus. None of these buses authenticates its traffic by default, so every component that can transmit on a bus is implicitly trusted.

Fault detection, isolation, and recovery (FDIR) software monitors telemetry points

against limits and responds hierarchically: it retries, switches to a redundant unit, disables a function, or, as a last resort, enters *safe mode*. In safe mode the spacecraft sheds nonessential loads, points its solar arrays at the Sun, keeps a low-rate TT&C link, and waits for ground intervention. Safe mode is designed to make the spacecraft survivable and *commandable*. It often runs a minimal, simpler software image, and that image and its command handling need the same scrutiny as the nominal mode.

Onboard software is updated by uploading files (for example with CFDP), checking their integrity, staging them in non-volatile memory, and switching over under ground control, with a fallback image. Platform flight-software patches are often sent over the TT&C link; given the pass durations of [Example 2.3](#) and TT&C data rates of tens to hundreds of kbps, a multi-megabyte image then needs several contacts. On software-defined satellites, in-orbit reconfiguration of the payload can be frequent, and its much larger files are typically transferred over the high-rate payload or feeder link, while TT&C still carries control, authorization, status monitoring, and fallback [22]. Either way, patches take time to roll out across a large fleet, and an in-progress rollout leaves the fleet running mixed versions.

Two trends define the security of this layer. The first is *IT/OT convergence*: cloud-hosted MOCs, GSaaS APIs, and CI/CD pipelines connect mission operations (operational technology) to enterprise and third-party information technology, and the JPL case shows how an ordinary enterprise weakness can reach systems that support flight. The second is a *long-lived onboard attack surface*. A satellite flies for 5 to 15 years with software and cryptography largely fixed at design time, with limited ability to patch, no physical access, and buses that trust every device connected to them, so a flaw that escapes the ground pipeline may persist for the life of the vehicle. [Chapter 12](#) treats secure boot, update signing, onboard intrusion detection, and verification of flight software.

2.8 The Onboard Control Loop

The attitude determination and control system (ADCS), together with orbit guidance, navigation, and control (GNC), keeps a satellite pointed and on station. It is the place where software decisions become physical motion, which is why security analysis of cyber-physical effects returns to it repeatedly. The standard reference is Markley and Crassidis [87]. This section describes the loop from sensors through estimation and control to actuators, and then marks where adversarial effects can enter it.

2.8.1 Sensors and actuators

Attitude sensors differ in accuracy, rate, and failure modes, and a practical ADCS combines several of them. *Star trackers* match imaged star patterns against a catalog and give full three-axis attitude to arcsecond accuracy, but they can be blinded by the Sun, Moon, or Earth and update only a few times per second. *Sun sensors* give a coarse Sun direction and are used in safe mode. *Magnetometers* measure the geomagnetic field vector, which, combined with a field model, gives

coarse attitude in LEO. *Gyroscopes* (rate sensors) measure angular velocity at a high rate but drift. *GNSS receivers* provide position, velocity, and time for orbit determination and onboard time-tagging.

Actuators turn torque and thrust commands into motion. *Reaction wheels* exchange angular momentum with the spacecraft for fine pointing; they saturate under persistent disturbance torques and must periodically be “desaturated.” *Magnetorquers* produce torque by pushing against the geomagnetic field and are used for momentum dumping and detumbling in LEO. *Thrusters*, which on HELIOS are Hall-effect electric thrusters, change the orbit for orbit raising, station keeping, collision avoidance, and end-of-life disposal.

2.8.2 Estimation, control, and mode logic

Attitude, represented by a unit quaternion q , and angular velocity ω evolve according to

$$\dot{q} = \frac{1}{2} q \otimes \begin{bmatrix} 0 \\ \omega \end{bmatrix}, \quad J\dot{\omega} = -\omega \times (J\omega + h_w) + \tau_c + \tau_d, \quad (2.20)$$

where $\otimes$ is quaternion multiplication, J is the inertia matrix, h_w is the wheel momentum, τ_c is the control torque, and τ_d collects disturbances (gravity gradient, drag, solar pressure, and magnetic torques). An estimator, usually a *multiplicative extended Kalman filter* (MEKF) descended from Kalman’s original filter [88], propagates the state with gyro measurements and corrects it with star-tracker, Sun-sensor, and magnetometer measurements. At each correction the filter compares a measurement with its prediction; the difference, called the *innovation*, is normally small and statistically well characterized, and measurements whose innovations fall outside a gate are rejected. A controller, often a quaternion-feedback proportional-derivative law $\tau_c = -K_p \delta q_v - K_d \omega$ with momentum management, then computes torque commands, where δq_v is the vector part of the attitude-error quaternion and K_p, K_d are gain matrices. Mode logic switches among detumble, Sun-pointing (safe), Earth-pointing, and maneuver modes, and ground commands set the mode, targets, and parameters. Figure 2.17 assembles these elements into a single loop.

The loop in Fig. 2.17 is where cyber and signal effects become physical, and each marked entry point has a distinct character. A false-data injection at A2 that keeps every innovation within the estimator’s gates can bias the attitude estimate without triggering FDIR, which is why slowly growing biases are harder to detect than abrupt ones. Spoofed GNSS at A5 corrupts onboard navigation and time, and dazzling a star tracker forces the estimator to coast on drifting gyros. A manipulated thruster command at A3 changes the orbit, an effect that cannot be reversed without spending propellant. Because the loop runs autonomously between ground contacts, an effect can also grow for many minutes before the ground sees the telemetry that would reveal it; the pass durations of Section 2.2 bound that blind interval. Chapter 15 develops resilient control, safe-mode design, and autonomy under attack on this model.

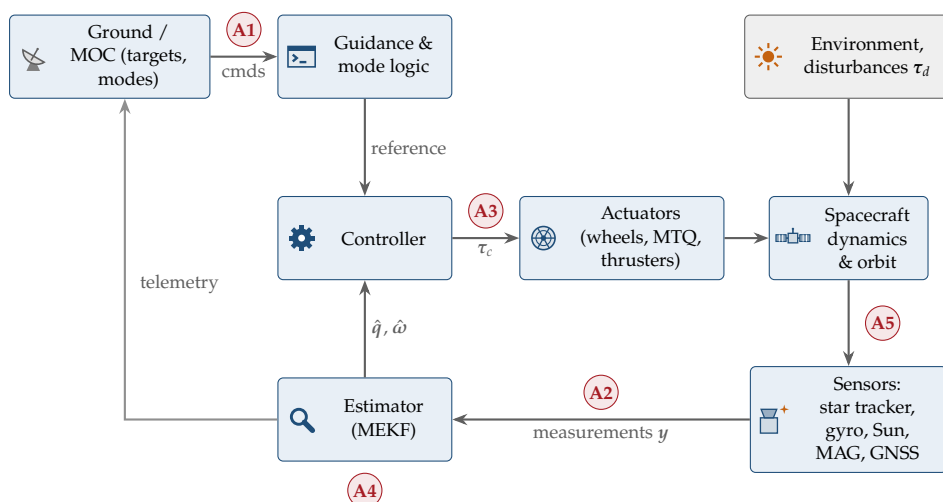


Figure 2.17. The onboard attitude and orbit control loop and five points where adversarial effects can enter: (A1) the command channel (forged, replayed, or legitimately signed but malicious commands); (A2) measurement data in software (false-data injection); (A3) actuator commands (manipulated torque or thrust); (A4) the estimator itself (corrupted parameters, biases, or code); and (A5) the physical sensor input (GNSS spoofing, blinding a star tracker).

2.9 The Space Mission Lifecycle

Security is not a property a satellite acquires at launch. Decisions made a decade before launch fix its cryptography, and ground stations may be decommissioned while old satellites still accept commands. The lifecycle is therefore the time axis of security analysis, complementing the layer and segment axes used so far.

2.9.1 Phases and critical periods

NASA's project lifecycle [89, 90] divides a mission into a *formulation* stage, comprising Pre-Phase A (concept studies), Phase A (concept and technology development), and Phase B (preliminary design and technology completion), and an *implementation* stage, comprising Phase C (final design and fabrication), Phase D (system assembly, integration and test, launch and checkout), Phase E (operations and sustainment), and Phase F (closeout). Figure 2.18 lays out these phases together with the security concerns characteristic of each. Commercial constellations compress and repeat the phases. HELIOS replenishes about 120 satellites a year in batches of about 30, so it is in Phases C through F *simultaneously*.

Within Phases D and E, operators distinguish several critical periods. *Assembly, integration, and test* (AIT) is when flight hardware, software, and often cryptographic keys come together in a facility with many people and much test equipment. *Launch* places the satellite in the custody of the launch provider and, for rideshares, alongside other customers' payloads. The *launch and early orbit phase* (LEOP) covers separation, detumbling, first acquisition of signal, deployment of arrays and antennas, and, for HELIOS, orbit raising with electric propulsion from a

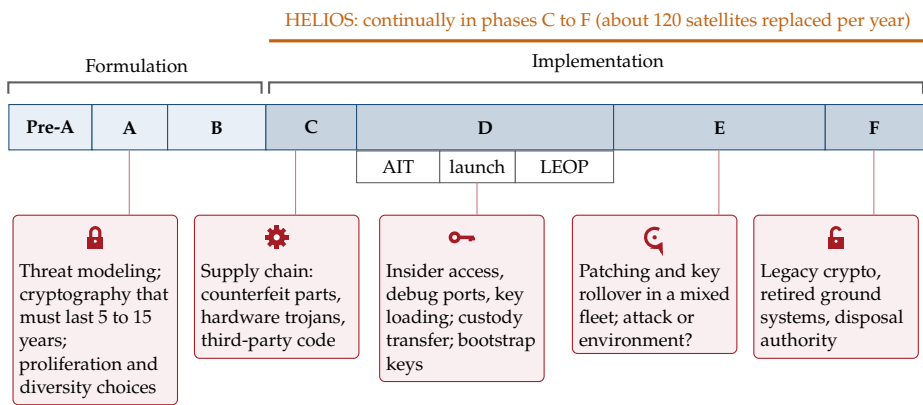


Figure 2.18. The mission lifecycle as a security timeline. Phase widths are schematic. Red boxes list representative security concerns for each period; weaknesses introduced on the left, such as a key-loading procedure or an enabled debug port, persist through every phase to the right. A proliferated constellation such as HELIOS occupies Phases C through F at all times.

low insertion orbit. During LEOP, contact is intense, often through partner ground networks, and the spacecraft is in its most fragile configuration. *Commissioning* checks out the bus and payload before entry into service.

2.9.2 End of life and disposal

End of life requires passivation (venting propellant and discharging batteries) and disposal. In the United States, the Federal Communications Commission adopted a rule in September 2022 that replaced the earlier 25-year guideline. Satellites that end their missions in or pass through LEO (below 2000 km) and dispose of themselves by uncontrolled atmospheric reentry must now reenter as soon as practicable and no more than five years after mission completion. The rule has a two-year grandfathering period, so it applies to satellites launched after 29 September 2024 [91]. Disposal is also a command-authority problem: the ability to deorbit a satellite is, by construction, the ability to end its mission, and it must remain available, and protected, after the rest of the ground system has moved on.

Seen across the lifecycle, each phase offers its own opportunities to an adversary, and weaknesses introduced early, such as a weak key-loading procedure at AIT, a debug port, or an unauthenticated safe-mode command, persist for the life of the satellite. Proliferated constellations change the picture in two opposing ways. Short design lives and continuous replenishment let an operator upgrade cryptography and software fleet-wide within a few years, whereas a GEO satellite is bound to design-time choices for fifteen years. The price is that some fraction of a proliferated fleet is continually in the exposed AIT, launch, and LEOP periods, so those periods must be defended as a routine operation rather than as a one-time event. [Chapter 16](#) develops lifecycle and supply-chain security.

2.10 Summary

This chapter built a quantitative model of a space system's nominal behavior, because security analysis cannot distinguish an attack from physics without one. Orbital motion is predictable and public: vis-viva, Kepler's third law, and J_2 precession give the 95.44 min period of HELIOS Shell A, the 97.63° sun-synchronous inclination of Shell B, and ground tracks that an observer with a public element set can compute. Access geometry turns those orbits into contact windows, with a 927 km footprint radius at 25° elevation and TT&C passes of at most about 8 min, and the same windows are available to operators and adversaries alike. Walker-delta constellations with +Grid inter-satellite links give proliferated LEO systems graceful degradation, but they also give them a regular, analyzable network structure.

A space system consists of ground, link, space, and user segments, and its security depends on the trust boundaries at their interfaces, most of which are open RF media or systems administered by third parties. The link budget $C/N_0 = \text{EIRP} - L_{\text{fs}} - L_a + G/T - k_B$ reproduces the HELIOS reference values: C/N_0 falls from 101.6 dB-Hz at nadir to 95.4 dB-Hz at the coverage edge, cutting clear-sky throughput by a third, while the TT&C downlink closes with $E_b/N_0 \approx 27$ dB and about 17 dB of margin. Link margin is the defender's budget against interference, and adaptive coding and modulation spends it gracefully, which can also mask a low-level attack. A GEO system pays about 36 dB of extra path loss relative to LEO and must recover it with larger antennas and higher power.

Above the physical link, protocols each protect at most one layer. CCSDS frames gain authentication and encryption only when the optional SDLS sublayer is used, DVB-S2X carries no native link security, and the scheduling and routing functions of DTN and software-defined networking are themselves control channels. Ground computing is converging with cloud and enterprise IT, while onboard computing is long-lived, hard to patch, and built on buses that trust every connected device. The onboard control loop is where these cyber and signal effects become physical, through five distinct entry points at commands, measurements, the estimator, actuators, and the sensors themselves.

Finally, security concerns shift over the mission lifecycle, from threat modeling and supply-chain integrity in design and fabrication, through the exposed periods of integration, launch, and early orbit, to patching in operations and command authority at disposal. A proliferated constellation lives in all of these periods at once. Together, these baselines supply the nominal performance Q_0 against which degradation, detection, and recovery are measured throughout the rest of the book.

Exercises

Exercise 2.1 [C] Explain why orbital predictability benefits both an adversary and a defender. Give one concrete example of each, drawing on access windows, SDA, and maneuver planning, and describe one way an operator can reduce the adversary's advantage without spending propellant.

Exercise 2.2 [A] (a) Compute the sun-synchronous inclination for circular orbits at 400 km and 800 km. (b) HELIOS Shell A regresses at about -4.51° per day. What inclination at 540 km would give zero nodal drift, and why is it unsuitable for broadband service? (c) Explain qualitatively how an eccentric orbit changes (2.3).

Exercise 2.3 [A] Derive (2.5) through (2.7) from the law of sines and the law of cosines applied to the triangle in Fig. 2.6. Show that $d \rightarrow h$ as $\varepsilon \rightarrow 90^\circ$, and find the slant range to the horizon ($\varepsilon = 0^\circ$) for HELIOS Shell A.

Exercise 2.4 [A] For HELIOS Shell B (560 km) and a TT&C station with $\varepsilon_{\min} = 10^\circ$, compute $\lambda_{\max}$, the footprint radius, and the maximum pass duration using (2.9). Using (2.10), find the pass duration when the ground track misses the station by a central angle of 10° . If a patch of 4 MB must be uploaded at 32 kbps with 20 % protocol overhead, how many such passes are needed?

Exercise 2.5 [A] Compare Walker-delta constellations $53^\circ:528/24/1$ and $53^\circ:528/48/1$. For each, compute satellites per plane, in-plane spacing, node spacing, and intra-plane ISL chord length at 540 km. Discuss which design is more robust to the loss of a single launch batch that populates one plane, and which is more expensive to deploy.

Exercise 2.6 [A] A HELIOS terminal at 25° elevation experiences 4 dB of rain attenuation, which also raises its system noise temperature so that G/T drops by 1.5 dB. Recompute C/N_0 and C/N , select the most efficient MODCOD from Table 2.4 with a 1 dB margin, and compute the throughput. What fraction of the clear-sky nadir throughput remains?

Exercise 2.7 [A] Using (2.19), compute the ratio J_0/N_0 at which (a) the HELIOS TT&C downlink of Example 2.6 loses its margin against a required E_b/N_0 of 9.6 dB, and (b) the Ku user link at 25° must drop from 16APSK 3/4 to QPSK 3/4. Discuss what these numbers say about which link an operator should monitor most closely for low-level interference.

Exercise 2.8 [A] Starting from $C_{\text{Sh}} = B \log_2(1 + C/N)$, derive the minimum E_b/N_0 as a function of spectral efficiency ρ in (2.18), and show that it tends to $\ln 2$ as $\rho \rightarrow 0$. For each MODCOD selected in Example 2.5, compute its distance in dB from the Shannon bound.

Exercise 2.9 [C] An operator states: “Our telecommands are protected by SDLS with AES-GCM, so our command path is secure.” Using Table 2.5 and Fig. 2.9, list at least four attacks on the command path that this statement does not address, and name the layer and segment of each.

Exercise 2.10 [C] HELIOS leases six of its eight TT&C sites from a GSaaS provider. Identify the new trust boundaries this creates compared with an all-owned network, and propose technical and contractual controls for each. Which controls still protect the satellite if the provider’s own network is compromised?

Exercise 2.11 [C] Relate the JPL incident in Section 2.7 to the HELIOS ground architecture. Which HELIOS interface in Table 2.3 is most analogous to the

unsegmented gateway, and what monitoring would have detected an unauthorized device?

Exercise 2.12 [A] A satellite's gyro has an uncompensated bias of $0.01^\circ \text{ h}^{-1}$. (a) If star-tracker updates are unavailable for 30 min, for example because the tracker is blinded, how large an attitude error accumulates? (b) Suppose instead that an adversary injects a slowly growing bias into the gyro data at point A2 of Fig. 2.17 while star-tracker updates continue. Explain qualitatively how the estimator's innovation sequence could reveal the attack, and why a slowly growing bias is harder to detect than a step.

Exercise 2.13 [C] A GEO satellite launched fifteen years ago still accepts commands authenticated with a key-management scheme designed in its Phase B. Its operator has retired the original ground equipment. Using Fig. 2.18, discuss the security risks of the end-of-life phase and propose a decommissioning checklist.

Exercise 2.14 [P] **Chapter Lab: orbits, passes, and link budgets.** (a) Using the Python `sgp4` or `skyfield` library, download a current public TLE for a real LEO satellite (for example, the International Space Station or an Earth-observation satellite) from CelesTrak, and propagate it for 48 hours. Plot its ground track. (b) For a ground station of your choice with $\varepsilon_{\min} = 10^\circ$, compute all passes (rise, culmination, set, maximum elevation, duration). Compare the longest pass with (2.9). (c) Write a link-budget calculator implementing (2.12) through (2.16) and reproduce the table in Example 2.5 and the results of Example 2.6. (d) Using a circular orbit at 540 km and 53° inclination as a stand-in for a HELIOS satellite, compute C/N as a function of time over a pass above a terminal at 40° N latitude, apply the MODCOD selection rule of Example 2.5, and plot throughput versus time. Report the time-averaged throughput of the pass. (e) Briefly discuss how TLE age affects the accuracy of your pass predictions, and what that implies for an adversary relying only on public data.

Exercise 2.15 [P] Simulate HELIOS Shell A as an ideal Walker-delta constellation using two-body propagation with J_2 nodal precession. On a latitude grid from -70° to 70° , compute over one day the time-averaged number of satellites above 25° elevation and the fraction of time with at least one. Then remove all satellites of one plane and recompute. Plot both results versus latitude and interpret them in terms of graceful degradation.

Notes and Further Reading

Orbital mechanics. Vallado [39] is the standard reference for astrodynamics, with algorithms for coordinate systems, propagation, perturbations, and orbit determination. Curtis [40] is a gentler, example-rich introduction for students. Montenbruck and Gill [43] cover precise orbit models and determination. The history and details of SGP4 and the TLE format are in Spacetrack Report No. 3 [41] and its revision by Vallado et al. [42], which is also the source of the widely used open-source implementations. Walker's paper [46] remains the classic source on symmetric constellations.

Mission engineering. *Space Mission Engineering: The New SMAD* [38] covers the whole system (orbits, coverage, subsystems, ground systems, operations, and cost) and is the best single reference for the segment architecture of [Section 2.4](#). The NASA Systems Engineering Handbook [89] and NPR 7120.5 [90] define the lifecycle phases of [Section 2.9](#).

Satellite communications. Maral, Bousquet, and Sun [51] and Pratt and Allnutt [52] are comprehensive texts on link budgets, propagation, access, and payloads. Sklar [53] covers modulation, coding, and error rates. Del Portillo et al. [48] compare the architectures of real LEO broadband constellations, including ISLs and throughput estimates, and Handley [47] analyzes low-latency routing over LEO ISLs. Kodheli et al. [62] survey modern satellite communications, and Lin et al. [78] introduce 3GPP NTN. Rain and gaseous attenuation models are in ITU-R Recommendations P.618, P.838, and P.676 [58, 59, 60].

Protocols and standards. CCSDS Blue Books are freely available from the CCSDS website. Start with the overview Green Book [61], then the data link protocols [64, 65, 66, 67], the Space Packet Protocol [68], and SDLS and its extended procedures [70, 71]. The CCSDS security Green Book [72] explains the security rationale. DVB-S2 and S2X are specified in ETSI EN 302 307 [55, 56]. DTN is specified in RFC 4838, RFC 9171, and RFC 9172 [74, 75, 76]. [Chapter E](#) collects these standards.

Computing and control. The core Flight System [82] and F' [83] are open source and are good starting points for hands-on study of flight software. Markley and Crassidis [87] is the standard modern text on attitude estimation and control.

Security perspectives. Pavur and Martinovic [50] survey six decades of satellite cyber incidents and argue that the ground and link segments have been the dominant targets. Willbold et al. [84] give an experimental security analysis of real satellite firmware. Pavur et al. [73] demonstrate eavesdropping on maritime VSAT traffic. Huang et al. [49] study the security of constellation-wide operations over ISLs. NIST IR 8270 and IR 8401 [80, 79] and Space Policy Directive-5 [34] give policy-level guidance for commercial and government operators. The Starlink loss of February 2022 is analyzed in the space-weather literature, for example by Fang et al. [45].

Security and Resilience: Concepts and Metrics

Learning Objectives

- ▶ Distinguish security, safety, sustainability, and mission assurance, and give formal definitions of reliability, robustness, survivability, fault tolerance, resilience, and reconstitution, stating the quantitative question each one answers.
- ▶ State the security properties that matter at each layer of a space system, and model a space system as a networked cyber-physical system in which attacks enter through identifiable channels.
- ▶ Relate the resilience goals of anticipate, withstand, recover, and adapt to the architectural resilience attributes and to defense in depth and zero trust.
- ▶ Compute resilience-curve metrics (depth, time to recover, loss area, normalized resilience) for a mission performance function $Q(t)$, and compute availability and expected loss from a continuous-time Markov degradation-recovery model.
- ▶ Evaluate network resilience with giant-component size and algebraic connectivity under random and targeted node removal.
- ▶ Compare expected, conditional value-at-risk, and worst-case resilience, and explain why the ranking of architectures can depend on the choice.

Words such as *secure*, *reliable*, *robust*, *survivable*, and *resilient* are used loosely in policy documents, requirements specifications, and engineering reviews, and often interchangeably. The looseness is costly. A requirement that a constellation be “resilient” cannot be verified until it states against which disturbances, over what horizon, and with what measure of mission performance. A design that is highly reliable against random part failures may collapse under a single targeted action, and a design optimized for its expected performance may be the worst choice when an adversary chooses the scenario. This chapter supplies the shared vocabulary and the quantitative machinery that the rest of the book relies on. It defines each concept precisely, states the mathematical question that each concept answers, and develops the families of metrics that turn the word “resilience” into

	Ground	Link	Space	User
Mission/strategic				
Cyber				
Physical/orbital				
Link & network				
Signal/RF				

Figure 3.1. Layer map for this chapter. The chapter’s primary home is the mission/strategic row (dark cells), where resilience is measured as preserved mission performance across all four segments. Every other cell is covered secondarily (light cells), because the definitions, security properties, and metrics of the chapter apply at each layer and segment.

numbers that can be computed, compared, and optimized.

The chapter moves from concepts to metrics. Sections 3.1 and 3.2 separate security, safety, sustainability, and mission assurance, and then define reliability, robustness, survivability, fault tolerance, resilience, and reconstitution. Sections 3.3 and 3.4 state the security properties at each layer and model a space system as a cyber-physical system. Sections 3.5 and 3.6 present resilience goals, architectural attributes, defense in depth, and zero trust. The final four technical sections quantify resilience: the resilience curve and its metrics (Section 3.7), Markov degradation-recovery models (Section 3.8), network metrics (Section 3.9), and risk-averse and worst-case resilience (Section 3.10).

The book organizes security problems on a grid of *layers* (mission/strategic, cyber, physical/orbital, link and network, and signal/RF) and *segments* (ground, link, space, and user). Figure 3.1 places this chapter on that grid. Unlike the layer-specific chapters, it occupies the whole mission row: resilience is ultimately judged by the service delivered to users, whatever layer the disturbance enters through.

HELIOS Running Example

HELIOS is a *notional* proliferated low-Earth-orbit (LEO) broadband and positioning, navigation, and timing (PNT) augmentation constellation created for teaching; it is not modeled on a real operator. It has 600 satellites, 40 gateways, and eight telemetry, tracking, and command (TT&C) sites, six of which are leased from a single ground-station-as-a-service provider.

During an architecture review, the program office presents three slides. The first reports a predicted TT&C network availability of 0.99998, computed from independent site outages. The second reports that the constellation “degrades gracefully” because the loss of each individual satellite reduces coverage by less than one percent. The third asks the board to choose

between buying more on-orbit spares and adding owned TT&C sites, and ranks the options by expected annual service loss.

A reviewer from the security office raises three questions. What happens to commanding if the leasing provider is compromised rather than merely unlucky? Which satellites would an adversary remove first, and how fast does the network fragment then? Would the ranking survive if the board cared about the worst plausible month rather than the average one? This chapter builds the tools needed to answer each question.

The notation follows the list at the front of the book; in this chapter t_d and t_r denote the disruption time and the time at which recovery is complete, and symbols that appear only inside one example are defined where they are used.

3.1 Security, Safety, Sustainability, and Mission Assurance

Four words describe what the owner of a space system wants to protect and why: security, safety, sustainability, and mission assurance. They overlap heavily in practice, since a single maneuver can serve all four, yet they differ in the source of harm they address, in who bears the harm, and in the evidence that demonstrates success. Keeping them apart is the first step toward metrics that can be verified.

3.1.1 Four concerns and their sources of harm

Definition 3.1 (Security). *Security* is the protection of a system, its data, and the services it provides against *intentional* actions by adversaries. In the dependability tradition it is the concurrent existence of confidentiality, integrity, and availability, where confidentiality means the absence of unauthorized disclosure of information, integrity means the absence of improper alteration, and availability means readiness for correct service to authorized users [92].

The defining feature of security is an intelligent source of harm. An adversary observes the defender, chooses the time and place of action, and adapts after each attempt. This is why security analysis cannot be reduced to failure statistics: the probability that a vulnerability is exploited is not a property of the hardware but of the adversary's intent, capability, and knowledge.

Definition 3.2 (Safety). *Safety* is freedom from unacceptable risk of harm to people, property, or the environment caused by the system, whether through accidents, random failures, or design errors. In the dependability taxonomy it is the absence of catastrophic consequences on users and the environment [92].

Safety differs from security in two ways. Its hazards are usually unintentional, and the harm it addresses often falls on parties outside the system: a spacecraft that reenters uncontrollably endangers people on the ground, and a collision creates debris that endangers other operators. Safety arguments therefore rest

on hazard analysis and failure probabilities, whereas security arguments rest on threat models.

Definition 3.3 (Space sustainability). *Space sustainability* is the ability to continue conducting space activities indefinitely while preserving the orbital environment, the radio spectrum, and equitable access for future users. The guidelines for the long-term sustainability of outer space activities adopted by the United Nations Committee on the Peaceful Uses of Outer Space (COPUOS) in June 2019 give the internationally agreed formulation of this idea: activities should meet the needs of present generations while preserving the outer space environment for future generations [93].

Sustainability extends the time horizon and the set of stakeholders beyond those of safety. Its characteristic harms are externalities: a debris-generating event, whether accidental or deliberate, degrades an orbital regime for every operator for decades.

Definition 3.4 (Mission assurance). *Mission assurance* is the ability of an organization to deliver the mission functions its users depend on, with acceptable performance, across the full range of conditions it may face, whatever their cause. The U.S. Department of Defense taxonomy for the space domain distinguishes three contributors to mission assurance: *defensive operations*, which act against a threat; *reconstitution*, which restores capability after loss; and *resilience*, which is built into the architecture itself [94].

Mission assurance is the unifying objective. It is indifferent to cause: a lost service is a lost service whether it follows a jamming campaign, a solar storm, or a software fault. Security, safety, and sustainability are each necessary for mission assurance, and none is sufficient.

3.1.2 Where the concerns overlap

Figure 3.2 places representative concerns in the regions defined by the three narrower concepts. Some concerns are exclusive to one concept. Authentication of telecommands is purely a security measure, casualty-risk analysis for reentering hardware is a safety measure, and end-of-life disposal and spectrum coordination belong to sustainability. The overlaps are more instructive. A malicious thruster command is both a security and a safety problem, because the adversary exploits a safety-critical function. Debris from a destructive anti-satellite (ASAT) test is a security event with a lasting sustainability cost. Collision avoidance and the sharing of space domain awareness (SDA) data serve safety and sustainability together. The triple intersection, a hijacked maneuver capability, shows why space systems resist the separation of these disciplines that is customary in terrestrial engineering: flight software that commands propulsion is simultaneously a security boundary, a safety interlock, and a guardian of the shared orbital environment.

The overlap has a practical consequence for metrics. Safety engineering has mature probabilistic tools, such as fault trees and failure-rate data, that invite

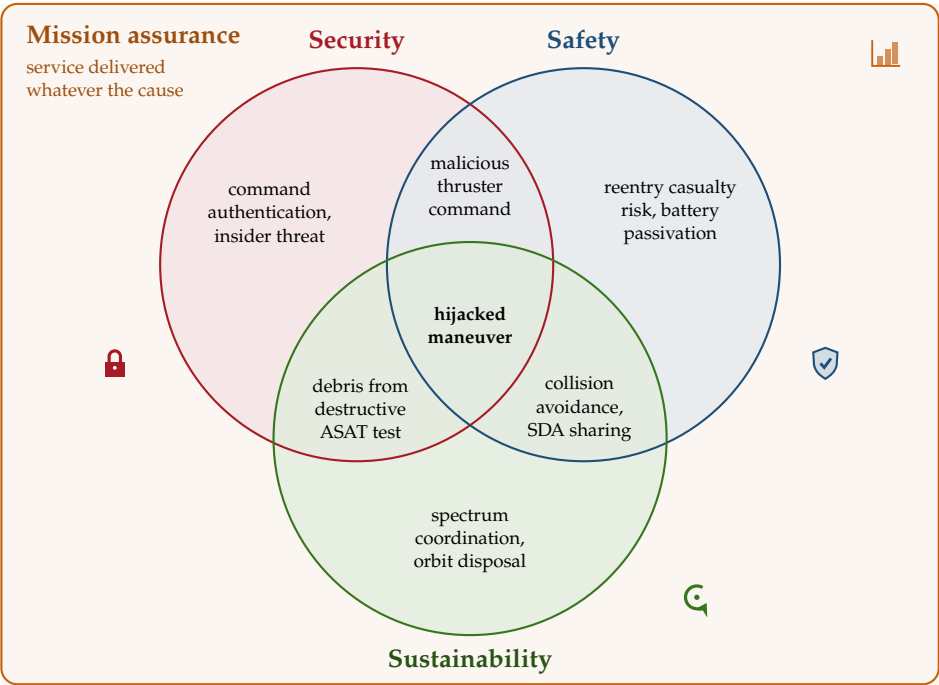


Figure 3.2. Security, safety, and sustainability as overlapping concerns inside mission assurance, with example concerns in each region. The central region holds the case that matters most for space: an adversary who seizes a maneuver function creates a security incident, a safety hazard, and a sustainability harm at once.

reuse for security. They can be reused only with care, because the probabilities that safety analysis treats as fixed become decision variables of the adversary in security analysis. The remainder of the chapter keeps this distinction explicit: stochastic models describe nature and accidents, while worst-case and game formulations describe adversaries. Mission assurance, measured through the performance function $Q(t)$ introduced in Section 3.7, is the common scale on which both are judged.

3.2 Reliability, Robustness, Survivability, Fault Tolerance, Resilience, and Reconstitution

Six related concepts describe how a system behaves when something goes wrong. Each answers a different question, uses different mathematics, and is verified with different evidence. We define them in the order in which they relax assumptions about the disturbance: reliability assumes random failures and asks only whether the system works; robustness admits bounded uncertainty; survivability admits deliberate damage; fault tolerance adds internal mechanisms that mask faults; resilience admits loss and measures the trajectory of recovery; and reconstitution addresses the restoration of capability that the architecture could not absorb.

3.2.1 Reliability

Definition 3.5 (Reliability). Let $\tau \geq 0$ be the random time to failure of a system that is operating at $t = 0$. The *reliability function* is $R(t) = \mathbb{P}(\tau > t)$, the probability that the system performs its required function without failure over $[0, t]$ under stated conditions [95]. The *hazard rate* is $h(t) = -\dot{R}(t)/R(t)$, and the *mean time to failure* is $\text{MTTF} = \int_0^\infty R(t) dt$.

For a constant hazard rate $h(t) = \lambda$, the lifetime is exponential,

$$R(t) = e^{-\lambda t}, \quad \text{MTTF} = 1/\lambda, \quad (3.1)$$

which is the standard model for electronic parts during their useful life. Wear-out mechanisms, such as battery degradation or propellant depletion, give increasing hazard rates and are commonly modeled with Weibull lifetimes $R(t) = \exp[-(t/\eta)^\beta]$ with shape $\beta > 1$. System reliability follows from the structure of the system. A series system of n independent components has $R_s(t) = \prod_i R_i(t)$, a parallel system has $1 - \prod_i (1 - R_i(t))$, and a k -out-of- n system of identical independent components, which works when at least k components work, has

$$R_{k|n}(t) = \sum_{j=k}^n \binom{n}{j} R(t)^j (1 - R(t))^{n-j}. \quad (3.2)$$

For repairable systems the analogous quantity is the *availability*, the probability that the system is operating at a given time, which Section 3.8 develops. Reliability answers the question: *with what probability does the system function over a given period when failures are random and independent?* The answer depends critically on independence, which is the first assumption an adversary violates.

3.2.2 Robustness

Definition 3.6 (Robustness). Let $J(\mathcal{A}, \delta)$ be a performance index of architecture $\mathcal{A}$ when an uncertain parameter δ takes a value in an *uncertainty set* $\mathcal{U}$ that contains the nominal value δ_0 . The architecture is *robust* at level $J_{\min}$ if $\inf_{\delta \in \mathcal{U}} J(\mathcal{A}, \delta) \geq J_{\min}$. The *robustness margin* is the size of the largest uncertainty set, for example the largest radius ρ of $\mathcal{U}_\rho = \{\delta : \|\delta - \delta_0\| \leq \rho\}$, for which the requirement holds.

Robustness answers the question: *does performance remain acceptable for every perturbation within a stated bound?* Its mathematics is that of uncertainty sets, sensitivity analysis, and worst-case design. For small perturbations the local sensitivity $\partial J / \partial \delta$ at δ_0 indicates which parameters matter; link margins, control-loop stability margins, and the H_∞ norm of a disturbance-to-error map are familiar examples of robustness measures. Robustness is *static* in the sense that it concerns the instantaneous performance level, not the time needed to return to it, and *bounded* in the sense that no guarantee is made outside $\mathcal{U}$.

3.2.3 Survivability

Definition 3.7 (Survivability). Let C be the set of components of a system and let a *damage event* remove a subset $D \subseteq C$. Let $\Phi(D) \in \{0, 1\}$ indicate whether the

essential mission function is still delivered after the loss of D . The system *survives* D if $\Phi(D) = 1$. Against random damage with distribution $\mathbb{P}_D$, survivability is $\mathbb{P}(\Phi(D) = 1)$; against an adversary who may remove an arbitrary set of at most m components, the *survivability order* is the largest m such that $\min_{|D| \leq m} \Phi(D) = 1$.

Survivability answers the question: *is the essential mission still delivered after deliberate or hostile damage?* The worst-case form connects directly to graph theory. If Φ requires a connected path between two nodes of a network, the survivability order is one less than the minimum vertex cut separating them. The difference between the probabilistic and the worst-case forms is the difference between error tolerance and attack tolerance, a distinction that [Section 3.9](#) quantifies for constellation networks.

3.2.4 Fault tolerance

Definition 3.8 (Fault tolerance). A system is *fault tolerant* if it continues to deliver correct service in the presence of faults, by means of redundancy together with fault detection, isolation, and recovery (FDIR) [92]. The effectiveness of FDIR is summarized by the *coverage* $c \in [0, 1]$, the conditional probability that a fault is detected and the system is successfully reconfigured given that the fault occurs.

Fault tolerance is a mechanism rather than an outcome, and coverage is its key parameter. Consider the cold-spares onboard computer of a HELIOS satellite. With a primary unit of failure rate λ , an unpowered spare assumed not to fail in storage, and switchover coverage c , the reliability of the pair is

$$R_{cs}(t) = e^{-\lambda t} (1 + c \lambda t). \quad (3.3)$$

With perfect coverage the spare doubles the MTTF; with $c = 0$ the spare is useless. Security enters through coverage: an adversary who can suppress fault detection, or trigger a switchover to a spare running the same compromised software, sets the effective coverage to zero. Fault tolerance answers the question: *can internal faults be masked before they become service failures?*

3.2.5 Resilience

The preceding concepts judge a system at a point in time or by a binary outcome. Resilience judges the whole trajectory of mission performance through a disruption. Zhu and Başar [27] argue that robustness and resilience are dual rather than nested properties: robustness is designed against an uncertainty set specified in advance, whereas resilience concerns the response once a disturbance has left that set, and the two call for different game-theoretic formulations. Let $Q(t; \mathcal{A}, \pi, w) \geq 0$ denote mission performance at time t under architecture $\mathcal{A}$, defender policy π (detection, response, rerouting, and recovery rules), and disturbance or attack w , and let $Q_0 > 0$ be nominal performance. The attack set decomposes by layer, $\mathcal{W} = \mathcal{W}_{\text{sig}} \times \mathcal{W}_{\text{net}} \times \mathcal{W}_{\text{phys}} \times \mathcal{W}_{\text{cyb}}$, so that a campaign across layers is a tuple of signal, network, physical, and cyber actions.

Definition 3.9 (Resilience functional). Over a horizon $[0, T]$, the *normalized resilience* of $(\mathcal{A}, \pi)$ against a single disturbance w is

$$R(\mathcal{A}, \pi; w) = \frac{1}{T} \int_0^T \frac{Q(t; \mathcal{A}, \pi, w)}{Q_0} dt,$$

and the *worst-case normalized resilience* over the attack set $\mathcal{W}$ is

$$R(\mathcal{A}, \pi; \mathcal{W}) = \inf_{w \in \mathcal{W}} \frac{1}{T} \int_0^T \frac{Q(t; \mathcal{A}, \pi, w)}{Q_0} dt. \quad (3.4)$$

Resilience answers the question: *how much mission performance is preserved, and how quickly is it restored, when disruptions succeed?* Following NIST systems security engineering guidance, we organize the trajectory into four goals or phases: to *anticipate, withstand, recover from, and adapt to* adversity [26]. The functional equals one when performance stays at or above nominal throughout and zero when the mission is lost for the whole horizon under some attack. When Q may exceed Q_0 after adaptation, we cap the integrand at one unless the benefit of over-performance is intended to offset earlier losses. Section 3.7 develops the metrics derived from the trajectory, and Section 3.10 replaces the infimum with risk measures when disturbances are better described probabilistically.

3.2.6 Reconstitution

Definition 3.10 (Reconstitution). *Reconstitution* is the restoration of a capability after losses that exceed what the fielded architecture can absorb, by deploying replacement assets, activating alternative providers, or re-establishing destroyed ground infrastructure [94]. Its principal metric is the *reconstitution time* $T_{rc} = \inf\{t \geq t_d : Q(t) \geq q^* Q_0\}$ for a target fraction q^* , together with its cost.

Reconstitution answers the question: *how long and how much does it take to rebuild what was lost?* For HELIOS, reconstitution after the loss of satellites means phasing an on-orbit spare into the gap, which takes about seven days per plane, or calling up a replacement launch, with replacement satellites operational some 60 to 75 days after the loss. For ATLAS-G, the geostationary contrast system, the answer is leased transponders restoring roughly 30% of capacity after two weeks and a full replacement after about three years. Reconstitution time is thus dominated by logistics and industrial capacity rather than by engineering within the fielded system.

Figure 3.3 summarizes the six concepts. The placement is schematic, since the boundaries depend on the mission, but the ordering is informative. Moving up the severity axis replaces probabilistic arguments by worst-case ones, and moving right along the horizon axis replaces instantaneous measures by integrals over time. Resilience occupies the center because it links both directions: it measures the cumulative consequence of severe disturbances, and it is bounded below by robustness and fault tolerance, which set the depth of the initial drop, and above by reconstitution, which sets how quickly lost capacity can return.

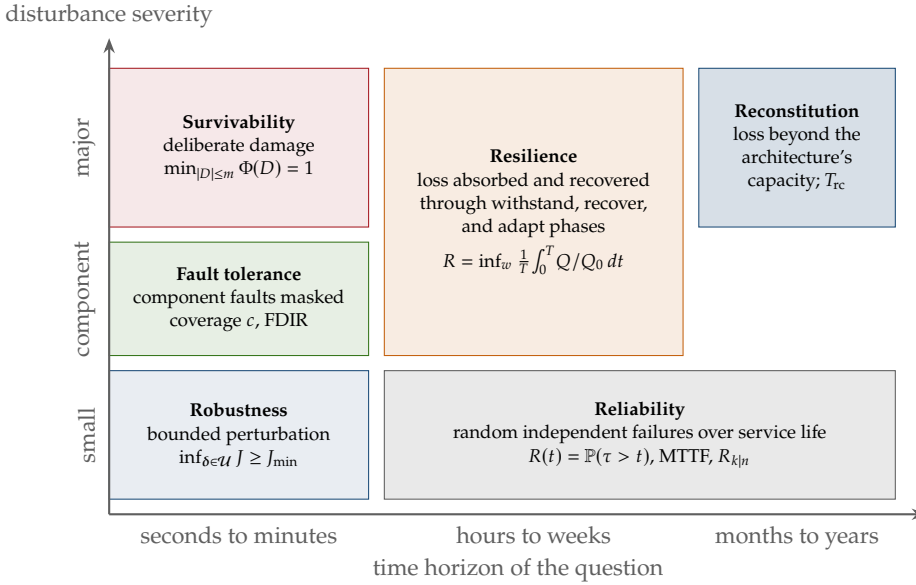


Figure 3.3. The six concepts placed on a plane of disturbance severity and the time horizon of the question they answer. Robustness and fault tolerance act within seconds to minutes against small or component-level disturbances; reliability integrates random failures over the service life; survivability concerns the immediate outcome of deliberate damage; resilience spans the hours to weeks of withstanding and recovering; reconstitution rebuilds losses beyond the architecture's capacity over months to years.

Example 3.1 (Reliability versus survivability of the HELIOS TT&C network). HELIOS commands its fleet through eight TT&C sites: two owned high-latitude sites and six sites leased from one ground-station-as-a-service provider. Full commanding cadence requires at least $k = 4$ sites. Each site is unavailable with probability 0.05, independently of the others. We compare the availability of the network with its survivability.

With independent outages, (3.2) with $R = 0.95$, $n = 8$, and $k = 4$ gives

$$A_{4|8} = \sum_{j=4}^8 \binom{8}{j} 0.95^j 0.05^{8-j} = 0.999985,$$

an unavailability of 1.5×10^{-5} , or about eight minutes per year. In the worst-case sense of Definition 3.7, the network survives the loss of an arbitrary set of four sites, so its survivability order counted in sites is $m = 4$. Both numbers are reassuring, and both assume that the unit of loss is a single site. If instead a single action, such as the compromise of the provider's shared scheduling system, removes all six leased sites at once, only the two owned sites remain and the requirement $k = 4$ fails with certainty. Counted in adversary actions, the survivability order is zero. Even without an adversary, a common-cause provider outage with probability $q = 0.01$ reduces availability to approximately $(1 - q) A_{4|8} \approx 0.990$, an unavailability three orders of magnitude larger than the independent estimate.

The binary view also hides a gradation. If commanding capacity scales as $\min(n_{\text{up}}/4, 1)$, the two owned sites still deliver half of the required cadence, so the mission degrades rather than fails. High reliability computed under independence thus says little about survivability against a single well-chosen action, and a graded performance measure, rather than a binary one, is what resilience analysis requires.

3.3 Security Properties by Layer

A security requirement is meaningful only when it names the property to be protected and the object that must have it. The classical triad of confidentiality, integrity, and availability is necessary for space systems but not sufficient. Commands must be *authentic* as well as unaltered, actions by operators and suppliers must be *attributable*, control loops and collision-avoidance maneuvers depend on *timely* delivery, and users of positioning and timing services need a quantified guarantee that a hazardous error will be flagged. This section defines these properties and maps them to the layers of a space system.

3.3.1 Properties

Let M be a message, command, data object, or signal, and let a system process it on behalf of a set of authorized principals. *Confidentiality* holds when no unauthorized party learns M , or, in the information-theoretic sense, when the mutual information between M and an eavesdropper's observation is negligible. *Integrity* holds when every unauthorized modification of M is either impossible or detected. *Availability* holds when authorized principals obtain correct service when they need it; operationally it is measured as the fraction of time, or the probability at a random instant, that service is delivered. *Authenticity* holds when the receiver can verify the claimed origin of M ; on the telecommand uplink this is the property that defeats command injection, and it is typically provided by message authentication codes or authenticated encryption. *Non-repudiation* holds when a principal cannot credibly deny having originated M , which requires digital signatures and trustworthy logs rather than shared keys, and which underpins the attribution of insider actions and of software updates in the supply chain.

Two further properties are specific to cyber-physical and navigation services. *Timeliness* holds when M is delivered within a deadline D with high probability, $\mathbb{P}(\text{delay} \leq D) \geq 1 - \varepsilon$. A correct but late collision-avoidance command is a failed command. *PNT integrity* is the ability of a positioning or timing service to warn users within a specified time when its output should not be trusted. It is quantified by an *alert limit* AL , a *time to alert* T_A , and an *integrity risk* P_{IR} :

$$\mathbb{P}(\|e_t\| > AL \text{ for longer than } T_A \text{ without an alert}) \leq P_{\text{IR}}, \quad (3.5)$$

where e_t is the position or time error. Integrity in this sense is distinct from cryptographic integrity. An authenticated navigation message proves origin, but a replayed or delayed authentic signal can still produce a hazardous error, so PNT integrity must be protected at both the message and the signal level.

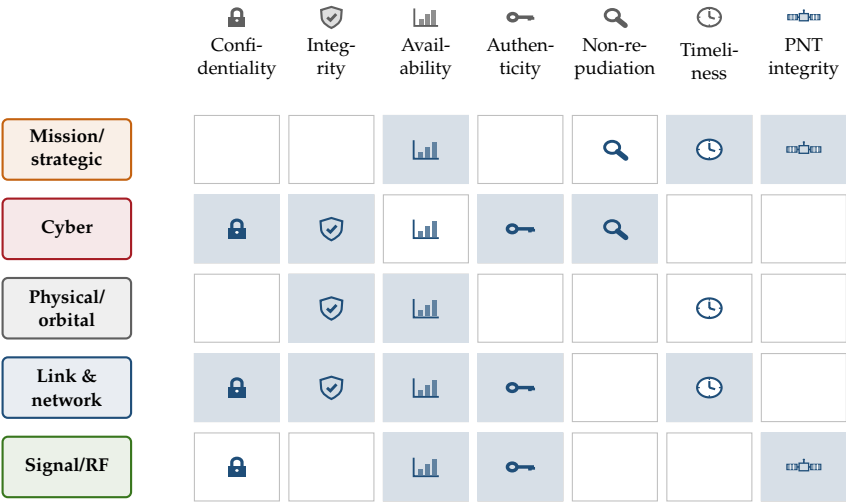


Figure 3.4. Security properties mapped to the layers of a space system. Shaded cells with icons mark properties that are primary at a layer; unshaded cells with icons mark secondary relevance. Availability appears at every layer, which is why mission performance $Q(t)$ is ultimately an availability-type measure; authenticity is primary wherever commands or signals can be forged, and PNT integrity links the signal layer directly to the mission.

3.3.2 The layer map of properties

Figure 3.4 maps the seven properties to the five layers. At the *signal/RF* layer, the dominant concerns are availability against jamming, authenticity against spoofing of navigation and timing signals, and PNT integrity; confidentiality at this layer is limited to low-probability-of-intercept waveforms and physical-layer secrecy. At the *link and network* layer, confidentiality, integrity, and authenticity are provided cryptographically, for example by authenticated encryption of telecommand frames, while availability and timeliness depend on routing and on resistance to flooding. The *physical/orbital* layer is concerned with the integrity of the spacecraft as a physical object, with the availability of its function in the presence of debris, space weather, or deliberate damage, and with the timeliness of maneuvers. The *cyber* layer carries the full triad for ground systems, flight software, and terminals, together with authenticity of software updates and non-repudiation of operator and supplier actions. At the *mission/strategic* layer, the properties are those that users experience: availability of service, timeliness of delivery, the trustworthiness of positioning and timing, and the attributability of hostile acts, which conditions deterrence.

The map has two uses. As a requirements checklist, it prevents the common error of specifying confidentiality for a link that mainly needs availability and authenticity. As an analytical device, it identifies cross-layer dependencies. For example, a timing error at the signal layer (a PNT integrity failure) can invalidate certificate validity checks at the cyber layer, which depend on correct time. Every property in the figure contributes, directly or through such dependencies, to the availability of mission performance, and this is why the rest of the chapter

measures resilience through $Q(t)$.

3.4 Space Systems as Cyber-Physical Systems

A space system is a *cyber-physical system*: computation and communication close feedback loops around physical processes. Attitude and orbit control, thermal and power management, beam steering, and clock steering are all loops in which a sensor measures a physical state, software computes a decision, and an actuator changes the state. Many of these loops pass through networks, either on board or across the space-ground link, so that a disturbance at the signal, link, or cyber layer can end as a physical consequence. A compact state-space model makes these pathways explicit and lets us say precisely where each class of attack enters.

3.4.1 A networked control model with attack channels

Let $\mathbf{x}_k \in \mathbb{R}^n$ be the physical state at time step k (for example, attitude, orbit, or clock bias), $\mathbf{u}_k \in \mathbb{R}^m$ the control input, $\mathbf{y}_k \in \mathbb{R}^p$ the measurement, and $\mathbf{w}_k$ and $\mathbf{v}_k$ process and measurement noise. The nominal model is

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k), \quad \mathbf{y}_k = h(\mathbf{x}_k) + \mathbf{v}_k. \quad (3.6)$$

An estimator produces $\hat{\mathbf{x}}_k$ from past measurements, a controller computes $\mathbf{u}_k = \kappa(\hat{\mathbf{x}}_k)$, and the commands and measurements travel over networks. To represent adversarial action we augment the model as

$$\begin{aligned} \mathbf{x}_{k+1} &= f(\mathbf{x}_k, \gamma_k^u(\mathbf{u}_k + \mathbf{a}_k^u), \mathbf{w}_k + \mathbf{a}_k^x), \\ \mathbf{y}_k &= \gamma_k^y(h(\mathbf{x}_k) + \mathbf{v}_k + \mathbf{a}_k^y), \end{aligned} \quad (3.7)$$

where $\mathbf{a}_k^y$ is a *sensor attack* (spoofed navigation signals, falsified telemetry), $\mathbf{a}_k^u$ is an *actuator or command attack* (an injected or altered telecommand), $\mathbf{a}_k^x$ is a *physical attack* on the plant (a kinetic, directed-energy, or co-orbital effect, or an environmental extreme), and $\gamma_k^u, \gamma_k^y \in \{0, 1\}$ indicate whether command and measurement packets are delivered, so that a *denial-of-service attack* forces them to zero. A fifth channel is the integrity of the software itself: a compromised estimator, controller, or model changes κ or the estimator's equations directly. The formulation follows the attack-space view of secure control, in which each attack is characterized by the resources it needs (which channels it can access and what model knowledge it has) and by its effect on the loop [96, 97].

Figure 3.5 draws the loop for ground-in-the-loop control, such as orbit maintenance commanded from the mission operations center (MOC). For onboard loops, such as attitude control, the estimator and controller move into the space segment and the network links shrink to the spacecraft data bus, but the five channels remain. The model clarifies two points. First, an attack need not be large to be harmful: an adversary who knows the loop can shape $\mathbf{a}_k^y$ so that the detector's residual stays small while the estimate drifts. Second, the consequence of an attack is determined by the closed loop, not by the attack alone, so that security metrics for cyber-physical systems must be expressed in terms of physical outcomes such as state error, fuel consumed, or service lost.

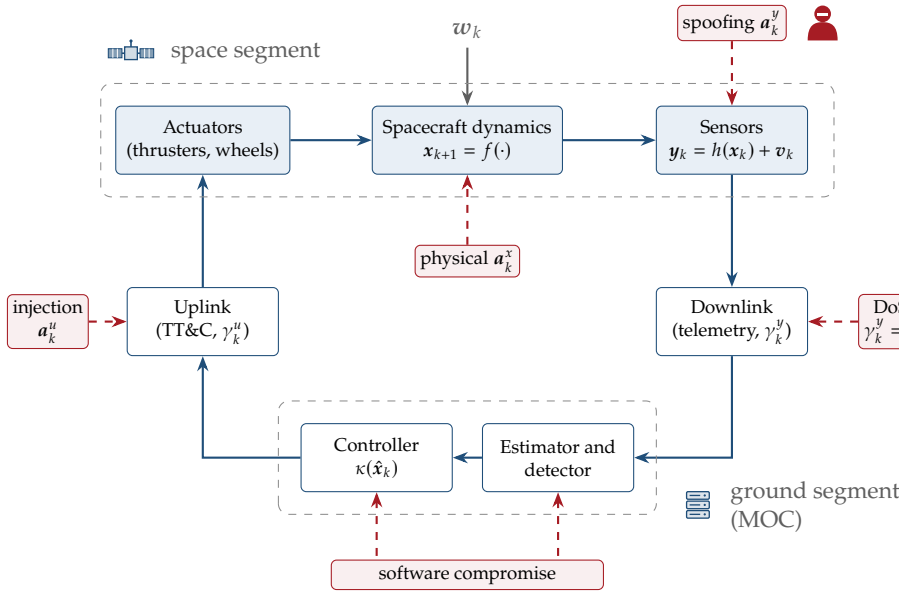


Figure 3.5. A space system as a networked control loop, with the five channels through which attacks enter (red). Sensor spoofing and telemetry denial act on the measurement path, command injection on the uplink, physical effects on the dynamics, and software compromise on the estimator and controller. Each channel corresponds to a term of Eq. (3.7).

3.4.2 Stealthy drift in a timing loop

Example 3.2 (Ramp attack on the HELIOS clock-steering loop). Each HELIOS satellite steers its onboard clock, which underlies the timing signal of the PNT augmentation service, toward a reference derived from time-transfer measurements. We model the clock bias as a constant x over the attack window and the measurement as $y_k = x + a_k$, with one update per second. The estimator is the fixed-gain filter $\hat{x}_{k+1} = \hat{x}_k + L(y_k - \hat{x}_k)$ with $L = 0.1$, and a detector raises an alarm when the residual $r_k = y_k - \hat{x}_k$ exceeds $\eta = 30$ ns; these values are illustrative assumptions for this example. We ask how large a bias the measurement path can accumulate without triggering the detector, and ignore noise.

Let $\varepsilon_k = \hat{x}_k - x$. Then $\varepsilon_{k+1} = (1 - L)\varepsilon_k + La_k$ and $r_k = a_k - \varepsilon_k$. A step bias $a_k = b$ produces $r_0 = b$ at onset, so every step larger than η is detected immediately. A ramp $a_k = \delta k$, by contrast, drives the error toward $\varepsilon_k = \delta k - \delta/L$, so the residual settles at the constant value $r_k \rightarrow \delta/L$ while the estimate follows the ramp. The ramp remains below the threshold whenever $\delta < L\eta = 3$ ns per second. With $\delta = 2.5$ ns s⁻¹, the residual does not exceed 25 ns, and after one hour the steered clock is offset by $\delta \cdot 3600 - \delta/L \approx 8.98$ μ s, far beyond typical timing-service tolerances.

A residual threshold alone therefore bounds the *rate* of an undetected attack, not its accumulated effect. The defensive implications are a rate-of-change monitor on the estimate, a cumulative test on the residual mean, and above all an independent time reference, which converts a single-channel estimation problem

into a consistency check across diverse sources.

The example previews a pattern that recurs throughout the book. Detection mechanisms designed for random faults set thresholds on instantaneous statistics, whereas a strategic adversary exploits the dynamics of the loop. The architectural remedy, diversity of sources and channels, is one of the resilience attributes of the next section.

3.5 Resilience Goals, Phases, and Attributes

Resilience becomes an engineering discipline when it is decomposed into goals that can be allocated, objectives that can be assessed, and design attributes that can be traded. Two sources supply the vocabulary used in this book. NIST's guidance on developing cyber-resilient systems provides goals and objectives that apply to systems of systems in general [26]. The U.S. Department of Defense resilience taxonomy for the space domain provides architectural attributes specific to constellations and their ground infrastructure [94]. The two are complementary: the first says what a resilient system must achieve over the course of an incident, and the second says how a space architecture can be shaped to achieve it.

3.5.1 Goals and objectives

NIST SP 800-160 Volume 2 defines four *cyber resiliency goals* [26]. To *anticipate* is to maintain a state of informed preparedness for adversity. To *withstand* is to continue essential mission functions despite adversity. To *recover* is to restore mission functions during and after adversity. To *adapt* is to modify mission functions or supporting capabilities in response to predicted changes in the technical, operational, or threat environment. The goals double as the phases of the resilience curve: anticipation precedes the disruption, withstanding sets the depth of the drop, recovery sets its duration, and adaptation changes the curve for the next event.

The same guidance refines the goals into eight *objectives*, which are more specific statements of what a system should achieve. *Prevent or avoid* precludes the successful execution of an attack or the realization of adverse conditions; *prepare* maintains realistic courses of action for anticipated adversity; *continue* maximizes the duration and viability of essential functions during adversity; *constrain* limits damage; *reconstitute* restores as much functionality as possible after adversity; *understand* maintains useful representations of dependencies and of the status of resources; *transform* modifies functions and processes to handle adversity more effectively; and *re-architect* modifies architectures for the same purpose. Each objective maps onto a feature of $Q(t)$. Prevent and constrain reduce the probability and the depth of a drop, continue raises the floor during the withstand phase, reconstitute shortens the recovery, and transform and re-architect change the curve of future events. Understand supports all of them, because recovery cannot be planned for dependencies that have not been mapped.

3.5.2 Architectural resilience attributes

The Department of Defense taxonomy adopts the definition of resilience in DoD Directive 3100.10: the ability of an architecture to support the functions necessary for mission success with higher probability, shorter periods of reduced capability, and across a wider range of scenarios, conditions, and threats, in spite of hostile action or adverse conditions [94]. The three clauses of this definition correspond to three features of the resilience curve: the probability that performance stays acceptable, the duration of the degraded period, and the breadth of the attack set $\mathcal{W}$ over which the first two hold. The taxonomy lists six architectural attributes that contribute to resilience, which we paraphrase from its definitions [94, 98].

Disaggregation separates dissimilar capabilities into separate platforms or payloads; separating tactical from strategic protected communications is the example the taxonomy gives, and it notes that disaggregation often serves resilience indirectly, by simplifying systems so that other measures are easier to apply. *Distribution* uses a number of nodes, working together, to perform the same mission as a single node would; satellite navigation constellations are the canonical example, since no individual satellite or monitoring site is essential and losses degrade accuracy gracefully. *Diversification* contributes to the same mission in multiple ways, using different platforms, different orbits, or the systems of commercial, civil, or international partners, as when a user draws on several satellite navigation systems. *Protection* comprises active and passive measures that ensure systems provide the required quantity and quality of support across operating environments, from jam resistance and hardening to maneuverability, onboard countermeasures, and the self-awareness needed to characterize attacks. *Proliferation* deploys larger numbers of the same platforms, payloads, or systems to perform the same mission. *Deception* comprises measures that confuse or mislead an adversary about the location, capability, operational status, mission type, or robustness of a system. The taxonomy deliberately excludes alternatives in other domains, such as airborne relays, from space resilience: they contribute to mission assurance but do not make the space architecture itself more resilient.

Distribution and proliferation are easily confused. Distribution is a statement about *how a function is realized*: the mission is performed collectively, with graceful degradation as nodes are lost. Proliferation is a statement about *how many units of the same kind exist*: more are deployed than a minimal design would field. A constellation can be distributed without being proliferated (every satellite needed for full coverage) and proliferated without being distributed (multiple spares of a single monolithic satellite). The taxonomy also places two contributors to mission assurance outside the architecture: *defensive operations*, which interrupt an adversary's kill chain or provide warning, and *reconstitution*, which brings new assets on line to replenish lost functions after an attack or catastrophic event.

Figure 3.6 maps the attributes to the phases they principally support and contrasts the two reference architectures of the book. HELIOS is proliferated in the literal sense: its 528 primary-shell satellites leave room for one on-orbit spare per plane and a replenishment rate of about 120 satellites per year. It is distributed across its space and ground segments, and diversified across two

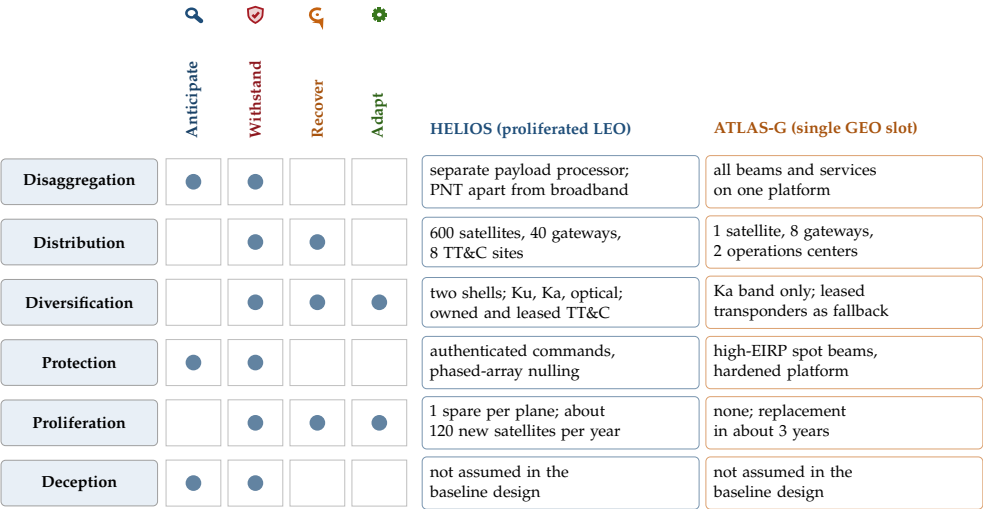


Figure 3.6. The six architectural resilience attributes, the resilience phases they principally support (dots), and how the notional HELIOS and ATLAS-G architectures realize them. Proliferated LEO architectures earn resilience mainly through distribution, diversification, and proliferation, which act in the withstand and recover phases; the single geostationary architecture relies on protection, which acts before and during an event but offers little once a loss has occurred.

orbital shells, three link technologies, and owned as well as leased TT&C sites. ATLAS-G, a single high-throughput satellite in one geostationary slot, is the “few, exquisite assets” contrast. Its resilience rests on protection: high link margins, a hardened platform, and careful operations. When protection fails, the absence of distribution and proliferation leaves reconstitution as the only recourse.

Two cautions keep the attribute view honest. First, attributes are not metrics. Proliferation does not guarantee resilience if every satellite shares one software image or one leased ground provider, as [Example 3.1](#) showed; common-mode dependencies can defeat distribution and proliferation at a single stroke. Second, attributes interact with cost and with the adversary’s response. A proliferated architecture changes the exchange ratio for kinetic attack but may enlarge the cyber attack surface through more terminals, more ground contacts, and more software updates. The metrics of [Sections 3.7 to 3.10](#) are what turn these qualitative trade-offs into comparisons.

3.6 Defense in Depth Across Layers and Zero Trust

No single protective measure is perfect, and a space system offers an adversary many layers through which to act. *Defense in depth* responds by placing several independent barriers between a threat and the mission, so that the failure of one barrier does not by itself produce a mission failure. *Zero trust* responds to a specific weakness of classical depth, the tendency to treat all assets inside a perimeter as trustworthy, by requiring every access to be authenticated and authorized on its own merits. Both principles apply across the layers of a space system, and both

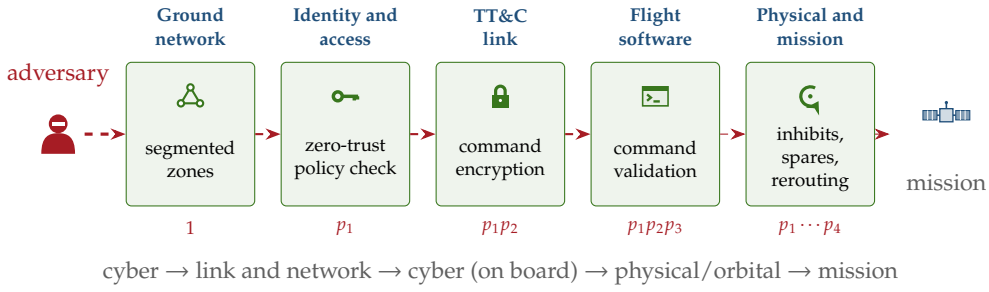


Figure 3.7. Defense in depth along a command-injection path from the ground network to the mission. Each barrier sits at a different layer, and the probability that the attack reaches each barrier (red labels beneath the barriers) shrinks multiplicatively only if the barriers fail independently. Shared keys, software, or suppliers collapse several barriers into one.

can be stated quantitatively.

3.6.1 Layered barriers and their independence

Consider an attack path that must defeat n barriers in sequence, and let p_i be the probability that barrier i is defeated given that the attacker has reached it. If the barriers fail independently, the probability that the attack reaches the mission is

$$P_{\text{succ}} = \prod_{i=1}^n p_i. \quad (3.8)$$

Four barriers with $p_i = 0.3$ each give $P_{\text{succ}} = 0.3^4 \approx 0.008$. The product form is the quantitative argument for depth, and it is also its main pitfall. Barriers that share a credential, a software component, a supplier, or a management network are not independent. If barriers 2 and 3 in this example rely on the same signing key, defeating one defeats both, the effective number of barriers drops to three, and P_{succ} rises to $0.3^3 = 0.027$, more than a threefold increase. Depth must therefore be measured by the number of *independent* barriers along the weakest path, which is a minimum-cut quantity on an attack graph rather than a count of security products. When the adversary adapts to the barriers it meets, defense in depth becomes a sequential game in which each layer is designed against the attacker's best continuation; Rass and Zhu [99] develop this view for advanced persistent threats.

Figure 3.7 draws such a path for command injection against HELIOS. The first barrier is the segmentation and monitoring of the terrestrial ground network, including the cloud region that hosts the primary MOC. The second is identity and access control, where a zero-trust policy check decides whether a request to issue commands is legitimate. The third is cryptographic: telecommands are protected with authenticated encryption on the TT&C link, so that forged or modified frames are rejected. The fourth is the flight software, which validates commands against allowed ranges, rates, and operating modes before execution. The fifth belongs to the physical and mission layers: hardware inhibits on propulsion, safe modes,

and, if all else fails, spares and rerouting that contain the consequence. The path crosses layers, which is what makes it hard to defend with measures designed for one layer, and it also means that each layer's defenses can compensate for another's weaknesses.

3.6.2 Zero trust

NIST SP 800-207 describes zero trust as a set of principles that move defenses away from static, network-based perimeters toward users, assets, and resources [100]. Its central premise is that no implicit trust is granted to an asset or a user account on the basis of physical or network location or of ownership. Authentication and authorization of both the subject and the device are discrete functions performed before a session to a resource is established, and access decisions are made per session by a *policy decision point* and enforced by a *policy enforcement point* placed in front of each resource. Decisions draw on dynamic information, such as the security posture of the requesting device and observed behavior, and the enterprise continuously monitors the integrity of its assets.

Space systems are natural candidates for zero trust because their perimeters have dissolved. The HELIOS MOC runs in a commercial cloud region; six of its eight TT&C sites belong to another company; terminal firmware and flight software arrive through a continuous-integration pipeline; and user terminals number in the hundreds of thousands. A perimeter model would treat a request arriving from a leased ground station as trusted because it arrives over a private circuit. A zero-trust model instead evaluates each command session on its own: which operator is authenticated, on which device, for which spacecraft and command class, at what time, and whether the pattern is consistent with the approved flight plan. Carried on board, the same idea implies that the flight software authenticates commands end to end, independently of the ground station through which they were relayed, and that payload and bus subsystems do not trust each other implicitly.

Zero trust and defense in depth interact directly with (3.8). Zero trust increases the number of independent barriers, because it prevents a single perimeter breach from granting access to all resources behind it, and it reduces the correlation between barriers, because each decision uses fresh evidence rather than inherited trust. Neither principle guarantees resilience, since both concern the prevention of successful attacks. The next sections turn to what happens when an attack or failure does succeed, which is measured through the trajectory of mission performance.

3.7 The Resilience Curve and Performance Functions

The resilience functional of Definition 3.9 compresses a trajectory into one number. Before compressing, it is useful to read the trajectory itself. The *resilience curve* plots normalized mission performance $Q(t)/Q_0$ against time through a disruption, and its geometry yields a family of metrics, each of which isolates one aspect of resilient behavior. This section defines the metrics, relates them to one another, discusses how to choose $Q(t)$ for space missions, and computes the metrics for a

HELIOS incident.

3.7.1 The resilience triangle and its descendants

The quantitative study of resilience curves began in earthquake engineering. Bruneau and colleagues measured the resilience of a community by the *quality* of its infrastructure, $Q(t) \in [0, 100]\%$, and defined the loss of resilience after an earthquake at t_0 as the area $\int_{t_0}^{t_1} [100 - Q(t)] dt$, where t_1 is the time of full recovery [28]. When quality drops instantly and recovers linearly, that area is a triangle, and the image of a *resilience triangle* has shaped the field since. The same framework identified four properties of resilient systems, the “4R”: *robustness*, the ability to withstand stress without loss of function; *redundancy*, the extent to which substitutable elements exist; *resourcefulness*, the capacity to identify problems and mobilize resources; and *rapidity*, the speed at which function is restored [28]. Robustness sets the depth of the triangle, rapidity sets its base, and redundancy and resourcefulness are the means by which both are improved.

Later work generalized the triangle to arbitrary time-dependent trajectories and to engineered systems. Henry and Ramirez-Marquez defined resilience at time t as the ratio of recovery achieved to loss suffered, $(Q(t) - Q(t_m))/(Q_0 - Q(t_m))$, where t_m is the time of maximum degradation [101], and Hosseini, Barker, and Ramirez-Marquez review the many deterministic and probabilistic variants that followed [102]. For cyber resilience, Linkov and Kott argue for the same performance-over-time view, organized by the phases of anticipate, withstand, recover, and adapt [29].

3.7.2 Metrics of the resilience curve

Let a disruption begin at t_d , let $Q(t)$ reach its minimum at t_m , and let recovery be complete at $t_r = \inf\{t > t_m : Q(s) \geq Q_0 \forall s \geq t\}$. We use the following metrics throughout the book.

Definition 3.11 (Resilience-curve metrics). The *depth* is $\Delta = 1 - \min_t Q(t)/Q_0$. The *time to recover* is $\text{TTR} = t_r - t_d$, and for a target fraction $q \in (0, 1)$ the *time to q -recovery* is $t_q = t'_q - t_d$, where t'_q is the first time after t_m at which $Q(t) \geq qQ_0$. The *loss area* over a horizon $[0, T]$ is

$$A_L = \int_0^T \left(1 - \frac{Q(t)}{Q_0}\right)^+ dt, \quad (3.9)$$

measured in units of time (hours of lost nominal service), and the *normalized loss* is $L_T = A_L/T$. The *rapidity* is the mean recovery rate $\rho = (Q_0 - Q(t_m))/(Q_0(t_r - t_m))$.

When performance stays at or below nominal, the single-scenario resilience of Definition 3.9 and the normalized loss are complementary, $R(\mathcal{A}, \pi; w) = 1 - L_T$. The metrics are related by simple bounds. Because the integrand in (3.9) is at most Δ and vanishes outside $[t_d, t_r]$,

$$L_T \leq \frac{\Delta \cdot \text{TTR}}{T}, \quad (3.10)$$

with equality for a rectangular loss, and $L_T = \Delta \cdot \text{TTR}/(2T)$ for Bruneau's triangle. The bound shows why no single metric suffices. Depth captures what users experience at the worst moment and matters when service below a threshold is useless, as for safety-of-life navigation. Time to recover captures how long the organization is under stress and matters for missions with deadlines. Loss area captures cumulative service lost and is the natural input to cost and to the resilience functional. Two systems with equal loss area can differ by an order of magnitude in depth, and a requirement stated in one metric does not constrain the others.

3.7.3 Choosing the performance function for space missions

The metrics are only as meaningful as the performance function $Q(t)$ on which they are computed. A good choice satisfies four conditions: it increases with the value delivered to users, it can be normalized by a nominal level, it aggregates across users and regions without hiding local losses, and it can be observed or estimated from telemetry. Five families recur in this book. *Capacity* measures, such as served throughput or the fraction of demand satisfied, suit broadband missions like HELIOS and ATLAS-G. *Coverage* measures, the fraction of a service area (weighted by population or demand) that is covered by at least k satellites, suit constellations for which geometry is the binding constraint; coverage-based performance functions also admit resilient control formulations in which satellites reconfigure to restore coverage after losses [103]. *PNT quality* measures, such as the fraction of users whose protection level is below the alert limit of (3.5), suit navigation and timing services and embed integrity directly. *Revisit* measures, such as the reciprocal of the maximum revisit time over targets of interest, suit Earth observation. *Command and control* measures, such as available contact minutes per spacecraft per day, describe the TT&C function on which all others depend.

Two refinements are often needed. First, a threshold transformation $\tilde{Q}(t) = \mathbb{1}\{Q(t) \geq q_{\min}Q_0\}$ converts a graded measure into a binary service indicator when partial service has no value; the time average of $\tilde{Q}$ is then an availability. Second, when the mission serves several user classes with different priorities, a weighted sum $Q(t) = \sum_c \omega_c Q_c(t)$ with weights ω_c expresses the mission's value judgments explicitly. Aggregation must be chosen with care, because a global average can mask the complete loss of service to a small but critical region, which is precisely the effect a regional jammer produces.

3.7.4 A worked resilience curve

Example 3.3 (Gateway intrusion: HELIOS versus ATLAS-G). A ground-network intrusion forces operators to isolate gateways for forensic re-imaging. We take $Q(t)$ to be served capacity as a fraction of demand, with $T = 24$ h and the disruption at $t_d = 1$ h. HELIOS has 40 Ka-band gateways; 12 are isolated, so capacity falls to $Q/Q_0 = 0.70$. After 30 minutes, the network operations center reroutes traffic over optical inter-satellite links to the remaining gateways, which absorbs half of the shortfall and raises performance to 0.85. Re-imaging begins at $t = 6$ h and returns the isolated gateways at a constant rate, restoring full capacity at $t_r = 18$ h.

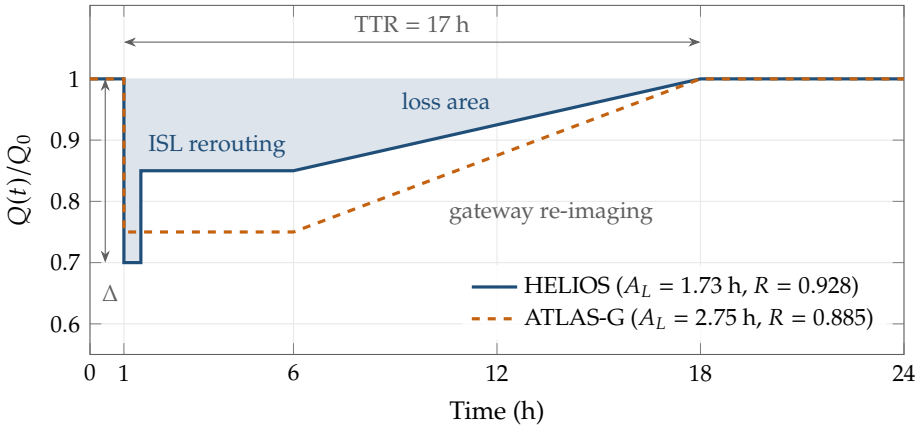


Figure 3.8. Resilience curves for Example 3.3. HELIOS (solid) has the deeper drop but recovers part of its capacity within 30 minutes by rerouting over inter-satellite links, so its shaded loss area is smaller than that of ATLAS-G (dashed). Both recover fully after 17 hours.

ATLAS-G has 8 gateways; the same intrusion isolates 2, capacity falls to 0.75, no inter-satellite rerouting exists, and re-imaging follows the same schedule. The rerouting fraction and the re-imaging schedule are assumptions for this example.

For HELIOS the loss area (3.9) consists of a rectangle during the first half hour, a rectangle during rerouted operation, and a triangle during re-imaging:

$$A_L = 0.30 \times 0.5 + 0.15 \times 4.5 + \frac{1}{2} \times 0.15 \times 12 = 0.15 + 0.675 + 0.90 = 1.725 \text{ h.}$$

Hence $L_T = 1.725/24 = 0.072$ and $R = 0.928$. The depth is $\Delta = 0.30$, the time to recover is $TTR = 17 \text{ h}$, and the time to 95% recovery is $t_{0.95} = 13 \text{ h}$. For ATLAS-G, $A_L = 0.25 \times 5 + \frac{1}{2} \times 0.25 \times 12 = 2.75 \text{ h}$, so $L_T = 0.115$ and $R = 0.885$, with $\Delta = 0.25$, $TTR = 17 \text{ h}$, and $t_{0.95} = 14.6 \text{ h}$. Numerical integration of both curves at one-second resolution reproduces these values.

HELIOS suffers the deeper initial drop, because it loses more gateways in absolute terms, yet it loses 37% less service in total, because inter-satellite rerouting converts part of the withstand phase into early recovery. Depth and loss area rank the two architectures in opposite orders, which illustrates the bound (3.10): a single metric cannot capture both what users experience at the worst moment and what the mission loses in total.

Figure 3.8 shows the two curves. The shape of the HELIOS curve, a sharp drop followed by a rapid partial recovery and then a slow ramp, is typical of architectures with automated reconfiguration and manual forensic recovery: the first minutes are decided by design, the following hours by operations. The case study that follows shows that real incidents exhibit the same structure, and that summary statistics can obscure it.

Space Security in Practice: The Galileo Service Outage of July 2019

In July 2019, Europe's Galileo satellite navigation system suffered an interruption of its navigation and timing services that lasted about six days [104]. The European GNSS Service Centre issued a notice of potential service degradation on 11 July at 14:45, a notice of service outage on 13 July at 20:15, and a notice of service recovery on 18 July at 08:20, while warning that some fluctuations might continue [105]. The incident originated in an equipment malfunction in the ground infrastructure that affected the calculation of the time and orbit predictions used to generate the navigation message; the satellites themselves were not at fault [105]. An independent analysis reported that the backup facility was not available at the time and that new equipment was being deployed during an upgrade [106], and the European Commission set up an independent inquiry board. The quarterly performance report later recorded that, because of the incident, the availability parameters for positioning and timing were met only 81% of the time in July [104].

Read as a resilience curve (Fig. 3.9), the event had a gradual onset of about 53 hours, a period of total loss of the open service of about 108 hours, and a recovery tail. Three lessons follow for metric design. First, depth dominated user experience: navigation messages that are stale or flagged unhealthy provide no service, so the performance function was effectively binary. Second, a monthly availability of 81% is consistent with roughly six days of loss in a 31-day month, but it cannot distinguish one contiguous six-day outage from many short ones, which have very different consequences for users. Third, the root cause was a common-mode dependency created during maintenance, when redundancy was temporarily absent, which is the kind of dependency that reliability estimates under independence overlook. Users of receivers that combined several constellations could fall back on other systems, an example of diversification acting at the user segment.

3.8 Stochastic Resilience Models

The resilience curve of Section 3.7 describes one disruption. Over a mission lifetime, disruptions recur at random times, recoveries take random durations, and a system may be struck again before it has recovered. Stochastic models describe this repeated cycle and yield long-run and transient versions of the resilience metrics. Continuous-time Markov chains are the workhorse: they are expressive enough to represent degraded modes, repair, and escalation, and simple enough to solve in closed form or with a matrix exponential [107].

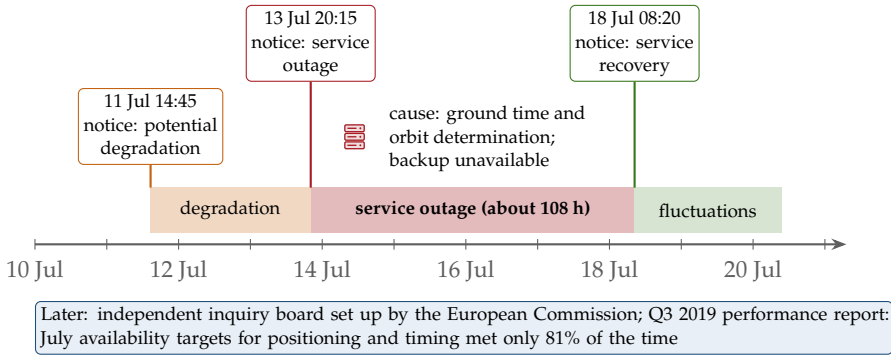


Figure 3.9. Timeline of the July 2019 Galileo outage read as a resilience curve. The colored band shows the phases of the event as announced in notices to users; the deep, contiguous outage phase dominates the loss even though the monthly availability statistic reports it only as a percentage.

3.8.1 Degradation-recovery chains and Markov rewards

Let the system occupy one of finitely many *service states* $s \in \mathcal{S}$, such as nominal, degraded, and failed, and let $X(t)$ be the state at time t . In a continuous-time Markov chain (CTMC) the time spent in each state is exponentially distributed and the next state depends only on the current one. The chain is specified by a *generator matrix* $G = [g_{ij}]$, whose off-diagonal entry $g_{ij} \geq 0$ is the transition rate from i to j and whose rows sum to zero. The row vector of state probabilities $p(t)$ satisfies the Kolmogorov forward equation

$$\dot{p}(t) = p(t) G, \quad p(t) = p(0) e^{Gt}, \quad (3.11)$$

and, for an irreducible chain, a unique stationary distribution π^* solves $\pi^* G = 0$ with $\pi^* \mathbf{1} = 1$.

Resilience enters by attaching a *reward* to each state, the normalized performance $r_s = Q(s)/Q_0$ delivered while the system is in state s . Collecting rewards in a column vector r gives three quantities that parallel the metrics of the resilience curve:

$$\mathbb{E}[Q(t)]/Q_0 = p(0) e^{Gt} r, \quad \bar{R}_T = \frac{1}{T} \int_0^T p(0) e^{Gt} r dt, \quad \bar{R}_\infty = \pi^* r. \quad (3.12)$$

The first is the expected resilience curve, the second is the expected resilience functional over a finite horizon, and the third is the long-run fraction of nominal performance. The expected loss per unit time in steady state is $1 - \bar{R}_\infty$. The *steady-state availability* with respect to a service threshold $q_{\min}$ is $A = \sum_{s: r_s \geq q_{\min}} \pi_s^*$.

Recovery times are first-passage times. Let $\mathcal{T} \subset \mathcal{S}$ be the set of states that have not yet recovered, and let $G_{\mathcal{T}\mathcal{T}}$ be the block of the generator restricted to them. The vector of mean times to reach the recovered set from each state in $\mathcal{T}$ is

$$m = (-G_{\mathcal{T}\mathcal{T}})^{-1} \mathbf{1}. \quad (3.13)$$

For the simplest two-state chain with failure rate λ and repair rate μ , these formulas reduce to the classical results $A = \mu/(\lambda + \mu) = \text{MTTF}/(\text{MTTF} + \text{MTTR})$ and $\text{MTTR} = 1/\mu$.

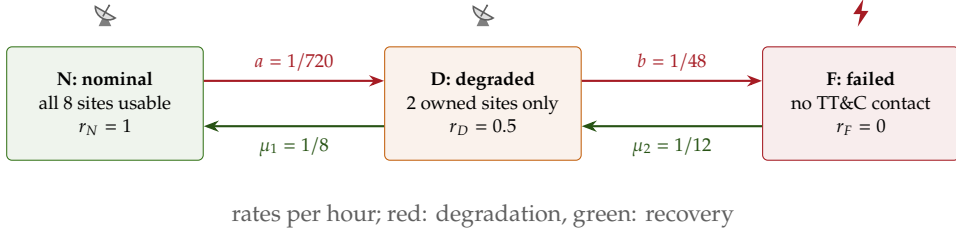


Figure 3.10. Three-state degradation-recovery model of the HELIOS TT&C network used in Example 3.4. A provider-wide outage or compromise moves the network from nominal to degraded; the loss of the owned sites while degraded leads to failure. Recovery proceeds one level at a time. Rewards r_s give the fraction of required commanding capacity available in each state.

3.8.2 A HELIOS TT&C example

Example 3.4 (Availability and expected resilience of the HELIOS TT&C network). We return to the TT&C network of Example 3.1 and model provider-wide events rather than independent site outages. In state N all eight sites are usable. A provider-wide outage or compromise, which we assume occurs on average once every 30 days, moves the network to state D, in which only the two owned sites remain and half of the required commanding capacity is available. From D, the provider restores service at rate $\mu_1 = 1/8$ per hour. While in D, a loss of the owned sites (weather, power, or a coordinated action) moves the network to F at rate $b = 1/48$ per hour, and from F one owned site is restored at rate $\mu_2 = 1/12$ per hour. All rates are assumptions for this example. Figure 3.10 shows the chain, whose generator, with states ordered (N, D, F), is

$$G = \begin{pmatrix} -1/720 & 1/720 & 0 \\ 1/8 & -(1/8 + 1/48) & 1/48 \\ 0 & 1/12 & -1/12 \end{pmatrix}, \quad r = \begin{pmatrix} 1 \\ 0.5 \\ 0 \end{pmatrix}.$$

The chain is a birth-death process, so its stationary distribution follows from the balance equations $a\pi_N^* = \mu_1\pi_D^*$ and $b\pi_D^* = \mu_2\pi_F^*$. They give $\pi_D^* = (8/720)\pi_N^*$ and $\pi_F^* = (12/48)\pi_D^*$, and normalization yields $\pi^* = (0.98630, 0.01096, 0.00274)$. The availability of at least partial commanding is $A = \pi_N^* + \pi_D^* = 0.99726$, and the long-run expected performance is $\bar{R}_\infty = \pi_N^* + 0.5\pi_D^* = 0.99178$, an expected loss of $0.00822 \times 8760 \approx 72$ hours of full commanding capacity per year. Applying (3.13) with $\mathcal{T} = \{D, F\}$ gives mean times to full recovery of $m_D = 10$ h and $m_F = 22$ h, which can be checked by hand from $m_D = 48/7 + \frac{1}{7}(12 + m_D)$. The mean time from N to total loss of contact is 5088 h, about 212 days.

The transient view is more informative for operators. Starting in D, the expected performance recovers to 0.72 after 6 hours, 0.92 after 24 hours, and 0.95 only after about 31 hours, and the expected resilience over the first day is $\bar{R}_{24} = 0.793$. Starting in F, the corresponding values are 0.25, 0.73, and 53 hours, with $\bar{R}_{24} = 0.427$. Figure 3.11 plots these curves. The long-run figures are excellent, yet a day that begins with a provider-wide compromise loses on average about five of 24 hours of commanding capacity, and one that begins with

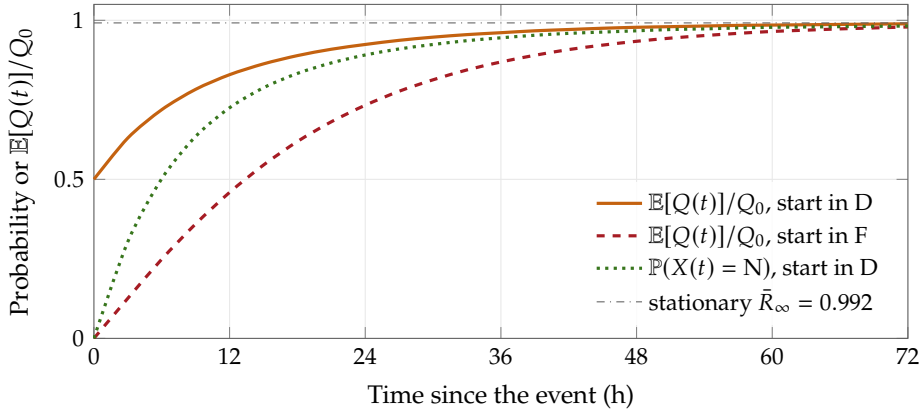


Figure 3.11. Transient expected performance of the TT&C chain of [Example 3.4](#), computed with the matrix exponential (3.11). Recovery from total loss of contact (dashed) lags recovery from the degraded state (solid) by about a day, although both approach the same stationary value.

total loss of contact loses nearly 14. Availability describes the typical year; the transient expected resilience describes the bad day, and the architecture decision in the vignette depends on both.

3.8.3 Beyond exponential durations

The exponential sojourn times of a CTMC are a modeling convenience, not a property of real recoveries. Forensic re-imaging of a ground system takes a nearly deterministic time once started; phasing a spare satellite into an orbital gap takes about a week; and attack campaigns often have a minimum preparation time. Two extensions preserve tractability. A *semi-Markov process* keeps the Markov property at transition epochs but allows arbitrary sojourn distributions with means $\bar{\tau}_i$. If ν is the stationary distribution of the embedded jump chain, the long-run fraction of time in state i is

$$\pi_i = \frac{\nu_i \bar{\tau}_i}{\sum_j \nu_j \bar{\tau}_j}, \quad (3.14)$$

so long-run availability and $\bar{R}_\infty$ depend on the sojourn distributions only through their means [107]. Transient quantities, such as the probability of still being degraded after 24 hours, do depend on the full distributions. A *phase-type* representation recovers a Markov model for them by replacing each non-exponential sojourn with a chain of exponential phases. An Erlang sojourn with k phases, each of rate $k/\bar{\tau}$, has mean $\bar{\tau}$ and coefficient of variation $1/\sqrt{k}$, so increasing k approaches a deterministic duration while keeping the model a CTMC with an enlarged state space. For the tail probabilities that drive risk-averse resilience in [Section 3.10](#), choosing such distributions to match the observed variance matters as much as matching the mean.

3.9 Network Resilience Metrics

A proliferated constellation is a network. Satellites relay traffic over inter-satellite links (ISLs), gateways connect the constellation to terrestrial networks, and coverage of a service area emerges from the collective geometry of many satellites. When satellites are lost, whether to failures, space weather, or hostile action, the question is not only how many remain but whether the remaining ones still form a network that can deliver the mission. Graph theory and percolation theory supply the metrics for this question [108].

3.9.1 Connectivity, algebraic connectivity, and percolation

Model the constellation as an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $N = |\mathcal{V}|$ satellites as nodes and ISLs as edges. The coarsest resilience metric is whether $\mathcal{G}$ is connected. A finer one is the *vertex connectivity* $\kappa(\mathcal{G})$, the minimum number of nodes whose removal disconnects the graph, which equals the worst-case survivability order of Definition 3.7 plus one when the mission requires global connectivity. Both are worst-case, combinatorial quantities. Two further metrics describe how connectivity degrades.

Definition 3.12 (Algebraic connectivity). Let A be the adjacency matrix of $\mathcal{G}$ and D the diagonal matrix of node degrees. The *Laplacian* is $L = D - A$, with eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$. The *algebraic connectivity* is $\lambda_2(L)$ [109].

Fiedler showed that $\lambda_2 > 0$ if and only if the graph is connected, and that $\lambda_2 \leq \kappa(\mathcal{G})$ for graphs that are not complete [109]. Algebraic connectivity is therefore a smooth, spectral lower bound on how hard the network is to cut. It also governs dynamics on the network: the convergence rate of distributed consensus, such as time synchronization or distributed orbit determination across ISLs, scales with λ_2 , and a small λ_2 signals a bottleneck even while the graph remains connected.

Definition 3.13 (Giant-component fraction). Let $\mathcal{G}_f$ be the graph that remains after a fraction f of the nodes has been removed according to a stated rule. The *giant-component fraction* $S(f)$ is the number of nodes in the largest connected component of $\mathcal{G}_f$, divided by the original number of nodes N .

The removal rule is the crux. Under *random* removal, the behavior of $S(f)$ is the subject of percolation theory: a large network retains a giant component until a critical fraction f_c is removed, beyond which it shatters into small fragments. For random graphs with arbitrary degree distributions, Callaway, Newman, Strogatz, and Watts gave exact solutions for $S(f)$ under random and degree-dependent removal [110]. Under *targeted* removal, the adversary chooses the most important nodes first. Albert, Jeong, and Barabási showed that networks with highly heterogeneous degrees are remarkably tolerant of random errors but fragile under attacks on their hubs [111]. The gap between the random and the targeted curves $S(f)$ is the network analogue of the gap between reliability and survivability.

Two features distinguish constellation networks from the networks studied in that literature. First, ISL topologies are nearly regular: in the common *+Grid*

pattern each satellite links to its two neighbors in the same orbital plane and to one satellite in each adjacent plane, so every node has degree four and there are no hubs to target. Importance is instead structural. Satellites whose removal cuts many shortest paths, measured by *betweenness centrality*, and whole planes, whose removal cuts the torus along a seam, are the natural targets. Second, removal need not be one node at a time. A common software image or a shared launch vehicle can remove correlated sets, and physical attacks on satellites in one plane or at one altitude are correlated by geometry.

3.9.2 From connectivity to coverage

Connectivity is a means; the mission is coverage and capacity. Let $\mathcal{P}$ be a set of service points weighted by demand ω_p , and let $n_p(t)$ be the number of satellites visible above the minimum elevation from point p at time t . The k -coverage performance function is

$$Q_k(t) = \sum_{p \in \mathcal{P}} \omega_p \mathbb{I}\{n_p(t) \geq k\}, \quad (3.15)$$

with $k = 1$ for basic service, $k = 4$ for stand-alone satellite positioning, and larger k where handover margin or interference rejection requires several satellites in view. Network metrics and coverage metrics complement each other. The fraction of satellite pairs that can still communicate is approximately $S(f)^2$, which bounds the traffic that can be relayed to distant gateways, while Q_k measures whether users see enough satellites at all. Losses can also be partly reversed by control: remaining satellites can adjust their phasing or orbits to close coverage gaps, which turns $Q_k(t)$ into the output of a feedback loop. Zhao and Zhu formulate this as autonomous and resilient coverage control of a LEO constellation facing space threats [103], a formulation in which the recovery phase of the resilience curve is designed rather than merely observed.

3.9.3 A HELIOS inter-satellite network example

Example 3.5 (Random versus targeted loss in the HELIOS ISL network). The HELIOS primary shell has 528 satellites in 24 planes of 22, each with four optical ISL terminals: two intra-plane and two cross-plane. We model its ISL network as the torus $C_{24} \square C_{22}$, the Cartesian product of a 24-cycle and a 22-cycle (Fig. 3.12). The model ignores the seam that a Walker-delta phasing introduces and the switching-off of cross-plane links at high latitude, both of which lower connectivity further. We compare $S(f)$ and λ_2 under three removal rules: uniformly random removal (averaged over 20 trials); an adaptive betweenness attack, which removes the five satellites of highest estimated betweenness, recomputes, and repeats; and a seam attack, which removes whole planes.

The Laplacian eigenvalues of a Cartesian product are the sums of the eigenvalues of its factors, and those of a cycle C_n are $2 - 2\cos(2\pi j/n)$. The intact network therefore has $\lambda_2 = 2 - 2\cos(2\pi/24) = 0.0681$, set by the longer cycle, and vertex connectivity $\kappa = 4$. Computing the remaining quantities numerically gives the curves of Fig. 3.13. Under random removal the network degrades gracefully: after

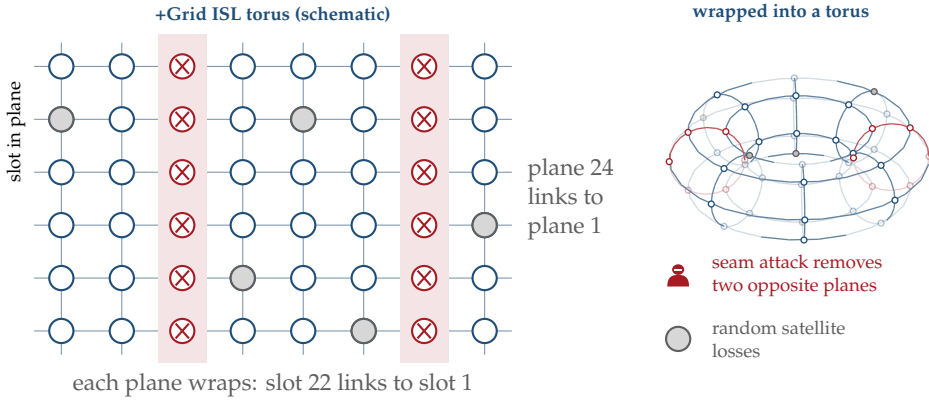


Figure 3.12. Schematic of the +Grid inter-satellite link topology of the HELIOS primary shell, drawn with 8 of the 24 planes and 6 of the 22 slots per plane. Each satellite links to its two in-plane neighbors (vertical) and to one satellite in each adjacent plane (horizontal), and both directions wrap around, forming the torus drawn on the right with the same losses. Random losses (gray) leave the grid connected; removing two opposite planes (red) cuts the torus into two halves.

losing 20% of the satellites, the giant component still holds $S = 0.80$ of the original nodes, which is nearly all of the survivors, and the giant component disappears only when about 45% to 50% are removed ($S = 0.39$ at $f = 0.44$ and $S = 0.13$ at $f = 0.52$), consistent with the site-percolation threshold of the square lattice. Under the adaptive betweenness attack, S falls to 0.57 at $f = 0.12$ and to 0.31 at $f = 0.20$. The seam attack is the most efficient: removing one plane ($f = 0.042$) leaves $S = 0.958$, but removing a second, opposite plane ($f = 0.083$) splits the torus into two halves and leaves $S = 0.458$.

Algebraic connectivity anticipates this collapse. Removing the first plane barely changes S , yet it lowers λ_2 from 0.0681 to 0.0186, a 73% reduction, because the torus has become a cylinder that a single further cut can separate. Under random removal, λ_2 declines steadily, falling below 0.01 at $f \approx 0.32$ while S is still 0.67. The difference between the random and targeted curves is what a reliability analysis misses: the same constellation tolerates 20% random attrition with little loss of connectivity, and loses more than half of its connectivity when 8% of its satellites are removed along a structural seam.

The example suggests how network metrics enter the resilience functional. If $Q(t)$ is taken as the fraction of satellite pairs that can still exchange traffic, approximately S^2 , then the seam attack reduces performance to about 0.21 at once, while random attrition of the same size reduces it only to about 0.84. The defensive implications follow the attributes of [Section 3.5](#): diversify the paths between planes (for example, by laser links that can reach two planes away), keep a second shell whose links cross the seams of the first, and treat whole planes, not individual satellites, as units of common-mode risk. [Chapter 7](#) develops routing and interdiction models on such graphs.

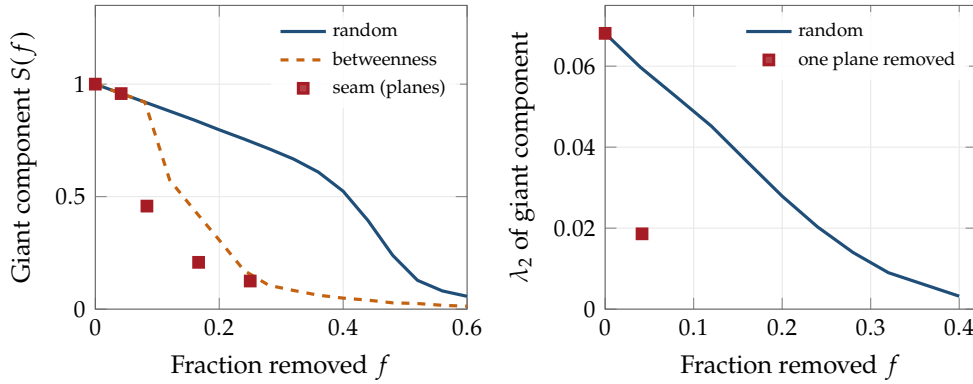


Figure 3.13. Connectivity of the HELIOS primary-shell ISL torus in [Example 3.5](#). Left: the giant-component fraction degrades gracefully under random loss but collapses under an adaptive betweenness attack and, most efficiently, under removal of whole planes. Right: algebraic connectivity falls by 73% when a single plane is removed, even though the giant component is barely changed, giving early warning of fragility.

3.10 Risk-Averse and Worst-Case Resilience

The resilience functional (3.4) takes the infimum over an attack set. The Markov models of [Section 3.8](#) take expectations over random events. These are two ends of a spectrum, and neither is right in every case. Expectations suit frequent, independent disturbances whose statistics are known, such as part failures and routine anomalies. Worst cases suit disturbances chosen by an adversary. Between them lie rare, severe events whose probabilities are uncertain and whose consequences are too large to be averaged away. Risk measures from financial engineering, above all conditional value-at-risk, interpolate between the two ends and give design problems that remain tractable.

3.10.1 From expectation to worst case

Let $\mathcal{W}$ be a finite or measurable set of scenarios, each a disturbance or attack w , and let the *loss* of architecture $\mathcal{A}$ with policy π in scenario w be the normalized loss of [Definition 3.11](#),

$$L(\mathcal{A}, \pi; w) = 1 - R(\mathcal{A}, \pi; w) \in [0, 1].$$

When scenarios are assigned a probability distribution $\mathbb{P}$, L is a random variable. Its mean gives the *expected resilience* $\mathbb{E}[R] = 1 - \mathbb{E}[L]$, which is the stochastic criterion. The *value-at-risk* at level $\alpha \in (0, 1)$ is the α -quantile of the loss, $\text{VaR}_\alpha(L) = \inf\{\ell : \mathbb{P}(L \leq \ell) \geq \alpha\}$: with probability α , the loss does not exceed it. Value-at-risk ignores how bad the losses beyond the quantile are, and it is not convex in the design, which makes it a poor objective.

Definition 3.14 (Conditional value-at-risk). The *conditional value-at-risk* of the loss at level $\alpha \in [0, 1)$ is

$$\text{CVaR}_\alpha(L) = \min_{c \in \mathbb{R}} \left\{ c + \frac{1}{1 - \alpha} \mathbb{E}[(L - c)^+] \right\}, \quad (3.16)$$

and the CVaR resilience is $R_\alpha^{\text{CVaR}} = 1 - \text{CVaR}_\alpha(L)$.

Rockafellar and Uryasev introduced this minimization formula and showed that the minimizing c is the value-at-risk, so that for continuous distributions CVaR_α equals the expected loss conditional on exceeding VaR_α [112]. They later extended the results to general distributions, including the discrete scenario distributions used in practice, and showed that CVaR is a coherent risk measure [113]. Its interpretation for resilience is direct: $R_{0.9}^{\text{CVaR}}$ is the average resilience over the worst 10% of outcomes.

Proposition 3.1 (Ordering of resilience criteria). *For every architecture, policy, and scenario distribution whose support is $\mathcal{W}$, and for $0 \leq \alpha \leq \alpha' < 1$,*

$$\inf_{w \in \mathcal{W}} R(\mathcal{A}, \pi; w) \leq R_{\alpha'}^{\text{CVaR}} \leq R_\alpha^{\text{CVaR}} \leq R_0^{\text{CVaR}} = \mathbb{E}[R].$$

For a finite scenario set with positive probabilities, R_α^{CVaR} equals the worst-case resilience once $1 - \alpha$ does not exceed the probability of the worst scenario.

Proof. With $c = \sup L$ the objective in (3.16) equals $\sup L$, so $\text{CVaR}_\alpha \leq \sup L$. For every c , $c + \mathbb{E}[(L - c)^+]/(1 - \alpha) \geq c + \mathbb{E}[(L - c)^+] \geq \mathbb{E}[L]$, so $\text{CVaR}_\alpha \geq \mathbb{E}[L]$, with equality at $\alpha = 0$. For fixed c the objective is nondecreasing in α , and so is its minimum. For a finite set, the tail of probability mass $1 - \alpha$ lies wholly within the worst scenario when $1 - \alpha$ is at most that scenario's probability, and the conditional expectation over that tail is the worst loss. Taking complements gives the stated ordering. $\square$

The ordering makes the choice of criterion a statement about the decision maker. A designer who maximizes $\mathbb{E}[R]$ is neutral to risk; one who maximizes R_α^{CVaR} is averse to the worst $1 - \alpha$ fraction of outcomes; and one who maximizes the worst case treats every scenario in $\mathcal{W}$ as one an adversary might choose. With S scenarios of probabilities p_s and losses $L_s(\mathcal{A})$, the CVaR design problem becomes

$$\min_{\mathcal{A} \in \mathcal{F}, c, z} \quad c + \frac{1}{1 - \alpha} \sum_{s=1}^S p_s z_s \quad \text{s.t.} \quad z_s \geq L_s(\mathcal{A}) - c, \quad z_s \geq 0, \quad (3.17)$$

over a feasible set $\mathcal{F}$ of architectures, which is a linear program whenever the losses are affine in the design variables [112]. The worst-case problem $\max_{\mathcal{A}, \pi} \min_{w \in \mathcal{W}} R(\mathcal{A}, \pi; w)$ is a robust optimization problem of the same form with z_s replaced by a single epigraph variable.

3.10.2 Comparing architectures under a scenario set

Example 3.6 (Expected, CVaR, and worst-case resilience of three architectures). We compare three architectures over a 30-day horizon, $T = 720$ h: HELIOS as specified; HELIOS-D, which adds owned TT&C sites and a second leasing provider so that no single provider carries most of the commanding capacity; and ATLAS-G. We consider six scenarios, one of which occurs in the 30-day window with the probabilities in Table 3.1. For each scenario and architecture we specify

Table 3.1. Scenario probabilities for a 30-day window and loss areas A_L (hours of lost nominal service) for [Example 3.6](#). The last four rows give the resilience criteria, with losses converted to $R = 1 - A_L/720$.

Scenario	Probability	HELIOS	HELIOS-D	ATLAS-G
Routine anomaly	0.560	0.06	0.06	0.06
Regional jamming	0.200	3.75	3.75	1.25
Ground-network intrusion	0.150	3.15	3.15	2.75
Leasing-provider compromise	0.060	27.00	1.92	0.36
Severe geomagnetic storm	0.027	10.08	10.08	0.09
Direct-ascent ASAT attack	0.003	24.96	24.96	604.80
$\mathbb{E}[R]$		0.9955	0.9976	0.9965
$R^{\text{CVaR}}_{0.90}$		0.9722	0.9915	0.9711
$R^{\text{CVaR}}_{0.95}$		0.9625	0.9883	0.9460
Worst case		0.9625	0.9653	0.1600

a piecewise-linear $Q(t)$: a drop of depth Δ , a hold, and a linear recovery. For example, regional jamming holds HELIOS at 85% for 24 hours, but ATLAS-G, with high-EIRP spot beams, only at 95%. A compromise of the leasing provider holds HELIOS at 75% for 72 hours followed by a 72-hour recovery, whereas HELIOS-D loses 8% for 12 hours. A direct-ascent anti-satellite attack removes three HELIOS satellites, costing 5% of capacity for the seven days needed to phase in a spare and 3% thereafter, but removes the single ATLAS-G satellite outright, leaving no service for 14 days and 30% of capacity from leased transponders afterward. The storm scenario costs HELIOS 6% for two days with a 10-day recovery. All profiles and probabilities are illustrative assumptions chosen to exhibit the trade-off, not estimates of real threat frequencies. Integrating each profile gives the loss areas in [Table 3.1](#).

The expected losses are $\sum_s p_s A_{L,s} = 3.22$, 1.72, and 2.53 hours, which give the expected resilience in the table. For CVaR at $\alpha = 0.9$, the worst 10% of probability mass for HELIOS consists of the provider compromise (0.06), the ASAT attack (0.003), the storm (0.027), and 0.01 of the jamming scenario, so $\text{CVaR}_{0.9} = (0.06 \cdot 27 + 0.003 \cdot 24.96 + 0.027 \cdot 10.08 + 0.01 \cdot 3.75) / (0.1 \cdot 720) = 0.0278$ and $R^{\text{CVaR}}_{0.9} = 0.9722$. The same computation, or the minimization (3.16) over c , gives the other entries.

By expected resilience, ATLAS-G ranks ahead of HELIOS, because its dominant loss is very unlikely and its exposure to jamming is smaller. By CVaR at either level, and by worst case, the order reverses, and the worst case of ATLAS-G is catastrophic. HELIOS-D ranks first under every criterion; it dominates because diversifying the ground segment removes HELIOS's worst scenario at small cost to the others. Note that the worst case of HELIOS-D is a different scenario from that of HELIOS: hardening one layer moves the infimum to another layer.

[Figure 3.14](#) shows the reversal. The example returns an answer to the third question of the vignette: a ranking by expected annual loss can favor an architecture whose bad month is catastrophic, and a board that cares about the tail should

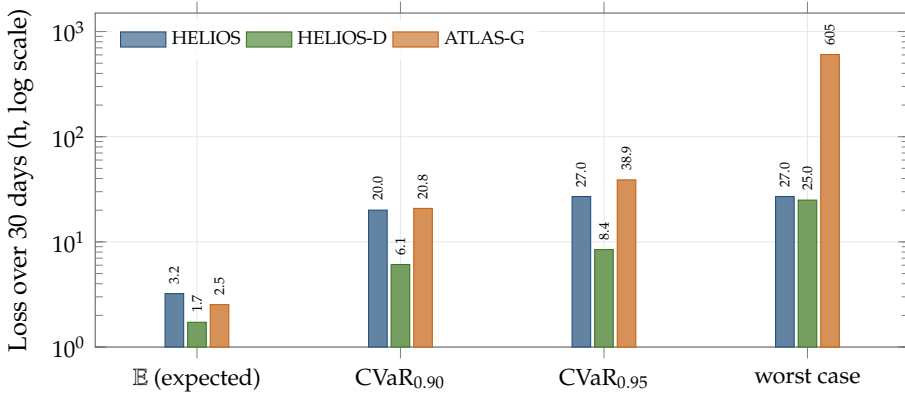


Figure 3.14. Loss under four criteria for the three architectures of [Example 3.6](#), on a logarithmic scale. ATLAS-G has a smaller expected loss than HELIOS but a larger tail loss, and its worst case is more than twenty times that of either HELIOS variant; the diversified HELIOS-D has the smallest loss under every criterion.

rank by CVaR or by the worst case over a credible scenario set.

3.10.3 When the adversary chooses the scenario

The worst-case functional treats the attack set as fixed and the attack as chosen after the design. An intelligent adversary does more: it observes the architecture and the defender’s policy, anticipates the defender’s response, and selects w to minimize resilience given that response. The infimum in (3.4) then becomes the inner problem of a sequential game, $\max_{\mathcal{A}, \pi} \min_{w \in \mathcal{W}} R(\mathcal{A}, \pi; w)$, in which the defender leads and the attacker follows; when both adapt during an incident, it becomes a dynamic zero-sum game. The example already shows the signature of strategic behavior: once HELIOS-D closes the ground-segment weakness, the worst case moves to a physical attack. In cyber-physical systems, where the adversary may act at the cyber layer to produce physical effects, Zhu and Başar develop a games-in-games principle that couples a cyber-layer security game with a physical-layer robust control game, so that cross-layer resilience is designed jointly [96], and Zhu and Xu give a book-length treatment of cross-layer design for secure and resilient cyber-physical systems [97]. [Chapter 4](#) introduces the strategic adversary models on which such formulations rest.

3.11 Summary

This chapter replaced loosely used words with precise concepts and computable metrics. Security protects against intentional harm, safety against harm to people and the environment, and sustainability against harm to the shared orbital and spectral commons; mission assurance is the cause-indifferent objective that all three serve. Six related concepts describe behavior under disturbance, and each answers a different question: reliability asks whether a system functions when failures are random and independent, robustness whether performance holds for

bounded perturbations, survivability whether the mission survives deliberate damage, fault tolerance whether internal faults are masked, resilience how much performance is preserved and how quickly it is restored, and reconstitution how long it takes to rebuild what the architecture could not absorb. The TT&C example showed how far these answers can diverge: a network with an availability of 0.99998 under independent outages had a survivability order of zero against a single action on a shared provider.

Security properties must be named per layer, and the classical triad of confidentiality, integrity, and availability must be extended with authenticity, non-repudiation, timeliness, and PNT integrity. Modeling a space system as a networked control loop located five attack channels, on sensors, commands, links, dynamics, and software, and the ramp-attack example showed that detectors tuned for faults bound only the rate of an undetected attack, not its accumulated effect. The NIST goals of anticipate, withstand, recover, and adapt, together with the architectural attributes of disaggregation, distribution, diversification, protection, proliferation, and deception, organize the design space. Defense in depth multiplies protection only when barriers are independent, and zero trust is the principle that keeps them so.

The quantitative core of the chapter is a hierarchy of resilience metrics built on a mission performance function $Q(t)$. The resilience curve yields depth, time to recover, loss area, and the normalized resilience functional; depth and loss area can rank architectures in opposite orders, as the gateway-intrusion and Galileo cases showed. Markov reward models turn recurring disruptions into availability, expected loss, mean recovery times, and transient expected resilience, which describes a bad day rather than a typical year. Network metrics, the giant-component fraction and algebraic connectivity, separate tolerance of random attrition from vulnerability to targeted and structural removal, and algebraic connectivity warns of fragility before connectivity is lost.

Finally, the choice of criterion matters as much as the model. Expected resilience, conditional value-at-risk, and worst-case resilience are ordered, and an architecture that is preferred on average, such as a single well-protected geostationary satellite, can be the worst choice in the tail. When the scenario is chosen by an adversary, the worst case becomes a game in which hardening one layer moves the attack to another. The later chapters of the book compute $Q(t)$ for specific layers and then compose the layers, using the vocabulary and metrics established here.

Exercises

Exercise 3.1 [C] For each of the following requirements, state which of the six concepts of [Section 3.2](#) it expresses and what evidence would verify it: (a) “the attitude control loop shall remain stable for inertia uncertainties of up to $\pm 15\%$ ”; (b) “the probability of losing the payload processor within five years shall not exceed 0.1”; (c) “the loss of an arbitrary pair of gateways shall not interrupt service”; (d) “90% of user capacity shall be restored within 12 hours of a ground-network intrusion”; (e) “a replacement satellite shall be operational within 75 days of a

loss". Rewrite one of them so that it also constrains a second concept.

Exercise 3.2 [C] Place the following concerns in the regions of Fig. 3.2 and justify each placement: a counterfeit radiation-hardened part in the supply chain; an uncontrolled reentry of a spent upper stage; a denial-of-service attack on a space-weather alert feed used for collision screening; a firmware update that disables end-of-life passivation. Identify one concern that you believe belongs outside mission assurance, and explain why.

Exercise 3.3 [A] A primary unit with lifetime $\tau_1 \sim \text{Exp}(\lambda)$ is backed by a cold spare that does not fail in storage, has the same lifetime distribution once switched on, and is switched in successfully with probability c . (a) Derive (3.3) by conditioning on the failure time of the primary. (b) Show that $\text{MTTF} = (1 + c)/\lambda$. (c) For the HELIOS onboard computer take $\lambda = 0.1$ per year and $c = 0.9$, and compute the five-year reliability with and without the spare. (d) An adversary can force a switchover to a spare that runs the same compromised software. Explain how this changes c and the answer to (c).

Exercise 3.4 [A] Extend Example 3.1 with a common-cause model: with probability q the leasing provider fails and all six leased sites are lost; otherwise sites fail independently with probability 0.05. (a) Derive the availability of the requirement $k = 4$ as a function of q . (b) Find the value of q at which the common-cause term contributes as much unavailability as independent failures. (c) Repeat for a design with four owned and four leased sites, and comment on the result.

Exercise 3.5 [A] In the clock-steering loop of Example 3.2, suppose the measurement bias follows $a_k = \frac{1}{2}\gamma k^2$. (a) Show that the residual r_k grows without bound and find its asymptotic slope. (b) With $L = 0.1$, $\eta = 30$ ns, and $\gamma = 0.01$ ns s⁻², estimate the time at which the detector first alarms and the clock offset accumulated by then. (c) Explain why a detector on the *rate of change* of the estimate, rather than on the residual, changes the class of stealthy attacks.

Exercise 3.6 [A] (a) Prove the bound (3.10) and the identity $R(\mathcal{A}, \pi; w) = 1 - L_T$ when $Q \leq Q_0$. (b) Construct two piecewise-linear resilience curves over $T = 24$ h with the same loss area of 2 h, one with depth 0.1 and one with depth 0.8, and give their times to recover. (c) Propose a scalar metric that combines depth and loss area, and state one property it should satisfy.

Exercise 3.7 [A] For the two-state chain with failure rate λ and repair rate μ , solve (3.11) in closed form starting from the failed state, and show that $\bar{R}_T = \mu/(\lambda + \mu) [1 - (1 - e^{-(\lambda + \mu)T})/((\lambda + \mu)T)]$ when the reward is one in the up state and zero otherwise. Compare $\bar{R}_{24}$ for $(\lambda, \mu) = (1/720, 1/8)$ and $(1/7200, 1/80)$ per hour, which have the same steady-state availability, and interpret the difference.

Exercise 3.8 [A] In Example 3.4, replace the exponential sojourn in state D by a deterministic duration with the same mean, keeping the exit probabilities to N and F unchanged. (a) Use (3.14) to show that the stationary probabilities are unchanged. (b) Compute the probability that the network is still in D twelve hours after entering it under both models. (c) Represent the deterministic sojourn

approximately by an Erlang distribution with four phases, write the enlarged generator, and compute the same probability numerically.

Exercise 3.9 [A] (a) Using the fact that the Laplacian eigenvalues of a Cartesian product are sums of those of the factors, show that $\lambda_2(C_{24} \square C_{22}) = 2 - 2 \cos(2\pi/24)$. (b) Removing one plane turns the torus into $P_{23} \square C_{22}$, where P_{23} is a path with 23 nodes whose Laplacian eigenvalues are $2 - 2 \cos(\pi j/23)$. Show that $\lambda_2 = 2 - 2 \cos(\pi/23) \approx 0.0186$, confirming [Example 3.5](#). (c) Show that the vertex connectivity of the torus is four, and give a set of four satellites whose removal isolates one satellite.

Exercise 3.10 [A] (a) For a discrete loss distribution with values $\ell_1 \geq \ell_2 \geq \dots \geq \ell_S$ and probabilities p_s , show that the minimum in (3.16) is attained at $c = \text{VaR}_\alpha(L)$ and that CVaR_α is the average of the worst $1 - \alpha$ probability mass, splitting a scenario if necessary. (b) Verify the entries for ATLAS-G in [Table 3.1](#) at $\alpha = 0.90$ and $\alpha = 0.95$. (c) Find the smallest probability of the ASAT scenario, with the routine-anomaly probability reduced by the same amount, at which ATLAS-G also ranks below HELIOS in expected resilience.

Exercise 3.11 [C] Sketch a zero-trust architecture for commanding HELIOS in the sense of NIST SP 800-207. Place policy decision points and policy enforcement points in the MOC, at the leased TT&C sites, and on board, and state the evidence each decision uses. Identify which barriers of [Fig. 3.7](#) become more independent as a result, and one new common-mode dependency that the zero-trust design itself introduces.

Exercise 3.12 [P] Write a Python function that takes a sampled performance series $Q(t_i)$ and returns depth, time to recover, time to q -recovery, loss area, normalized resilience, and rapidity as in [Definition 3.11](#). Test it on the two curves of [Example 3.3](#), and then add zero-mean Gaussian measurement noise with standard deviation 0.02. Propose and test a smoothing or hysteresis rule that keeps the estimated time to recover within 10% of its true value.

Exercise 3.13 [P] *Lab: HELIOS coverage resilience under random versus targeted satellite loss.* (a) Generate the HELIOS primary shell as a Walker-delta constellation 528/24/1 at 540 km altitude and 53° inclination, and a grid of ground points between -60° and 60° latitude weighted by area. (b) Using a minimum elevation of 25° , compute the time-averaged 1-coverage and 4-coverage performance functions (3.15) over one orbital period. (c) Remove satellites at random, by plane, and by a greedy rule that removes the satellite whose loss most reduces 1-coverage, and plot both coverage measures and the ISL giant-component fraction against the number removed. (d) Using on-orbit spares that close one gap per plane after seven days, compute the resilience functional over 30 days for the loss of 12 satellites under each removal rule, and discuss which attribute of [Fig. 3.6](#) most improves the worst case.

Exercise 3.14 [P] Formulate a small investment problem for HELIOS: a budget allows at most two of the options “one extra spare per plane”, “two additional owned TT&C sites”, and “a second leasing provider”, each of which scales down

the loss areas of specific scenarios in Table 3.1 by factors you specify and justify. Solve the problem with the linear program (3.17) for $\alpha \in \{0, 0.9, 0.95\}$ and with the worst-case criterion, using binary variables or enumeration, and report how the optimal portfolio changes with the criterion.

Notes and Further Reading

Concepts and taxonomies. The dependability taxonomy of Avižienis, Laprie, Randell, and Landwehr [92] remains the most careful treatment of the relations among reliability, availability, safety, integrity, and security, and of faults, errors, and failures; its vocabulary underlies the definitions of Sections 3.1 and 3.2. Rausand and Høyland [95] cover lifetime models, k -out-of- n structures, common-cause failures, and the availability of maintained systems. For the space domain, the Department of Defense resilience taxonomy [94] and the earlier Air Force Space Command white paper on disaggregation [114] set out the architectural attributes, and Console [98] gives an accessible NATO-oriented discussion of how they combine. The long-term sustainability guidelines of COPUOS [93] are the reference point for sustainability.

Resilience engineering and cyber resilience. NIST SP 800-160 Volume 2 [26] develops the goals, objectives, techniques, and design principles of cyber resiliency in far more detail than this chapter, including a mapping from techniques to objectives that is useful when allocating requirements. Linkov and Kott [29] and the chapters of their edited volume connect cyber resilience to the performance-curve view. Zero trust is specified in NIST SP 800-207 [100].

Resilience metrics. The resilience triangle and the 4R properties originate with Bruneau et al. [28]. Henry and Ramirez-Marquez [101] give time-dependent generic metrics, and the review of Hosseini, Barker, and Ramirez-Marquez [102] classifies deterministic, probabilistic, and structural resilience measures across engineering domains. Readers who wish to apply the curve metrics to a specific layer will find signal-layer resilience curves in Chapter 5 and physical-layer attrition and reconstitution models in Chapter 8.

Stochastic models. Trivedi [107] is a standard text for Markov and semi-Markov availability models, Markov reward models, and their numerical solution, including phase-type approximations. The general theory of continuous-time chains used in Section 3.8 is summarized in Chapter A.

Network resilience. Newman [108] covers percolation, robustness, and spectral graph theory at a level suitable for this book. The distinction between error and attack tolerance was popularized by Albert, Jeong, and Barabási [111], and Callaway et al. [110] give exact percolation results for random graphs. Algebraic connectivity is due to Fiedler [109]. For resilient coverage control of LEO constellations under space threats, see Zhao and Zhu [103].

Risk measures and strategic resilience. Conditional value-at-risk and its linear-programming formulation are due to Rockafellar and Uryasev [112, 113]. Game-theoretic formulations of robustness, security, and resilience for cyber-physical control systems, including the games-in-games principle for cross-layer design, are developed by Zhu and Başar [96] and in the monograph of Zhu and Xu [97]; [Chapter B](#) collects the game-theoretic background. Zhu and Başar [27] distinguish resilience from robustness in game-theoretic terms, and Zhu [37] surveys the foundations of cyber resilience at the confluence of game, control, and learning theories.

The Galileo outage. Official notices to users and the service-restoration announcement are available from the European GNSS Service Centre [105]. The quarterly performance report covering the outage is summarized in [104], and an independent technical reconstruction of the event is given by Hubert [106].

Threats, Risk, and the Space Kill Chain

Learning Objectives

- ▶ Classify threats to space systems by origin, mechanism, entry layer, effect layer, reversibility, attributability, and debris generation, and place them on a segment architecture.
- ▶ Describe the eight stages of the space kill chain, relate them to MITRE ATT&CK, SPARTA, and SPACE-SHIELD, and map a documented incident onto them.
- ▶ Model the kill chain as an absorbing Markov chain and compute campaign success probabilities, dwell times, and the sensitivity of risk to stage-level detection.
- ▶ Formalize bounded, stochastic, sparse, resource-limited, denial-of-service, Byzantine, adaptive, and strategic adversaries, and solve small zero-sum and Stackelberg security games.
- ▶ Quantify risk with attack-defense trees, attack graphs, Bayesian networks, and Monte Carlo loss-exceedance curves, and identify insecure control actions with STPA-Sec.
- ▶ Explain how NIST IR 8270, NIST IR 8401, the NIST Cybersecurity Framework 2.0, SPD-5, SPARTA, and ISO/IEC 27005 fit together in a risk-management process.

A space system is attacked by someone, through something, over time, and with consequences that can be weighed. This chapter builds the vocabulary and the models for each of those four elements. It begins with a layered taxonomy of threats and the actors that pose them, and places those threats on the architecture of a space system. It then follows the adversary through time along the eight-stage space kill chain and turns the chain into a stochastic process that the defender can interrupt. Because intelligent adversaries respond to defenses, the chapter next formalizes the adversary models used throughout the book and gives a first look at game-theoretic defense. The remaining sections assemble the quantitative toolkit of risk assessment (attack-defense trees and graphs, Bayesian networks, loss-exceedance curves, and STPA-Sec) and relate it to the principal government

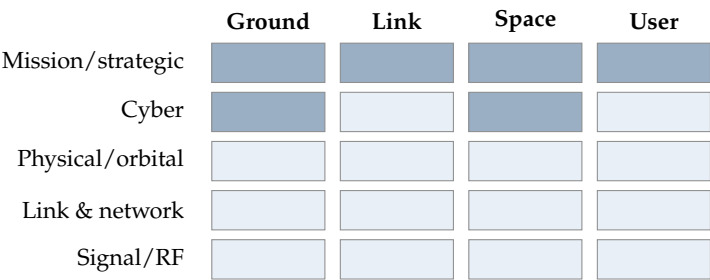


Figure 4.1. Layer map for this chapter. Dark cells mark the primary coverage: the cyber layer of the ground and space segments and the mission layer of every segment, where risk is measured. Light cells mark the remaining layers, which the threat taxonomy, the architecture threat map, and the kill chain treat at a survey level.

and industry risk frameworks.

The Space Security Stack is the book’s organizing framework: it partitions the security problems of a space system into five layers (signal/RF, link and network, physical/orbital, cyber, and mission/strategic) and four segments (ground, link, space, and user). Threats and risk are cross-cutting subjects, so this chapter touches every cell of the stack (Fig. 4.1). Its primary coverage is the cyber layer of the ground and space segments, where most kill chains begin and where attack trees and STPA-Sec are most mature, together with the mission layer, where risk is ultimately expressed as loss of service.

HELIOS Running Example

An alert at the leased station. HELIOS is the notional proliferated low-Earth-orbit (LEO) broadband and PNT-augmentation constellation used throughout this book; it is a teaching construct and is not modeled on a real operator. Its 600 satellites are commanded from a cloud-hosted mission operations center through two owned telemetry, tracking, and command (TT&C) sites and six sites leased from a ground-station-as-a-service provider.

Late on a Friday, the security operations center receives an anomalous-login alert on the provider’s customer portal: a valid HELIOS account, an unfamiliar network location, a successful second factor. Minutes later the flight-software telemetry of one satellite reports two rejected telecommands with invalid authentication codes.

The duty operator must decide whether these are two unrelated events or two stages of one campaign. The security lead wants to know how likely a compromise is, which defenses stand between the adversary and an accepted command, how long an intruder could remain undetected, and where the next unit of monitoring effort would buy the most risk reduction. Every question is quantitative, and each is answered by a model developed in this chapter.

Table 4.1. Chapter-specific notation. Symbols common to the book are listed in the front-matter Notation section.

Symbol	Meaning	Symbol	Meaning
$\ell_{\text{in}}, \ell_{\text{eff}}$	Entry layer; effect layer of a threat	$\tilde{w}$	Bound on the disturbance
α_i, δ_i	Per-step progression, detection probabilities	$\mathcal{A}_{\text{adv}}, J$	Adversary action set; adversary objective
λ_i, μ_i	Progression and detection rates (continuous time)	x_j, y_j	Defender coverage and attacker probability at target j
η_i	Probability that stage i is interdicted	p_v	Success probability of attack-tree node v
t, τ	Expected steps to absorption; dwell time	Λ_s	Annual frequency of loss scenario s
r, g	Reattack and abandonment probabilities	$\bar{F}_L(\ell)$	Exceedance probability $\mathbb{P}(L > \ell)$ of the annual loss
ε_i	Elasticity of success probability to δ_i		

Symbols common to the book are listed in the front-matter Notation section; Table 4.1 collects the symbols specific to this chapter. The HELIOS values used in the worked examples are restated in Table 4.2; parameters marked as assumptions are introduced for illustration in this chapter and describe no real system.

4.1 A Layered Threat Taxonomy

Risk analysis starts with a disciplined answer to the question of what can go wrong. For space systems the answer spans an unusually wide range of mechanisms, from a solar energetic particle event to a phishing e-mail, and from a jammer costing a few hundred dollars to a direct-ascent interceptor. A taxonomy is useful only if it groups threats in ways that predict how they must be analyzed and defended against. We therefore classify threats along two complementary axes: by *mechanism*, which determines the physics and the engineering of the threat, and by a small set of *attributes*, which determine where it enters the system, where it produces effects, and what the defender can do about it. We first fix the terms.

Definition 4.1 (Threat, vulnerability, exposure, and risk). A *threat* is a circumstance or event with the potential to adversely affect a mission, its assets, or its users. A threat is *adversarial* if it is caused by an actor with intent, and *natural* (or environmental) otherwise. A *vulnerability* is a weakness in a system, its procedures, or its implementation that a threat can exploit. *Exposure* is the degree to which a vulnerability is reachable by a threat. *Risk* is a measure of the extent to which the mission is threatened, expressed as a combination of the likelihood that a threat event occurs and succeeds and the magnitude of its consequence.

These definitions follow the risk-management usage of the NIST and ISO/IEC literature [35, 36]. The distinction between natural and adversarial threats is fundamental for modeling. A natural threat is indifferent to the defense: the rate of single-event upsets does not rise because an operator has added error-correcting

Table 4.2. HELIOS parameters used in this chapter. HELIOS is notional; entries marked (A) are assumptions introduced for the examples of this chapter.

Element	Value
Space segment	600 satellites: Shell A, 528 at 540 km, 53°; Shell B, 72 sun-synchronous at 560 km; illustrative unit cost \$2M
Onboard	Radiation-tolerant onboard computer (OBC) with cold spare; NASA cFS-based flight software; separate payload processor
TT&C	S-band; CCSDS Space Data Link Security (SDLS) authenticated encryption on telecommands
Ground segment	Cloud-hosted primary MOC and warm-standby backup; 2 owned and 6 leased (GSaaS) TT&C sites; 40 Ka-band gateways; NOC/SOC in the terrestrial core
Software supply	CI/CD pipeline builds signed flight-software and terminal-firmware updates
User segment	About 200,000 terminals with over-the-air firmware updates
Kill-chain rates (A)	Per-day progression and detection probabilities in Table 4.3
Loss scenarios (A)	Frequencies and severities in Table 4.5

memory. An adversarial threat is not indifferent: an intelligent adversary observes defenses and redirects effort to where they are weakest. Probability models calibrated from history are therefore appropriate for natural threats and for the non-strategic part of adversarial activity, whereas the strategic core of adversarial risk requires the game-theoretic reasoning of [Section 4.7](#).

4.1.1 Threat classes by mechanism

Open-source counterspace assessments group adversarial capabilities by the mechanism that produces the effect [8, 20, 115]. We adopt six adversarial classes and add a seventh, natural class, as drawn in [Fig. 4.2](#).

Electronic warfare (EW) acts on radio-frequency signals. *Jamming* is the deliberate emission of energy that raises the noise-plus-interference floor of a receiver until communication or navigation fails; *uplink* jamming targets the satellite receiver and so affects every user of a transponder or beam, whereas *downlink* jamming targets ground receivers within a local area. *Spoofing* transmits counterfeit signals that a receiver accepts as genuine, such as false navigation signals or forged telemetry, and *meaconing* rebroadcasts captured signals with a delay. *Eavesdropping* is the passive interception of signals whose content or metadata is insufficiently protected. *Cyber* threats act on data and on the software that processes it: intrusion into ground networks, malware, malicious or replayed telecommands, and compromise of flight or terminal software. *Directed-energy* threats deposit electromagnetic energy without mass impact: lasers can dazzle or damage optical sensors, and high-power microwave (HPM) sources can upset or damage electronics. *Kinetic* threats damage a target through impact or a fragmenting warhead. A *direct-ascent* (DA) anti-satellite (ASAT) weapon engages a satellite on its first pass after launch, while a *co-orbital* system is first placed in orbit and later approaches its target in a rendezvous and proximity operation (RPO).

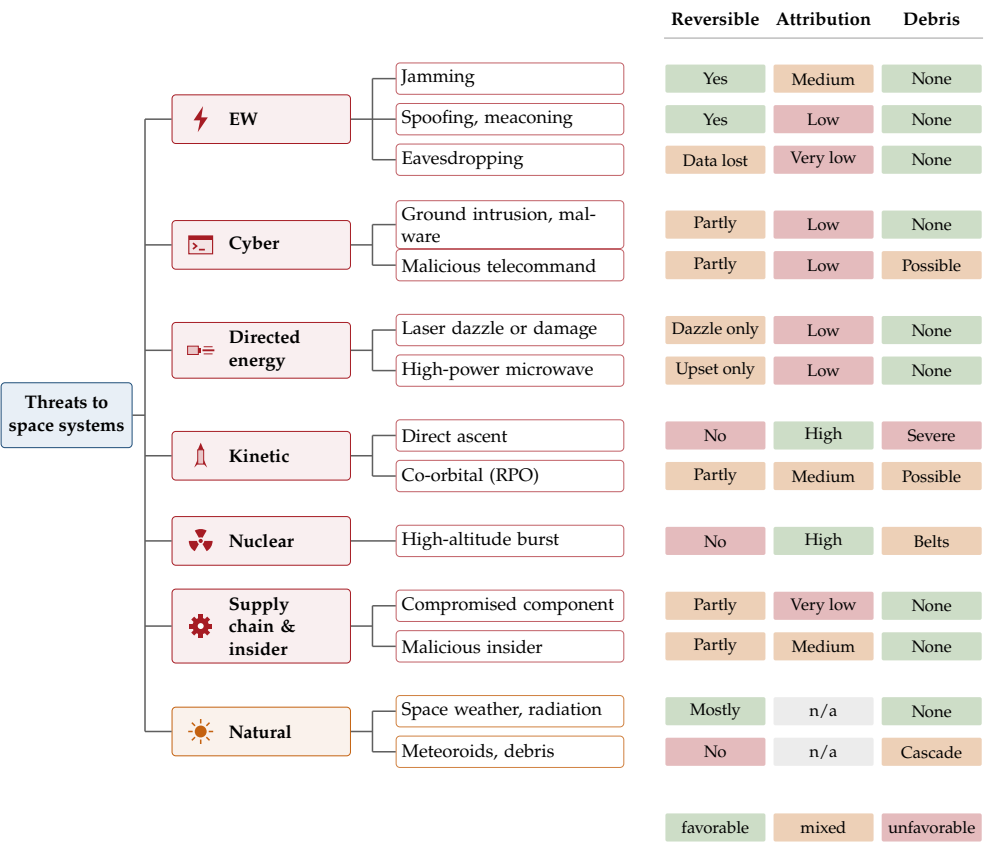


Figure 4.2. Taxonomy of threats to space systems by mechanism, with three effect attributes rated from the defender’s viewpoint (green favorable, orange mixed, red unfavorable). Six adversarial classes (red) sit above the natural class (orange). The most reversible and least attributable effects generate no debris, which makes them attractive for use below the threshold of open conflict; the most destructive effects are the easiest to attribute and the hardest to contain.

Nuclear detonation at high altitude produces prompt radiation, electromagnetic pulse, and long-lived artificial radiation belts that degrade every unhardened satellite passing through them. *Supply-chain and insider* threats, finally, act through trusted relationships: a component, software library, update pipeline, supplier, or employee is subverted before or during operations.

Natural threats form the seventh class. *Space weather* (solar flares, energetic particle events, and geomagnetic storms) causes single-event effects in electronics, surface and internal charging, and increased atmospheric drag in LEO. The *trapped radiation* environment accumulates total ionizing dose. *Meteoroids and orbital debris* threaten impact damage. *Thermal extremes* and eclipse cycling stress materials and power systems. Natural threats are included in a security taxonomy for two reasons. They consume the same margins that protect against adversarial effects, and they supply an adversary with cover: a satellite anomaly during a geomagnetic storm is plausibly natural, which delays both detection and attribution.

4.1.2 Threat attributes: entry, effect, reversibility, attribution, debris

Mechanism alone is a poor guide to defense. A jammer and a malicious telecommand can both deny a service, while two cyber intrusions can differ sharply in consequence depending on whether they stop at a billing server or reach a propulsion subsystem. We therefore attach five attributes to every threat.

Definition 4.2 (Threat profile). The *profile* of a threat records five attributes, written as the tuple $(\ell_{\text{in}}, \ell_{\text{eff}}, \rho, \kappa, \xi)$. Here ℓ_{in} is the *entry layer* at which the threat first acts on the system; ℓ_{eff} is the set of *effect layers* at which it produces consequences; ρ describes *reversibility*, that is, whether the system returns to full function once the threat stops acting; κ describes *attributability*, the confidence with which the defender can identify the responsible actor in time to respond; and ξ describes *debris generation*. A threat is *cross-layer* if ℓ_{eff} contains a layer different from ℓ_{in} .

The layers are those of the Space Security Stack: signal/RF, link and network, physical/orbital, cyber, and mission/strategic. Every threat that matters eventually has an effect at the mission layer, so we record the mission layer only when a threat acts there directly, as coercive signaling does, and otherwise read ℓ_{eff} as the technical layers through which the mission is harmed.

The right-hand columns of Fig. 4.2 rate each mechanism on reversibility, attribution, and debris. The ratings are generalizations from the open literature [8, 20] and individual systems vary, but a pattern is plain. Jamming, spoofing, dazzling, and most cyber effects stop when the attacker stops, are hard to attribute quickly, and create no debris. Destructive kinetic and nuclear effects are the opposite. The 2025 Secure World Foundation assessment accordingly observes that the counterspace capabilities actually used in current conflicts are non-destructive ones [8]. For risk analysis this means that reversible, deniable threats dominate expected frequency, whereas irreversible ones dominate the tail of the consequence distribution, a distinction to which the loss-exceedance analysis of Section 4.9 returns.

Figure 4.3 arranges representative threats by entry layer (rows) and effect layer (columns). Diagonal cells hold single-layer threats, the traditional province of specialist communities: anti-jam engineers, network defenders, flight dynamicists, and cyber analysts. Off-diagonal cells hold cross-layer threats, which exploit the seams between those communities. A spoofed timing signal (signal entry) that invalidates certificates in a ground network (cyber effect), a denial-of-service attack on a TT&C link (network entry) that prevents the upload of a collision-avoidance maneuver (physical effect), and a malicious command (cyber entry) that fires a thruster (physical effect) are all invisible to a defense that monitors only its own layer. Supply-chain and insider threats occupy a row of their own because their entry point is a trusted relationship rather than a technical layer, and they can surface in one or more columns.

Two consequences of the profile view shape the rest of the chapter. First, the defender's problem is not to protect each layer in isolation but to interrupt pathways, and a pathway is a sequence of stages in time as much as a set of cells in

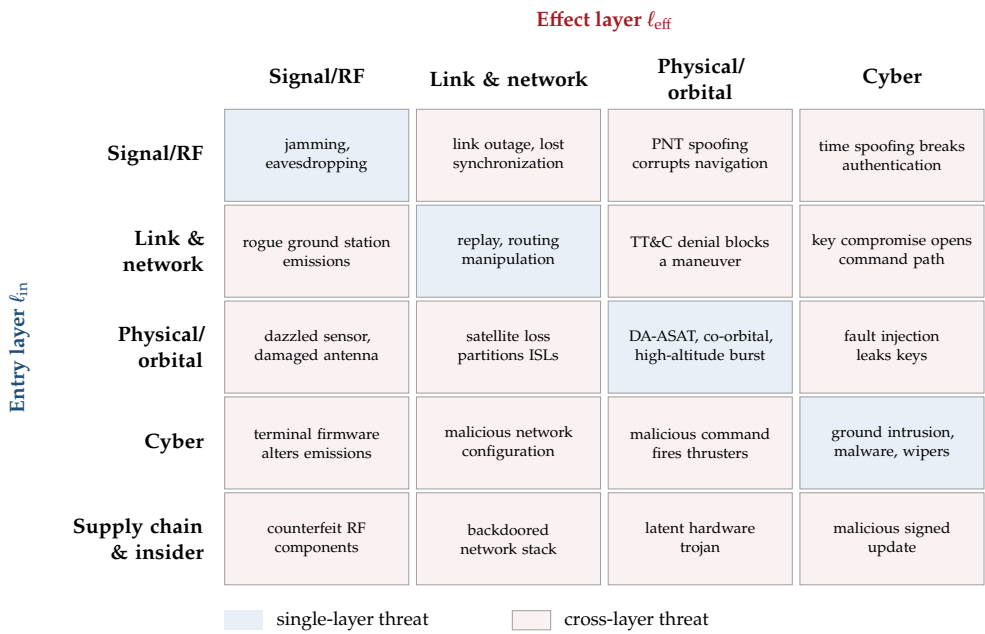


Figure 4.3. Entry layer versus effect layer for representative threats. Diagonal cells are single-layer threats that one specialist community can see end to end; every off-diagonal cell is a cross-layer pathway that crosses an organizational as well as a technical boundary. Supply-chain and insider threats enter through trusted relationships and can surface at one or more layers.

a matrix; this motivates the kill-chain model of [Section 4.4](#). Second, attributability and reversibility determine the adversary's incentives, and incentives cannot be captured by frequencies alone. A threat that is cheap, deniable, and reversible will be used more often than its technical severity suggests, which is why the actor and capability analysis of the next section precedes the assignment of probabilities.

4.2 Actors and Capabilities

A threat mechanism becomes a threat to a particular mission only when some actor has the capability to use it, the intent to do so, and an opportunity created by the target's exposure. Classical intelligence analysis expresses this as a qualitative product of *capability*, *intent*, and *opportunity*: if even one factor is absent, the threat is latent. The product form is a heuristic, not a probability law, but it clarifies which lever each defensive measure pulls. Diplomacy, deterrence, and norms of behavior act on intent; export controls, supply-chain assurance, and attribution act on capability; and system hardening, segmentation, and the reduction of exposed services act on opportunity. Only the last lever is fully under an operator's control, which is why the remainder of the chapter concentrates on exposure and its interruption.

Definition 4.3 (Capability tier). A *capability tier* is a class of actors with similar resources, technical sophistication, persistence, and tolerance for attribution. We distinguish six tiers: (T1) major spacefaring states with full-spectrum counterspace programs; (T2) other states with selected counterspace capabilities; (T3) state-sponsored or state-tolerated proxies, including advanced persistent threat (APT) groups; (T4) financially motivated criminal groups; (T5) hacktivists and ideologically motivated groups; and (T6) insiders. Security researchers and authorized red teams form a seventh, non-hostile category whose published work reveals exposure before hostile actors exploit it.

The tiers differ less in which mechanisms they can reach than in the reliability, scale, and persistence with which they can apply them. Open-source assessments by the Secure World Foundation [8] and the Center for Strategic and International Studies [20], together with the 2022 assessment of the U.S. Defense Intelligence Agency [115], describe a small group of states that have tested destructive direct-ascent ASAT weapons, a larger group developing rendezvous and proximity capabilities that could be used for inspection or interference, and a much larger population of actors able to jam or spoof satellite signals. These assessments draw on public evidence of uneven quality and some of their judgments are contested; we use them only to characterize which classes of capability are plausibly available at each tier, not to rank or accuse particular states.

Destructive kinetic tests are the best-documented tier-1 activity because they create tracked debris. Four states have destroyed their own satellites in tests: the United States in 1985, China in 2007, India in 2019, and Russia in 2021 [7, 116, 117, 118]. Tier-1 and tier-2 states are also assessed to field ground-based jammers and to pursue directed-energy systems [8, 115]. Cyber capability is the most widely distributed. The 2022 attack on the KA-SAT network, publicly

		EW	Cyber	Directed energy	Kinetic (DA, RPO)	Nuclear	Supply chain
T1 Major spacefaring states		demonstrated	demonstrated	assessed	demonstrated	assessed	demonstrated
T2 Other states		assessed	assessed	limited	limited	not assessed	limited
T3 State-sponsored proxies		assessed	demonstrated	not assessed	not assessed	not assessed	assessed
T4 Criminal groups		limited	assessed	not assessed	not assessed	not assessed	limited
T5 Hacktivists		limited	limited	not assessed	not assessed	not assessed	not assessed
T6 Insiders		limited	assessed	not assessed	not assessed	not assessed	assessed
Researchers, red teams		limited	assessed	not assessed	not assessed	not assessed	limited

Figure 4.4. Capability tier versus threat class, summarizing open-source assessments [8, 20, 115] at a deliberately coarse level. “Demonstrated” denotes capability shown in tests or operations, “assessed” capability reported as fielded or in development, and “limited” capability confined to low-end or opportunistic use. Destructive capabilities are confined to states, whereas EW and cyber capability reach every tier; the row below the dashed line shows authorized researchers, whose published findings reveal exposure.

attributed by the European Union, the United States, and the United Kingdom to Russia [13], showed that a tier-1 actor can achieve large-scale effects against a commercial space service without touching a satellite [12]. Criminal ransomware groups routinely compromise the enterprise networks of aerospace suppliers, and although their intent is extortion, the access they obtain can be sold to or reused by more capable actors. Insiders hold the rarest and most valuable asset of all, legitimate access, and bypass most perimeter defenses by construction.

At the bottom of the capability scale, researchers have repeatedly shown how little specialized equipment exposure requires. Pavur and Martinovic eavesdropped on unencrypted satellite broadband traffic with consumer-grade equipment [73], and Willbold and colleagues found exploitable vulnerabilities in the firmware of several in-orbit satellites [84]. These results do not imply that low-tier actors can reach a well-defended mission, but they establish that the lower bound of the capability distribution is far lower in space than it was a generation ago, when ground stations and waveforms were proprietary and expensive.

Figure 4.4 summarizes the relationship between tiers and threat classes. The shading records the level of capability that open-source assessments attribute to each tier, and is deliberately coarse. The figure makes three structural points. The directed-energy, kinetic, and nuclear columns (destructive physical capabilities) are populated only by states. The cyber and EW columns are populated at every tier, which means that most adversarial risk for a commercial operator such as HELIOS lies there. And the insider row is short but intense: an insider needs little technical capability because the trusted role supplies it.

For the HELIOS operator, the tier analysis narrows the threat model. HELIOS is a commercial service whose users include government and emergency-services customers, so it is a plausible target for T1 to T3 actors during a crisis and a steady target for T4 extortion, while its large workforce and supplier base create

T6 exposure. The operator cannot influence the intent of these actors and has little influence on their capability. Its task is to make the opportunities they need expensive and short-lived, and to measure how much each defensive investment achieves. The next section makes opportunity concrete by placing threats on the HELIOS architecture.

4.3 The Architecture Threat Map

The taxonomy says what threats exist and the actor analysis says who can use them. The architecture threat map says *where* they act. A space-enabled service is a coupled system of four segments. The *ground segment* comprises TT&C ground stations, gateway Earth stations, the mission operations center (MOC) that commands the spacecraft, the terrestrial core network, and the network operations and security operations centers (NOC/SOC). The *link segment* carries TT&C, feeder, and user links between ground and space, inter-satellite links (ISLs) between spacecraft, and, for HELIOS, a PNT broadcast. The *space segment* comprises the satellites together with their onboard operations and maintenance (O&M) endpoints: the software functions that accept commands, report telemetry, support tracking, and install updates. The *user segment* comprises terminals of many kinds (fixed, maritime, aeronautical, and mobile) and the government, commercial, emergency-service, and public communities that depend on them.

Figure 4.5 draws the HELIOS instance of this architecture and overlays both threat families at the points where they enter the system. Each adversarial threat carries a tag naming the security property it attacks: confidentiality [C], the protection of information from unauthorized disclosure; integrity [I], the protection of information and commands from unauthorized modification or origin; and availability [A], timely and reliable access to the service. Natural threats are drawn across the top because they act on every segment at once.

Reading the map by segment reveals where exposure concentrates. In the ground segment, the TT&C sites are the physical gateways of the command path. The six HELIOS sites leased from a ground-station-as-a-service (GSaaS) provider are reached through the provider's customer portal, so credential theft there is an integrity threat to commanding even though the portal is outside the operator's own network. The MOC and the NOC/SOC are exposed to unauthorized access and exploitation, and the core network, which also carries customer traffic, is exposed to malware and supply-chain compromise of its equipment and software. In the link segment, uplinks are exposed to jamming, command links to impersonation and injection, and user and feeder links to eavesdropping; the PNT broadcast is exposed to spoofing and meaconing at the receiver. In the space segment, ISLs can in principle be intercepted or disrupted, and the satellites themselves are exposed to physical threats. In the user segment, the over-the-air update channel is an integrity and availability exposure of the whole terminal population, as the KA-SAT incident demonstrated for a different system [12].

The C/I/A tags also organize the defense. Confidentiality exposures (telemetry, payload, and user data) are addressed by end-to-end encryption and key

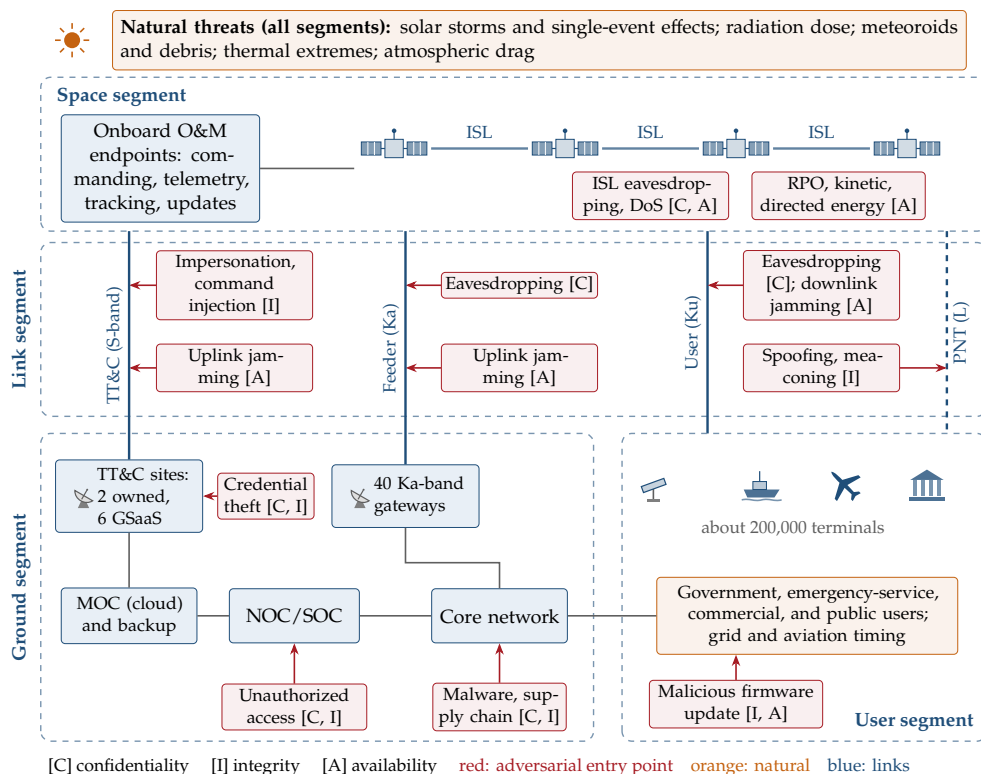


Figure 4.5. Architecture threat map for HELIOS. The four segments are coupled by TT&C, feeder, user, PNT, and inter-satellite links; adversarial threats (red) are drawn at the interfaces where they enter and tagged with the property they attack, while natural threats (orange) act on every segment. Most entry points lie in the ground, link, and user segments, and none of the red tags can be neutralized by protecting a single node.

management. Integrity exposures (unauthorized commands, falsified data, and malicious firmware) are addressed by message authentication, including the CCSDS Space Data Link Security (SDLS) protocol on HELIOS telecommands [70], and by secure boot and signed updates. Availability exposures (jamming, denial of service, and environmental disruption) are addressed by anti-jam waveforms, alternative paths and bearers, redundancy, and plans for reconstitution. These are the same three families of control that recur in the space-specific guidance of NIST IR 8270 and IR 8401 [35, 36].

The map carries a lesson that no single-node analysis delivers: security is a property of the whole architecture. The command path to a HELIOS satellite runs from a cloud console through the MOC, across a terrestrial network to a leased antenna, and over an RF link to an onboard command handler. An encrypted, authenticated RF link does not protect a command that is correctly authenticated because it was issued from a compromised console, and a hardened MOC does not protect against a malicious update signed by a compromised build pipeline. Each red tag in Fig. 4.5 is an entry point, but mission effects require a path from an entry point to a critical function, and paths cross segment boundaries. The map is therefore static in two senses: it shows where an adversary can begin but not how it proceeds, and it shows entry points but not their conditional dependencies. The kill chain of the next section supplies the temporal view, and the attack trees and graphs of Section 4.8 supply the dependency view.

4.4 The Space Kill Chain

Serious attacks are campaigns, not events. The adversary that disabled tens of thousands of KA-SAT modems in 2022 had first to find an exposed appliance, obtain access through it, reach a trusted management segment, prepare a destructive payload, and issue commands at a chosen moment [12]. Each of these steps took time, left some trace, and could in principle have been interrupted. Intrusion analysts formalized this observation as the *kill chain*, an ordered sequence of phases that an adversary must complete to achieve an objective; Hutchins, Cloppert, and Amin showed that defenders who analyze intrusions phase by phase can detect and disrupt campaigns earlier and learn from partial failures [30]. The book adapts the idea to space systems as an eight-stage chain that spans every layer of the Space Security Stack.

Definition 4.4 (Space kill chain). The *space kill chain* is the ordered sequence of stages (1) Reconnaissance, (2) Resource development, (3) Initial access, (4) Execution & persistence, (5) Lateral movement, (6) Mission effects, (7) Assessment & reattack, and (8) Recovery & resilience. Stages 1 to 7 are adversary stages; stage 8 is the defender's stage, in which service is restored, capacity reconstituted, and the system adapted. A stage is not tied to a particular layer of the Space Security Stack, and successive stages need not share a layer.

Figure 4.6 draws the chain with, for each stage, representative activities, the adversary's typical goal, the segments most often targeted, and the defensive controls that most directly interrupt it. The figure reads left to right on the top

row and right to left on the bottom row. The segment chips G, L, S, and U mark the ground, link, space, and user segments; a filled chip indicates that the stage commonly acts on that segment.

4.4.1 The stages in depth

In *reconnaissance* the adversary builds a picture of the target from sources that are mostly public: orbital elements and registration data, spectrum filings, observed emissions, the locations of ground sites, the public-facing services of operators and their suppliers, and information about personnel. For a space system much of this picture is unavoidably public, because orbits are observable and spectrum use is coordinated internationally. Indicators are correspondingly weak, and are limited to scanning of exposed services, unusual interest in staff through social media, or observed approaches by other spacecraft. The defender's leverage lies in reducing what is exposed: operational security, emission control where the mission allows it, and periodic audits of the internet-facing attack surface.

In *resource development* the adversary acquires the means to act: exploits, malware, stolen or purchased credentials, leased or rogue ground infrastructure, software-defined radios and jammers, or, at the high end of the capability scale, directed-energy systems and co-orbital spacecraft. This stage takes place largely on the adversary's own infrastructure, so direct detection by the operator is rare. It is nonetheless the stage at which threat-intelligence sharing, credential-leak monitoring, and supplier vetting act, and at which the capability tiers of [Section 4.2](#) are most visible.

Initial access establishes a foothold in a trusted path. The path may be cyber, as with a vulnerable virtual private network (VPN) appliance, a phished operator account, or a misused GSaaS customer credential; it may be RF, as with an attempt to inject commands on an unauthenticated uplink; or it may be physical, as with an inspector satellite positioned near its target. Indicators include anomalous logins, authentication failures, rejected telecommands, and unexpected conjunctions. Controls are multi-factor authentication, prompt patching, and cryptographic authentication of commands, so that possession of an RF transmitter alone does not confer command authority.

In *execution and persistence* the adversary turns access into action and arranges to keep it. Characteristic techniques are the issuing of legitimate commands for malicious ends, the installation of implants or malicious updates, the modification of onboard tables, the abuse of safe modes or maintenance states in which protections are relaxed, and, in the RF domain, jamming or spoofing timed to support other stages. Command allow-listing, secure boot and signed software, integrity monitoring of flight software, and anomaly detection on telemetry are the principal controls.

Lateral movement extends control from the initial foothold toward the functions the adversary actually wants: from an enterprise network into a mission network, from a ground system into the spacecraft bus or payload, or across ISLs to other satellites. In a proliferated constellation, lateral movement through shared software or shared keys is the mechanism by which a single compromise could scale to many spacecraft. Segmentation, isolated management planes, per-satellite

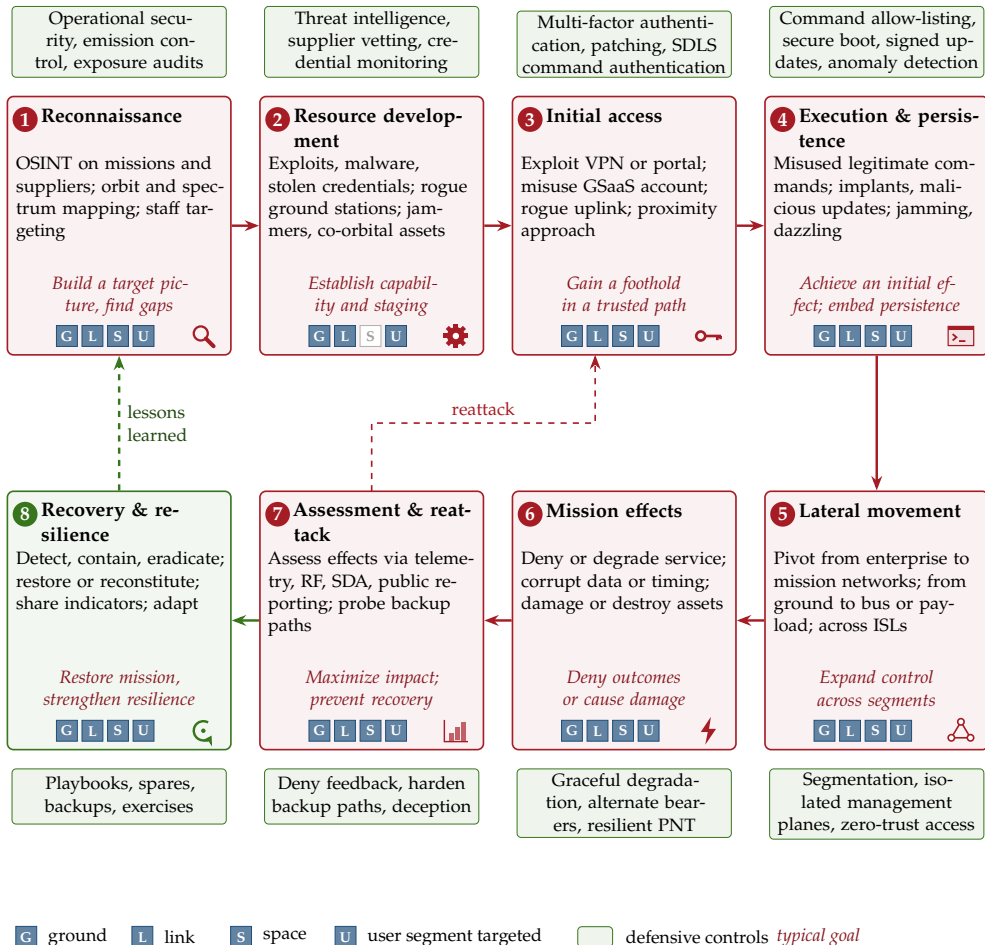


Figure 4.6. The eight-stage space kill chain. Red cards are adversary stages (activities, typical goal in italics, targeted segments), and green boxes are the defensive controls that most directly interrupt each stage. The dashed red arrow marks reattack after assessment; the final stage belongs to the defender, whose lessons feed the next cycle of anticipation. Nearly every stage can involve every segment, which is why the chain must be defended across the whole architecture.

keys, and zero-trust access, in which every request is authenticated and authorized regardless of network location, are the controls.

Mission effects are the purpose of the campaign: to deny, degrade, deceive, damage, or destroy a space-enabled service. Effects may be reversible, such as a jamming-induced outage or corrupted data, or irreversible, such as a wiped terminal, a depleted propellant tank, or a destroyed satellite. At this stage detection is usually immediate, because the effect is visible, so the defender's controls shift from prevention to limitation: graceful degradation, alternate bearers and paths, resilient timing, and the capacity to operate through.

In *assessment and reattack* the adversary evaluates its effect through telemetry and RF observation, space domain awareness (SDA), and public reporting, and decides whether to sustain, extend, or repeat the campaign, often by probing the backup paths the defender turns to. Denying the adversary feedback, hardening backup paths before they are needed, and deception all raise the cost of this stage. Finally, *recovery and resilience* is the defender's stage: detection, containment, and eradication; restoration of service; reconstitution of lost capacity; the sharing of indicators with other operators; and adaptation of the architecture. Its duration and completeness determine the area lost under the performance curve, and its lessons feed the defender's next cycle of anticipation.

Two properties distinguish the space kill chain from its enterprise ancestor. First, stages switch layers. Reconnaissance of RF emissions (signal layer) can locate a ground station whose remote-access service (cyber layer) is then exploited to issue a maneuver command (physical layer) that degrades a service (mission layer). Second, the chain is not strictly linear. Detection may evict an adversary from stage 5 back to stage 3, reattack loops from stage 7 to earlier stages, and a patient adversary may dwell in stage 4 for months. Both properties argue for a quantitative model in which stage transitions are uncertain and the defender's controls act on transition probabilities. Before building that model we relate the chain to the catalogs of adversary behavior that practitioners use.

4.4.2 Relation to ATT&CK, SPARTA, and SPACE-SHIELD

Three public knowledge bases catalog adversary behavior as *tactics* (the adversary's tactical goal, such as gaining initial access) refined into *techniques* (how the goal is achieved). MITRE ATT&CK for enterprise networks is the most widely used; its current release lists fifteen tactics, from Reconnaissance and Resource Development through Initial Access, Execution, Persistence, Privilege Escalation, Stealth, Defense Impairment, Credential Access, Discovery, Lateral Movement, Collection, Command and Control, and Exfiltration to Impact [119]. SPARTA (Space Attack Research and Tactic Analysis), maintained by The Aerospace Corporation, adapts the idea to spacecraft with nine tactics: Reconnaissance, Resource Development, Initial Access, Execution, Persistence, Defense Evasion, Lateral Movement, Exfiltration, and Impact, and pairs its techniques with countermeasures [31]. The European Space Agency's SPACE-SHIELD matrix covers the space segment and its communication links with fourteen tactics that follow the pre-2026 ATT&CK structure [120]. Tactic lists are revised between releases, so an analysis should record the version it uses.

	MITRE ATT&CK v19 (15)	SPARTA (9)	ESA SPACE-SHIELD (14)
1 Reconnaissance	Reconnaissance	Reconnaissance	Reconnaissance
2 Resource development	Resource Development	Resource Development	Resource Development
3 Initial access	Initial Access; Cre- dential Access	Initial Access	Initial Access; Cre- dential Access
4 Execution & persistence	Execution; Persistence; Priv- ilege Escalation; Stealth; Defense Impairment	Execution; Persis- tence; Defense Evasion	Execution; Persis- tence; Privilege Escala- tion; Defense Evasion
5 Lateral movement	Discovery; Lateral Movement; Command and Control	Lateral Movement	Discovery; Lateral Movement; Command and Control
6 Mission effects	Collection; Exfiltration; Impact	Exfiltration; Impact	Collection; Exfiltration; Impact
7 Assessment & reattack	<i>no tactic</i>	<i>no tactic</i>	<i>no tactic</i>
8 Recovery & resilience	<i>outside scope</i>	<i>countermeasure catalog</i>	<i>outside scope</i>

Figure 4.7. Alignment of the eight-stage space kill chain (rows) with the tactics of MITRE ATT&CK (version 19), SPARTA, and ESA SPACE-SHIELD (columns). The catalogs agree on stages 1 to 6 and differ in granularity; none has a tactic for assessment and reattack, and recovery belongs to defender frameworks rather than adversary catalogs.

Figure 4.7 aligns the three catalogs with the eight stages. Stages 1 to 6 are covered by all three, with the catalogs differing mainly in granularity: what the kill chain treats as one stage of execution and persistence, ATT&CK divides into five tactics. Two gaps are instructive. None of the catalogs has a tactic corresponding to assessment and reattack, because they describe individual intrusions rather than campaigns that adapt to observed effects; and none covers recovery, which lies outside an adversary catalog by construction and is addressed instead by defender frameworks such as the Respond and Recover functions of the NIST Cybersecurity Framework [121]. The kill chain is therefore not a replacement for the catalogs but a coarser, campaign-level index into them, suited to the stochastic modeling of the next section.

Example 4.1 (Mapping the KA-SAT incident onto the kill chain). The public record of the attack on the KA-SAT network on 24 February 2022 is summarized in Viasat’s incident overview [12] and in the SentinelLabs analysis of the AcidRain wiper [23]. The task is to place each documented fact on a stage and to note which stages the record leaves blank.

Reconnaissance is not publicly documented, but the campaign presupposes identification of a misconfigured VPN appliance and knowledge of the network’s management architecture. Resource development is represented by AcidRain, a wiper built to erase modems and routers, which SentinelLabs assessed with medium confidence to share developmental similarities with an earlier VPNFilter plugin [23]. Initial access was remote access through the misconfigured VPN appliance [12]. Lateral movement took the adversary into the trusted management segment of the network. Execution consisted of legitimate, targeted management commands sent simultaneously to a large number of residential modems, followed

by destructive commands that overwrote key data in flash memory [12]; persistence was not needed for a one-shot destructive effect and is not reported. Mission effects were the loss of connectivity for tens of thousands of modems and, as a third-party consequence, the loss of remote monitoring for about 5,800 wind turbines in Germany [24]. Assessment and reattack are not publicly documented. Recovery required shipping nearly 30,000 replacement modems [12], and the turbine operator needed weeks to restore connectivity [24].

The mapping shows that every stage before mission effects took place in the ground and user segments, with the satellite acting only as a transparent relay, and that the defender's decisive opportunities lay at stages 3 and 5, where authentication and segmentation apply. It also shows the characteristic asymmetry of destructive cyber effects: execution took minutes, while recovery in stage 8 took weeks and a logistics operation.

4.5 The Kill Chain as a Stochastic Process

The kill chain becomes a quantitative tool once its stage transitions are treated as uncertain. At each stage the adversary may progress, may be detected and evicted, or may remain, and none of these outcomes is certain for either side. The natural model is a Markov chain whose transient states are the adversary stages and whose absorbing states are the two ways a campaign can end: with the effect achieved, or with the adversary interdicted. The model is simple enough to solve in closed form, rich enough to capture reattack loops, and directly answers the questions posed in the chapter's opening vignette: how likely a campaign is to succeed, how long an intruder dwells, and which stage most rewards added detection. The standard reference for the mathematics is Kemeny and Snell [122].

4.5.1 An absorbing Markov chain model

Definition 4.5 (Absorbing Markov chain). A discrete-time Markov chain on a finite state space is *absorbing* if it has at least one absorbing state (a state that, once entered, cannot be left) and if an absorbing state can be reached from every state. The remaining states are *transient*. Ordering the m transient states before the k absorbing states, the transition matrix takes the canonical form

$$P = \begin{pmatrix} Q & R \\ 0 & I_k \end{pmatrix}, \quad (4.1)$$

where $Q \in \mathbb{R}^{m \times m}$ holds transition probabilities among transient states and $R \in \mathbb{R}^{m \times k}$ holds transition probabilities from transient to absorbing states.

For the kill chain, the transient states are the adversary stages $1, \dots, 7$ and the absorbing states are E (*effect achieved*) and X (*interdicted or evicted*), after which the defender's recovery stage begins. One time step is one day. In stage $i \in \{1, \dots, 5\}$ the adversary progresses to stage $i + 1$ with probability α_i , is detected and evicted with probability δ_i , and otherwise remains. In stage 6 the effect is delivered with probability α_6 , and with probability δ_6 the defender blocks it, in which case the adversary moves to stage 7 to assess the failure. In stage 7 the adversary reattacks

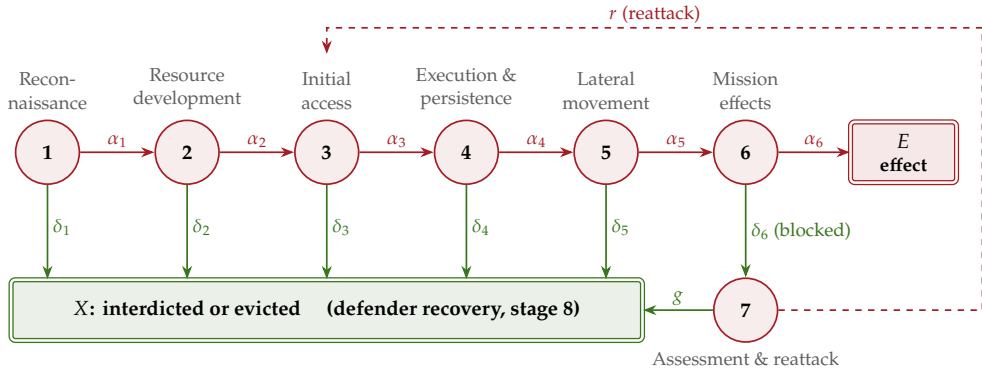


Figure 4.8. The kill chain as an absorbing Markov chain with one-day steps. Red arrows are adversary progress, green arrows are detection, blocking, or abandonment, and the dashed arrow is reattack from assessment back to initial access. Self-transitions, with probabilities $1 - \alpha_i - \delta_i$ and $1 - r - g$, are omitted for clarity; E and X are absorbing.

by returning to stage 3 with probability r , abandons the campaign with probability g , and otherwise continues to assess. The structure is drawn in Fig. 4.8.

The quantities of interest follow from classical results.

Proposition 4.1 (Fundamental matrix, absorption, and time to absorption). *For an absorbing chain in canonical form (4.1), the matrix $I - Q$ is invertible and the fundamental matrix*

$$N = (I - Q)^{-1} = \sum_{n=0}^{\infty} Q^n \quad (4.2)$$

has entries N_{ij} equal to the expected number of visits to transient state j starting from transient state i . The probability of absorption in absorbing state a starting from i is the entry B_{ia} of

$$B = NR, \quad (4.3)$$

and the expected number of steps before absorption starting from i is the i th entry of $t = N\mathbf{1}$.

Proof. Let $(X_n)_{n \geq 0}$ be the chain, let $\mathcal{T}$ be the set of transient states, write $\mathbb{P}_i$ and $\mathbb{E}_i$ for probability and expectation given $X_0 = i \in \mathcal{T}$, and let $\tau = \min\{n : X_n \notin \mathcal{T}\}$ be the time of absorption.

Powers of P . Because the lower-left block of (4.1) is zero and the lower-right block is I_k , induction on n shows that P^n has upper-left block Q^n , lower-left block $\mathbf{0}$, and lower-right block I_k . Hence $(Q^n)_{ij} = \mathbb{P}_i(X_n = j)$ for $i, j \in \mathcal{T}$, and since absorbing states cannot be left, the i th row sum of Q^n equals $\mathbb{P}_i(X_n \in \mathcal{T}) = \mathbb{P}_i(\tau > n)$.

Convergence and invertibility. Fix $i \in \mathcal{T}$. Some absorbing state is reachable from i , so a path of positive probability leads from i to an absorbing state; deleting its cycles leaves a path that visits each transient state at most once and therefore has at most m steps. Hence $\mathbb{P}_i(\tau \leq m) > 0$, and $c = \max_{i \in \mathcal{T}} \mathbb{P}_i(\tau > m) < 1$. Every row sum of Q^m is at most c . For nonnegative matrices A and C whose row sums are at most c_1 and c_2 , the row sums of AC are at most $c_1 c_2$; hence the row sums

of $Q^{qm} = (Q^m)^q$ are at most c^q , and those of Q^n , which do not increase with n because Q is substochastic, are at most $c^{\lfloor n/m \rfloor}$. Thus $0 \leq (Q^n)_{ij} \leq c^{\lfloor n/m \rfloor}$, so $Q^n \rightarrow 0$ and $\sum_n Q^n$ converges entrywise. Letting $K \rightarrow \infty$ in $(I - Q) \sum_{n=0}^K Q^n = I - Q^{K+1}$ gives $(I - Q)N = I$, so the square matrix $I - Q$ is invertible with inverse N . The same bound gives $\mathbb{P}_i(\tau > n) \leq c^{\lfloor n/m \rfloor} \rightarrow 0$, so $\tau < \infty$ almost surely.

Expected visits. The number of visits to $j \in \mathcal{T}$ is $V_j = \sum_{n \geq 0} \mathbb{1}\{X_n = j\}$, counting time 0. By monotone convergence, $\mathbb{E}_i V_j = \sum_{n \geq 0} \mathbb{P}_i(X_n = j) = \sum_{n \geq 0} (Q^n)_{ij} = N_{ij}$.

Absorption probabilities. Since $X_0 = i$ is transient, $\tau \geq 1$. For an absorbing state a , the event $\{X_\tau = a\}$ occurs if and only if there are $n \geq 0$ and $j \in \mathcal{T}$ with $X_n = j$ and $X_{n+1} = a$, and then necessarily $n = \tau - 1$ and $j = X_{\tau-1}$. The events $\{X_n = j, X_{n+1} = a\}$ are therefore disjoint over (n, j) and their union is $\{X_\tau = a\}$. By the Markov property,

$$B_{ia} = \sum_{n \geq 0} \sum_{j \in \mathcal{T}} \mathbb{P}_i(X_n = j) R_{ja} = \sum_{j \in \mathcal{T}} N_{ij} R_{ja} = (NR)_{ia}.$$

Expected time to absorption. For each n , $X_n \in \mathcal{T}$ if and only if $n < \tau$, so $\tau = \sum_{n \geq 0} \mathbb{1}\{X_n \in \mathcal{T}\} = \sum_{j \in \mathcal{T}} V_j$. Taking expectations gives $t_i = \mathbb{E}_i \tau = \sum_{j \in \mathcal{T}} N_{ij} = (N\mathbf{1})_i$. $\square$

Two derived quantities make the model interpretable. The *interdiction probability* of stage $i \leq 5$ is the probability that the stage ends in eviction rather than progress,

$$\eta_i = \frac{\delta_i}{\alpha_i + \delta_i}, \quad (4.4)$$

and the expected *sojourn* in stage i is $1/(\alpha_i + \delta_i)$ days. Writing $\sigma = \alpha_6/(\alpha_6 + \delta_6)$ for the probability that an attempted effect succeeds and $\rho = r/(r + g)$ for the probability that a blocked adversary eventually reattacks, a first-step argument over the reattack loop gives the campaign success probability from stage 1 in closed form:

$$P_E = B_{1E} = \frac{\sigma \prod_{i=1}^5 (1 - \eta_i)}{1 - (1 - \sigma) \rho \prod_{i=3}^5 (1 - \eta_i)}. \quad (4.5)$$

The numerator is the probability of passing every detection opportunity once and then succeeding; the denominator accounts for the geometric series of reattacks, each of which must pass stages 3 to 5 again. Without reattack ($\rho = 0$), the chain reduces to the product of stage survival probabilities that underlies the intuition of defense in depth.

4.5.2 A HELIOS campaign model

To make the model concrete we consider a campaign against the HELIOS command path of the kind anticipated in the opening vignette: an adversary seeks to obtain command authority over satellites through the ground segment. Table 4.3 lists per-day progression and detection probabilities. They are illustrative assumptions for a patient, capable adversary and a moderately mature defender, chosen so that sojourn times (days to weeks per stage) are consistent with the order of magnitude of dwell times reported in the incident-response literature; they are not calibrated

Table 4.3. Kill-chain parameters for a notional campaign against the HELIOS command path (assumptions), with derived quantities. Time step one day; stage 6 “detection” means the effect is blocked and the adversary moves to stage 7, and stage 7 has reattack probability $r = 0.20$ and abandonment probability $g = 0.10$.

	Stage	α_i	δ_i	η_i	Sojourn (days)	N_{1i} (days)
1	Reconnaissance	0.10	0.005	0.048	9.5	9.5
2	Resource development	0.05	0.005	0.091	18.2	17.3
3	Initial access	0.08	0.02	0.200	10.0	9.9
4	Execution & persistence	0.20	0.03	0.130	4.3	3.4
5	Lateral movement	0.10	0.04	0.286	7.1	4.9
6	Mission effects	0.50	0.30	0.375	1.25	0.61
7	Assessment & reattack	n/a	n/a	n/a	3.3	0.61

to a real operator. The table also lists the derived interdiction probabilities (4.4), mean sojourn times, and the expected number of days the campaign spends in each stage when it starts in stage 1, which is the first row of N .

Example 4.2 (HELIOS campaign success versus stage-level detection). Using the parameters of Table 4.3, compute the probability that a campaign starting in reconnaissance achieves its effect, the expected campaign duration, and the uniform per-stage interdiction probability needed to hold the success probability to 0.10.

Solution. Building $\mathbf{Q} \in \mathbb{R}^{7 \times 7}$ and $\mathbf{R} \in \mathbb{R}^{7 \times 2}$ from the table and applying Theorem 4.1 gives $B_{1E} = 0.307$ and $B_{1X} = 0.693$; the closed form (4.5) reproduces the same value, with stage survival product $\prod_{i=1}^5 (1 - \eta_i) = 0.430$, $\sigma = 0.625$, and $\rho = 2/3$. The expected time to absorption is $t_1 = 46.3$ days, most of it spent in resource development (17.3 days), where the defender has little visibility. From an established foothold in stage 3 the success probability rises to $B_{3E} = 0.355$. Without reattack the success probability would be $0.430 \times 0.625 = 0.269$, so the reattack loop adds almost four percentage points.

To study detection investment, set $\eta_i = \eta$ for all $i \leq 5$ and keep σ and ρ fixed. Figure 4.9 plots $P_E(\eta)$ from (4.5). The HELIOS baseline is equivalent to a uniform $\eta = 0.160$. Holding P_E to 0.10 requires $\eta = 0.318$, that is, roughly doubling the per-stage interdiction probability; if the adversary could not reattack, $\eta = 0.307$ would suffice.

The success probability of a multi-stage campaign falls geometrically in the per-stage interdiction probability, so moderate detection at every stage is far more effective than strong detection at one; reattack erodes this benefit most when detection is weak, where the two curves in the figure separate.

4.5.3 Continuous time and sensitivity

A daily time step is convenient but arbitrary. In continuous time the adversary progresses out of stage i at rate λ_i and is detected at rate μ_i , so the sojourn in stage i is exponentially distributed with mean $1/(\lambda_i + \mu_i)$. Collecting the rates among

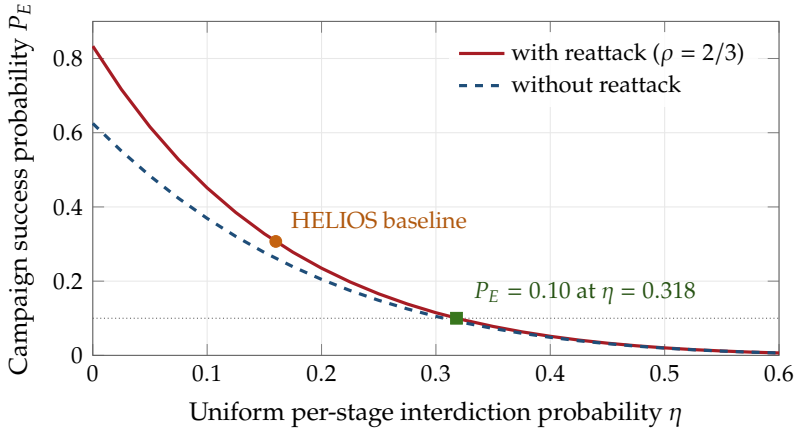


Figure 4.9. Campaign success probability as a function of a uniform per-stage interdiction probability, from (4.5) with $\sigma = 0.625$. The orange dot is the uniform-equivalent HELIOS baseline and the green square the detection level that holds success to 10%. Reattack matters most when detection is weak.

transient states in the matrix $G_{\mathcal{T}}$, whose diagonal entries are minus the total exit rates, and the rates into absorbing states in $G_{\mathcal{A}}$, the absorption probabilities, expected times to absorption, and distribution of the time to absorption τ are

$$B = (-G_{\mathcal{T}})^{-1} G_{\mathcal{A}}, \quad t = (-G_{\mathcal{T}})^{-1} \mathbf{1}, \quad \mathbb{P}(\tau > s) = \mathbf{e}_i^{\top} \exp(G_{\mathcal{T}} s) \mathbf{1}, \quad (4.6)$$

where $\mathbf{e}_i$ is the indicator of the starting stage. The last expression is a *phase-type* distribution. The interdiction probabilities become $\eta_i = \mu_i / (\lambda_i + \mu_i)$, so absorption probabilities depend only on ratios of rates and (4.5) holds unchanged. If the rates per day are numerically equal to the daily probabilities of Table 4.3, the continuous-time model reproduces $B_{1E} = 0.307$ and $t_1 = 46.3$ days exactly; only the shape of the dwell-time distribution changes slightly. The continuous-time form is preferable when detection is driven by events such as alerts and audits rather than by a daily review cycle, and it is the form used by the detection-rate models of intrusion response.

Defenders rarely ask for the success probability alone; they ask where the next unit of detection effort should go. Sensitivity analysis answers that question. Because detection probabilities differ by orders of magnitude across stages, a scale-free measure is useful.

Definition 4.6 (Detection elasticity). The *elasticity* of the campaign success probability with respect to the detection probability of stage i is

$$\varepsilon_i = \frac{\delta_i}{P_E} \frac{\partial P_E}{\partial \delta_i}, \quad (4.7)$$

the percentage change in P_E caused by a one-percent increase in δ_i .

For a stage outside the reattack loop the elasticity has a simple form. Differentiating (4.5), only the factor $1 - \eta_i = \alpha_i / (\alpha_i + \delta_i)$ depends on δ_i , and its logarithmic

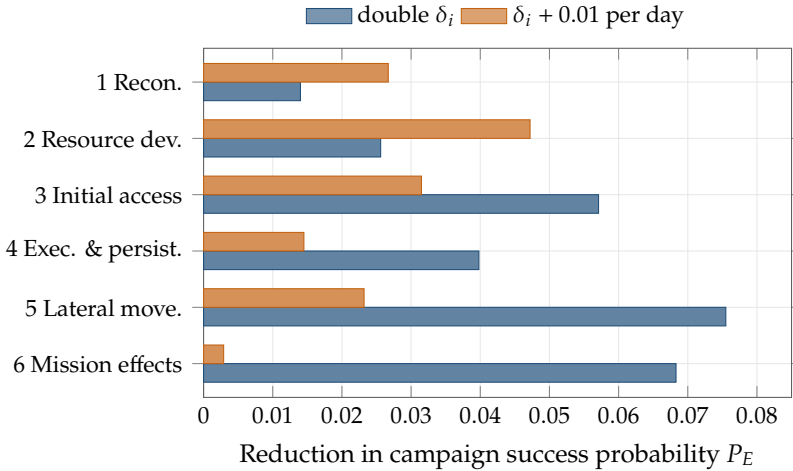


Figure 4.10. Reduction in HELIOS campaign success probability (baseline 0.307) from improving detection at one stage at a time. Proportional improvements pay most at lateral movement and mission effects; additive improvements pay most at long-dwell early stages, where the adversary gives the defender many daily chances.

derivative with respect to $\log \delta_i$ is $-\delta_i/(\alpha_i + \delta_i)$, so $\varepsilon_i = -\eta_i$ for $i = 1, 2$. For stages 3 to 5, which are traversed again on each reattack, the magnitude of the elasticity exceeds η_i .

Example 4.3 (Where detection investment buys the most). For the HELIOS campaign of Table 4.3, rank the stages by the reduction in P_E obtained from (i) a proportional improvement that doubles δ_i , and (ii) an additive improvement that raises δ_i by 0.01 per day, such as one new analytic that catches the stage's activity on one day in a hundred.

Solution. Numerical differentiation of B_{1E} gives elasticities of -0.05 , -0.09 , -0.23 , -0.15 , -0.33 , and -0.29 for stages 1 to 6, confirming $\varepsilon_1 = -\eta_1$ and $\varepsilon_2 = -\eta_2$. Recomputing B with each change applied in turn gives the reductions in Fig. 4.10. Doubling detection buys the most at lateral movement (P_E falls by 0.076, from 0.307 to 0.231), followed by mission effects (0.068) and initial access (0.057). An additive +0.01 per day buys the most at resource development (0.047), followed by initial access (0.032) and reconnaissance (0.027), and very little at mission effects (0.003).

The two rankings differ because an additive improvement is worth most where the adversary dwells longest: resource development lasts 18 days on average, so a small daily detection probability accumulates many chances. A proportional improvement is worth most where detection is already effective and the stage lies in the reattack loop. For HELIOS, detection at resource development is largely outside the operator's reach and depends on shared threat intelligence, so the operator's own best investments are at initial access and lateral movement, where authentication monitoring and segmentation apply.

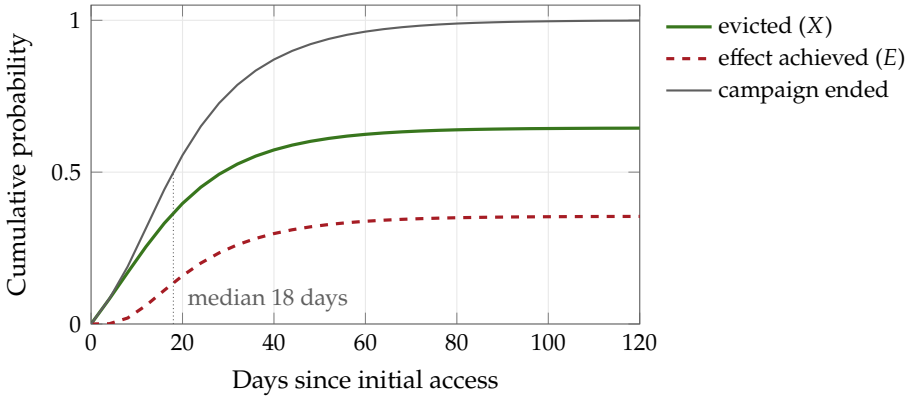


Figure 4.11. Cumulative probability that a HELIOS campaign starting at initial access has ended by eviction, by success, or either, as a function of dwell time. Half of campaigns end within 18 days, but the success curve keeps rising after most evictions have occurred, which is the tail that faster detection truncates.

Example 4.4 (Dwell-time distribution after initial access). The *dwell time* τ is the time from the adversary's first foothold (entry into stage 3) until the campaign ends in either E or X . For the HELIOS campaign, find its mean, spread, median, and the probability that an adversary is still active after 30 and 60 days.

Solution. Starting from stage 3, $t = \mathbf{N1}$ gives $\mathbb{E}[\tau] = 22.5$ days, and the second-moment identity $\mathbb{E}[\tau^2] = \mathbf{e}_3^\top (2\mathbf{N} - \mathbf{I})\mathbf{t}$ gives a standard deviation of 17.1 days. Propagating the distribution day by day, $\pi_{k+1} = \pi_k \mathbf{P}$, yields the cumulative absorption curves in Fig. 4.11: the median dwell time is 18 days and 90% of campaigns have ended by day 45. The probability that the adversary is still active is 0.241 at 30 days and 0.037 at 60 days. The campaign ends in eviction with probability 0.645 and in success with probability 0.355; conditional on each outcome, the mean dwell time is 20.3 days before eviction and 26.4 days before success.

Successful campaigns take longer than evicted ones, because success requires passing every stage and eviction can occur at each of them. A defender that shortens the time to detect at stages 3 to 5 therefore both raises the eviction probability and truncates the long right tail from which successful campaigns are drawn.

The Markov model has limits that motivate the rest of the chapter. Its transition probabilities are fixed, whereas an intelligent adversary chooses its path in response to the defenses it observes, shifting effort toward the stages the defender neglects. Its stages are a single chain, whereas real campaigns branch across alternative access paths whose success probabilities combine. And it measures probability, not consequence. The following sections address each limit in turn: adversary models and games for adaptivity, attack trees and graphs for branching, and loss distributions for consequence.

4.6 Adversary Models

Every quantitative security claim is a claim relative to an adversary model. A controller certified to reject disturbances of bounded energy, an estimator proven to tolerate two faulty sensors, and a consensus protocol proven correct with one malicious participant are each secure against a precisely stated class of adversaries and against no other class. Much confusion in the security of cyber-physical systems, space systems included, arises from importing a guarantee proved under one model into a setting governed by another. This section states the principal models formally, in a common notation, and pairs each with a space example.

We use a generic discrete-time system with state x_k , control input u_k , and measurement y_k ,

$$x_{k+1} = f(x_k, u_k) + w_k, \quad y_k = h(x_k) + v_k + a_k, \quad (4.8)$$

where w_k is a disturbance or actuation attack, v_k is sensor noise, and $a_k \in \mathbb{R}^p$ is an attack on the p measurements. The system may be a spacecraft's attitude loop, an orbit-determination filter fed by several ground stations, or a network of satellites exchanging time or routing information. An adversary model specifies the set of signals the adversary may choose and the information it may use to choose them.

4.6.1 Non-strategic and constrained adversaries

The *bounded* model allows every disturbance within a norm ball, $\|w_k\| \leq \bar{w}$ for all k , or, in energy form, $\sum_k \|w_k\|^2 \leq \bar{w}^2$. It is the model of robust and H_∞ control, whose guarantees hold against the worst admissible disturbance. Its virtue is that it requires no probability model; its weakness is that it treats a jammer's power limit, a thruster misalignment, and a malicious command the same way, and gives no special treatment to an attack that is small in norm but aimed at a structurally weak direction. A space example is the design of an attitude controller that must hold pointing against every disturbance torque below a bound that includes solar pressure, gravity gradient, and residual magnetic torques.

The *stochastic* model draws w_k from a known distribution independent of the defense, such as Gaussian noise or a Poisson process of single-event upsets. It supports expected-value reasoning, reliability analysis, and the Markov model of [Section 4.5](#). It is appropriate for natural threats and for the aggregate behavior of many unsophisticated actors, and inappropriate whenever the adversary chooses its actions after observing the system.

The *sparse* model allows the adversary to corrupt at most q of the p measurement channels, with arbitrary magnitude and with no assumption on the corrupted values:

$$\|a_k\|_0 \leq q \quad \text{for all } k, \quad \text{with a fixed but unknown support.} \quad (4.9)$$

Fawzi, Tabuada, and Diggavi showed that for linear systems the state can be reconstructed exactly despite such attacks if and only if the system remains observable after the removal of an arbitrary set of $2q$ sensors, and connected the resulting decoder to error correction and compressed sensing [123]. The model fits a constellation whose orbit determination fuses ranging from several ground

stations, some of which may be compromised or spoofed: the number of stations, not the size of the corruption, determines how many compromised stations can be tolerated.

The *resource-limited* model of Teixeira and colleagues characterizes an adversary by three resources: its *model knowledge* (how much of f and h it knows), its *disclosure resources* (which signals it can read), and its *disruption resources* (which signals it can modify), together with a budget such as $\sum_k \|a_k\|^2 \leq E$ [124]. Different points in this resource space yield denial-of-service, replay, bias-injection, and stealthy zero-dynamics attacks. The model makes explicit that a stealthy attack on a spacecraft control loop requires both model knowledge and access to specific channels, and it prices defenses by the resources they deny.

The *denial-of-service* or *switching* model allows the adversary to interrupt communication over a set of time intervals Ξ , constrained in frequency and duration. In the formulation of De Persis and Tesi, the number $n(\tau, t)$ of interruptions and their total length $|\Xi(\tau, t)|$ over every window $[\tau, t]$ satisfy

$$n(\tau, t) \leq \nu_0 + \frac{t - \tau}{\tau_D}, \quad |\Xi(\tau, t)| \leq \kappa + \frac{t - \tau}{T_D}, \quad (4.10)$$

and stability of the closed loop can be guaranteed when interruptions are sufficiently infrequent and short relative to the loop's dynamics [125]. The model fits intermittent jamming of a TT&C or ISL link over which a control or station-keeping loop is closed.

4.6.2 Byzantine, adaptive, and strategic adversaries

The *Byzantine* model allows up to f of n participants in a distributed protocol to behave arbitrarily, including sending contradictory information to different peers. Lamport, Shostak, and Pease showed that agreement with unsigned messages is achievable if and only if $n \geq 3f + 1$ [126]. In a constellation, the model applies to distributed time transfer, routing-state dissemination, and collaborative anomaly detection, where a compromised satellite or ground node can lie selectively; the bound quantifies how much redundancy a protocol needs.

An *adaptive* or *learning* adversary selects $a_k = \pi_k(\mathcal{H}_k)$ as a function of the history $\mathcal{H}_k$ of its own actions and observed outcomes, and updates the policy π_k over time, for example by a bandit or reinforcement-learning rule. A reactive jammer that learns a frequency-hopping pattern, or a campaign that probes backup paths after an initial effect, is adaptive. Adaptive adversaries break the stationarity assumption on which detection thresholds and Markov transition probabilities are calibrated.

A *strategic* adversary chooses its action to optimize an objective that depends on the defender's policy π :

$$a^* \in \arg \max_{a \in \mathcal{A}_{\text{adv}}} J(a, \pi), \quad (4.11)$$

where $\mathcal{A}_{\text{adv}}$ is the adversary's feasible action set, which may itself be bounded, sparse, or resource-limited, and J measures the adversary's gain, for instance the defender's loss of mission performance. When the adversary observes π before

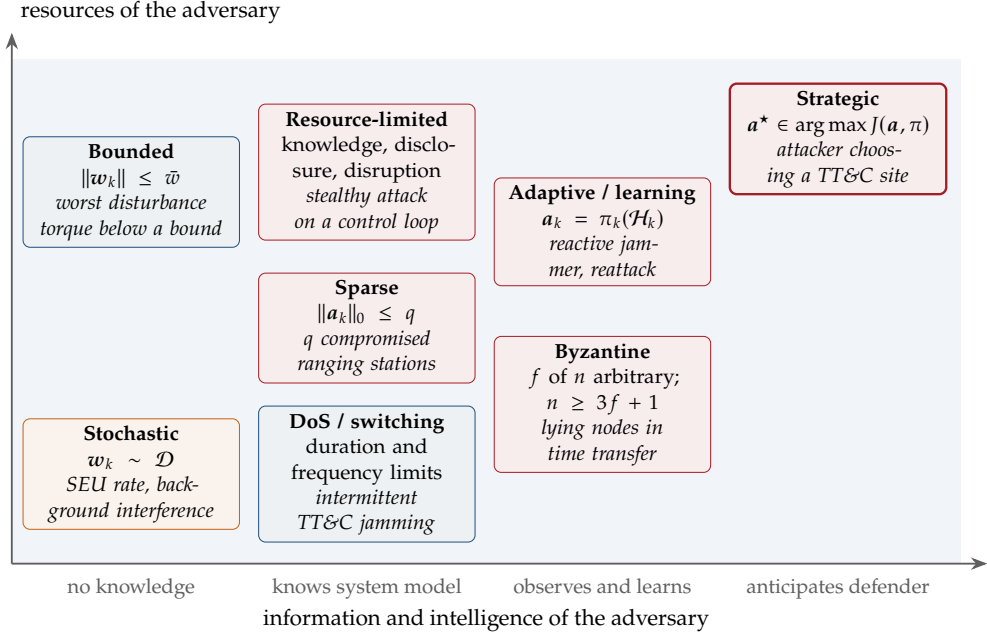


Figure 4.12. Adversary models arranged by the information and intelligence the adversary brings (horizontal) and the resources it commands (vertical), each with a space example. Orange marks the natural-threat model, blue the models of robust and resilient control, and red the models in which the adversary exploits system structure. Guarantees proved toward the lower left do not transfer toward the upper right.

acting, (4.11) is its best response in a leader-follower game; when both choose simultaneously, the relevant solution is an equilibrium. Figure 4.12 arranges the eight models by the information and intelligence the adversary brings and by the resources it commands.

4.6.3 Choosing a model

The models are not interchangeable, and the choice among them is itself a design decision. A controller that tolerates every bounded disturbance torque may fail against a sparse attack of arbitrary magnitude on one gyroscope; an anomaly detector tuned against natural telemetry faults says little about an adversary who knows the detector; and a protocol that tolerates random message loss may not tolerate one Byzantine participant. The ordering in Fig. 4.12 rests on a simple dominance property.

Proposition 4.2 (Worst-case dominance). Let $Q(a, \pi)$ be a performance measure of the defended system under adversary action $a \in \mathcal{A}_{\text{adv}}$ and defender policy π . For every probability distribution $\mathcal{D}$ supported on $\mathcal{A}_{\text{adv}}$,

$$\min_{a \in \mathcal{A}_{\text{adv}}} Q(a, \pi) \leq \mathbb{E}_{a \sim \mathcal{D}} [Q(a, \pi)]. \quad (4.12)$$

Proof. For every $a \in \mathcal{A}_{\text{adv}}$ we have $\min_{a'} Q(a', \pi) \leq Q(a, \pi)$; taking expectations

over $\mathcal{D}$ preserves the inequality. $\square$

A guarantee against a strategic adversary with action set $\mathcal{A}_{\text{adv}}$ therefore implies the same guarantee against every stochastic or non-adaptive adversary confined to that set, but not conversely. The price of the stronger guarantee is conservatism: designing against the worst case in a large set can be expensive or impossible. Practical analysis therefore pairs models with threats. Natural threats and the background of opportunistic activity are modeled stochastically. Physical and RF limits are expressed as bounds and budgets. Compromise of a subset of components is expressed as sparsity or Byzantine behavior. The deliberate choice of target and timing by a capable actor is modeled strategically. The next section develops the simplest strategic models in enough detail to compute a defense.

4.7 Strategic Adversaries and Game Models: A First Look

Probabilistic risk analysis estimates how often threat events occur and how much they cost. Against natural hazards this is exactly the right question, because the hazard does not respond to the defense. Against an intelligent adversary it can be the wrong question. If an operator estimates from past incidents that most intrusion attempts target its most exposed service and moves its monitoring there, a capable adversary observes the change and moves elsewhere. The historical frequency that justified the allocation is then no longer the frequency the operator faces. Risk estimates for strategic threats are endogenous: they depend on the defense they are meant to inform. Game theory is the mathematics of decisions whose outcomes depend on the choices of others, and it replaces the fixed attack distribution of probabilistic analysis with an adversary who optimizes against the defense. This section introduces the two simplest game models used in security; a systematic treatment of games is beyond the scope of the chapter.

4.7.1 Zero-sum matrix games

Definition 4.7 (Zero-sum matrix game). A two-player zero-sum matrix game is given by an $m \times n$ matrix A . The defender chooses a row $k \in \{1, \dots, m\}$, the attacker simultaneously chooses a column $j \in \{1, \dots, n\}$, and the defender loses (and the attacker gains) A_{kj} . A *mixed strategy* is a probability vector over rows, $x \in \Delta_m = \{x \geq 0 : \mathbf{1}^\top x = 1\}$, or over columns, $y \in \Delta_n$, and the expected loss is $x^\top A y$. A pair (x^*, y^*) is a *saddle point* (equilibrium) if neither player can improve its expected payoff by deviating unilaterally.

The central fact about such games is von Neumann's minimax theorem: every finite zero-sum game has a saddle point in mixed strategies, and

$$v = \min_{x \in \Delta_m} \max_{y \in \Delta_n} x^\top A y = \max_{y \in \Delta_n} \min_{x \in \Delta_m} x^\top A y. \quad (4.13)$$

The common value v is the *value* of the game. The defender's minimax strategy guarantees an expected loss of at most v whatever the attacker does, and it does not require the defender to predict the attacker's choice. It is computed by the

linear program

$$\min_{x, v} v \quad \text{subject to} \quad A^\top x \leq v\mathbf{1}, \quad \mathbf{1}^\top x = 1, \quad x \geq 0. \quad (4.14)$$

Randomization is essential. A deterministic defense is predictable, and a predictable defense can be avoided; the mixed strategy makes the defender's allocation uncertain to the attacker in exactly the proportions that leave the attacker no profitable target.

4.7.2 Stackelberg security games

Two features of real security problems are not captured by the zero-sum matrix game. First, defenses are usually deployed before attacks and can be observed, at least statistically, by a patient adversary who conducts reconnaissance; the defender *commits* and the attacker *responds*. Second, the attacker's gains need not equal the defender's losses: an attacker may value a target for its symbolic or intelligence value, or may be deterred from a target by the risk of attribution, independently of the damage it would do. The Stackelberg security game, developed and deployed in physical-security applications by Tambe and colleagues [127, 128], captures both.

Definition 4.8 (Stackelberg security game). There are n targets. The defender chooses a *coverage* vector $x \in [0, 1]^n$ with $\mathbf{1}^\top x \leq m$, where x_j is the probability that target j is protected by one of m defensive resources. The attacker observes x and attacks one target. If target j is attacked, the expected utilities are

$$U_d(j, x) = x_j U_d^c(j) + (1 - x_j) U_d^u(j), \quad U_a(j, x) = x_j U_a^c(j) + (1 - x_j) U_a^u(j), \quad (4.15)$$

where superscripts c and u denote covered and uncovered, with $U_d^c(j) \geq U_d^u(j)$ and $U_a^c(j) \leq U_a^u(j)$. A *strong Stackelberg equilibrium* is a coverage x^* and attacker response j^* such that j^* maximizes $U_a(\cdot, x^*)$, ties being broken in the defender's favor, and x^* maximizes $U_d(j^*(x), x)$ over all feasible coverages.

The equilibrium can be computed by solving one linear program per target. For each candidate target j , the defender maximizes $U_d(j, x)$ over feasible x subject to $U_a(j, x) \geq U_a(j', x)$ for all $j' \neq j$; the best of the n solutions is the equilibrium. In a zero-sum security game, where $U_a = -U_d$ target by target, the Stackelberg coverage coincides with the minimax coverage, so commitment neither helps nor hurts. In general-sum games the leader's payoff from optimal commitment is at least its payoff in every simultaneous-move equilibrium [129], and it can be strictly greater, because the leader can exploit what it knows about the follower's preferences.

Example 4.5 (Monitoring the HELIOS TT&C sites). HELIOS commands its satellites through two owned high-latitude TT&C sites, O1 and O2, and through six sites leased from a GSaaS provider and reached through the provider's customer portal, which we treat as a single target G. The security operations center has one team capable of intensive monitoring (threat hunting, session review, and rapid credential revocation) and must decide how to divide that team's attention,

$x = (x_{O1}, x_{O2}, x_G)$ with $x_{O1} + x_{O2} + x_G = 1$. As illustrative assumptions, a compromise of an owned site is worth a loss of 10 units to HELIOS and succeeds with probability 0.20 if unmonitored and 0.05 if monitored; a compromise via the GSaaS portal is worth 6 units (commands still pass SDLS authentication, but contacts and schedules can be disrupted) and succeeds with probability 0.50 unmonitored and 0.15 monitored. Find the defender's minimax allocation, and then its Stackelberg allocation when the attacker's preferences differ from the defender's losses.

Solution. The expected losses when a target is attacked are 2.0, 2.0, and 3.0 unmonitored and 0.5, 0.5, and 0.9 monitored. In the zero-sum game in which the defender monitors one target and the attacker attacks one, the rows of A are (0.5, 2, 3), (2, 0.5, 3), and (2, 2, 0.9). Solving (4.14) gives $x^* = (11/57, 11/57, 35/57) \approx (0.193, 0.193, 0.614)$ and the value $v = 1.711$. At x^* an attack on each target yields the same expected loss, and the attacker's equilibrium mix is $y^* = (7/19, 7/19, 5/19)$. Naive allocations do worse: monitoring only the most exposed target G invites an attack on an owned site with loss 2.0, and an even split invites an attack on G with loss 2.3.

Suppose now that intelligence assessments indicate that the adversary values the owned sites at 8 units and the GSaaS path at 6, and regards an attack on an owned site as costing an additional 0.2 units in attribution risk. The attacker's utilities are (1.4, 1.4, 3.0) unmonitored and (0.2, 0.2, 0.9) monitored. Solving the three linear programs of Definition 4.8 gives the Stackelberg coverage $x^S = (5/54, 5/54, 22/27) \approx (0.093, 0.093, 0.815)$, at which the attacker is indifferent among targets with utility 1.289 and, breaking the tie in the defender's favor, attacks G. The defender's expected loss is 1.289. Had the defender used the zero-sum allocation x^* against this attacker, the attacker would attack G for a loss of 1.711. The two cases are plotted in Fig. 4.13.

Randomized monitoring that leaves the adversary indifferent outperforms allocations driven by exposure alone; and knowledge of the adversary's preferences, here its aversion to attribution at owned sites, lets the defender concentrate coverage and cut expected loss by a quarter. The Stackelberg allocation is fragile to tie-breaking, so in practice the defender shifts a small margin of coverage toward the owned sites.

Single-shot games abstract away the time structure that the kill chain makes explicit. Advanced persistent threats (APTs) unfold over many stages, and at each stage the defender acts on incomplete information about whether an intrusion is under way and what kind of adversary is behind it. Zhu and Rass model such campaigns as sequences of games, one per phase, whose outcomes feed the next [130], and Huang and Zhu formulate proactive APT defense in cyber-physical systems as a dynamic game of incomplete information in which the defender updates its belief about the attacker's type from observed behavior, as the Bayesian network of Section 4.9 does for a single alert [131]. Rass and Zhu use the same sequential structure to design layered defenses that remain effective as the campaign advances [99]. These multi-stage models connect the stochastic kill chain of Section 4.5, in which transition probabilities are fixed, with

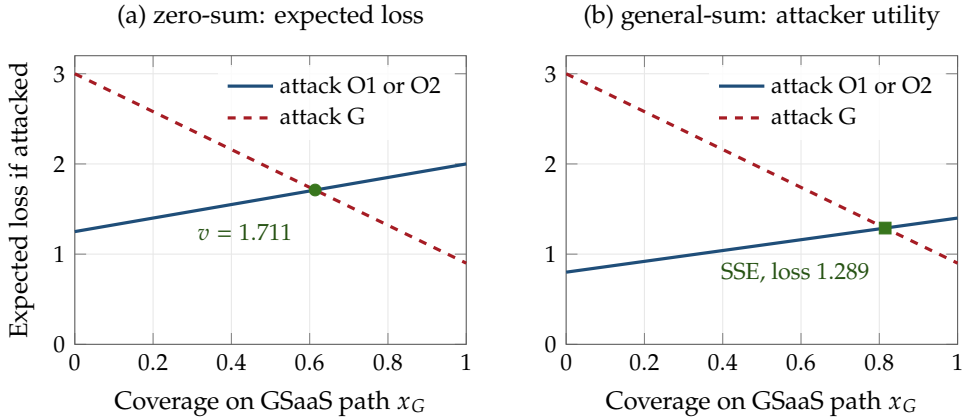


Figure 4.13. The HELIOS TT&C monitoring game with the remaining coverage split equally between the owned sites. (a) In the zero-sum game the minimax coverage equalizes the expected loss across targets. (b) With the attacker’s own preferences, the Stackelberg coverage equalizes the attacker’s utility instead, at a higher coverage of the GSaaS path and a lower defender loss.

the strategic models of this section, in which they are chosen. [Chapter B](#) develops the underlying game theory in full.

4.8 Attack Trees and Attack Graphs

The kill chain models a campaign as a single sequence of stages. Real adversaries have choices: an intruder seeking command authority over a HELIOS satellite might steal a portal credential, subvert a software update, or inject a command over RF, and would pursue whichever path is cheapest. Attack trees and attack graphs represent these alternatives explicitly and let the defender reason about the combination of controls that closes them. They are the workhorse models of structured threat analysis and appear throughout the space-security literature, from CubeSat threat models to mission-centric cyber assessments [50].

4.8.1 Attack trees and probability propagation

Definition 4.9 (Attack tree). An *attack tree* is a rooted tree whose root is an attacker goal and whose internal nodes are refinements of that goal into subgoals. An OR node is achieved when *at least one* of its children is achieved; an AND node is achieved when *all* of its children are achieved. Leaves are elementary attack steps. If each leaf v is assigned a success probability p_v and leaf successes are independent, the success probability of a node propagates upward by

$$p_{\text{AND}} = \prod_{c \in \text{children}} p_c, \quad p_{\text{OR}} = 1 - \prod_{c \in \text{children}} (1 - p_c). \quad (4.16)$$

Attack trees were introduced by Schneier as a systematic way to enumerate how a goal can be reached [132] and given formal semantics by Mauw and Oostdijk [133]. The independence assumption in (4.16) is a modeling choice, not a fact:

shared credentials, shared software, or a common operator couple leaves whose compromise is then correlated, and the propagation formula bounds rather than equals the true probability. Other attributes propagate through the same tree with different combinators: minimum cost of attack uses sum at AND nodes and minimum at OR nodes, and the least-skill or shortest-time attributes propagate similarly. The tree thus answers not only how likely the goal is but which single path an optimizing attacker would take.

Definition 4.10 (*Attack-defense tree*). An *attack-defense tree* augments an attack tree with *defense* nodes attached to attack nodes. A countermeasure attached to a leaf multiplies the leaf's success probability by a residual factor, or, in the binary case, sets it to a small value if the control is present. Attack and defense nodes may alternate, so that an attacker step defeats a defense that itself defends against a lower step.

Attack-defense trees, formalized by Kordy and colleagues [134], let the analyst evaluate the root probability before and after a set of controls and so rank controls by the risk reduction they deliver, which is the calculation that connects a threat model to a security investment.

Example 4.6 (*Attack-defense tree for an unauthorized HELIOS command*). The attacker's goal is to have a HELIOS satellite accept and act on an unauthorized telecommand within a year. Figure 4.14 models three disjoint access paths as an OR decomposition: a ground intrusion into the command chain, a subverted software update, and direct RF command injection. The ground path is an AND of obtaining ground access and defeating command authorization; ground access is itself an OR over a stolen GSaaS credential, a stolen MOC cloud credential, and an insider. The leaf probabilities in Table 4.4 are annual assumptions for a capable adversary; the residual factors show the effect of five controls: multi-factor authentication (MFA) on portals, privileged-access management (PAM) on the cloud console, command allow-listing with two-person review, code signing with independent build review, and CCSDS SDLS authentication of telecommands.

Propagating (4.16) with no controls gives a root success probability of 0.366: the RF path alone contributes 0.30, because an unauthenticated uplink accepts a forged command once the attacker can transmit with the right geometry. Enabling SDLS reduces the RF path from 0.30 to 0.0003 and the root to 0.094, the single most effective control because it closes a whole path. Adding the four ground and supply-chain controls reduces the root to 0.015. Removing SDLS from the fully controlled configuration returns the root to 0.31, confirming that authenticated commanding is load-bearing: without it, the other controls address paths that the attacker would not need to use.

The calculation ranks controls by marginal risk reduction rather than by cost or coverage, and it exposes a structural point that a checklist would miss: because the paths are combined by OR, the root probability is dominated by whichever path the controls leave open, so a defense is only as strong as its weakest path.

Table 4.4. Leaf probabilities and control residuals for the attack-defense tree of Fig. 4.14 (annual, assumptions). “With control” is the leaf probability when the named control is present.

Leaf	Elementary step	No control	With control	Control
g_1	Steal GSaaS portal credential	0.15	0.04	MFA
g_2	Steal MOC cloud credential	0.10	0.04	PAM
g_3	Insider provides access	0.02	0.02	none
a	Defeat command authorization	0.30	0.10	Allow-list, two-person
s_1	Compromise CI/CD pipeline	0.05	0.05	none
s_2	Malicious build passes review	0.40	0.10	Code signing, review
r_1	Obtain uplink and geometry	0.30	0.30	none
r_2	Command accepted without MAC	1.00	0.001	SDLS authentication

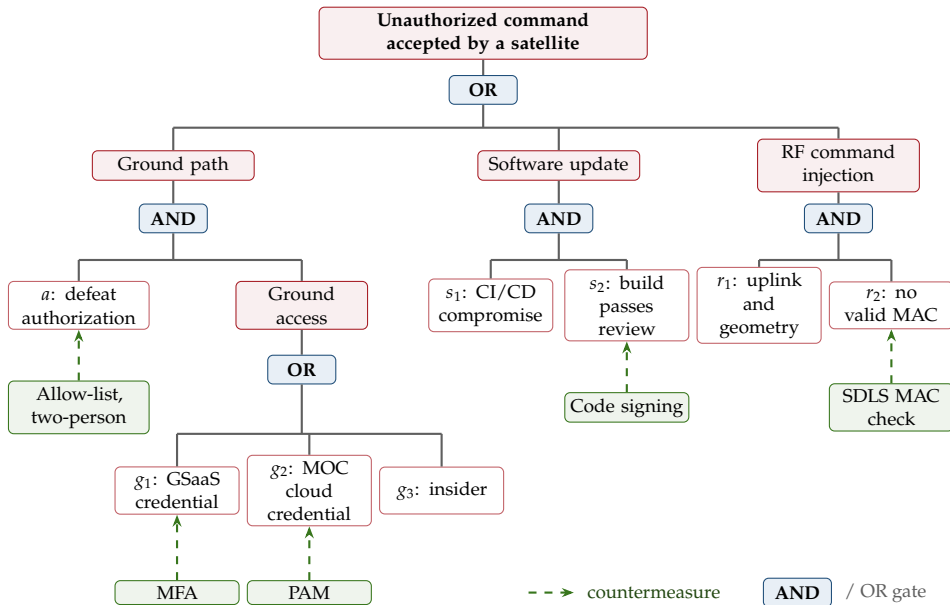


Figure 4.14. Attack-defense tree for acceptance of an unauthorized telecommand by a HELIOS satellite. Three access paths (ground, software supply, RF) are combined by an or gate; each is an AND of its preconditions. Green dashed edges are the five countermeasures of Table 4.4. The root probability falls from 0.366 with no controls to 0.094 with SDLS alone and 0.015 with all five.

4.8.2 Attack graphs and hardening

An attack tree fixes the goal and enumerates ways to reach it. An *attack graph* instead takes a model of a network (hosts, services, connectivity, and known vulnerabilities) and computes the reachable set of adversary states, so that new paths appear automatically as the model changes. Tools such as MulVAL derive attack graphs from vulnerability scans and network configuration by logical inference [135], and the formal generation of attack graphs from a system model goes back to Sheyner and colleagues [136].

Definition 4.11 (Attack graph). An *attack graph* is a directed graph $G = (V, \mathcal{E})$ whose vertices are adversary states (for example, a privilege held on a host) and whose edges are atomic exploits, each enabled by a precondition and establishing a postcondition. A weight $c(e) \geq 0$ on edge e measures the effort, cost, or negative log-probability of the exploit. An *attack path* from an entry state to a goal state is a directed path, and its weight is the sum of its edge weights.

Two graph computations guide hardening. The *shortest path* from an entry to the goal is the easiest route the adversary can take; if edge weights are $c(e) = -\log p_e$, the shortest path is the most probable single path, and minimizing it lower-bounds the defender's task. The *minimum cut* is the cheapest set of edges whose removal disconnects every entry from the goal; by the max-flow min-cut theorem it identifies the smallest set of controls that blocks all paths, which is the graph analogue of the observation in Example 4.6 that an or of paths is only as strong as its weakest path. Hardening then becomes an interdiction problem: choose a set of edges to remove, within a budget, to maximize the shortest remaining path or to raise the min-cut. For the HELIOS command example, the RF edge carries near-unit probability without SDLS, so it lies on the min cut; enabling SDLS removes it and forces the adversary onto the longer ground and supply paths, exactly the shift the tree calculation quantified. Attack-graph interdiction and its game-theoretic extensions recur throughout the later chapters on network and cross-layer security.

Attack trees, attack graphs, and the kill chain are complementary views of the same campaign. The kill chain orders the campaign in time; the attack tree decomposes one goal into alternative and joint preconditions; the attack graph derives reachable paths from a system model. All three feed the same downstream question, which is how much loss the residual risk implies, and that is the subject of the next section.

4.9 Probabilistic Risk Quantification

The models so far produce probabilities of events: a campaign succeeds, a command is accepted, a target is attacked. A risk decision needs more than a probability; it needs a distribution of consequences against which to size defenses and, for a commercial operator, against which to buy or forgo insurance. This section develops two probabilistic tools that turn event probabilities into decision-relevant quantities: Bayesian networks, which fuse uncertain evidence into a posterior belief, and Monte Carlo loss simulation, which produces a

loss-exceedance curve.

4.9.1 Bayesian networks for evidence fusion

The opening vignette poses a diagnostic problem: given an anomalous-login alert and two rejected telecommands, how likely is a genuine compromise? A Bayesian network answers such questions by encoding conditional dependences among uncertain variables and computing posteriors as evidence arrives.

Definition 4.12 (Bayesian network). A *Bayesian network* is a directed acyclic graph whose nodes are random variables and whose edges encode conditional dependence, together with a conditional probability table for each node given its parents. The joint distribution factorizes as $\mathbb{P}(X_1, \dots, X_n) = \prod_i \mathbb{P}(X_i \mid \text{pa}(X_i))$, where $\text{pa}(X_i)$ is the set of parents of X_i . Given evidence on a subset of variables, the posterior over each other variable is obtained by conditioning on that evidence.

Figure 4.15 models the vignette with five binary variables: a targeted *campaign* C is active this month; the GSaaS *credential* K is compromised; an *alert* A fires on the portal; an unauthorized *command* U is attempted; and a *telemetry* anomaly T (rejected-command counters) is observed. The structure encodes that a compromised credential is the common cause of both the login alert and the unauthorized command, and that the telemetry anomaly is a consequence of the command attempt. With the conditional probabilities shown, the prior probability that the credential is compromised in a given month is 0.039. The alert alone raises the posterior $\mathbb{P}(K \mid A)$ to 0.55. The alert together with the telemetry anomaly raises it to $\mathbb{P}(K \mid A, T) = 0.88$, and the posterior that a campaign is active is $\mathbb{P}(C \mid A, T) = 0.69$. An alert with normal telemetry, by contrast, leaves $\mathbb{P}(K \mid A, \neg T) = 0.47$. The two weak signals of the vignette combine into a strong one, which is precisely the inference the duty operator must make, and the network makes explicit which combination of observations warrants an aggressive response.

4.9.2 Loss-exceedance curves from Monte Carlo simulation

A posterior probability of compromise is an input to a risk decision, not the decision itself. To size defenses and to compare their value, the operator needs the distribution of annual loss. The FAIR approach to information risk factors each risk scenario into a *frequency* of loss events and a *magnitude* per event, and aggregates scenarios by simulation [137, 138]. The result is a loss-exceedance curve.

Definition 4.13 (Loss-exceedance curve). Let the annual loss be the sum $L = \sum_s \sum_{i=1}^{N_s} X_{s,i}$, where scenario s produces $N_s \sim \text{Poisson}(\Lambda_s)$ events per year, each of severity $X_{s,i}$ from a distribution F_s , independently across scenarios and events. The *loss-exceedance curve* is the complementary cumulative distribution

$$\bar{F}_L(\ell) = \mathbb{P}(L > \ell), \quad (4.17)$$

and the *annualized loss expectancy* is $\text{ALE} = \mathbb{E}[L] = \sum_s \Lambda_s \mathbb{E}[X_s]$.

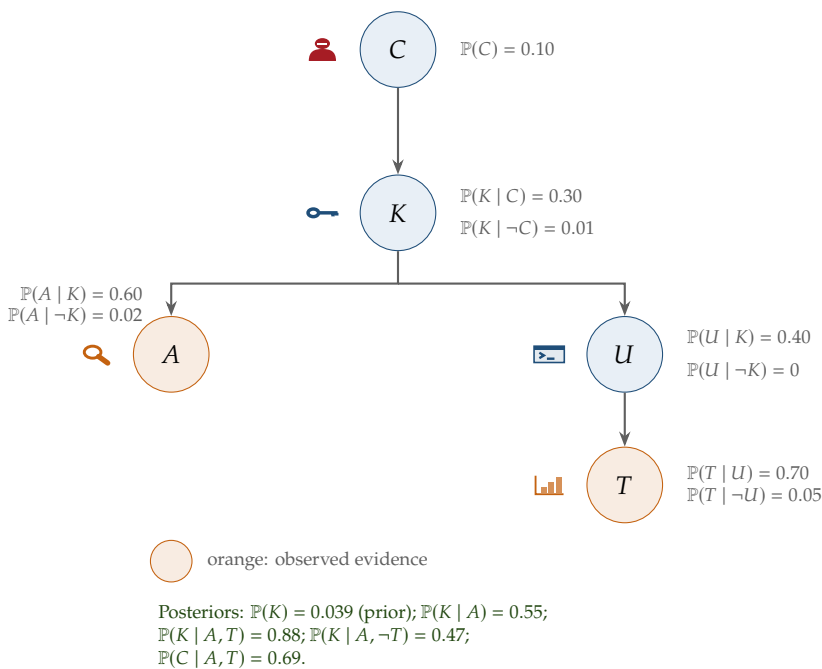


Figure 4.15. Bayesian network for the opening vignette. A compromised credential K is the common cause of the login alert A and the unauthorized command U , and the telemetry anomaly T follows from U . Observing A and T together (orange) raises the posterior probability of compromise to 0.88, whereas the alert alone gives 0.55.

Table 4.5. Loss scenarios for the HELIOS loss-exceedance example (assumptions). Frequency Λ_s is events per year; severity is lognormal with the stated median (\$M) and log-standard-deviation ς .

Scenario	Baseline			Hardened		
	Λ_s	median	ς	Λ_s	median	ς
RF interference outage	4.0	0.15	1.0	3.0	0.10	1.0
Ground intrusion, ops disruption	0.30	4.0	1.2	0.12	2.5	1.2
Ransomware on enterprise IT	0.50	1.5	1.0	0.25	1.0	1.0
Malicious command, satellite loss	0.03	20.0	1.0	0.005	15.0	1.0
Compromised firmware update	0.05	15.0	1.3	0.02	8.0	1.3

The frequency-severity split matches the threat structure of this chapter: reversible, deniable threats (Section 4.1) are frequent with modest severity, whereas destructive threats are rare with heavy tails. When severities are lognormal, the aggregate has no closed form, and Monte Carlo simulation is the standard method: draw N_s for each scenario, draw that many severities, sum, and repeat over many simulated years to estimate (4.17) and its quantiles.

Example 4.7 (HELIOS loss-exceedance with and without controls). Table 4.5 defines five loss scenarios for HELIOS with illustrative annual frequencies and lognormal severities in millions of U.S. dollars, in a baseline (current controls) and a hardened configuration that adds the controls of Example 4.6 and improved monitoring. Estimate the loss-exceedance curve, the annualized loss expectancy, and the 99th-percentile annual loss in each configuration.

Solution. Simulating 4×10^5 years for each configuration gives the curves in Fig. 4.16. The annualized loss expectancy falls from \$7.4M to \$2.0M, in agreement with the closed-form $\sum_s \Lambda_s \mathbb{E}[X_s]$. The probability of a year with more than \$10M of loss falls from 0.16 to 0.034, and the 99th-percentile annual loss from \$84M to \$24M. The tail is dominated by the two rare, high-severity scenarios (satellite loss and a bad firmware update), so the controls that most reduce the tail are those that lower the frequency of destructive commands and firmware compromise, which are the same SDLS and code-signing controls that dominated the attack-defense tree.

The expected loss is reduced by about a factor of four, but the tail is reduced by more, so the value of the controls is understated by the expected-loss figure alone. An operator comparing the cost of the controls with the reduction in the 99th-percentile loss makes a risk-transfer decision, such as how much cyber insurance to carry, that the expected value cannot inform.

The loss-exceedance curve is the quantitative bridge from the threat models of this chapter to the resilience metrics of the book, in which the same losses are expressed as area lost under the mission performance function $Q(t)$. What the probabilistic view does not capture is how an unsafe interaction between correctly functioning components can produce loss without a single component being compromised. That gap is the subject of the next section.

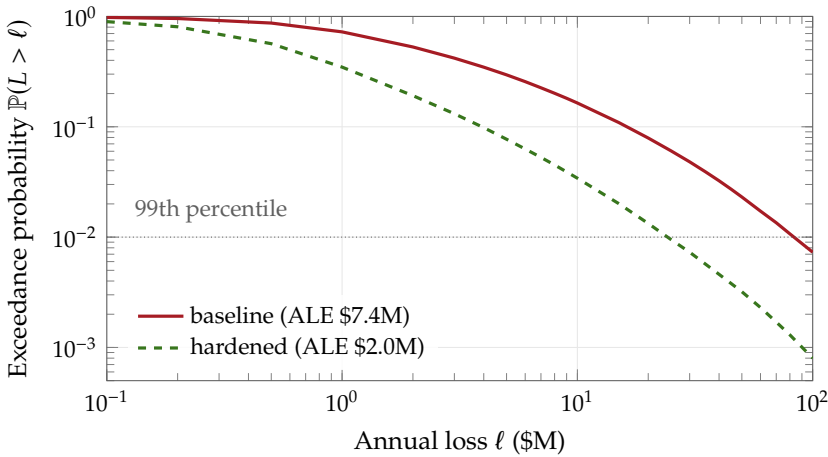


Figure 4.16. Loss-exceedance curves for HELIOS with baseline and hardened controls, from Monte Carlo simulation of the scenarios in Table 4.5. Both axes are logarithmic. The hardened configuration lowers the whole curve, and lowers the tail (the region above the dotted 99th-percentile line) proportionally more than the body, because it targets the rare destructive scenarios.

4.10 STPA-Sec: A Control-Theoretic View of Security

Attack trees, attack graphs, and the kill chain all begin from the adversary: they enumerate threats, paths, or stages, and ask how a component is compromised. This is powerful when the loss is caused by a compromised component, but it can miss losses that arise from unsafe interactions among components that are each behaving as designed. A satellite placed in a maintenance mode that disables command authentication, a ground procedure that uploads a maneuver based on a state estimate corrupted by a spoofed signal, or a race between two controllers issuing conflicting commands can all produce loss without a single component being breached. System-Theoretic Process Analysis for Security (STPA-Sec) analyzes such losses by modeling the system as a hierarchical control structure and asking how control actions can be unsafe or insecure [139]. It extends STPA, itself built on the systems-theoretic accident model of Leveson [140], from safety to security by treating an adversary as one cause among several of a hazardous control action.

Definition 4.14 (Control structure and unsafe/insecure control action). A *control structure* is a hierarchy of controllers connected to controlled processes by *control actions* (downward) and *feedback* (upward). Each controller acts on a *process model*, its belief about the state of the process below it. A *control action* is *unsafe* or *insecure* (a UCA) if, in some context, issuing it, not issuing it, issuing it too early or too late, or stopping it too soon leads to a hazard. A UCA arises whenever the controller’s process model diverges from the true process state, whether through a natural fault, a design flaw, or an adversary who manipulates feedback or the process.

Figure 4.17 draws a control structure for HELIOS spacecraft operations, from

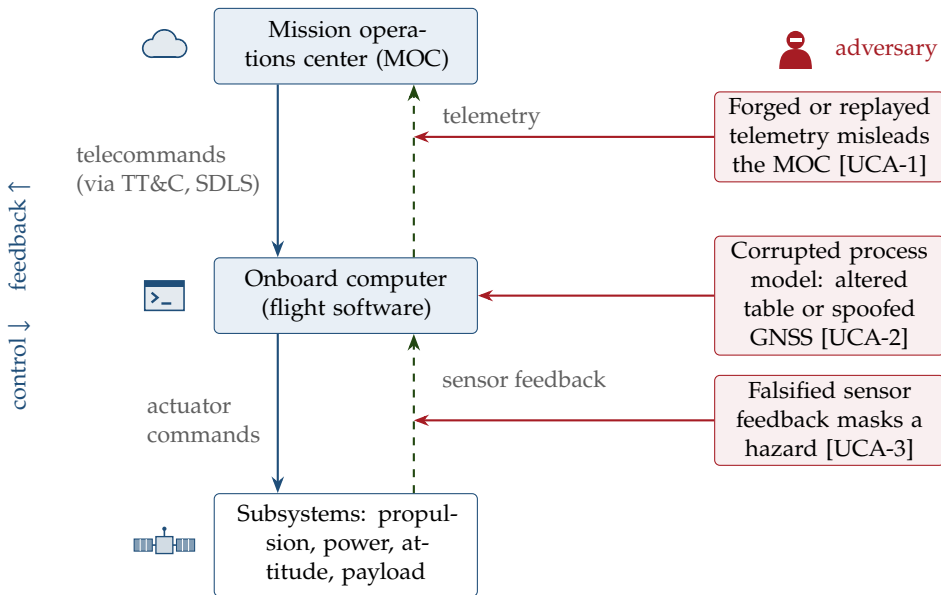


Figure 4.17. STPA-Sec control structure for HELIOS spacecraft operations. Solid blue arrows are control actions, dashed green arrows are feedback, and red arrows are adversary manipulations of the loop. Each manipulation makes a controller act on a wrong process model, producing an unsafe or unsecure control action (UCA) even though the controller itself is not compromised.

the mission operations center through the TT&C link to the onboard computer and the subsystems it controls. Reading the diagram as a control problem, rather than as a network to be defended, directs attention to the feedback paths: the MOC commands on the basis of telemetry, and the onboard computer acts on the basis of sensor data and stored tables. An adversary who corrupts feedback, whether by spoofing a sensor, replaying old telemetry, or altering an onboard table, causes a correctly functioning controller to issue a hazardous control action. This is the cross-layer pathway of [Section 4.1](#) seen from the control side: the entry is at the signal or cyber layer, but the loss is a physical action taken by an uncompromised controller.

The analysis proceeds by listing, for each control action, the four ways it can be unsafe, and then asking what could cause each. [Table 4.6](#) does this for the “execute maneuver” and “authenticate command” actions. The value of the table is that it generates security requirements that a threat-first analysis can miss. The requirement that the onboard computer reject a maneuver command whose resulting trajectory violates a stored keep-out constraint, for example, defends against a forged command, a corrupted table, and an operator error alike, because it constrains the control action by its effect rather than by its origin. STPA-Sec thus complements the adversary-centric models: the attack tree finds the paths an adversary takes to issue a command, and STPA-Sec finds the commands that must not be executed, whatever their origin.

Table 4.6. Selected unsafe or insecure control actions (UCAs) for two HELIOS control actions, with a security requirement addressing each. Contexts and requirements are illustrative.

Control action	Unsafe/unsecure in context	Security requirement
Execute maneuver	Executed when it violates a keep-out or de-orbit constraint (forged command or corrupted ephemeris)	Onboard check rejects every maneuver whose predicted trajectory violates a stored constraint, independent of command origin
Execute maneuver	Not executed when a validated collision-avoidance maneuver is due (TT&C denial)	Autonomous safe maneuver if a validated conjunction is unresolved before a deadline
Authenticate command	Provided (accepted) for a command lacking a valid SDLS tag (authentication disabled in a maintenance mode)	Maintenance modes retain command authentication; mode changes are themselves authenticated and logged
Authenticate command	Provided for a replayed command with a stale counter	Monotonic anti-replay counter enforced across mode changes and resets

4.11 Risk Frameworks

The models of this chapter are analytical instruments. In practice they are used within a governance framework that fixes the process (identify, analyze, treat, and monitor risk), the vocabulary, and the division of responsibility. Several such frameworks apply to space systems, and their relationship is a frequent source of confusion because they operate at different levels. This section places the principal frameworks on a common map so that a HELIOS security program can use each where it fits, rather than treating them as competitors.

At the most general level, ISO/IEC 27005 gives a domain-independent process for managing information-security risk: establish context, identify and analyze risk, evaluate it against criteria, treat it by mitigation, transfer, avoidance, or acceptance, and monitor the residual [141]. It says how to run a risk process but not what a space system’s risks are. The NIST Cybersecurity Framework (CSF) 2.0 organizes security activities into six functions, Govern, Identify, Protect, Detect, Respond, and Recover, and provides a common language for describing a program’s posture and target profile [121]. The kill chain of Section 4.4 maps onto four of these: the Protect and Detect functions supply the controls that interrupt stages 1 to 5, Respond covers mission effects and the defender’s stage 8 begins there, and Recover completes it.

At the space-specific level, two NIST interagency reports translate the general guidance to space. NIST IR 8270 introduces cybersecurity for commercial satellite operations, describing the ground, space, link, and user segments and the threats

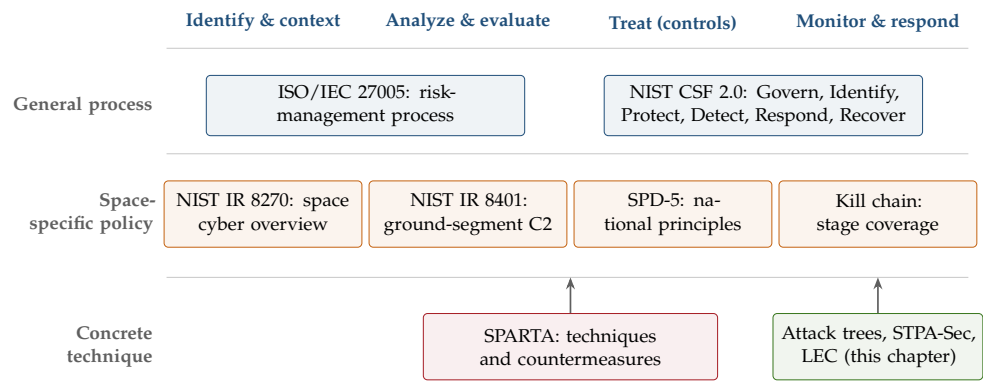


Figure 4.18. The principal risk frameworks arranged by abstraction level (rows) and by the risk-management step they principally serve (columns). General processes (blue) span the whole cycle; space-specific policy (orange) scopes assets, principles, and stage coverage; concrete techniques (red and green) supply countermeasures and the quantitative models of this chapter. Each framework occupies a distinct place, and a complete program uses all of them.

to each [35]. NIST IR 8401 applies the CSF specifically to the satellite ground segment, mapping command-and-control functions to CSF categories [36]. Space Policy Directive 5 states national-level principles for space-system cybersecurity, including protection of command chains, supply-chain risk management, and the physical protection of ground systems [34]. These documents say what to protect and to what principles, but they are not catalogs of concrete techniques and countermeasures. That role is filled by SPARTA, which enumerates space-specific adversary techniques and, for each, candidate countermeasures [31], and which therefore connects the kill chain and attack trees of this chapter to specific controls.

Figure 4.18 arranges these frameworks on two axes: their level of abstraction, from general process to concrete technique, and the risk-management step they principally serve. The figure makes the division of labor explicit. A HELIOS risk program would run the ISO/IEC 27005 or CSF process, use NIST IR 8270 and IR 8401 to scope space-specific assets and controls, apply SPD-5 principles as constraints, and draw concrete techniques and countermeasures from SPARTA, checking coverage against the kill chain. No single framework does all of this, and using one where another belongs, for instance treating SPARTA as a risk process or CSF as a control catalog, produces gaps that the mapping is meant to prevent.

4.12 Two Case Studies

The models of this chapter are illustrated by two documented episodes at opposite ends of the adversarial spectrum: a hostile intrusion into a satellite-data network, and an authorized on-orbit hacking competition. Each is drawn from public reporting, and contested facts are hedged.

Space Security in Practice: The 2014 NOAA Satellite-Data Network Intrusion

In late September 2014 an intrusion compromised computer systems of the U.S. National Oceanic and Atmospheric Administration (NOAA), including systems supporting the ingest and dissemination of satellite weather data. According to reporting in *The Washington Post*, the intrusion reached NOAA through internet-facing web systems, and NOAA responded in October by sealing off some services, which it publicly described at the time as unscheduled maintenance; NOAA confirmed weeks later that it had been the target of an attack [142]. Public attribution to a specific actor was made by a member of Congress rather than by NOAA, and should be treated with caution.

Placed on the kill chain, the episode is an initial-access and lateral-movement campaign in the ground segment: the entry was a public-facing web application (initial access), the effect was a disruption of satellite-data dissemination to downstream users (mission effect), and the satellites themselves were not involved. It illustrates two themes of the chapter. First, the entry layer and effect layer were both in the ground segment, and a satellite operator's most exposed surface is often its terrestrial IT rather than its spacecraft. Second, delayed disclosure lengthened the assessment and recovery stages. A later U.S. Department of Commerce Office of Inspector General audit found longstanding weaknesses in NOAA's information-security program and reported that they prolonged the disruption of satellite-data dissemination [143]. In the loss-exceedance language of [Section 4.9.2](#), the incident is a ground-intrusion scenario whose severity was driven by the downstream dependence of weather-data users, not by physical damage.

Space Security in Practice: Moonlighter and the 2023 Hack-A-Sat Competition

Not every adversarial test is hostile. The Hack-A-Sat competition, run by the U.S. Air Force and Space Force, invites teams to find and exploit vulnerabilities in representative space systems under controlled rules, so that weaknesses are discovered by defenders before adversaries exploit them. For the 2023 event, The Aerospace Corporation built *Moonlighter*, a small satellite designed as a purpose-built, on-orbit cyber test range. Public reporting states that Moonlighter, a three-unit CubeSat, was launched to the International Space Station on a resupply mission in June 2023 and deployed into orbit the following month, and that the competition final was held in August 2023 at the DEF CON conference, where finalist teams attempted authorized exploitation of the on-orbit spacecraft; an Italian team was reported as the winner [144, 145].

The exercise maps onto the researcher tier of [Section 4.2](#) and onto the recovery-and-adaptation stage of the kill chain: authorized red teams reveal exposure so that operators can remove it, feeding the anticipation of the next cycle. It also exemplifies a principle that recurs in the security-research literature on satellites [84, 50]: on-orbit and hardware-in-the-loop testing surfaces vulnerabilities that paper analysis and ground emulation miss, because the real system includes timing, radio, and operational constraints that a model omits. For an operator such as HELIOS, the lesson is that a purpose-built test capability and an authorized adversarial program are themselves controls, acting at the reconnaissance and resource-development stages by closing the exposures a hostile actor would otherwise find first.

4.13 Summary

This chapter assembled the vocabulary and the models with which the rest of the book reasons about threats and risk. A threat was characterized not only by its mechanism, drawn from the seven classes of electronic warfare, cyber, directed energy, kinetic, nuclear, supply-chain and insider, and natural hazards, but by a profile of entry layer, effect layers, reversibility, attributability, and debris generation. That profile explained why reversible, deniable, debris-free threats dominate the frequency of adversarial activity while destructive threats dominate its consequences, and why security is a property of the whole architecture: mission effects require a path from an entry point to a critical function, and those paths cross the seams between segments and between the specialist communities that defend them.

The adversary was then followed through time along the eight-stage space kill chain, from reconnaissance to the defender's recovery, and mapped onto the MITRE ATT&CK, SPARTA, and SPACE-SHIELD catalogs. Treating the chain as an absorbing Markov chain turned qualitative defense-in-depth into calculation: the fundamental matrix yields campaign success probabilities, dwell-time distributions, and the sensitivity of risk to detection at each stage, and it showed that moderate detection spread across every stage beats strong detection at one, that additive detection pays most where the adversary dwells longest, and that faster detection at the middle stages truncates the long tail from which successful campaigns are drawn. Because a fixed transition model cannot represent an adversary who adapts, the chapter formalized eight adversary models, from bounded and stochastic to Byzantine and strategic, and established that a guarantee proved against a strategic adversary dominates one proved against a stochastic adversary in the same action set but not conversely. Game models made the strategic case computable: a small zero-sum game and its Stackelberg refinement showed that randomized monitoring that leaves the adversary indifferent outperforms allocation by exposure, and that knowledge of the adversary's preferences lets the defender concentrate coverage.

The quantitative risk toolkit followed. Attack-defense trees decomposed a goal into alternative and joint preconditions and ranked controls by the risk

they remove, showing that an OR of paths is only as strong as its weakest path and that authenticated commanding was the single load-bearing control for the HELIOS example. Attack graphs recast hardening as an interdiction problem on a reachable-state graph. Bayesian networks fused weak alerts into a strong posterior, and Monte Carlo loss-exceedance curves expressed residual risk as a distribution whose tail, not only whose mean, sizes defenses and risk transfer. STPA-Sec added the control-theoretic view that finds losses caused by unsafe interactions among uncompromised components, generating requirements that constrain a control action by its effect rather than by its origin. Finally, the principal governance frameworks were placed on a single map by abstraction level and risk-management step, so that ISO/IEC 27005, the NIST Cybersecurity Framework, NIST IR 8270 and IR 8401, SPD-5, and SPARTA each occupy a distinct and complementary place. Together these models turn the opening vignette's questions, how likely a compromise is, what stands in the adversary's way, how long an intruder can dwell, and where the next unit of effort buys the most, into quantities a defender can compute and act upon.

Exercises

Exercise 4.1 [C] Classify each of the following threats by its entry layer and its principal effect layer, and state whether it is cross-layer: (a) a reactive jammer on the HELIOS user downlink; (b) a spoofed PNT timing signal that invalidates certificates in a ground network; (c) a malicious telecommand that fires a thruster; (d) a hardware trojan in an onboard processor procured through the supply chain. For each, name one attribute from the threat profile of [Definition 4.2](#) that most distinguishes it from the others.

Exercise 4.2 [C] Explain why probabilities estimated from historical incident frequencies can be misleading for a strategic adversary but appropriate for natural threats. Illustrate with the HELIOS TT&C monitoring game of [Example 4.5](#): what happens to the historical distribution of attacks if the defender reallocates monitoring to the most frequently attacked site?

Exercise 4.3 [C] The kill chain has no tactic corresponding to “assessment and reattack” in MITRE ATT&CK, SPARTA, or SPACE-SHIELD, and none of the three covers recovery. Explain why, in terms of what those catalogs are designed to describe, and argue whether the omissions are a weakness of the catalogs or a consequence of their scope.

Exercise 4.4 [C] State the eight adversary models of [Section 4.6](#) and, for each, give a space example different from the one in the text. Then explain, using [Theorem 4.2](#), why a controller certified against a bounded disturbance need not be secure against a sparse attack of the same energy.

Exercise 4.5 [A] Consider a five-stage kill chain with no reattack, in which the defender interdicts each stage independently with probability η and the final effect succeeds with probability σ . (a) Write the campaign success probability $P_E(\eta)$.

(b) Show that the elasticity $d \log P_E / d \log(1 - \eta)$ equals the number of stages, and interpret this. (c) Find the uniform η that holds P_E to 0.05 when $\sigma = 0.6$.

Exercise 4.6 [A] For the absorbing chain of [Section 4.5.1](#) with transient block Q and absorbing block R , prove that $B = NR$ is row-stochastic, that is, each row of B sums to one. Interpret the result in terms of the campaign outcomes, and state what it implies about $B_{iE} + B_{iX}$ for every starting stage i .

Exercise 4.7 [A] Derive the closed form (4.5) for the campaign success probability with reattack by a first-step analysis of the reattack loop. Begin by writing the probability of success conditional on reaching stage 3, then account for the geometric series of returns from stage 7 to stage 3, each of which must re-traverse stages 3 to 5.

Exercise 4.8 [A] In the attack-defense tree of [Example 4.6](#), treat the three access paths as an or with success probabilities p_{ground} , p_{supply} , and p_{rf} . (a) Show that if one of the paths has probability near one, controls on the other two barely change the root. (b) Using the leaf values of [Table 4.4](#), compute the root probability if SDLS is deployed but the ground and supply controls are omitted, and compare with the fully controlled value. (c) Explain the min-cut interpretation of the result.

Exercise 4.9 [A] A defender covers three targets with one monitoring resource. The losses if a target is attacked are $\ell_1 = \ell_2 = 2$ and $\ell_3 = 3$ when uncovered, and 0.5, 0.5, and 0.9 when covered. (a) Set up the linear program (4.14) for the zero-sum game. (b) Verify that the minimax coverage equalizes the expected loss across targets, and find the value. (c) Explain why no pure strategy is optimal here.

Exercise 4.10 [A] Using the Bayesian network of [Fig. 4.15](#), compute $\mathbb{P}(K | A, \neg T)$ and $\mathbb{P}(K | \neg A, T)$ by hand, and reconcile the two with the intuition that the alert and the telemetry anomaly are conditionally independent given K . Which single additional observation would most raise the operator's confidence in a compromise, and why?

Exercise 4.11 [A] An STPA-Sec analysis of a HELIOS maneuver identifies the unsafe control action “execute maneuver when it violates a keep-out constraint.” List the four ways (provided when unsafe, not provided when needed, wrong timing, stopped too soon) in which the “authenticate command” action for a firmware update could be unsafe or insecure, and propose one security requirement for each that constrains the action by its effect.

Exercise 4.12 [P] Reproduce the loss-exceedance analysis of [Example 4.7](#). Using the scenario parameters of [Table 4.5](#), draw Poisson event counts and lognormal severities, sum per simulated year over at least 2×10^5 years, and plot the baseline and hardened loss-exceedance curves on logarithmic axes. Report the annualized loss expectancy, the probability of a year exceeding \$10M, and the 99th-percentile loss for each configuration, and verify the annualized loss expectancy against the closed form $\sum_s \Lambda_s \mathbb{E}[X_s]$ with $\mathbb{E}[X_s] = m_s e^{c_s^2/2}$.

Exercise 4.13 [P] **Chapter lab: kill-chain simulation and hardening.** Implement the absorbing Markov model of [Section 4.5](#) with the parameters of [Table 4.3](#).

(a) Build Q and R , compute $N = (I - Q)^{-1}$, $B = NR$, and $t = N\mathbf{1}$, and reproduce B_{1E} , the expected dwell time, and the dwell-time distribution of [Example 4.4](#). (b) Compute the detection elasticity (4.7) for each stage by numerical differentiation, and reproduce the ranking of [Example 4.3](#). (c) You have a budget that can raise the detection probability of one stage by 0.05; determine which stage minimizes B_{1E} , and compare with the stage chosen by an additive +0.01 criterion. (d) Add a second absorbing model in continuous time using (4.6) and verify that the absorption probabilities match the discrete-time model.

Exercise 4.14 [P] Implement the attack-defense tree of [Example 4.6](#) as a function of a set of enabled controls, using the leaf values of [Table 4.4](#) and the propagation rules (4.16). (a) Reproduce the root probabilities for no controls, SDLS only, and all controls. (b) Compute the marginal risk reduction of each control added to the SDLS-only baseline, and rank them. (c) Relax the independence assumption by making g_1 and g_2 positively correlated (for example, a shared identity provider), and describe qualitatively how the root probability and the control ranking change.

Exercise 4.15 [P] Solve the HELIOS TT&C monitoring game of [Example 4.5](#) numerically. (a) Set up and solve the zero-sum linear program for the minimax coverage and value, and confirm $x^* \approx (0.193, 0.193, 0.614)$. (b) Compute the strong Stackelberg equilibrium for the general-sum case by solving one linear program per target, and confirm the coverage and defender loss reported in the text. (c) Perform a sensitivity analysis: vary the attacker's attribution penalty at the owned sites from 0 to 0.5 and plot the equilibrium coverage of the GSaaS path.

Notes and Further Reading

Threats and counterspace capabilities. The open-source picture of counterspace capabilities is maintained in two annual assessments: the Secure World Foundation's *Global Counterspace Capabilities* [8] and the Center for Strategic and International Studies' *Space Threat Assessment* [20]. The U.S. Defense Intelligence Agency's *Challenges to Security in Space* gives a governmental view [115]. For satellite cybersecurity specifically, the systematization of Pavur and Martinovic [33, 50] and the experimental firmware analysis of Willbold and colleagues [84] are good entry points, and Falco's early survey framed the field [32].

The kill chain and technique catalogs. The intrusion kill chain originates with Hutchins, Cloppert, and Amin [30]. The related technique catalogs are MITRE ATT&CK for enterprise networks [119] and its space-specific counterparts, The Aerospace Corporation's SPARTA [31] and the European Space Agency's SPACE-SHIELD [120]. Tactic lists are revised between releases; the versions used here are recorded in the citations, and a current analysis should consult the live matrices.

Absorbing Markov chains. The fundamental-matrix results of [Theorem 4.1](#) are classical and are developed in full by Kemeny and Snell [122]. The continuous-time and phase-type extensions are standard in applied probability. The application to intrusion campaigns, with per-stage detection and eviction, follows the spirit of the mission-centric risk models proposed for space systems in the NASA/JPL literature surveyed in the notes to [Chapter 3](#).

Adversary models. The sparse-attack secure-estimation framework is due to Fawzi, Tabuada, and Diggavi [123]; the resource-limited adversary taxonomy to Teixeira and colleagues [124]; the denial-of-service switching model to De Persis and Tesi [125]; and the Byzantine bound $n \geq 3f + 1$ to Lamport, Shostak, and Pease [126]. These are developed for control systems in Chapter 15, and their space applications recur throughout Parts III and IV.

Games and strategic adversaries. The zero-sum and Stackelberg security games of Section 4.7 are treated at length by Tambe [128], with the exact Bayesian Stackelberg algorithm of Paruchuri and colleagues [127]; the value of commitment to mixed strategies is analyzed by von Stengel and Zamir [129]. Manshaei and colleagues survey game-theoretic models across network security and privacy [146], and Rass, Schauer, König, and Zhu develop game-theoretic risk management for critical infrastructures [147]. Multi-stage games for advanced persistent threats, which unify the stochastic kill chain with strategic play, are developed by Zhu and Rass [130] and by Huang and Zhu [131]. Chapter B gives the full game-theoretic background, and jamming games appear first in Chapter 5.

Attack trees, graphs, and STPA-Sec. Attack trees are due to Schneier [132] with formal semantics from Mauw and Oostdijk [133]; attack-defense trees are formalized by Kordy and colleagues [134]. Automated attack-graph generation goes back to Sheyner and colleagues [136], with the logic-based analyzer MulVAL from Ou, Govindavajhala, and Appel [135]. STPA-Sec extends Leveson's systems-theoretic accident model [140] to security through the work of Young and Leveson [139].

Risk quantification and frameworks. The FAIR approach to loss-exceedance analysis is given by Freund and Jones [137] and by Hubbard and Seiersen [138]. The governance frameworks are ISO/IEC 27005 [141], the NIST Cybersecurity Framework 2.0 [121], NIST IR 8270 [35] and IR 8401 [36], and Space Policy Directive 5 [34]; the CCSDS Space Data Link Security protocol referenced throughout is specified in [70].

Case studies. The KA-SAT/AcidRain incident is documented by Viasat [12], SentinelLabs [23], and the third-party effects in [24]; the 2014 NOAA intrusion is reported in [142] and audited in [143]; the Moonlighter test satellite and 2023 Hack-A-Sat competition are described in [144, 145]. Public attributions of the first two remain partly contested and are reported here with hedging.

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